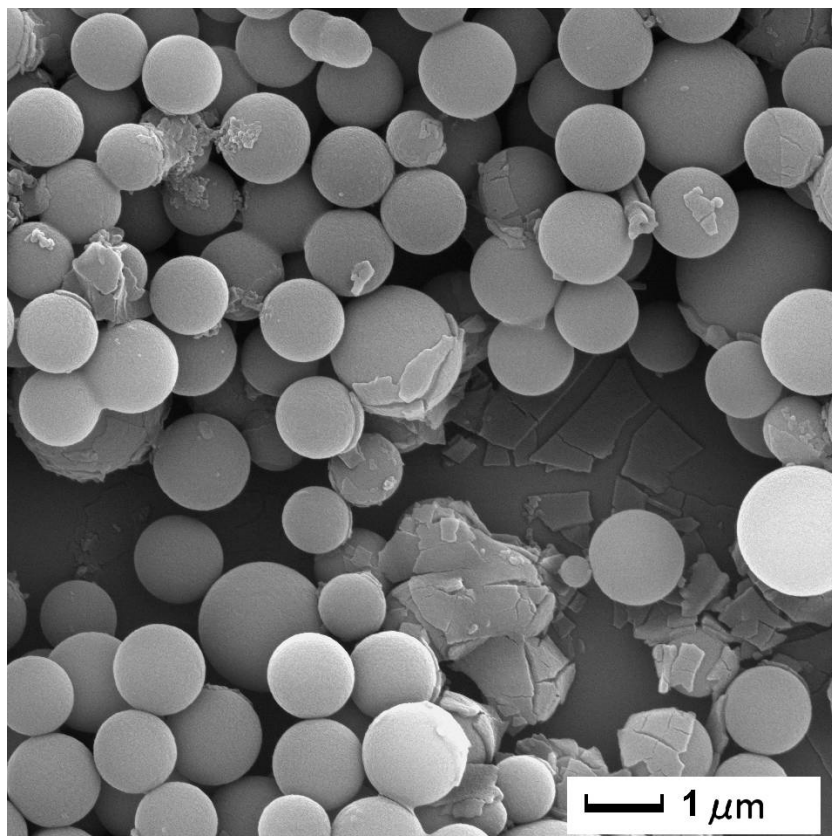


Giulia Ischia

# **Sustainable conversion of biomass wastes via hydrothermal processes: fundamentals and technology**







Doctoral School in Civil, Environmental and Mechanical Engineering  
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Giulia Ischia

# **Sustainable conversion of biomass wastes via hydrothermal processes: fundamentals and technology**

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*Nobody ever figures out what life is all about,  
and it doesn't matter. Explore the world.  
Nearly everything is really interesting if you  
go into it deeply enough.*

Richard Feynman

*Look, I don't want to wax philosophic, but I  
will say that if you're alive you've got to flap  
your arms and legs, you've got to jump  
around a lot, for life is the very opposite of  
death, and therefore you must at very least  
think noisy and colorfully, or you're not alive.*

Mel Brooks



# Abstract

In a worldwide context where the community has to make giant leaps forward to contain the catastrophic consequences of climate change, we need to face the discordant “How do we power our economies?” with green and circular solutions instead of hiding behind the hypocrisy of fossil fuels. Biomass is a renewable, abundant, and cheap source: it can trigger a shift towards a zero-carbon emission economy, in which it substitutes fossil fuels for the production of energy and materials. Among the strategies to valorize biomass, hydrothermal processes are green pathways for producing biofuels and bio-based materials. However, research has yet to fill several gaps to make these processes ready for industrial scaling and spreading.

Therefore, this thesis provides new insights around hydrothermal processes, touching several scientific areas: from in-depth research around the thermochemical fundamentals, unavoidable pillars tools for a broad understanding, to the engineering of new sustainable and biorefinery concepts.

*Fundamental research* tries to answer “What’s happening during hydrothermal processes?”. Indeed, the complexity behind the conversion is enormous: understanding the fundamentals behind the process is a mandatory step to untangle the complexity and optimize the processes. Therefore, we deduced chemical pathways and kinetics behind the hydrothermal decomposition of several biomasses through proper experimental activities and analysis techniques (like liquid chromatography, solid spectroscopy, microscopy, and calorimetry) over a broad hydrothermal spectrum, including carbonization and liquefaction. Particular attention was given to the mechanisms behind the formation of the fraction deriving from the self-condensation of the liquid products, the so-called secondary char: this solid phase consists of spherical particles of nano/micrometric size with specific properties that make the hydrochar appealing as a base for advanced carbon materials. Therefore, we investigated the fascinating optical properties and morphology of secondary char derived from glucose. Also, we studied the reaction pathway and developed a kinetic model for a holistic approach to the problem. Then, continuing the investigation of

reaction pathways, we investigated the mechanisms from mild to harsh conditions, and therefore from the predominant presence of solid (the hydrochar) to oily phases (the bio-crude). Always inside the fundamentals, we faced the thermodynamics of the phenomenon by measuring, through different techniques using a high-pressure differential scanning calorimeter, the enthalpy of reactions involved during hydrothermal carbonization.

Regarding the *engineering of the process*, we explored the optimization of hydrothermal conversion through coupling with solar energy. Indeed, despite their potential as green routes for the conversion of biomass, we have to remember that hydrothermal conversion is energy-intensive. We focused on two possible configurations: direct and indirect coupling. In particular, we demonstrated the effectiveness of a prototype involving the direct coupling between a solar concentrator and a hydrothermal carbonization reactor: solar energy fully covers the energy needs, making the process “zero-energy”. Then, to move towards bigger scale concepts, we performed a techno-economic analysis for future scenarios involving the indirect coupling among the technologies and we designed a possible lab-scale apparatus.

Finally, expanding the observation scale, we investigated the integration of hydrothermal carbonization with *subsequent treatments*. The process can be a pivot to concertize a biorefinery concept merging the management of biomass waste and the valorization of by-products. In particular, we focused on subsequent thermal and chemical treatments to make the hydrochar effective as a solid biofuel (indeed, by itself, the hydrochar suffers from fuel segregation, exhibiting a too high reactivity at low temperatures) and the recovery of chemicals.

# List of Publications

## Peer-reviewed

- **G. Ischia**, M. Cazzanelli, L. Fiori, M. Orlandi, A. Miotello, *Exothermicity of hydrothermal carbonization : Determination of heat profile and enthalpy of reaction via high-pressure differential scanning calorimetry*, Fuel. 310 (2022) 122312.  
<https://doi.org/10.1016/j.fuel.2021.122312>.
- G. Gonnella, **G. Ischia**, L. Fambri, L. Fiori. *Thermal Analysis and Kinetic Modeling of Pyrolysis and Oxidation of Hydrochars*, Energies. 15 (2022) 950. <https://doi.org/10.3390/en15030950>.
- **G. Ischia**, L. Fiori, L. Gao, J.L. Goldfarb, *Valorizing municipal solid waste via integrating hydrothermal carbonization and downstream extraction for biofuel production*, Journal of Cleaner Production. 289 (2021) 110067.  
<https://doi.org/10.1016/j.jclepro.2021.125781>.
- **G. Ischia**, L. Fiori, *Hydrothermal Carbonization of Organic Waste and Biomass: A Review on Process, Reactor, and Plant Modeling*, Waste and Biomass Valorization 12 (2021) 2797–2824.  
<https://doi.org/10.1007/s12649-020-01255-3>.
- **G. Ischia**, D. Castello, M. Orlandi, A. Miotello, L.A. Rosendahl, L. Fiori, *Waste to Biofuels through Zero-Energy Hydrothermal Solar Plants: Process Design*, Chemical Engineering Transaction. 80 (2020) 2020. <https://doi.org/10.3303/CET2080002>.
- **G. Ischia**, M. Orlandi, M.A. Fendrich, M. Bettonte, F. Merzari, A. Miotello, L. Fiori, *Realization of a solar hydrothermal carbonization reactor: A zero-energy technology for waste biomass valorization*, Journal of Environmental Management. 259 (2020) 110067 <https://doi.org/10.1016/j.jenvman.2020.110067>.

# List of selected awards

## International

- ***Best Young Researcher Award*** at the 8<sup>th</sup> International Conference on Engineering for Waste and Biomass Valorisation, released by Springer Nature. 2021.
- ***Fulbright scholarship*** (Visiting Student Researcher) to spend 6 months at Cornell University (USA) in 2021. Founded by Fulbright Commission together with Cesaretti Foundation.
- ***Joseph Breen Memorial Fellowship*** from American Chemical Society Green Chemistry Institute (ACS-GCI). 2020.
- ***Fellowship*** from EIT-Climate for the participation at the 4-weeks summer school The Journey, August in Tallinn (Estonia), Lund (Sweden), and Budapest (Hungary). 2019.

## National

- ***Award honor*** from the town of Arco (Italy) for the excellence in the academic career. December 2021.

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# Chapter 1

## Introduction

### 1.1 The climate crisis

The severity of the impacts of climate change on the planet and human health is increasing at bewildering levels, making climate change one of the most significant challenges of the 21<sup>st</sup> century. The worldwide community needs to take urgent actions to avoid catastrophic consequences and finally face the discordant “how do we power our economies?” with real green solutions instead of hiding behind the hypocrisy of fossil fuels. As scientists, we have the responsibility and duty to conceive and engineer new green processes to produce energy and materials from non-fossil sources.

Human activities are estimated to have caused approximately 1.0°C (with a likely range of 0.8 to 1.2°C) of global warming above pre-industrial levels [1]. As detailed explained in the IPCC report 2021 [2], the impacts of global warming are catastrophic. Increase in mean temperature in most land and ocean regions, increase of the hot extremes in most inhabited regions, heavy precipitations and

probability of drought and precipitation deficits in several regions, an increase of the sea level of 0.26 to 0.77 m by 2100 for 1.5°C of global warming, irreversible impacts on biodiversity and ecosystems, risks to health, livelihoods, food security, water supply, human security, and economic growth [2].

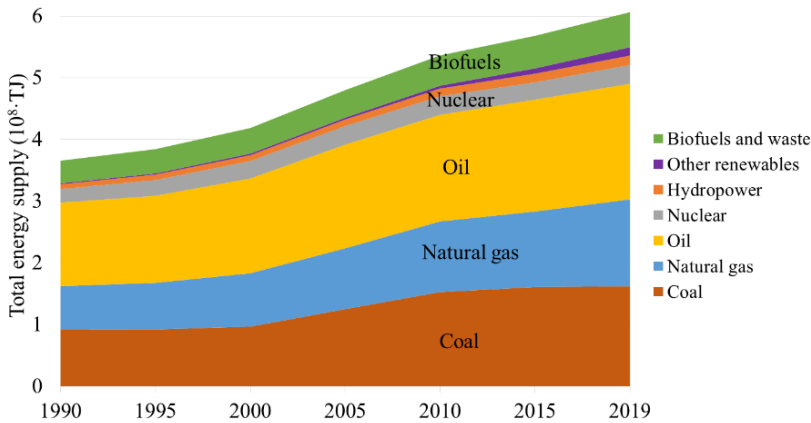
Unless CO<sub>2</sub> emissions will be kept below zero, the warming-induced is expected to persist for decades or centuries causing further long-term changes in the climate system [3]. To avoid/contain climate catastrophe, 196 countries in the 2015 Paris agreement pledged to limit global warming to 1.5°C above pre-industrial levels and well below 2°C, introducing the new focus of net-zero emissions [4].

In this framework, the energy sector, dominated by fossil fuels, is responsible for almost three-quarters of the emissions [5]. Also, with a constantly growing population (estimated to grow by 2 billion people by 2050), the IEA report 2021 [6] affirmed that *today energy sector is not capable to meet challenges*. Occupying around 81% of the energy demand, fossil fuels continue to dominate the global primary energy demand, as shown in Figure 1.1. For example, coal accounts for around 36.7% of global electricity generation and is crucial in industries such as iron and steel [6]. To contain global warming, the growing trends of fossil fuels need to decline rapidly: the IPCC report projects that to limit to a 1.5°C warming, in 2050, coal has to represent around 1-7% of the primary energy use, while oil and methane should decline of 39-77% and 13-62%, respectively. This goal implies that a large number of fossil fuels will not be extracted, with implications on producers, investors, and banking [7]. Despite the limits, the future of fossil fuel is uncertain, affected by the inertia of the system and the need to introduce clean technologies with a parallel retirement of fossil fuel infrastructure [7]. Ledurer et al. [8] already demonstrated how fossil fuel will probably exceed the 1.5°C carbon budget.

In few words, the global situation is disastrous, the costs for inaction are immense, and the low-emission revolution is long overdue: we need to take immediate actions. The World Energy Outlook 2021 [5] identifies four key strategies trying to avoid catastrophes: (1) push for clean electrification, (2) focus on energy efficiency, (3) cut methane emission from fossil fuel operations, (4) boost to clean energy innovation and spread of these new technologies.

In this context, biomass can play a crucial role in the transition from a fossil-based to a biomass-based economy, where *sustainably*

*produced, renewable natural sources in the form of biomass will replace fossil fuel materials.*



**Figure 1.1.** World total energy supply, by source [6].

## 1.2 Bioenergy in the climate crisis

In 2020, global primary energy demand for bioenergy was almost 65 EJ, 90% of which is solid biomass [9]. The efficient and sustainable usage of bioenergy (referred to as “modern bioenergy”, far from the “traditional use” of biomass that is often harmful) falls in the UN Sustainable Development Goal 7 (*Affordable and clean energy*) and the net-zero energy scenarios play an essential role in sectors where other carbonization options are not available. The GBEP (an intergovernmental organization) produced a set of indicators to assess the sustainability of biomass, accompanied by an ISO standard (13065:2015 Sustainability criteria for bioenergy) [10]. Example of parameters are life-cycle GHG emissions, soil quality, land use, and harvest levels.

Modern technologies can convert biomass to solid, liquid, and gaseous forms that can find an end-use as (1) *biofuels for transports*, (2) *power generation* (for heat, electricity, and co-generation), (3) *biomass-based materials* [10].

Using *biofuels* is a primary strategy to decarbonize the transport sector, complementary to electrification and hydrogen. In the IEA roadmap [9], the consumption of biofuel should supply 12% of the

global transport demand in 2030, with 45% of biofuels deriving from biomasses not competing with food crops (like wastes, residues, or dedicated crops), versus the 7% of 2020. The European Union [11], the United States [12], and Brazil [13] have codified criteria for biofuel sustainability by introducing GHG thresholds. Rather than road transport, where electrification is predicted to spread thanks to the falling price of batteries and hydrogen, biofuels are thought to have a crucial role in the decarbonization of the aviation industry. Meanwhile, in shipping, biofuels will have to be competitive with green ammonia to diffuse.

Regarding *power generation*, on a cost-basis, biomass is less competitive than wind and solar technologies [14]. However, it still plays an important role. Indeed, it can be a complement source: being dispatchable, it can provide energy when needed (for example during peak hours). Biofuels have the advantage of being used in existing infrastructure: biomethane can be used in existing pipelines, drop-in liquid biofuels in the existing oil distribution network, and solid biofuels in co-firing applications. Also, solid biofuels are still an effective transitional solution for coal plants, whose disposal is controversial. Indeed, from one side, countries are moving towards a phase-off of coal plants. Germany agreed on shutting down all coal plants by 2038 [15], and 46 nations including big power plants users (like Indonesia, Vietnam, Poland, Ukraine, and South Korea) have signed the *Global Coal to Clean Power Transition Statement* at the COP26 of 2021 for a phase-out of the coal plant by 2050 [16]. On the other side, notably big coal consumers like China, India, the United States, and Australia, did not sign the accord. Therefore, solid biofuels will still be an answer to decarbonization, especially for their usage in co-firing applications. Finally, power generation from biomass, like forest or agricultural residues or municipal solid waste, can help in reducing waste.

Then, the area of *bio-based materials* is estimated to have high value for the future of biomass use [14]. Bio-based materials are the products deriving from biomass or processes that use biomass [17]. Indeed, they represent the highest value that biomass can have in a net-zero context (instead of mere use as energy) and are an alternative resource in petroleum-derived products, including plastics and chemicals. Using biomass could disengage from the usage of fossils, from which we are fully dependent. For example, plastics are responsible for more than 1

billion tons of CO<sub>2</sub> equivalent per year, equivalent to the global fossil fuels used for aviation or shipping [18].

### **1.3 Biomass, biomass waste, and circular economy**

The supply of biomass derives mainly from three sources: forestry, agriculture, and waste. The current yearly biomass supply in Europe is 10.2 EJ, 70% of which is woody biomass, mainly consisting of harvested wood from forests, industrial by-products, and residues [14]. Then, the remaining 30% is equally divided between agricultural crops, like energy crops and agricultural residues (containing both food and non-food crops) and waste (including paper waste, wood waste, food waste, sludge) [14]. Similarly, in the U.S., around 50% of the biomass supply is based on wood and wood waste, and biomass covers around 5% of the total primary energy use [19].

In the future, expecting the spread of circular economy systems and an improvement of waste valorization technologies, the waste stream is expected to cover higher percentages in the supply [14]. *Residues and wastes* like the organic fraction of municipal solid waste, wet wastes from agriculture, sewage sludge, organic liquid effluents, animal wastes. The usage of organic fraction of wastes and sewage sludge as source stands in a “waste-to-energy” pathway that can combine the management of waste with circular economy principles. Indeed, using biomass waste for the production of bioenergy and bioproducts has gained rising attention in the last few decades [20], merging waste management strategies and the principles of circular economy: biorefineries offer the potential to upgrade biowaste to products with a higher value. In March 2020, the European Commission adopted the new circular economy action plan, announcing initiatives to prevent waste and keep resources as long as possible in the economy [21]. The worldwide community is facing the issue of waste generation from anthropogenic activities, strained by the constant growth of the global population, expected to count from the current 7 billion to over 9 billion by 2050 [22]. There is an urgent need to develop sustainable waste management strategies, minimizing landfilling and uncontrolled disposal. According to the World Bank, about 2.01 billion tons/year of municipal solid waste are produced worldwide, expected to rise due to the globalization of non-developed countries [23]. The organic fraction

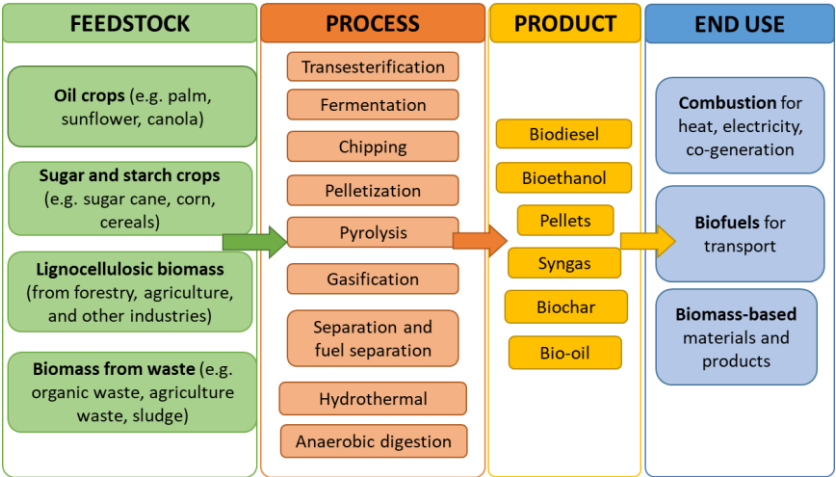
is estimated to occupy between 44 and 46% of the global production of municipal solid waste [23]. The exact definition of biowaste varies among the countries. For example, the Waste Framework Directive (WFD) defines *bio-waste as biodegradable garden and park waste; food and kitchen waste from households, restaurants, caterers and retail premises; and comparable waste from food-processing plants* [24]. In this context, converting biomass waste to bio-products and bioenergy represents a sustainable strategy to manage wastes. The common treatment methods for biowaste are composting and anaerobic digestion. In parallel to biological processes, thermochemical conversion processes are a viable alternative to treat biowaste, requiring shorter time processes and avoiding the management of the biological community [25].

### 1.4 Bioenergy technologies

Regarding bioenergy technologies, some are mature while others that can be crucial in sustainability are not yet developed or commercialized. Accelerating their development is highly needed in a zero-net emissions vision. Generally, processing biomass is needed to optimize the product and the efficiency of the pathway. Figure 1.2 shows a general scheme of the pathway biomass to bio-products, passing through valorization stages. In particular, among the many ways to categorize technologies, the IEA technology roadmap [10] distinguishes three categories according to the level in the pathway.

1. *Preparation*, necessary to make the material homogenous, easier to handle, and with a higher energy density. In general, these technologies are widely developed. Examples include drying, size reduction, pelletization, torrefaction.
2. *Pretreatment*, to change the physical and chemical structure of the biomass with the aim of obtaining a product more suitable for the conversion or as an end product. Pre-treatments like anaerobic digestion or biogas upgrading are widely developed, pyrolysis and gasification are still at an early market stage, while hydrothermal processes are at a prototype stage.
3. *Conversion*, to produce heat, electricity, transport fuels, and other products. Large-scale combustion systems and cogeneration are widely diffused, while co-firing is still at a demonstration level. Regarding transport biofuels, the production of ethanol and

biodiesel from sugar and oil crops are commercially developed technologies. The upgrading of pyrolysis oil for co-processing with crude oil is at an early market, while hydrothermal liquefaction is at a prototype level.



**Figure 1.2.** Pathway for biomass to the final use, scheme adapted from How2Guide by IEA and FAO [26].

## 1.5 Hydrothermal processes

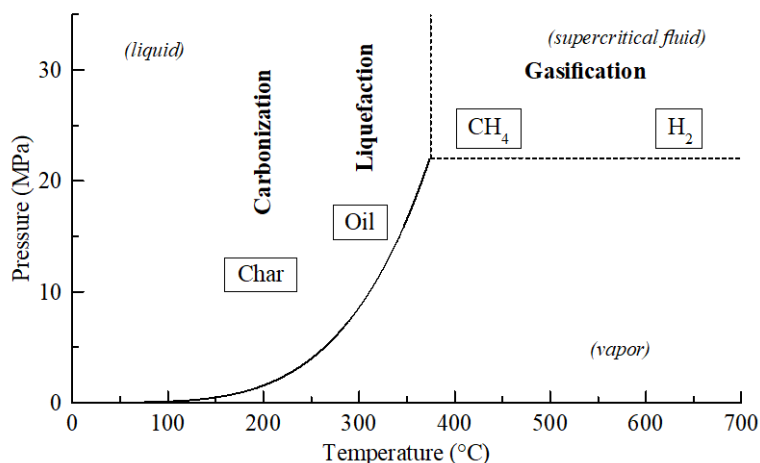
### 1.5.1 Overview

Even if known for around 100 years, the research around hydrothermal processes has acquired a rising interest in the last 20 years [27]. The diffusion of hydrothermal processes was initially pushed by the oil crisis for the production of biofuels (in solid, liquid, or gas forms), while today they are conceived in a more general concept that involves the usage of biomass as a carbon source for a variety of purposes, from biofuels to the production of bio-based materials and chemicals. Also, they find applications as management and valorization strategies for wet biomass wastes, becoming an alternative to landfilling and traditional waste strategies [28].

As the name suggests, hydrothermal processes occur in high-temperature and high-pressure water, which is at a subcritical or supercritical state. Under high temperature and pressure, water drastically changes its properties to ambient conditions, acting as a



solvent for biomass and catalyzing acid or base reactions. Therefore, biomass is converted into a char, oil, or gas phase. Hydrothermal processes can be distinguished in three regions depending on the operating conditions, visualized on the phase diagram of water (Figure 1.3). Hydrothermal carbonization (HTC) and liquefaction (HTL) occupy the subcritical region. HTC occurs at mild conditions (180–250°C) and favors the production of the solid phase (the so-called hydrochar), while HTL occurs at harsher conditions (250–373°C) and leads to the formation of the bio-oil. Then, supercritical water gasification (SCWG) occupies the supercritical region (above 373°C), in which biomass or organics are gasified as carbon dioxide and hydrogen or methane.



**Figure 1.3.** Hydrothermal regions on the phase diagram of water.

The interest in hydrothermal processes raised at significant levels only in the early years of 2000, driven by the necessity of substituting fossil fuels and developing new waste management approaches. Figure 1.4 clearly shows this trend, with the predominance of HTC in the current literature. Some generalities on hydrothermal processes are reported here below.

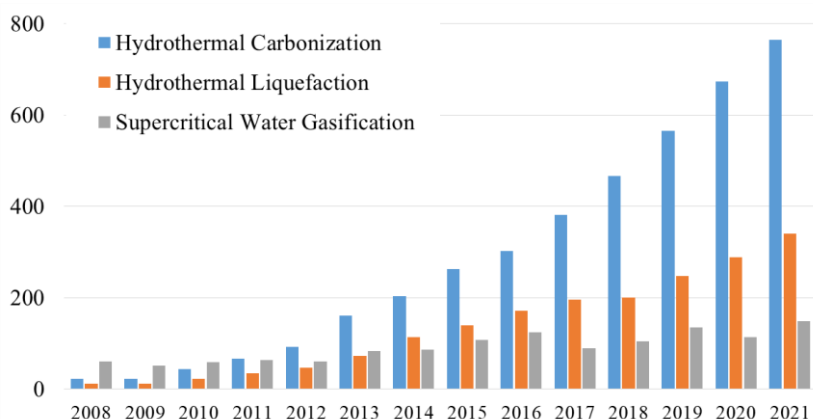
*Hydrothermal carbonization (HTC)* has been the object of great interest from researchers in the last years for fuel and non-fuel applications (like the production of advanced biomass-based materials) as well as waste management strategies. The mild operating conditions

of HTC limit liquefaction and gasification reactions, converting the biomass into a char phase (the hydrochar). Along with the hydrochar, HTC produces an aqueous phase rich in organics and some gas. The hydrochar finds a variety of applications like solid biofuel and base for the production of advanced carbon materials (like supercapacitors). More details around HTC will be given in the next section and through the entire thesis.

*Hydrothermal liquefaction* (HTL) aims to convert biomass into tarry oil (bio-crude). Indeed, in these conditions, biomass is broken into a water-insoluble fraction (the bio-oil) with an oxygen content lower than the starting feedstock and with carbon-carbon bonds. The oily phase is accompanied by water-soluble compounds and a certain amount of solid and gaseous phases. The interest around HTL has seen a rapid increase in the last years for biofuel applications and the production of a platform of chemicals (like hydroxymethylfurfural, HMF, as polymer precursor, or levulinic acid [27]). Compared to other thermochemical processes (like fast pyrolysis), it has the advantage of producing an oil with a lower oxygen content (the oxygen is dissolved in the aqueous phase) [29]. Feedstock with a high lipid content like algae is particularly suitable for HTL [28]. In general, the oil obtained by HTL needs to be upgraded to be used as fuel (the viscosity is too high and the hydrogen content too low), and techniques like hydrogenation have been investigated. In this field, the pilot plant realized in Denmark (Aalborg) allowed reaching significant progress in this sector [30].

*Supercritical water gasification* (SCWG) exploits the change of properties of water in the supercritical region, like the drastic decrease in dielectric constant, density, and ion dissociation constant [31]. Under supercritical conditions, water dissolves the biomass into a gas phase. Indeed, compounds soluble in the subcritical region (like organic acids and phenols) are here converted into the gas phase. A certain amount of solid phase is also produced when operating in the low-temperature region. In the near-critical region (around 400°C), the gas phase is mainly composed of CH<sub>4</sub> and CO<sub>2</sub>, while at higher temperatures (600-700°C), the main products are H<sub>2</sub> and CO<sub>2</sub> [32]. The process can occur in the presence of a catalyst or not [33]. SCWG occupies a small area of research, with around 100 papers published per year. Especially due to poor competition with hydrogen derived from fossil fuels and methane, industrial applications are still far from being achieved [27]. There are two main large pilot plants, one in Germany (the VERENA

plant [33]) and one in Japan [34], with a capacity of 2 and 1 ton/day, respectively.



**Figure 1.4.** Number of publications in Scopus of hydrothermal carbonization, hydrothermal liquefaction, and supercritical water gasification.

### 1.5.2 The fascinating properties of water

Water represents the unquestioned star of hydrothermal processes. Thanks to its unique properties under elevated temperatures and pressure, it can act as a non-polar solvent, enabling it to degrade biomass and synthesize materials through a green route.

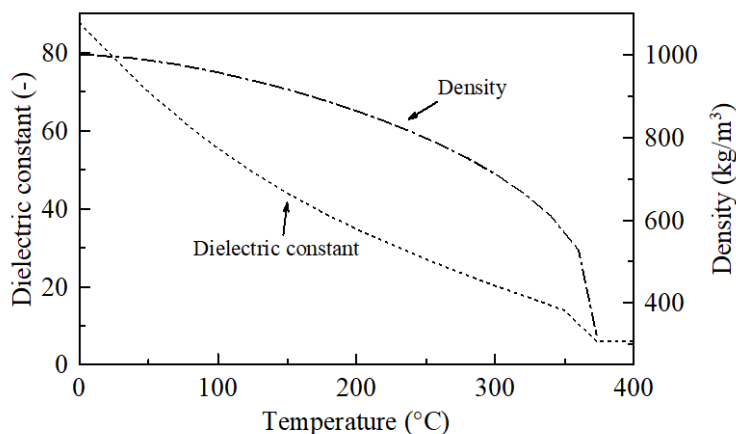
Having studied the properties of water under sub- and supercritical conditions in the second half of the 20<sup>th</sup> century (it is worth remembering the extensive research carried out by Franck [35–38]), opened the field to investigate its use as a reaction medium. Later, the reviews by Akiya and Savage [39] or Kruse and Djnus [40,41] represent some encompassing compendia summarizing the chemistry of water under hydrothermal conditions.

Even in the subcritical region, water at high temperatures shows very different properties from those at ambient conditions (as shown in Figure 1.5). It macroscopically behaves as a non-polar solvent, for example enabling the formation of C-C bonds, but keeping at the same time its polar structure, making it able to interact with ions and to attack chemical bonds [40].

Under hydrothermal conditions, the network of hydrogen bonds of water becomes weaker, increasing the mobility of the single molecules

and their capacity to structure around solutes [40]. These aspects favor the participation of water in reaction pathways not occurring at ambient conditions. Indeed, water can participate in the activation complex of some reactions (like hydrolysis, aldol condensation, and decarboxylation), catalyzing them [42].

For example, by increasing temperature (below the critical point), water increases its ionic product, which results in some order of magnitude higher than at ambient conditions [43]. This aspect occurs because the self-dissociation of water is endothermic. Therefore, the increment in concentration of  $\text{H}_3\text{O}^+$  and  $\text{OH}^-$  ions makes water behave as a catalyst-precursor for acid or base reactions. The relative dielectric constant drastically decreases with temperature, triggering ionic reactions. For example, it is 78.5 at ambient conditions, while at  $300^\circ\text{C}$  is equal to 20 [43], coming closer to those of organic solvents like methanol of 32.6 or methylene chloride of 9.1 [44]. Thanks to the lower dielectric constant and density than ambient conditions, small organic compounds become progressively more soluble in hot compressed water (while they are completely miscible in supercritical water), otherwise poorly miscible at ambient conditions.



**Figure 1.5.** Variations of the relative dielectric constant and density with temperature (data from Ref. [43]).

### 1.5.3 Hydrothermal carbonization

Despite the early studies from Bergius in 1913 [45], hydrothermal carbonization (HTC) was neglected for almost one century, coming

back “in style” in the early 2000s. HTC mimics in a few hours the natural coalification of biomass (occurring on timescales from hundreds to million years) and leads to the formation of a coal-like material (the hydrochar) together with dissolved organics in the liquid phase. The renewed interest founds on exploring HTC as a synthesis pathway for the production of carbon-based materials, the recovery of chemicals and energy starting from a renewable source like biomass. HTC has the advantage of operating at relatively mild conditions (180-250°C, 10-50 bar), making the process technologically easier to handle than liquefaction and gasification. This aspect, combined with the advantages provided by hydrothermal processes themselves, and among all, the possibility of treating wet and heterogeneous biomasses, made HTC particularly appealing for the treatment of biomass wastes.

During HTC biomass undergoes a complex series of reactions, including hydrolysis, dehydration, decarboxylation, aromatization, and polymerization. Hydrochar forms through both a solid-solid conversion (the “primary char”) directly from the parent material and a liquid-solid conversion (the “secondary char”) from the condensation and polymerization of compounds dissolved in the liquid phase. In parallel to the hydrochar, biomass degrades into the liquid phase into a wide variety of chemicals (like HMF, carboxylic acids, and phenol derivatives). The resulting hydrochar is characterized by a reduced content of oxygen and hydrogen than the starting feedstock, with properties resembling those of lignin-like coal [46] and, thanks to its hydrophobicity and homogeneity, easily separable from the liquid phase [47]. The hydrochar found a wide range of applications, like solid biofuels (its higher heating value ranges around 28-30 MJ/kg versus the original feedstock of 16-18 MJ/kg [48]), soil amendment, a base for activated carbon for waste-water treatment and advanced carbon materials.

Overall, many aspects characterize HTC and the approaches to differentiate it are disparate, like the effect of the process conditions on the hydrochar (mainly temperature, residence time, and solid load), the hydrochar applications, investigation of different feedstock, integration of HTC with other processes, recovery of chemicals, and techniques to upgrade the hydrochar.

Among the ways HTC can be subdivided, we can identify three main areas of application that advanced the knowledge of this process, which

are (1) production of carbon materials, (2) treatment of wastes, (3) recovery of chemicals. Here below, the details.

The *production of carbon materials* has seen its initial development thanks to the works carried out by Titirici et al. [49–52], who introduced the concept of “sustainable carbons” remarking the importance of a “back to black” in which carbon materials are produced through green routes like HTC. Therefore, HTC is thought of as a green synthesis route, where simple and abundant substrates, like simple carbohydrates, can be a base for the production of advanced carbon materials. Indeed, the resulting hydrochar has interesting properties that make it an interesting precursor. For example, the fraction of hydrochar formed from the liquid phase consists of carbonaceous spheres [53,54], whose porosity can be tuned using natural templates or activation processes. Also, they can easily be post-functionalized thanks to the presence of oxygenated groups on their surface and modified by combining them with other components (for example with inorganic nanoparticles to form composites) [49]. Also, HTC can be combined with post-processes like templating or chemical activation, enabling the possibility of tuning properties like porosity. Possible applications of these carbon materials include water treatment and energy-storage applications, electrocatalysis. Moreover, the recent discovery of carbon dots (i.e. a carbon material with diameters lower than 10 nm), opened the potential to many applications like in photocatalysis and photovoltaic.

During HTC, biomass dissolves into the liquid phase into a *range of chemicals*, which own their value and can represent a valuable by-product in a biorefinery concept. During HTC, biomass dissolves in a large number of chemicals inside the liquid phase, which can be divided mainly into carboxylic acids, ketones, furan derivatives, and phenols. The ratio and concentrations depend on the type of feedstock involved. Among these, 5-hydroxymethylfurfural (5-HMF) received great attention for its potential as a bio-based precursor for the production of plastics, biofuels, and chemicals [52]. Then, research-oriented on optimizing the production of other chemicals typical of HTC, like levulinic acid, formic acid, furfural, and lactic acid. In particular, lactic acid can be a promising chemical for the synthesis of biodegradable plastics and the synthesis of chemicals [27].

The other wide field of HTC is that of *biomass waste treatment*. As mentioned previously, finding alternatives to treat biomass waste has

seen a huge development in the last years triggered by waste management strategies. In parallel to circular economy values, treating waste has the advantage of making the overall process more economically advantageous, since it benefits from the income of treating the waste. This enabled the recent spreading of companies and startups (like TerraNova® [55], SunCoal [56], Ingelia [57], Antaco [58], and CarboRem [59]), which apply HTC to mostly treat wastes. In this field, HTC is particularly suitable to treat substrates characterized by a high content in water (70-90%), which are difficult to treat and require expensive pre-drying stages. Examples of common substrates are food wastes, sewage sludge, anaerobic digestion digestate, agricultural wastes, animal manure, and municipal wastes [48]. Attention is oriented on integrating HTC with already existing treatment strategies, like anaerobic digestion: HTC has advantages like reducing the volume of digestate and improving the biogas production by the recycling of the HTC liquor [60]. Given the heterogeneity and high variability of the feedstock, the treatment of wastes complicates the process. Indeed, these substrates are complicated by the presence of components rare in lignocellulosic feedstock, like proteins (very common is sewage sludge since they derive from bacteria) and metals, which can alter the standard formation mechanism of hydrochar. Anyway, the presence of nutrients like N and P, which get concentrated during the process, opened the field to their possible recovery [61].

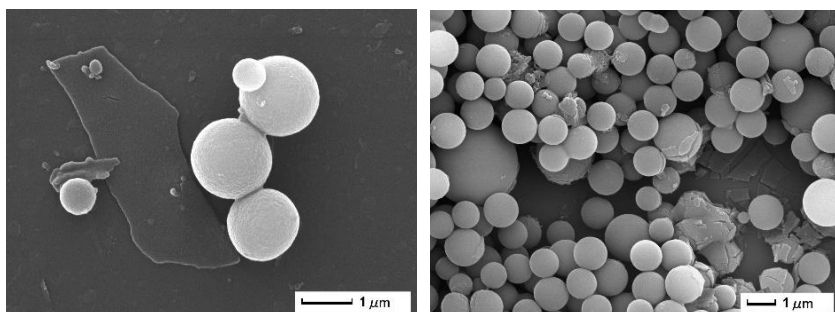
### 1.6 Aims and structure of the thesis

This thesis aims to provide some new insights around hydrothermal processes, touching several scientific areas: from in-depth research around the thermochemical fundamentals, unavoidable pillars tools for a broad understanding, to the engineering of new sustainable solutions and some holistic biorefinery concepts. Hydrothermal carbonization represents the core of the research, while hydrothermal liquefaction is marginally covered.

*Fundamental research* (chapter 2) includes insights into the chemistry, kinetics, and thermodynamics characterizing the hydrothermal conversion. We tried to answer “What’s happening during the hydrothermal conversion?” by deducing the chemical pathways and kinetics behind the conversion of several biomasses over

a broad hydrothermal spectrum, including carbonization and liquefaction. In particular:

- Attention was given to the mechanisms behind the formation of the solid phase deriving from the self-condensation and polymerization of the soluble products (the so-called “secondary char”), which exhibits fascinating properties. Indeed, consisting of spherical particles with a nano-micrometric size and having interesting properties like easy functionalization and tunability, secondary char can be a base for producing advanced green carbon materials. Therefore, we investigated the mechanisms behind the formation of the secondary char (Figure 1.6 shows an image of the produced carbonaceous spheres).
- To explore the hydrothermal mechanisms and kinetics, we investigated the pathways behind the transition from hydrothermal carbonization and liquefaction of several biomasses. In particular, we focused on the characterization of the liquid phase by following the evolution of main compounds over temperature and time.
- We faced the thermodynamics of hydrothermal carbonization. The process is exothermic, meaning that energy is released during decomposition. Therefore, we observed the effects of temperature and substrate on exothermicity. The knowledge of the magnitude and the trends of the heat profiles can be beneficial both for the overall understanding of the HTC mechanisms and for accurate energy analysis of the process.



**Figure 1.6.** Carbonaceous spheres produced from the hydrothermal carbonization of glucose, acquired through SEM in this thesis.



Then, as a junction between fundamentals and application, we reviewed the *modeling* (chapter 3) approaches used for HTC. Indeed, modeling is an unavoidable step for the industrialization of the technology. We explained the current state of the art of modeling, highlighting strengths and gaps to guide future research to further improvements. The analysis was divided into three subsystems: the process (involving kinetics and statistical models), the reactor itself (with heat transfer and computational models), the HTC plant (with techno-economic analysis), and the life cycle of the overall process.

Regarding the *engineering of the process* (chapter 4), the focus was given on how to improve sustainability. Indeed, there is a certain lack of research around the optimization of the technological process itself, and aspects like the energy balance and efficiency are often neglected. Therefore, pursuing a zero-energy process, we proposed the coupling of the process with concentrated sunlight, which covers the energy needs. We approached this problem by investigating two plant schemes: a direct and an indirect one (Figure 1.7 shows the direct experimental apparatus). Then, to move towards possible larger-scale systems, we performed a techno-economic analysis on a possible indirect system and a sizing of a possible lab-scale apparatus.



**Figure 1.7.** The solar-hydrothermal apparatus at the University of Trento investigated in this thesis.

Finally, expanding the observation scale, we investigated the *integration* (chapter 5) of hydrothermal carbonization with subsequent treatments. Indeed, the process can be a pivot to concertize a biorefinery

concept merging the management of biomass waste and the valorization of by-products. In particular, we focused on subsequent thermal and chemical treatments to make the hydrochar effective as a solid biofuel (indeed, by itself, the hydrochar suffers from fuel segregation, exhibiting a too high reactivity at low temperatures) and the recovery of chemicals.

Summarizing, the thesis is divided into 6 chapters:

- **Chapter 2** explores the *fundamentals* of hydrothermal processes. The chapter tries to answer the question “What’s happening during the hydrothermal conversion?” focusing on the kinetics and formation of secondary char, the pathways behind the transition from hydrothermal carbonization to liquefaction, and the thermodynamics,
- **Chapter 3** reviews the state of the art of the current *modeling* tools used for hydrothermal carbonization, highlighting strengths and weaknesses.
- **Chapter 4** reports the integration of hydrothermal processes with *concentrated sunlight* to realize a zero-energy process. Focus is given to the details of an experimental apparatus where a hydrothermal reactor is directly coupled with a parabolic dish concentrator and to the techno-economic analysis and sizing of an indirect system.
- **Chapter 5** investigates some biorefineries concepts to make the hydrochar obtained from organic wastes effective as a solid biofuel through post-treatments.
- **Chapter 6** reports the *conclusions* and *future developments*.

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## Chapter 2

# Around the fundamentals

### Overview

The complexity behind the hydrothermal conversion of biomass is enormous. The intricacy spreads on different levels, from the variability of the substrates to reaction chemistry involving the interaction of biomass with hot pressurized liquid water, characterized by properties and chemistry completely different from ambient conditions.

Biomass is an obnoxious substrate: heterogeneous, seasonal, degradable, and with a highly variable composition. Even the chemistry of simple carbohydrates and lignocellulosic biomass, studied for over 20 years, is still poorly understood. And to increase the complexity, the suitability of hydrothermal processes for heterogeneous substrates combined with their potential for waste management has opened the field to a set of biomasses up to a few years ago ignored, from algae to sewage sludge and organic wastes. So, apart from the classic constituent (cellulose, hemicellulose, and lignin), already difficult to understand, we add a multitude of other constituents like oils, proteins, organics,



metals, which further complicate the mechanisms. Each component owns its structural and chemical properties and interacts through heterogeneous reactions with the other ones and chemicals. Also, the hydrothermal environment has a complex chemistry. Under high temperature and pressure, water macroscopically behaves as a non-polar solvent and keeps at the same time its polar structure, interacting with ions and catalyzing acid/base reactions. Then, this overall complexity has limited the investigation of other important aspects of thermochemistry like thermodynamics and the chemical kinetics, pillars of modeling, and the comprehensive understanding of the process.

Understanding the fundamentals behind the conversion is a mandatory step to untangling the complexity. Starting from simple biomasses can help answer some basic questions around decomposition and be a starting point for progressively more intricate problems. Also, apart from being the building blocks of biomass, simple carbohydrates are the substrates for the synthesis of carbon materials, as demonstrated by the extensive research of the Titirici's group in the last decade around hydrothermal carbonization. Therefore, studying simple substrates can help in clarifying the process itself and investigate deeper hydrothermal processes as a synthesis route.

In this context, the first two sections of this chapter try to address the question “what’s happening during hydrothermal carbonization?” by adopting a kinetic approach on simple substrates like glucose, cellulose, lignin, and wood. Details on the composition of the solid and liquid phases were investigated. Section 2.1 dedicates to secondary char (the solid fraction deriving from the condensation and polymerization of organics dissolved into the liquid phase), focusing on the reaction pathway, kinetics, and morphology. Then, section 2.2 investigates the transition from mild to harsh conditions (from carbonization to liquefaction) by analyzing the hydrochars and the aqueous phase composition through “real-time” analysis. Then, the last section of this chapter (section 2.3) dedicates to the thermodynamics of hydrothermal carbonization and the measure of the amount of heat released during the process.

## Section 2.1

# Kinetics, reaction pathways, and characterization of secondary char from glucose

This section is based on a research project from our collaboration with Graziano Guella of the Physics Department of the University of Trento, who helped us with NMR analysis. Many thanks to Nicola Bazzanella for the precious support for the acquisition of SEM images.

In this section, I will explore the chemistry behind the hydrothermal carbonization of glucose and the fascinating properties of the carbon micro and nanospheres obtained from this simple substrate.

### 2.1.1 Abstract

A great complexity characterizes reaction pathways and kinetics of hydrothermal carbonization (HTC), requiring investigations of the mechanisms starting from the fundamentals. Intending to contribute to the rudiments of HTC, this work provides a systematic insight into the decomposition of glucose during HTC. The interest in glucose stems from its nature. Indeed, it is a model compound, the monomeric unit of starch and cellulose, a substrate for producing advanced carbon materials, and a precursor of the so-called secondary char. The conversion of glucose was approached holistically, focusing on the kinetics and the characterization of the solid and liquid phases. To do this, A 1.1 M solution of glucose was hydrothermally carbonized at four

temperatures (180, 220, 250, and 270°C) and six residence times (0, 0.5, 1, 3, 6, and 8 h). Mass yields were measured, while the hydrochars and the liquid phase were characterized through several techniques (like elemental analysis, scanning electron microscopy, Raman spectroscopy, and nuclear magnetic resonance). Results show that hydrochar show substantial variations with time only at 180°C. At higher temperatures, the solid yield stabilizes at around 47-50%, while the carbon content at 67-70%. The evolution of liquid intermediates in time (like HMF, formic, and levulinic acids) was quantified and used to calibrate a higher-order kinetic model. Then, regarding the morphology, hydrochars show the typical nano and microspheres (with a low degree of crystallinity) as well as photo-luminescent properties.

### 2.1.2 Introduction

Engineering sustainable processes and materials starting from renewable and abundant sources can represent a new horizon to replace the usage of fossil fuels. So far, HTC has acquired considerable attention for its ability to manage and valorize wet heterogeneous biomasses, like organic wastes, and as a synthesis route for the preparation of carbon-based materials starting from simple abundant biomasses, like carbohydrates [1].

The chemistry behind the hydrothermal conversion is enormous, with an intricacy that spreads on different levels, from the variability of the substrates to reaction chemistry involving the interaction of biomass with hot pressurized liquid water, characterized by properties and chemistry completely different from ambient conditions. Every component, with its own physical and chemical structure, interacts with the others, making the overall understanding of the conversion harsh. However, elaborating a general reaction mechanism for biomass during HTC is crucial to control the production of carbon materials and optimize the process [2]. During HTC, biomass undergoes a series of reactions in the water medium, including hydrolysis, dehydration, decarboxylation, condensation, polymerization, and aromatization [3]. The final products can be divided into the solid phase (consisting of primary char, secondary char, and carbon dots), liquid species (water-soluble compounds and oils), gases (mainly CO<sub>2</sub> with traces of CO and CH<sub>4</sub> [4]). In particular, the primary char derives from the solid-solid conversion of biomass, while what we will refer here to as “secondary char” is from the condensation and polymerization of the dissolved

organics in the liquid phase derived from the primary biomass [5]. Meanwhile, the aqueous phase mainly consists of organic acids, sugars, furans, phenols [6]. Carbon dots are quasi-spherical particles with a diameter lower than 10 nm. Their a formation mechanism still completely understood [7] and they have interesting properties like photoluminescence that make them a potential candidate for applications like cell imaging [8], ions detection [9], or catalyst [10].

Deriving from the polymerization of aqueous products, secondary char has a different structure from primary char (similar to the parent biomass) and is composed of microspheres. Thanks to some interesting properties like easy tunability, functionalization, and modification [7,11], secondary char has acquired interest as a potential base for the production of advanced carbon materials. However, until now, literature has not been uniform in defining secondary char, with a little intersection between the sectors for biomass waste and those of the production of advanced materials or general understanding of HTC. At the beginning of the diffusion of HTC, secondary char was referred to as “humins” [12–15], re-calling the portion of the insoluble fraction of humus from soil or peat in an aqueous solution, or “coke” [16]. In the field of advanced materials, it has been recently called “hydrothermal carbon” [7] or just “hydrochar” or “spherical carbon” [17,18]. Then, works dealing with real biomasses (also from our group) identified it as the “extractable-fraction” from the hydrochar through chemical extraction techniques [19–24]. The reason behind this latter approach is practical and stems from the intention of removing it from the hydrochar. Indeed, apart from the mere characterization, secondary char can confer an undesired high reactivity lower the ignition temperature of the hydrochar, making it difficult to use as a solid biofuel [19,20,23,25]. However, it is worth noticing that defining secondary char through extraction makes it dependent on the extraction method. Therefore, even if the sector of waste treatment requires less precision than that of advanced material, we suggest paying attention to not confusing the methodologies. In this work, we will use the term “secondary char” referring to the products deriving from the liquid-solid conversion.

From the first studies of HTC, simple carbohydrates (like glucose, fructose, pentose, cellulose) have been used to both produce carbons and understand the reaction pathways. Cheap and abundant, carbohydrates can be an answer to green chemistry and the replacement of fossil fuels for producing materials and platforms of chemicals. Being the building blocks of biomass, their study can help in bringing

clarity and building the basis for understanding the complex reaction pathways behind HTC. In this regard, pioneering works like those by Sevilla and Fuertes [26], Titirici and Antonietti [4], or Knežević et al. [21], are pivot for the overall understanding. Then, literature has been investigating the usage of carbohydrates for the production of carbon microspheres [18,27,28] and the recent carbon nanodots [29,30], as well as the recovery of chemicals from the liquid phase [31,32].

In this context, with this work, we aim to provide a holistic approach around secondary char to contribute in both the fundamentals and in the characterization of the potential carbons obtained. Trying to pursue precision, we start from scratch by adopting pure glucose as a substrate for HTC. Glucose is the most abundant sugar unit of biomass, the main product of the acid hydrolysis of lignocellulosic biomass [2], a model compound, and a substrate for producing advanced materials. Being glucose already dissolved in water, the obtained hydrochar consists only of secondary char. The approach is systematic: we hydrothermally carbonized glucose at several temperatures (180, 220, 250, and 270°C) and residence times (0, 0.5, 1, 3, 6, and 8 h) and characterized both the solid and the liquid phase.

### 2.1.3 Materials and methods

#### *Hydrothermal carbonization of glucose*

D-(+)-Glucose (Sigma-Aldrich, purity  $\geq 99.5$  %) was hydrothermally carbonized in a 50 mL stainless steel batch reactor [33]. The reactor was filled with a 30 mL solution of glucose and distilled water 1.11 M (corresponding to a biomass to water mass ratio of 0.25), a value that ensures the complete dissolution of glucose already at ambient temperature [34]. Tests were performed at different temperatures (180, 220, 250, and 270 °C) and residence time (0, ½, 1, 3, 6, and 8 h), this last assumed to start once reached the HTC set-point temperature after the heating phase. Each test was performed at least twice. Table 1 sums up the HTC operating conditions. For each test, the HTC reactor was loaded with  $30.00 \pm 0.01$  g of a solution of glucose plus distilled water. Before starting the run, pure nitrogen was flushed inside the HTC reactor to guarantee an inert atmosphere. After the reaction time passed, the reactor was soaked in cold water. Then, the volume of gas produced in the reaction was measured using a cylinder filled with water and placed downstream of the reactor, while the HTC liquid fraction was

separated from the solid phase through a qualitative filter at 1  $\mu\text{m}$  (Whatman Grade 1). The solid phase (the hydrochar) was recovered and dried overnight at 105 °C in a ventilated oven. Then, the solid yield (SY) was computed in percentage terms as the ratio between the mass of the dried hydrochar ( $m_{\text{hc,dry}}$ ) and that of the starting glucose ( $m_0$ ):

$$\text{SY} = \frac{m_{\text{hc,dry}}}{m_0} \cdot 100 \quad (2.1)$$

The gas yield (GY) was computed by assuming a gas composed of sole  $\text{CO}_2$ , as well supported by the literature [35,36], while the liquid yield (LY) was computed by difference as 100-SY-GY.

**Table 2.1.** HTC operating conditions.

|                                       |                                         |
|---------------------------------------|-----------------------------------------|
| Initial concentration                 | Glucose to water ratio of 0.25 (1.11 M) |
| Temperature                           | 180, 220, 250, 270°C                    |
| Residence time                        | 0, 0.5, 1, 3, 6, 8 h                    |
| Heating time                          | 20-45 min                               |
| Cooling time (to ambient temperature) | 2-3 min                                 |

### *Hydrochar composition*

The elemental composition of hydrochars was determined via a LECO 628 Elemental Analyzer equipped via ASTM D-5375 (CHN). Oxygen was computed by difference. Each test was performed twice and if values differed by more than 2% then a third trial was run. The energy content was gauged by measuring the higher heating values (HHVs) using an IKA C200 calorimeter (CEN/ TS 14918 standard, dry basis). Values were compared with theoretical values ( $\text{HHV}_{\text{th}}$ ) computed through the Dulong's formula:

$$\text{HHV}_{\text{th}} = 32.79 \cdot \text{C}\% + 150.4 \cdot \left( \text{H}\% - \frac{\text{O}\%}{8} \right) + 4.97 \cdot \text{O}\% \quad (2.2)$$

### *Morphology, Raman, and fluorescence*

A further amount of solid fraction (not accounted in the solid yield given its negligible weight) was recovered by centrifuging the liquid phase (obtained after filtration) at 2000 rpm with a Thermo scientific SL 40 FR apparatus for large volumes. The “centrifuged hydrochar” was washed with distilled water and dried overnight at 50 °C. Then,

both hydrochars obtained through filtration and centrifugation were analyzed.

The morphology was investigated through JEOL JSM-7001F Field Emission Scanning Electron Microscopy (SEM), at an electron beam energy of 5-15 keV. Size distribution was evaluated by acquiring images in Scanning Transmission Electron Microscopy (STEM) at 30 keV, using a custom-made sample holder to adapt the SEM apparatus for STEM mode. Then, images were processed using the ImageJ software.

Amorphous vs crystalline structure and photoluminescence were investigated with a Jobin-Yvon Horiba LabRam Aramis confocal micro-Raman system (confocal hole 1000  $\mu\text{m}$ , slit width 100  $\mu\text{m}$ , and grating 1800 lines/mm). The excitation source was a laser with either 532 or 633 nm wavelength, and the signal was collected with an air-cooled multichannel CCD in the range of 100–8000  $\text{cm}^{-1}$ . Incident laser power was limited using neutral density filters to avoid damaging the samples.

### *Liquid analysis*

NMR analysis was performed to determine the composition of the liquor obtained from the HTC at 180°C at the different residence times. In particular, the concentrations of glucose, HMF, levulinic acid (LA), formic acid (FA), and acetic acid (AA) were investigated.

One-dimensional  $^1\text{H}$ -NMR spectra were acquired at 300 K on a Bruker Avance spectrometer (Bruker Biospin, Karlsruhe, Germany) equipped with a Broadband Inverse (BBI) probe and operating at a proton frequency of 400.13 MHz. A single pulse sequence (zg) with a 60° flip angle, without water pre-saturation, was used. All  $^1\text{H}$ -NMR measurements were carried out with a spectral width of 11 ppm (–1 to 10 ppm), a relaxation delay of 10s, an acquisition time of 7.5s, 256 transients collected with 64k data points following four dummy scans without sample spinning. Before the analysis, samples were prepared by adding liquor (previously filtered through Whatman 0.45 mm membrane filters) and 0.250 mL of  $\text{D}_2\text{O}$  containing 0.05% w/w of 3-trimethylsilyl- propionic-2,2,3,3- $\text{d}_4$  acid sodium salt (TSP). TSP was used both as a reference standard ( $\delta\text{H} = 0.00$  ppm) and as a calibration standard ( $\text{C}(\text{TSP}) = 3.2$  mM). To validate the TSP concentration, a 0.05% w/w known amount of pure (<99%) tryptophan (Trp) was added to the NMR tube containing 0.72 mL of  $\text{D}_2\text{O}$ . Then, the TSP

concentration was determined as  $C(\text{TSP}) = 3.20 \pm 0.02 \text{ mM}$  by the relative area integration of the TSP and Trp signals.

Spectra were analyzed both by TopSpin 3.5 (Bruker BioSpin) and by MestReNova 14.1.2 software package (Mestrelab Research S.L., Santiago de Compostela, Spain). Before Fourier transformation, all spectra were apodized by exponential decay function producing linear broadening of 1.0 Hz. After Fourier transforms, automatic phasing was applied, and baselines were corrected by polynomial baseline correction. The relevant peaks for any species produced during the hydrothermal degradation of glucose were initially manually integrated and quantified relative to the TSP peak in each spectrum. Metabolites were quantified based on the ratio of the integral of the metabolite peak to the TSP peak previously adjusted to account for the ratio of metabolite and TSP protons giving rise to the signals chosen by integration (1:9). The calculation equation of quantitative Nuclear Magnetic Resonance (qNMR) for the *i*-analyte concentration is as follows:

- $C(\text{HMF}) = [(\text{Peak area of selected peak of HMF})/(\text{Peak area of TSP})] \cdot 9/1 \cdot C(\text{TSP})$
- $C(\text{FA}) = [(\text{Peak area of selected peak of FA})/(\text{Peak area of TSP})] \cdot 9/1 \cdot C(\text{TSP})$
- $C(\text{LA}) = [(\text{Peak area of selected peak of LA})/(\text{Peak area of TSP})] \cdot 9/2 \cdot C(\text{TSP})$
- $C(\text{AA}) = [(\text{Peak area of selected peak of AA})/(\text{Peak area of TSP})] \cdot 9/3 \cdot C(\text{TSP})$
- $C(\text{glucose}) = [(\text{Peak area of anomeric protons of } \alpha+\beta \text{ epimers})/(\text{Peak area of TSP})] \cdot 9/1 \cdot C(\text{TSP})$

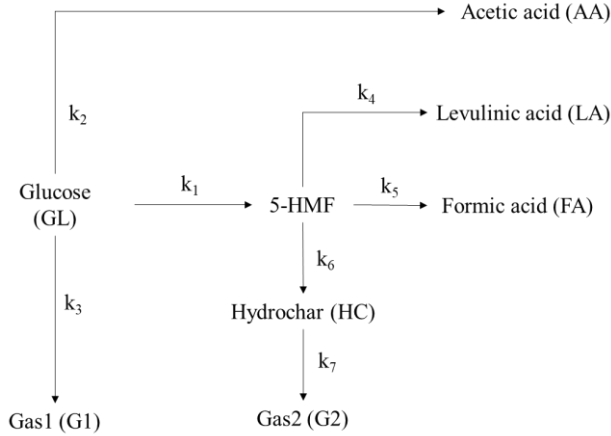
To improve the accuracy of peak integrations, peak deconvolution was also performed using the spectral tool GSD, as implemented in MestReNova software, allowing automatic peak integration of the deconvoluted peak.

### *Reaction kinetics*

The reaction rate constants were computed using the reaction pathway shown in Figure 2.1, mathematically described through the system of ordinary differential equations 2.3-2.9 (ODEs) that contains the derivatives of the dependent variables (the mass *i*-concentration  $C_i$ ) with respect to the single independent variable (time *t*). Concentrations



were refer in g/L. Those of hydrochar and gas were directly computed starting from the mass yields, while those of the compounds in the liquid phase (glucose, 5-HMF, acetic acid AA, levulinic acid LA, and formic acid FA) were obtained from molar concentrations and their molecular weights. All reactions were assumed first-order, apart from the polymerization of HMF into hydrochar, characterized instead by a variable reaction order  $n$ .



**Figure 2.1.** Reaction pathway used to kinetically model the HTC of glucose.

$$\frac{d[GL]}{dt} = -k_1[GL] - k_2[GL] - k_3[GL] \quad (2.3)$$

$$\frac{d[HMF]}{dt} = k_1[GL] - k_4[HMF] - k_5[HMF] - k_6[HMF]^n \quad (2.4)$$

$$\frac{d[HC]}{dt} = k_6[HMF]^n - k_7[HC] \quad (2.5)$$

$$\frac{d[AA]}{dt} = k_2[GL] \quad (2.6)$$

$$\frac{d[Gas]}{dt} = k_3[GL] + k_7[HC] \quad (2.7)$$

$$\frac{d[LA]}{dt} = k_4[HMF] \quad (2.8)$$

$$\frac{d[FA]}{dt} = k_5[HMF] \quad (2.9)$$

Since the ODE system does not have any explicit analytic solution, we adopted a numerical method based on an optimization technique consisting of evaluating all the terms of the system of equations containing derivative terms while testing the goodness of fit with experimental data. To facilitate the computation, several optimizations were performed by keeping the value of  $n$  fixed within the same optimization:  $n$  was varied in the range 0.5 – 2 in the different

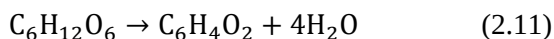
optimizations. Unknown parameters ( $k_{1-7}$ ) were estimated by combining a multi-step algorithm with a minimization function. The code, developed in Python, simultaneously solves the ODE system and minimizes the difference between the predicted and the experimental concentrations. This approach requires setting the initial conditions at time zero (set immediately after the heating phase), the time-length of interest (0-8 h), and a set of initial guess values of  $k_{1-7}$ . The error function  $E(k,n)$  was defined as the sum of the differences between the experimental and predicted concentrations at different times as:

$$E(k,n) = \sum_i (C_i^{\text{exp}} - C_i^{\text{mod}})^2 \quad (2.10)$$

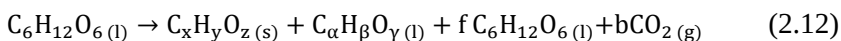
Then, the Python function ‘odeint’ was used to solve the ODE system, while the global minimum of error function  $E(k,n)$  was found by applying the basin-hopping attraction algorithm using the Python function ‘basinhopping’.

### *Reaction stoichiometry*

The maximum conversion yield can be computed by assuming: zero production of carbon dioxide and zero dissolution of compounds in the liquid phase. It bases on a model that considers hydrochar as a polymer made of HMF units condensed with a molecule of water. Clearly, this is a strong simplification, both for the process itself (no gas and no dissolution) and the solid produced (hydrochar as a clean, straight chain of repeated HMF unit where ramifications are not considered). No other structural modifications like cross-linking, cyclization, dehydration, and oxidation are considered [37]. The ending formula is reported in Equation 2.11. The molecular weight of the final carbon is 108.1 g/mol. This value could be overestimated (and therefore the moles of products underestimated), because the functionalization of the outer layers of carbonaceous, which should decrease the final mass, are not considered [37]. Considering this model, a maximum conversion of 60% can be achieved.



Meanwhile, a generic decomposition formula can be written by considering the formation of carbon dioxide and the dissolution of organics in the liquid phase (this considers also the water released).



$f$  is the mass fraction of unreacted glucose remained after the reaction, known from experimental data. The hydrochar coefficients ( $x$ ,  $y$ , and  $z$ ) were computed starting from ultimate analysis and solid yields, where  $g_i/g_{HC}$  was converted to  $mol_i/mol_{HC}$  through the molecular weight and the atomic mass unit of the hydrochar. The gas stoichiometric coefficient ( $b$ ) was computed through the gas yield and the molecular weight of  $CO_2$  as  $\frac{GY \cdot PM_{gluc}}{PM_{CO_2}}$ . Then, the liquor stoichiometric coefficients ( $\alpha$ ,  $\beta$  and  $\gamma$ ) were computed by difference. The molecular weights of hydrochar and the liquid phase (comprised of dissolved organics and water) were also computed.

### 2.1.4 Results and discussion

#### *Mass yields and hydrochar composition*

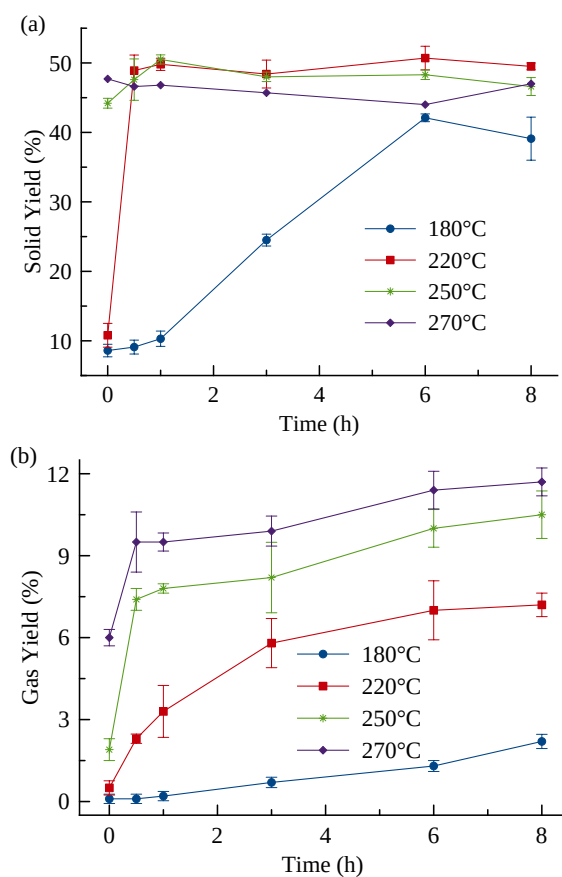
As shown in Fig. 2.2a, only at 180 °C the residence time significantly affects the solid yield. At the early stages of the process, glucose is dissolved into water and, as reactions proceed, progressively forms hydrochar through condensation and polymerization of intermediates produced during the process [38]. Indeed, during the first reaction hour, the solid yield increases from 8.6 to 10.3 % and then up to its maximum of 42.1 % at 6 h. Meanwhile, at 220, 250, and 270°C, the conversion significantly accelerates. The solid yield reaches its maximum value in less than 30 min and then stabilizes along the residence time. The inherent nature of glucose can explain this phenomenon. Being completely dissolved at the start of conversion, glucose is "easily accessible" to water molecules, whose ability to catalyze reactions (such as dehydration and polymerization) is enhanced by temperature. Therefore, at temperatures of 250 and 270°C, reactions leading to hydrochar formation are fast and occur already during the heating phase; at 220°C, they occur during the first half hour of residence time. Moreover, above 180°C, solid yields slightly vary with temperature and range between 44.0 and 50.7 % along the residence time, meaning that most of the conversion occurs at the early stages of the process. In this range, the 220°C case shows the highest values (48.4 - 50.7 %) compared to the other temperatures. The increasing extent of

gasification with temperature (Fig. 2.2b) can explain this aspect: the increment at high temperatures of reactions leading to gas formation compensates for the progressive hydrochar formation, flattening the solid yield curves. Thus, at 8 h, the gas yield passes from 2.1 to 11.7 % moving from 180 to 270°C, and its production mainly occurs at the beginning of the process (before 1 h), highlighting the larger effect of temperature than time on the conversion kinetic. Similarly, gasification reactions explain the peculiar decreasing trend of the solid yields at 180 °C from 6 to 8 h, during which the formation of hydrochar undergoes completion and gas production predominates. Although to a lesser extent than gasification, even liquefaction reactions contribute to reducing the solid formation. This typical “flattening aspect” completely differs from common biomasses, not dissolved in water and that form hydrochar through bulk reactions (solid-solid conversion) and polymerization/condensation of organics dissolved in the liquid phase (liquid-solid conversion) [39]. Indeed, before the HTC reactions start to occur (i.e. before the transient heating step begins) the solid yield is 100 % for non-dissolved biomasses and 0 % for glucose. Then, during the HTC conversion, hydrochar progressively forms, increasing the solid yield. In this regard, literature reports that the maximum theoretical achievable solid yield is 60 % computed by considering the glucose conversion into hydrochar accompanied by the release of four water molecules [40]. The reactions involved are the glucose dehydration into HMF and its condensation into hydrochar as  $\text{Glucose} \rightarrow \text{HMF} + 3\text{H}_2\text{O} \rightarrow \text{Hydrochar} + \text{H}_2\text{O}$  [37]. This theoretical limit is due to an unavoidable mass loss, intrinsic to the process of hydrochar formation when starting with glucose. Indeed, it does not account for any gas production and dissolution of organic species inside the liquid phase, aspects largely present during HTC. Indeed, our maximum measured yield is 50.7 % (220 °C 6 h).

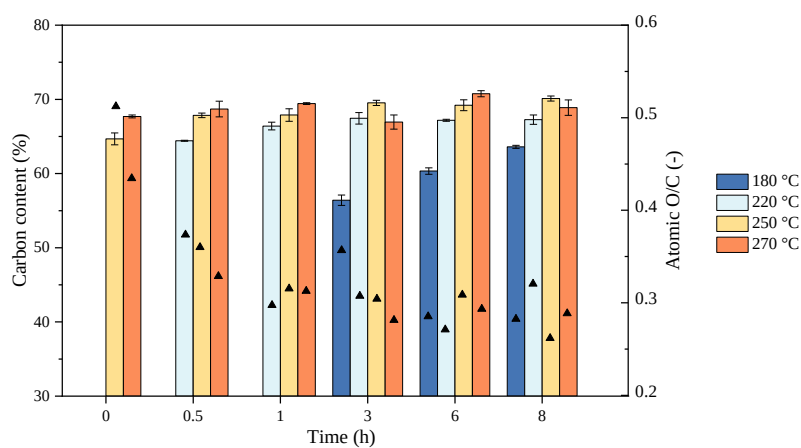
Meanwhile, the carbon content (Fig. 2.3 and Table 2.2) shows similar behavior to the solid yield, with an evident variation with time only at 180°C where values increase from 56.4 to 63.6 % passing from 3 to 8 h, respectively. At 180, 220, and 250°C, the only sharp change in carbon content occurs at the early stages of HTC, probably due to a high extent of dehydration reactions occurring at the beginning of the process [40]. At 270°C, values fluctuate, reaching their maximum of 70.8 % at 6 h. As for the yield, this behavior can be attributed to the higher thermal loss of carbon and oxygen due to gasification and liquefaction reactions. Then, since the hydrogen content remains almost constant over time, every increase in the carbon content is accompanied by a corresponding

oxygen decrease, till to a minimum value of 24.7 % (270°C, 6 h). Besides, higher temperatures and longer residence times lead to more carbonized hydrochars, with low O/C and H/C atomic ratios (Fig. 2.3).

Table 2.2 shows also HHV values, both the experimental and theoretical ones computed through Doulong's correlation. Not surprisingly, values increase with the severity of the process, passing from 15.5 MJ/kg of pure glucose up to values in the range of 27.1-27.3 MJ/kg for the hydrochars produced at 270 °C and residence times of 1-8 h. Overall, theoretical values resemble the experimental ones at 180, 220, and 250°C and slightly underestimate those at 270°C.



**Figure 2.2.** Effect of HTC temperature and residence time on (a) solid yields and (b) gas yields.



**Figure 2.3.** Effect of HTC temperature and residence time on carbon content (bars) and O/C atomic ratios (triangles).

**Table 2.2.** Mass yields, ultimate analysis, and HHV of hydrochars (standard deviations < 1.5%, for mass yields, 0.5% for ultimate, 1.2% for HHV).

| T<br>(°C) | Time<br>(h) | Mass Yields (%) |      |      | Ultimate Analysis<br>(wt.%) |     |      | HHV<br>(MJ/kg) |      |
|-----------|-------------|-----------------|------|------|-----------------------------|-----|------|----------------|------|
|           |             | SY              | GY   | LY   | C                           | H   | O*   | Exp            | Th   |
| Glucose   |             | -               | -    | -    | 40.0                        | 6.7 | 53.3 | 15.5           | 15.8 |
| 180       | 0           | 8.6             | 0.1  | 91.3 | -                           | -   | -    | -              | -    |
|           | 0.5         | 9.1             | 0.1  | 90.8 | -                           | -   | -    | -              | -    |
|           | 1           | 10.3            | 0.2  | 89.4 | -                           | -   | -    | -              | -    |
|           | 3           | 24.5            | 0.7  | 74.8 | 56.4                        | 5.1 | 38.5 | 19.0           | 20.8 |
|           | 6           | 42.1            | 1.3  | 56.6 | 60.3                        | 4.7 | 35.0 | 22.8           | 22.0 |
|           | 8           | 39.1            | 2.1  | 66.2 | 63.6                        | 4.7 | 31.7 | 23.8           | 23.6 |
| 220       | 0           | 10.8            | 0.5  | 88.7 | -                           | -   | -    | -              | -    |
|           | 0.5         | 48.9            | 2.3  | 48.8 | 64.4                        | 4.7 | 30.9 | 23.3           | 23.8 |
|           | 1           | 49.8            | 3.3  | 46.9 | 66.4                        | 4.5 | 29.1 | 24.6           | 24.5 |
|           | 3           | 48.4            | 5.8  | 45.8 | 67.4                        | 4.5 | 26.7 | 25.0           | 25.2 |
|           | 6           | 50.7            | 7.0  | 42.3 | 67.2                        | 4.6 | 28.2 | 24.9           | 25.0 |
|           | 8           | 49.5            | 7.2  | 43.2 | 67.3                        | 4.7 | 28.0 | 25.5           | 25.2 |
| 250       | 0           | 44.2            | 1.9  | 54.0 | 64.7                        | 4.6 | 30.8 | 24.3           | 23.8 |
|           | 0.5         | 47.6            | 7.4  | 45.0 | 67.9                        | 4.3 | 27.8 | 24.6           | 24.9 |
|           | 1           | 50.5            | 7.8  | 41.6 | 67.9                        | 4.6 | 27.5 | 25.7           | 25.3 |
|           | 3           | 48.0            | 8.2  | 43.7 | 69.5                        | 4.4 | 26.1 | 25.8           | 25.8 |
|           | 6           | 48.3            | 10.0 | 41.7 | 69.2                        | 4.5 | 26.3 | 25.8           | 25.8 |
|           | 8           | 46.6            | 10.5 | 42.8 | 70.1                        | 4.6 | 25.3 | 26.7           | 26.3 |
| 270       | 0           | 47.7            | 6.0  | 46.3 | 67.7                        | 4.4 | 27.9 | 25.1           | 25.0 |
|           | 0.5         | 46.6            | 9.5  | 43.9 | 68.7                        | 4.4 | 26.9 | 25.4           | 25.5 |
|           | 1           | 46.8            | 9.5  | 43.7 | 69.4                        | 4.4 | 26.2 | 27.1           | 25.8 |
|           | 3           | 45.7            | 9.9  | 44.4 | 66.9                        | 4.4 | 28.6 | 27.3           | 24.7 |
|           | 6           | 44.0            | 11.4 | 44.6 | 70.8                        | 4.5 | 24.7 | 27.1           | 26.6 |
|           | 8           | 47.0            | 11.7 | 41.3 | 68.9                        | 4.6 | 26.5 | 27.1           | 25.9 |

\*O computed by difference.

*Liquor composition kinetics*

Fig. 2.4a shows the variations, at 180°C, of the mass percentages of the major components as identified and quantified through <sup>1</sup>H-NMR analysis, i.e., hydroxymethylfurfural (HMF), formic acid (FA), levulinic acid (LA), and acetic acid (AA). Differently, Fig. 2.4b shows the mass percentage of (unreacted) glucose and (formed) hydrochar, the former measured through <sup>1</sup>H-NMR analysis, the latter corresponding to the hydrochar yield.

Before 3-4 h of treatment, HMF is the main detected compound. The HMF concentration grows in the first hour, reaching 2.5 %, and then slowly decreases with time, but remaining detectable after 8 h of treatment (1.2 %). HMF is a known intermediate produced during HTC that self-condensates into a dimer or oligomer forming hydrochar [41]. Also, HMF degrades to organic acids. To the growing stage of HMF before 1 h corresponds a slight increase in the formation of hydrochar, indeed mostly not collectible through filtration. Then, once HMF starts to decrease, the hydrochar yield shows a rapid increase. Therefore, still in the early stage of the process, even if with low yield, HMF polycondensates and, as time passes, undergo cross-link forming furan-like structures. Indeed, the intra-reactivity of some functional groups, such as hydroxy and aldehyde groups, is accompanied by an increase in volume and hydrophobicity, eventually leading to precipitation in the form of nanoparticles that progressively increase in diameter [42]. Both nanocarbons and microspheres were also detected in our hydrochars, as shown from photoluminescence and SEM analyses. In general, HMF forms through two possible pathways: (1) direct dehydration of glucose and (2) aldose-ketose isomerization of glucose to fructose and subsequent dehydration [43,44]. Previous research demonstrated that the dehydration of fructose by Brønsted acids dominates between the two reactions due to the lower reactivity and selectivity towards HMF of glucose than that of fructose [45], probably due to the lower abundance of acyclic glucose (open chain) than that of acyclic fructose under hydrothermal conditions [46,47]. This lower reactivity of glucose compared to fructose can explain the high amount of unreacted glucose measured, as shown in Fig. 2.4b: after 6 h, there is still about 12.2 % of unreacted glucose, value, which remains quite stable also at 8 h of residence time.

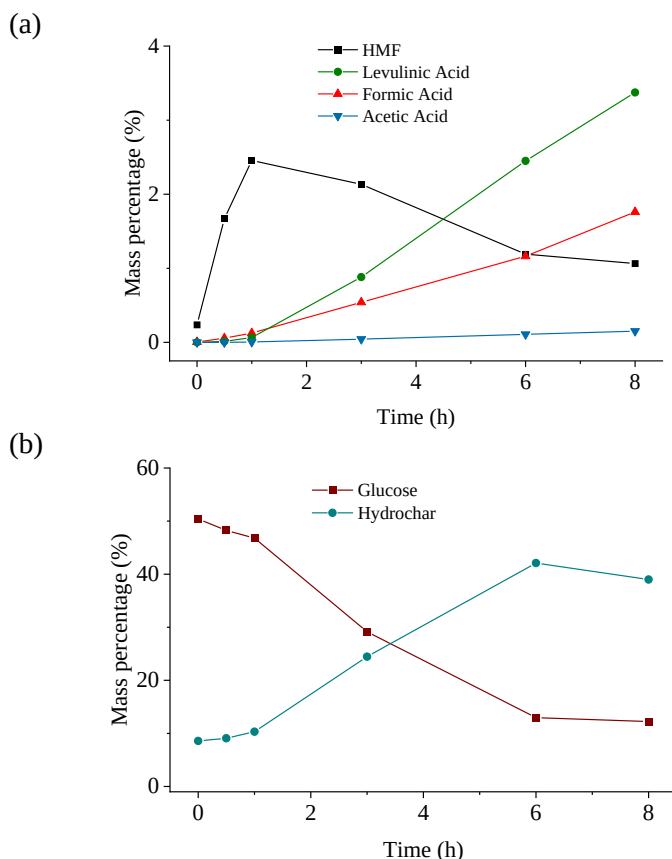
After an initial time lag of 1 h, with a process mainly characterized by the production of HMF, the concentration of carboxylic acids (FA, LA, AA) starts to be more relevant. They increase over time, reaching around 5.3 % altogether at 8 h. Among them, at the early stages of the conversion, only FA is formed, while LA and AA appear later (after 1 hour). These organic acids are end-products of the rehydration process of HMF. In particular, LA and FA are produced via  $\text{HMF} + 2\text{H}_2\text{O} \rightarrow \text{LA} + \text{FA}$  [12,37]. FA was demonstrated to act as a catalyzer for the dehydration of fructose into HMF and therefore plays a critical role in promoting the growth of hydrochar [47]. Conversely, LA does not act as a catalyzer. The mass ratio between LA and FA progressively decreases from 10.3 to 1.3 at the end of the treatment, indicating that



LA is consumed by other side reactions. These could include the reactions of LA with the hydrochar, as supported by previous observations based on  $^{13}\text{C}$  NMR characterization. Indeed, these demonstrated that LA can react with hydroxyl groups ( $-\text{OH}$ ) present on the material surface, resulting in an embedded molecule inside the solid phase and as copolymerized compounds [48]. Meanwhile, previous studies demonstrate that hydrochar does not embed FA.

Similar to LA and FA, the concentration of AA increases over time, and its presence becomes relevant only at long residence times (more than 6 h), highlighting its slow formation kinetics. Its production can derive from two different pathways: (1) fructose fragmentation, involving decarboxylation or decarbonylation of lactic acid produced from the oxidation of acetaldehyde derived from glucose; and (2) oxidation of HMF and furfurals [37,49]. Even if the oxidation of HMF was demonstrated to play a minor role with respect to fructose fragmentation, its partial oxidation could contribute to the decrement of the hydrochar yield from 6 to 8 h together with decarboxylation and decarbonylation reactions occurring at the surface of the solid phase.

Overall, the rest of the undetected liquid fraction can be attributed to the finest fraction of hydrochar not recovered through filtration (dispersed into the liquid phase) and other traces of compounds and intermediates dissolved into the liquid phase (like fructose and furfural).



**Figure 2.4.** Quantitative time dependence of major compounds present in the liquor during HTC at 180 °C: (a) HMF, formic, levulinic, and acetic acid; (b) glucose and hydrochar. Mass percentage indicates the mass of the i-compound divided by the initial mass of glucose loaded into the HTC reactor.

### *Kinetic parameters*

The kinetic model was applied to the 180°C case, since it shows the most variegate kinetics. To make the model solvable, the kinetic pathway was lumped using empirical mass concentrations expressed in g/L. Using a mass concentration instead of a molar one, enables to disengage in the hydrochar molecular weight, which is often assumed constant and equal to 108 g/mol [50,51]. The assumption behind derives from considering the hydrochar produced only from the theoretical reaction of polymerization of HMF  $\rightarrow$  Hydrochar + H<sub>2</sub>O. This is a strong assumption: during kinetics, hydrochar undergoes other reactions like aromatization and gasification, affecting the molar

composition. The stoichiometric calculations reported in the next section remark this point, showing theoretical molecular weights of the hydrochars comprised between 44 and 75.8 g/mol. Besides, no solid fraction was collected before 1 h, aspect that would force to hypothesize its molecular weight and increase approximations.

Table 2.3 summarizes the reaction rate constants at different  $n$ . We can notice that the orders of magnitude of  $k$  are stable at different  $n$ . With  $k$  of around  $10^{-5} \text{ min}^{-1}$ , the decomposition of glucose together with the formation of levulinic and formic acid are the fastest reactions occurring. The formation of acetic acid from glucose ( $k_2$  of  $10^{-8}$ ) dominates over the degradation of levulinic acid ( $k_3$  of  $10^{-7}$ ). Meanwhile, reactions leading to the gas phase are the slowest ( $k_{3-7}$  between  $10^{-7}$  and  $10^{-6} \text{ min}^{-1}$ ), with that deriving from hydrochar faster than that from glucose. As a comparison, we can notice that the decomposition of HMF (given by the sum of  $k_4$ ,  $k_5$ , and  $k_6$ ) equals  $4.7 \cdot 10^{-4} \text{ min}^{-1}$ , value similar to that found by Jung et al. [27] of  $3.9 \cdot 10^{-4} \text{ min}^{-1}$  for the fructose degradation.

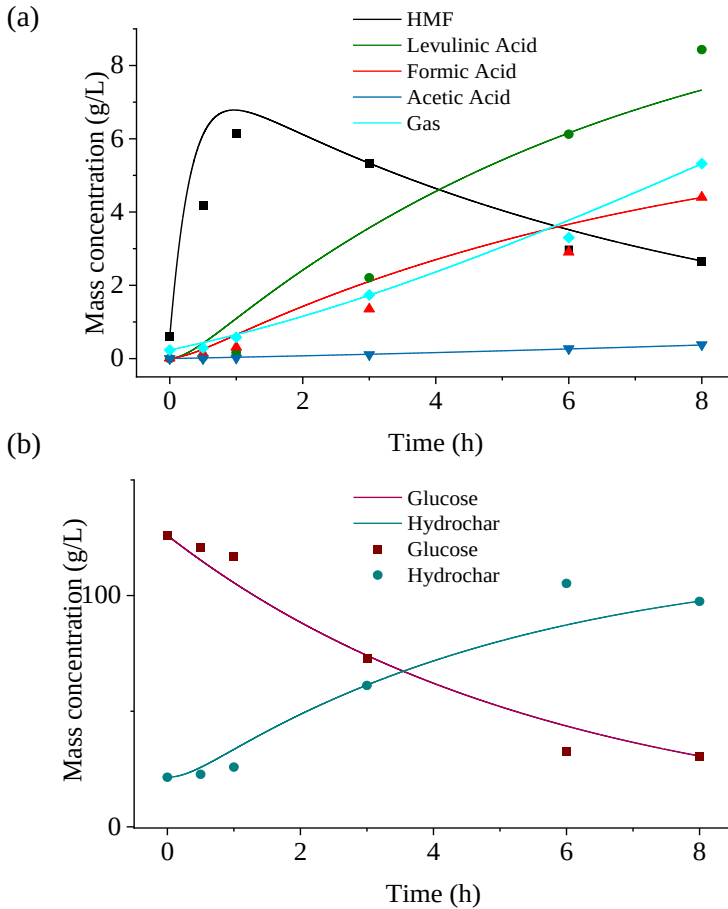
**Table 2.3.** Predicted reaction rate constants ( $k_i$ ) of the HTC of glucose at  $180^\circ\text{C}$  ( $\text{min}^{-1}$ ), at different  $n$ .

|       | $n=0.5$             | $n=1$               | $n=1.3$             | $n=1.5$             | $n=1.7$             | $n=2$               | $n=2.5$             |
|-------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| $k_1$ | $5.0 \cdot 10^{-5}$ | $4.9 \cdot 10^{-5}$ | $4.9 \cdot 10^{-5}$ | $4.8 \cdot 10^{-5}$ | $4.9 \cdot 10^{-5}$ | $4.8 \cdot 10^{-5}$ | $4.9 \cdot 10^{-5}$ |
| $k_2$ | $1.3 \cdot 10^{-8}$ | $6.3 \cdot 10^{-8}$ | $8.0 \cdot 10^{-8}$ | $1.6 \cdot 10^{-7}$ | $1.8 \cdot 10^{-7}$ | $1.8 \cdot 10^{-7}$ | $1.5 \cdot 10^{-7}$ |
| $k_3$ | $3.2 \cdot 10^{-7}$ | $5.4 \cdot 10^{-7}$ | $5.4 \cdot 10^{-7}$ | $5.7 \cdot 10^{-7}$ | $3.9 \cdot 10^{-7}$ | $5.1 \cdot 10^{-7}$ | $7.7 \cdot 10^{-7}$ |
| $k_4$ | $5.1 \cdot 10^{-5}$ | $6.3 \cdot 10^{-5}$ | $5.7 \cdot 10^{-5}$ | $5.4 \cdot 10^{-5}$ | $5.7 \cdot 10^{-5}$ | $6.0 \cdot 10^{-5}$ | $5.7 \cdot 10^{-5}$ |
| $k_5$ | $2.3 \cdot 10^{-5}$ | $2.8 \cdot 10^{-5}$ | $3.3 \cdot 10^{-5}$ | $2.5 \cdot 10^{-5}$ | $3.0 \cdot 10^{-5}$ | $2.6 \cdot 10^{-5}$ | $2.8 \cdot 10^{-5}$ |
| $k_6$ | $1.4 \cdot 10^{-4}$ | $6.3 \cdot 10^{-4}$ | $3.8 \cdot 10^{-4}$ | $1.1 \cdot 10^{-4}$ | $2.1 \cdot 10^{-4}$ | $2.7 \cdot 10^{-4}$ | $5.0 \cdot 10^{-5}$ |
| $k_7$ | $2.3 \cdot 10^{-6}$ | $2.1 \cdot 10^{-6}$ | $2.1 \cdot 10^{-6}$ | $2.0 \cdot 10^{-6}$ | $2.2 \cdot 10^{-6}$ | $2.1 \cdot 10^{-6}$ | $1.5 \cdot 10^{-6}$ |
| $E$   | 73.2                | 67.9                | 67.2                | 68.2                | 67.7                | 67.4                | 68.7                |

The minimum of the error function was reached at an  $n$  of 1.3. Having a  $n > 1$  should theoretically agree with what expected from the nature of reactions involved. Indeed, polymerization reactions are characterized by a higher-order kinetics, meaning that their rate varies more than proportionally with respect to the reagent concentrations [52]. However, we can notice that the differences between the error functions for different  $n$  are minimal, with a decomposition rate from HMF to hydrochar ( $k_6$ ) that compensates  $n$ . Therefore, we cannot consider these data as fully meaningful for interpreting the chemistry of the process. Since the order of reaction is associated with the effect of reactant

concentration on the product formation, performing more tests at different initial concentrations will help in finding a precise couple of  $n$  and  $k_6$ . For the purpose of this paper, in which modeling is an addendum to the overall glucose decomposition, knowing already the order of magnitude of the reaction rates, is satisfactory. Future works will tackle the issue of predicting a proper  $n$  using different starting concentrations.

Figure 2.5 shows curves obtained at  $n=1.3$ , highlighting the good fit between experimental and predicted data.



**Figure 2.5.** Modeled (continue) and predicted (scattered) mass concentrations of compounds detected at 180°C: (a) HMF, formic, levulinic, acetic acid, and gas; (b) glucose and hydrochar.

### *Stoichiometric parameters*

Table 2.4 shows the stoichiometric coefficients computed from the elemental composition and yields. Following ultimate analysis, as

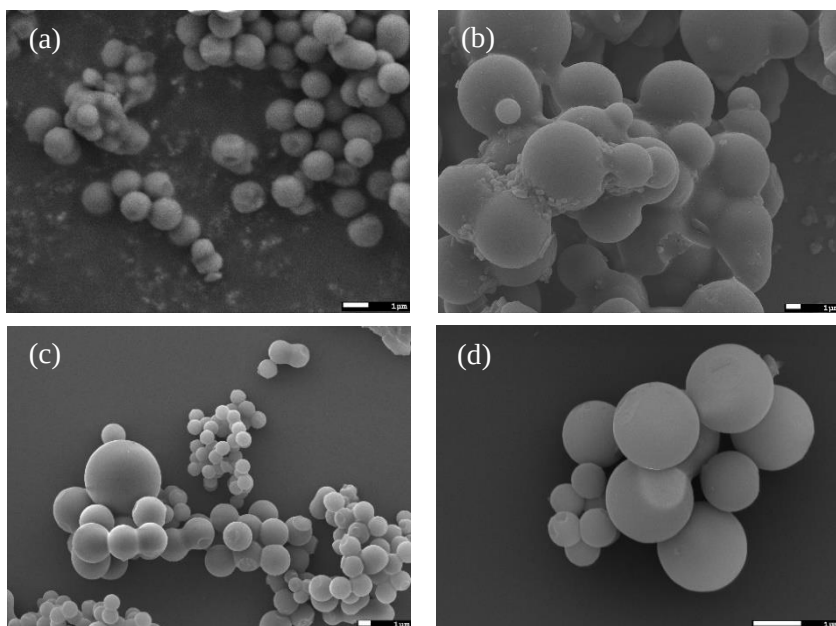
temperature increases, the ratio between carbon and oxygen increases consistent with decarboxylation and decarbonylation reactions. The MW reaches a value of 89.7 g/mol, much lower than the maximum theoretical one (108.1 g/mol). Indeed, the theoretical value only considers the condensation/polymerization of HMF, without accounting does not consider the gas production and other phenomena like functionalization of the outer layers. The discrepancy between experimental MW and theoretical is particularly relevant for modeling, where the MW directly appears inside the computations. Therefore, we suggest avoiding strong assumptions (like that of 108.1 g/mol) or trying to “skip” it using mass concentrations instead of moles.

**Table 2.4.** Stoichiometric coefficients of the generic HTC reaction of glucose (gas coefficient  $b < 0.4$ ).

| T<br>(°C) | Time<br>(h) | Hydrochar              |          |          |               | Liquor (organics + water)  |          |          |               |
|-----------|-------------|------------------------|----------|----------|---------------|----------------------------|----------|----------|---------------|
|           |             | moli/mol <sub>HC</sub> |          |          | MW<br>(g/mol) | moli/mol <sub>liquor</sub> |          |          | MW<br>(g/mol) |
|           |             | C<br>(x)               | H<br>(y) | O<br>(z) |               | C<br>(α)                   | H<br>(β) | O<br>(γ) |               |
| 180       | 3           | 2.1                    | 2.2      | 1.1      | 44.1          | 3.9                        | 9.8      | 4.9      | 134.8         |
|           | 6           | 3.8                    | 3.5      | 1.7      | 75.8          | 2.1                        | 8.5      | 4.2      | 101.9         |
|           | 8           | 3.0                    | 2.7      | 1.1      | 57.1          | 2.9                        | 9.3      | 4.7      | 119.3         |
| 220       | 0.5         | 4.7                    | 4.1      | 1.7      | 88.1          | 1.2                        | 7.9      | 4.1      | 87.9          |
|           | 1           | 5.0                    | 4.0      | 1.6      | 89.7          | 0.9                        | 8.0      | 4.1      | 84.4          |
|           | 3           | 5.0                    | 4.0      | 1.5      | 87.1          | 0.8                        | 8.0      | 4.0      | 82.5          |
|           | 6           | 5.1                    | 4.1      | 1.6      | 91.3          | 0.6                        | 7.9      | 3.8      | 76.2          |
|           | 8           | 5.0                    | 4.1      | 1.6      | 89.3          | 0.7                        | 7.9      | 3.8      | 77.9          |
| 250       | 0           | 4.3                    | 3.6      | 1.5      | 79.6          | 1.6                        | 8.4      | 4.3      | 97.2          |
|           | 0.5         | 4.8                    | 3.7      | 1.5      | 85.7          | 0.9                        | 8.3      | 3.9      | 81.1          |
|           | 1           | 5.1                    | 4.1      | 1.6      | 91.1          | 0.5                        | 7.9      | 3.8      | 75.0          |
|           | 3           | 5.0                    | 3.8      | 1.4      | 86.5          | 0.7                        | 8.2      | 3.9      | 78.8          |
|           | 6           | 5.0                    | 3.9      | 1.4      | 87.1          | 0.6                        | 8.1      | 3.8      | 75.2          |
|           | 8           | 4.9                    | 3.8      | 1.3      | 84.0          | 0.7                        | 8.2      | 3.8      | 77.2          |
| 270       | 0           | 4.8                    | 3.8      | 1.5      | 85.9          | 0.9                        | 8.2      | 4.0      | 83.5          |
|           | 0.5         | 4.8                    | 3.7      | 1.4      | 84.0          | 0.8                        | 8.3      | 3.8      | 79.0          |
|           | 1           | 4.9                    | 3.7      | 1.4      | 84.4          | 0.7                        | 8.3      | 3.8      | 78.7          |
|           | 3           | 4.6                    | 3.6      | 1.5      | 82.3          | 1.0                        | 8.4      | 3.7      | 80.0          |
|           | 6           | 4.7                    | 3.6      | 1.2      | 79.3          | 0.9                        | 8.4      | 3.8      | 80.4          |
|           | 8           | 4.9                    | 3.9      | 1.4      | 84.6          | 0.7                        | 8.1      | 3.6      | 74.5          |

### *Morphology of hydrochars*

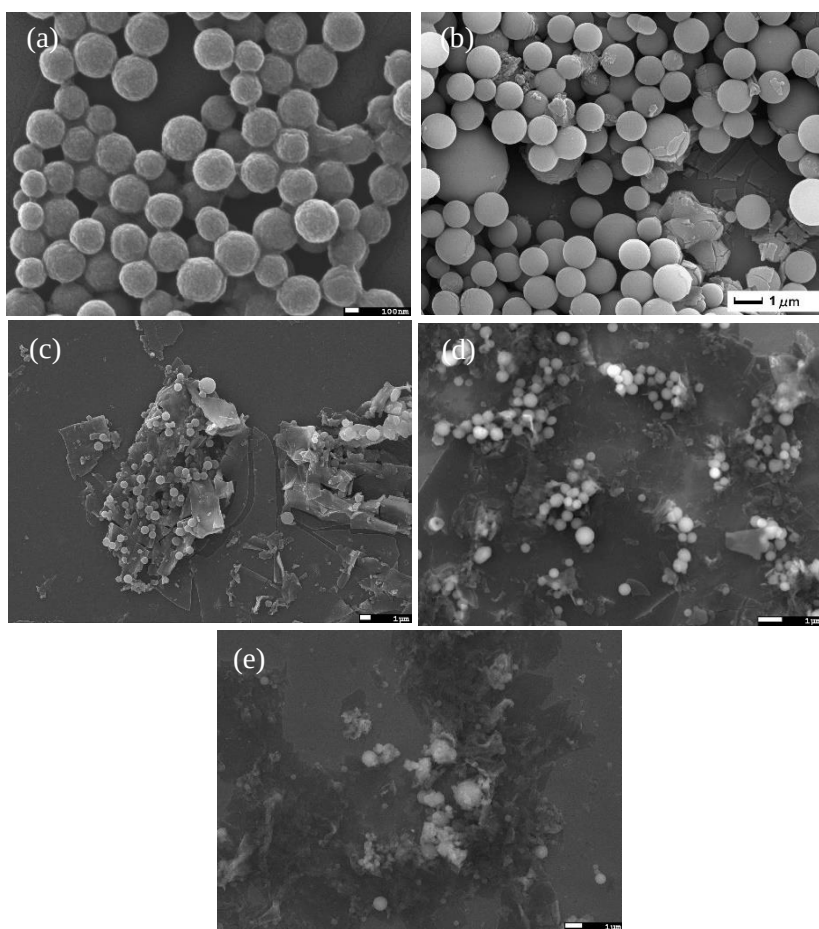
Figures 2.6 and 2.7 show SEM images of hydrochars at different operating conditions recovered through filtration and centrifugation (differing for the size of the recovered solid), respectively. Smooth spheres characterize both the filtrated and centrifuged hydrochars. Spheres mainly differ for their diameters: nanometer scale for the centrifuged and micrometer scale for the filtrate. Their presence agrees with many observations carried out by previous literature on glucose [7,40,53,54], as well as for other carbohydrates (like fructose, sucrose, xylose, starch, and cellulose [42]) or lignocellulosic biomasses [5,20]. Meanwhile, the main difference between the two is the presence of sheets of materials in centrifuged hydrochar, instead absent in the filtrate.



**Figure 2.6.** SEM images of hydrochars recovered through filtration of (a) 180°C 3 h, (b) 180°C 6 h, (c) 220°C 6 h, (d) 270°C 6 h.

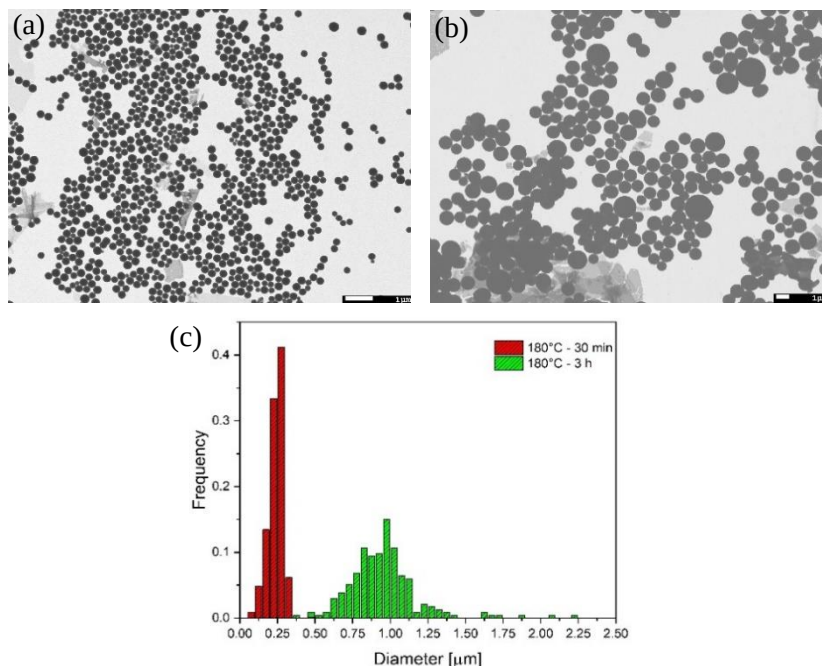
Filtrates (Figure 2.5) mainly present micrometer size particles. At temperatures higher than 220°C spheres tend to separate, be uniform, and partially fuse. Meanwhile, 180°C samples present some nanospheres (probably remained retained on the bigger size particles) and fused agglomerates at long residence times. Along with residence time, spheres grow until they reach a final size (depending on temperature) and then agglomerate among them. Nanospheres are not

detected above 220°C: since they are an early formation stage of the microspheres [40,42], 220°C is enough to cause the complete growth of nanospheres to larger diameters. Indeed, as soon as HMF undergoes cross-linking and dehydration, functional groups are lost, causing an increment in volume and a shift to hydrophobicity, causing the first precipitation of these nanoparticles in solution, which then aggregate and fuse among them [42]. At high temperatures, the water properties show huger variations than lower temperatures (like a higher dissociation into OH<sup>-</sup> and H<sub>3</sub>O<sup>+</sup> at higher temperatures), an aspect that affects the colloidal stability and surface tension of spheres [40,55,56]. Also, higher temperatures generally cause the decrease of functional groups and therefore facilitate the fusion of spheres and surface smoothing [56].



**Figure 2.7.** SEM images of hydrochars recovered through centrifugation: (a) 180°C 30 min, (b) 180°C 3 h, (c) 180°C 6 h, (d) 250°C 6 h, (e) 270°C 6 h.

Centrifuged samples (Figure 2.7), representing the finest fraction of solid-phase recovered, present an unexpected morphology, with both nano-microspheres and sheets of material. The number of spheres decreases with time and temperature: indeed, they grow in diameters, remaining trapped in the filter and leaving space to the only sheets. The diameter increment is very clear from STEM images (Figure 2.8): moving from 30 min to 3 h the average diameter increases from 0.25 to 1 mm. Meanwhile, the sheets increase as time and temperature increase, probably due to the complementary removal of microspheres (remaining on the filtrate). As abovementioned: the presence of this fraction is unexpected. SEM images suggest that these sheets could form at mild conditions as an envelope of the spheres (as clear at 180°C 30 min and 180°C 3 h), and then exfoliate forming sheets. These sheets could consist of graphite: even if most of the carbon is amorphous, we can not exclude their presence. Indeed, apart from the typical morphology of graphite, Raman spectra confirm a higher presence of graphite inside centrifuged samples rather than the filtrate one (Table 2.5). Up to now, graphite is known to be formed under harsher conditions (for example, during pyrolysis [7,57]). We will investigate this fact in future studies.



**Figure 2.8.** STEM images of hydrochars recovered through centrifugation of (a) 180°C 30 min, (b) 180°C 3 h, (c) diameter distribution of spheres.



As a general note, it is worth noticing that apart from residence time and temperature, comparison with literature requires particular attention to the initial concentration, which affects the final diameters. Indeed, it was observed that up to a certain concentration, less dilution generally causes spheres with bigger diameters that tend to aggregate and to form larger particles [54,56]. For example, Qui et al. [26] showed that a 0.1 M solution at 180°C resulted in spheres with average diameters below 0.2  $\mu\text{m}$  [42], while Sevilla and Fuertes [26] that a 0.5 M solution at 180°C leads to diameters 0.44  $\mu\text{m}$  and 1.0 M at 190°C to 1.4  $\mu\text{m}$ . Then, after a certain concentration, the size tends to remain constant, with maximum average diameters of around 5  $\mu\text{m}$  [58].

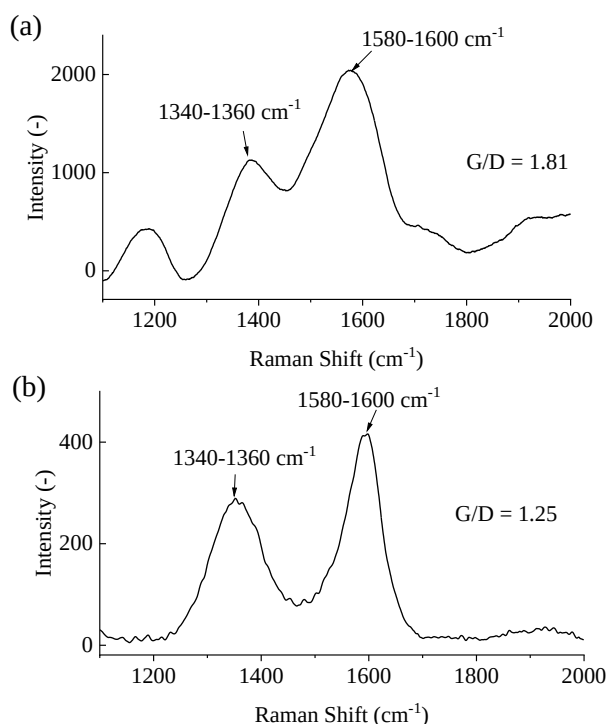
### *Raman spectra*

Examples of Raman spectra are reported in Figure 2.9. Spectra show two broad overlapping peaks, the G peak at 1580-1600  $\text{cm}^{-1}$  and the D peak at 1340-1360  $\text{cm}^{-1}$ . The G band can be assigned to the vibrational modes of  $\text{sp}^2$  carbon atoms originating from sheets of graphene defect-free, while the D band can be attributed to the breathing modes of rings or defects in graphitic lattices [59,60]. Even if graphene has a G band located at approximately 1580  $\text{cm}^{-1}$ , a graphitic structure is accepted to be negligible in hydrochars, as instead occurs for pyrolytic materials [59]. Indeed, graphitic materials are typically dominated by a G band, with G/D intensity ratios higher than 10 [59]. Meanwhile, our hydrochars show a G/D ratio of 1.1-2.3 (Table 2.5). Therefore, the structure of the hydrochar is mainly amorphous and disordered, with a tendency towards graphitization. Comparing centrifuged and filtrate samples, we can appreciate how the G/D ratios increase for the centrifuged hydrochars, highlighting a higher presence of crystallinity. As abovementioned, this aspect requires further investigations.

Indeed,  $^{13}\text{C}$  NMR analysis evidenced that hydrochars consist of a furan/arene structure, composed of cross-linked furan rings, deriving from HMF, and arene groups, identifying polyaromatic hydrocarbon clusters [48,61]. Brown et al. [59] proposed a model consisting of single furans and 2-3 arenes supported by NMR and infrared analysis. Moreover, the spheres have a core-shell structure, with a condensed and hydrophobic “hard” core surrounded by a “soft” shell rich in carboxylic and aldehydic groups [7,54,62].

**Table 2.5.** Ratios between the Raman peak intensities (G/D) of some centrifuged and filtrate hydrochars.

| Centrifuged |      | Filtrate  |      |
|-------------|------|-----------|------|
| 180°C ½ h   | 2.30 | 220°C 6 h | 1.09 |
| 180°C 3 h   | 1.81 | 250°C 6 h | 1.07 |
| 180°C 6 h   | 1.82 | 270°C 6 h | 1.18 |

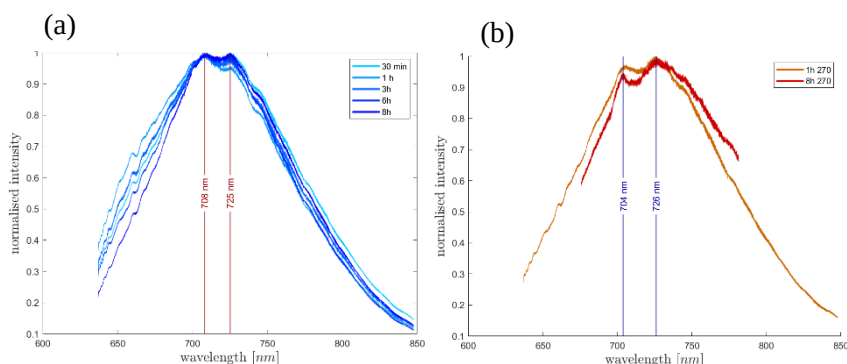
**Figure 2.9.** Raman spectra of (a) centrifuged hydrochar at 180°C 3 h, (b) filtrated hydrochars at 250°C 3 h.

### Photoluminescence

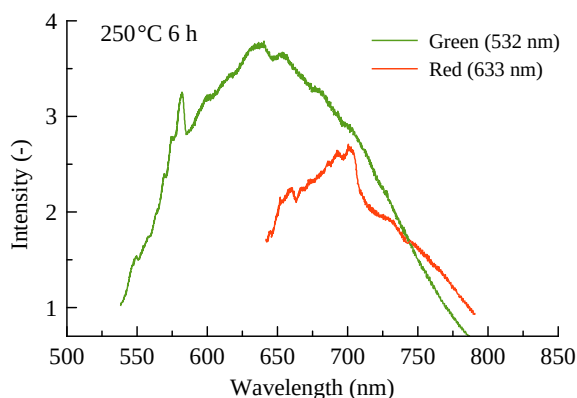
Hydrochars exhibit photoluminescence (PL), as shown in Fig. 10. PL is very probably caused by the presence of carbon dots, nanoparticles with sizes below 10 nm known to be produced during HTC [7,63,64].

Indeed, when excited at 532 nm (green light, Fig. 2.11), samples show PL spectra emission positioned in the orange-red region of the electromagnetic spectrum and a strong peak between 630 and 640 nm. Both HTC temperature and residence do not affect the peak position and shape of the spectra. However, we should notice that these spectra

are only qualitative and we cannot exclude a different intensity under different operating conditions. Indeed, Papaioannou et al. [64] demonstrated that longer HTC residence times (up to 12 h) cause a higher brightness level. The emission peak position shifts with the excitation wavelength (Fig. 2.10a-b), an aspect that can derive from both the different sizes of the carbon dots and their different emissive sites. Several works observed also this dependence [65–67].



**Figure 2.10.** Photoluminescence of centrifuged hydrochars at different times and at (a) 180°C and (b) 270°C. Emission source: 532 nm.



**Figure 2.11.** Effect of the emission source (532 and 633 nm) on photoluminescence of centrifuged hydrochar at 250°C 6 h.

### Reaction pathway

Merging what was observed through our experimental activity and the current literature, we can summarize the following reaction pathway (Fig. 2.12) and considerations:

(1) *Isomerization* of glucose to fructose: D-glucose is transformed into D-fructose and vice-versa, with their respective isomers (both open and closed forms) in equilibrium between each other [68].

(2) *Dehydration* of fructose and glucose to HMF dissolved in solution through a first-order reaction mechanism. The isomers of glucose and fructose undergo intermolecular condensation, losing three water molecules directly as  $\text{Glucose/fructose} \rightarrow \text{HMF} + 3\text{H}_2\text{O}$ . HMF is hydrophilic and therefore remains dissolved in solution. Indeed, -OH and two -H terminals of the furan ring are more likely to participate in direct reactions [69]. In our work, the maximum concentration of HMF is reached at 1 h of residence time at 180 °C.

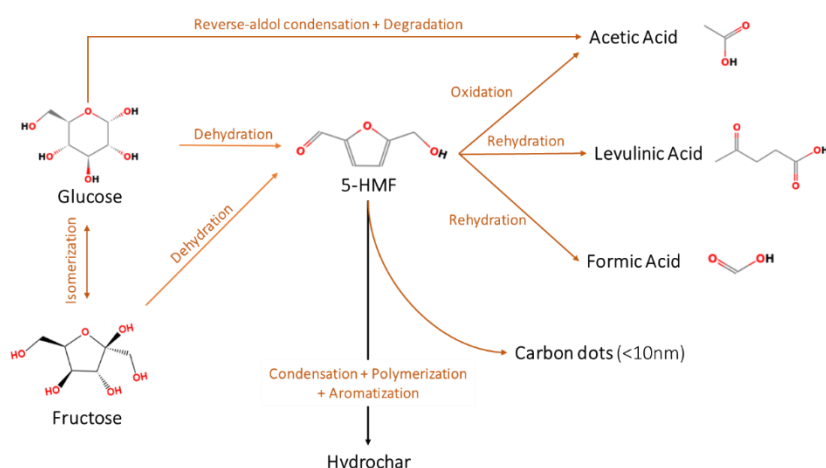
(3) *Rehydration* of HMF to form levulinic acid (LA) and formic acid (FA). These are produced from the furan ring cleavage of HMF [69].  $\text{HMF} + 2\text{H}_2\text{O} \rightarrow \text{LA} + \text{FA}$ . FA can catalyze the dehydration of fructose to HMF, while LA can react with the hydroxyl groups of hydrochar, reducing its growth rate and resulting in embedded in the structure. Meanwhile, LA molecules participate in the growth of carbonaceous spheres as building units. Also, they reduce the growth rate of spheres by replacing HMF molecules to react with hydroxyl groups (-OH) on the surface [69].

(4) *Production of acetic acid* (AA) involving (1) decarboxylation or decarbonylation of lactic acid produced from the oxidation of acetaldehyde derived from the reverse aldol condensation of glucose, (2) oxidation of HMF and furfurals [37,42,69].

(5) *Polymerization, condensation, and aromatization* of HMF to hydrochar. So far, the formation of the carbon spheres (both of micro and nanometric size) has been explained through different nucleation theories. The classical theory is based on the LaMer model, which explains the formation of spheres through outburst nucleation, occurring when the concentration of aromatic clusters reaches a critical super-saturation point, and the successive growth by molecular attachment [26]. Besides, Zhang et al. [70] developed a theory based on a hydrophobic ripening, according to which, during condensation, hydrophilic groups are lost, making the molecular cluster progressively hydrophobic and leading to its eventual precipitation into the solid form, producing primary particles: once formed, they progressively aggregate among them by the interaction between functional terminals to minimize the surface free energy.

(6) *Carbon dots formation*. Having a size below 10 nm, carbon dots represent a third phase grown between the solid microspheres and the liquid phase [7]. They confer photoluminescence properties to the

hydrochar and have crystalline domains. Also, they are formed by hydroxyl, carboxylic, epoxy, and carbonyl functional groups conferring a high solubility in water [71]. As for the carbon spheres, their formation has been explained through nucleation theories, in which soluble polymers (deriving from the polymerization and condensation of soluble chemicals) undergo aromatization. Indeed, when the concentration of aromatic clusters reaches a certain critical point, burst nucleation occurs and carbon dots grow [7]. Recently, Nicolae et al. [7] proposed an evolution mechanism involving the gas phase, in which polymeric nuclei formed at the beginning of the process act as hot spots for the formation of crystalline carbons.



**Figure 2.12.** Summarized reaction pathway of the HTC of glucose.

### 2.1.5 Conclusions

This work approached the HTC conversion of glucose holistically, from the reaction pathways to the kinetics and optical and morphological properties of final products. Glucose represents the monomeric unit of starch and cellulose, a substrate for producing advanced carbon materials, and a model compound to study the so-called secondary char.

A solution of glucose (1.1 M, 25 wt.%) was hydrothermally carbonized at several temperatures (180, 220, 250, and 270°C) and residence times (0, 0.5, 1, 3, 6, and 8 h). Analyses show that temperature highly affects the solid yield below 220°C, and at higher temperatures it stabilizes at 47-50%, indicating a conversion that undergoes

completion already during the heating phase. The evolution of intermediates (like HMF, formic, and levulinic acid) was monitored and used to calibrate a higher-order kinetic model. Hydrochars, which exhibit photoluminescence properties, show their characteristic nano and micro-spherical morphology, formed through condensation and polymerization. Hydrochars are mainly amorphous, with a higher crystallinity degree in the centrifuged phase. We observed the presence of sheets of material, maybe graphite: further studies are required to confirm their presence and understand their possible formation.

## Section 2.2

# From hydrothermal carbonization to liquefaction: evolution of compounds along the hydrothermal biomass reaction spectrum

This section is based on our joint research activity with Cornell University (USA), with the groups of Jillian Goldfarb and Jeff Tester. I am grateful to Alex Maag (Worcester Polytechnic Institute) for the precious mentoring and Hanif Sudibyo (Cornell University) for elemental and TOC analysis. Many thanks to Fulbright that financially supported me in the US.

What is reported here is part of a bigger project aimed to deduce some insights on the reaction mechanisms behind the transition from hydrothermal carbonization to liquefaction. I report some preliminary results of the effects of reaction conditions (temperature and time) on hydrochars and their aqueous phases.

### 2.2.1 Introduction and overview

Hydrothermal carbonization (HTC) and liquefaction (HTL) are efficient processing methods to convert biomass into bio-based products and waste valorization applications [72].

Until now, many studies have been carried out to understand the processes, observing the influence of operating parameters on the

conversion of different feedstock and focusing on the optimization of parameters to maximize the yields and qualities of the bio-based products [73]. Probably for simplicity and aims, research has often considered HTC and HTL separately, focusing on the production of the specific products [74]. Indeed, in a temperature spectrum of 180-373°C and pressure of 10-220 bar, the conversion shifts from the progressive carbonization to liquefaction of the biomass, translating in the predominant presence of solid (the hydrochar) to oily phases (the bio-crude). The complexity behind the chemistry is enormous, involving the biomass dissolution, the condensation/polymerization of organics into the solid phase or oily phase, as well as gasification, aromatization, and many other secondary reactions. Due to the vast number of compounds involved, some reaction pathways and reaction kinetics are still unknown [72]. Understanding the fundamental kinetics and reaction chemistry behind the transition between HTC and HTL could help for the overall understanding and for an optimization of the products, enabling to tune operating conditions according to the target. Moreover, kinetics data can be used in building kinetic models, a useful and unavoidable tool for the optimum design of reactors.

Therefore, with the aim of contributing to the fundamentals and kinetics of the hydrothermal conversion, this work encompasses an in-depth kinetic study into the transition from carbonization to liquefaction of three typical biomasses – cellulose, lignin, and wood chips. Studying model biomasses like cellulose and lignin can help to separate the effects of the single compounds, while wood chips, apart being a typical lignocellulosic biomass, in observing their interactions. The experimental method adopted approaches a “real-time” kinetic study, involving the drawing of the reaction slurry during hydrothermal operation. Analyses were performed on the solid and liquid phases (both aqueous and oily) obtained from the treatment of cellulose, lignin, and woodchips over a wide range of temperatures and residence times (220, 250, 280, 300, and 320°C, at 0, ¼, ½, 1 and 2 h). In particular, the composition and thermal stability of the hydrochars and composition of the liquid phase are here reported.



### 2.2.2 Material and methods

#### *Hydrothermal runs*

Hydrothermal runs were performed on microcrystalline cellulose powder (minimum 97% pure), lignin (Sigma Aldrich low sulfonate content alkali), and willow chips. Biomass was mixed with distilled water at a fixed mass biomass to water of 0.2. Then, around 30 g of slurry (biomass+water) was poured into a Parr Hydrothermal Reactor (300 mL, 350°C, 5000 psi). Before every tests, the reactor was flushed three times with pure nitrogen to make the environment inert. During the runs, the mixing was ensured through a stirrer at 300 rpm. Tests were performed at 190, 220, 250, 280, 300, and 320°C. To have a “real time analysis” the liquid phase was sampled at 0, 0.25, 0.5, 1, and 2 h, using a dip pipe with two valves positioned at 20% from the bottom of the reactor.

#### *Solid analysis*

Proximate analysis was performed on a TA Instruments Thermogravimetric Analyzer 5500. Samples (between 2 and 10 mg) were placed into a 90  $\mu$ L alumina crucible, heated in high purity nitrogen 10°C/min up to 110°C and held for 30 min to remove moisture, then at 10°C/min from 110 to 900°C and held for 30 min to determine volatile matter. Samples were then heated from 900 to 950°C under dry air and held for 30 min with loss attributed to fixed (oxidizable) carbon. Residual mass (non-oxidizable inorganics) was referred to as “ash”. The pyrolytic step was used to assess the thermal stability of the samples. The extent of sample converted at any time  $t$ ,  $x(t)$ , was computed as:

$$x(t) = \frac{m_{\text{dry}} - m_t}{m_{\text{dry}} - m_{\text{pyr}}} \quad (2.14)$$

where  $m_{\text{dry}}$  is the mass after the sample is held at 110°C (the start of pyrolysis),  $m_t$  the mass at any time  $t$ , and  $m_{\text{pyr}}$  the residual mass left at the end of the 30 min pyrolysis at 900°C. Derivative conversion curves (DTG) were constructed by plotting  $dx/dt$  versus temperature.

Carbon, hydrogen, and nitrogen contents in hydrochars were measured in triplicate using a CE-440 Elemental Analyzer (Exeter Analytical). To observe the effects of both temperature and time, the severity factor (SF) [3,7], which combines both, was introduced as a function of the residence time (s) and T the reaction temperature as:

$$SF = 50t^{0.2} \cdot e^{\left(\frac{3500}{T}\right)} \quad (2.15)$$

### *Liquid analysis*

The total organic carbon (TOC) of the aqueous phase was measured using an automated Shimadzu TOC-L. The detailed composition of the aqueous phase was investigated through a Shimadzu High-Performance Liquid Chromatography (HPLC) equipped with an Ultra Aqueous C18 Column. The column was operated at 30°C at a total flow rate of 1.5 mL/min. The mobile phase was a solution of 10 mM phosphoric acid at a pH of 2.6 and acetonitrile (90/10 in the first 10 min and 70/10 from 10 to 50 min).

The detection was performed through a UV-Vis Detector (SPD-20A) set at a wavelength of 210 nm. Table 2.7 shows the list of calibrated compounds at their retention time, subdivided into the groups of organic acids, furan, and phenol derivatives.

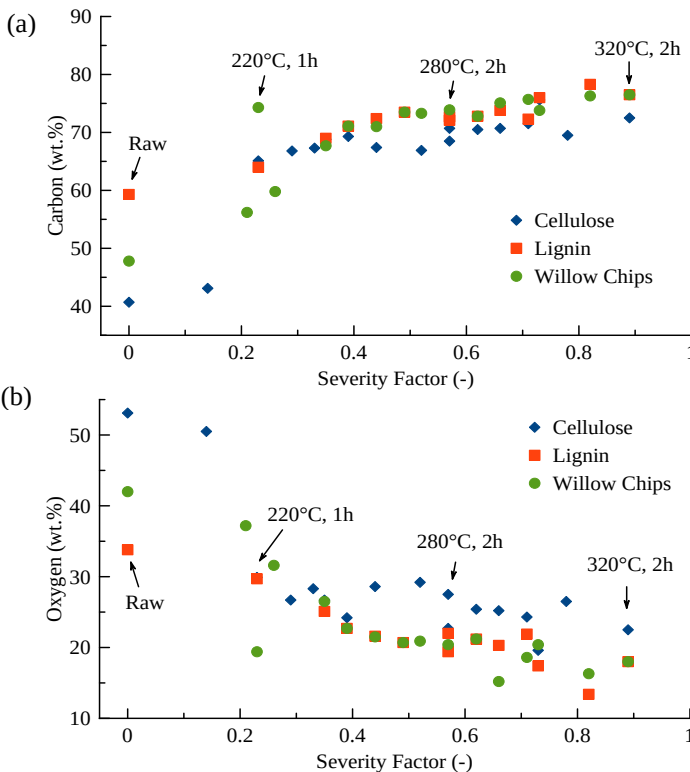
**Table 2.7.** List of compounds calibrated for HPLC analysis.

| RT<br>(min)                           | Compound         | RT<br>(min)    | Compound                 |
|---------------------------------------|------------------|----------------|--------------------------|
| <i>Carboxylic Acids</i>               |                  | <i>Ketones</i> |                          |
| 1.26                                  | Glycolic Acid    | 2.58           | 1,3 Cyclopentanedione    |
| 1.37                                  | Formic Acid      | 4.22           | 2-cyclopenten-1-one      |
| 1.49                                  | Lactic Acid      | 10.04          | 2-Methylcyclopenten-1one |
| 1.67                                  | Propionic Acid   | <i>Phenol</i>  |                          |
| 2.47                                  | Levulinic Acid   | 11.98          | Phenol                   |
| 2.74                                  | Acetic Acid      | 16.12          | 3-Methoxyphenol          |
| 5.65                                  | Butyric Acid     | 18.16          | 2,6 Dimethoxyphenol      |
| <i>Furan and furfural derivatives</i> |                  | 19.73          | 4-ethyl-2-methoxyphenol  |
| 2.36                                  | 2(5H) Furanone   | 22.05          | 2-Methoxy-4-Methylphenol |
| 3.36                                  | HMF              | 32.58          | 2,4 Dimethylphenol       |
| 5.418                                 | Furfural         |                |                          |
| 11.48                                 | 5-Methylfurfural |                |                          |

### 2.2.3 Results and discussion

#### *Evolution of hydrochar composition*

As the severity of the process increases (i.e. temperature and time), biomass undergoes a progressive volatile matter (VM) loss and aromatization. This clearly reflects in a change hydrochar composition, as shown in Figures 2.13 and 2.14. A complete summary of data is reported in the Appendix.



**Figure 2.13.** Effect of the severity factor on the (a) carbon and (b) oxygen contents of hydrochars from cellulose, lignin, and willow chips (standard deviations < 1.5%).

For all the biomasses, as temperature and time increase, VM and the oxygen content progressively decrease, while the fixed (FC) and elemental carbon (C) complementary increase. At the harshest investigated conditions, i.e. 320°C 2 h, the FC contents reach their maximum values of 63.2, 70.8, and 66.5% for cellulose, lignin, and

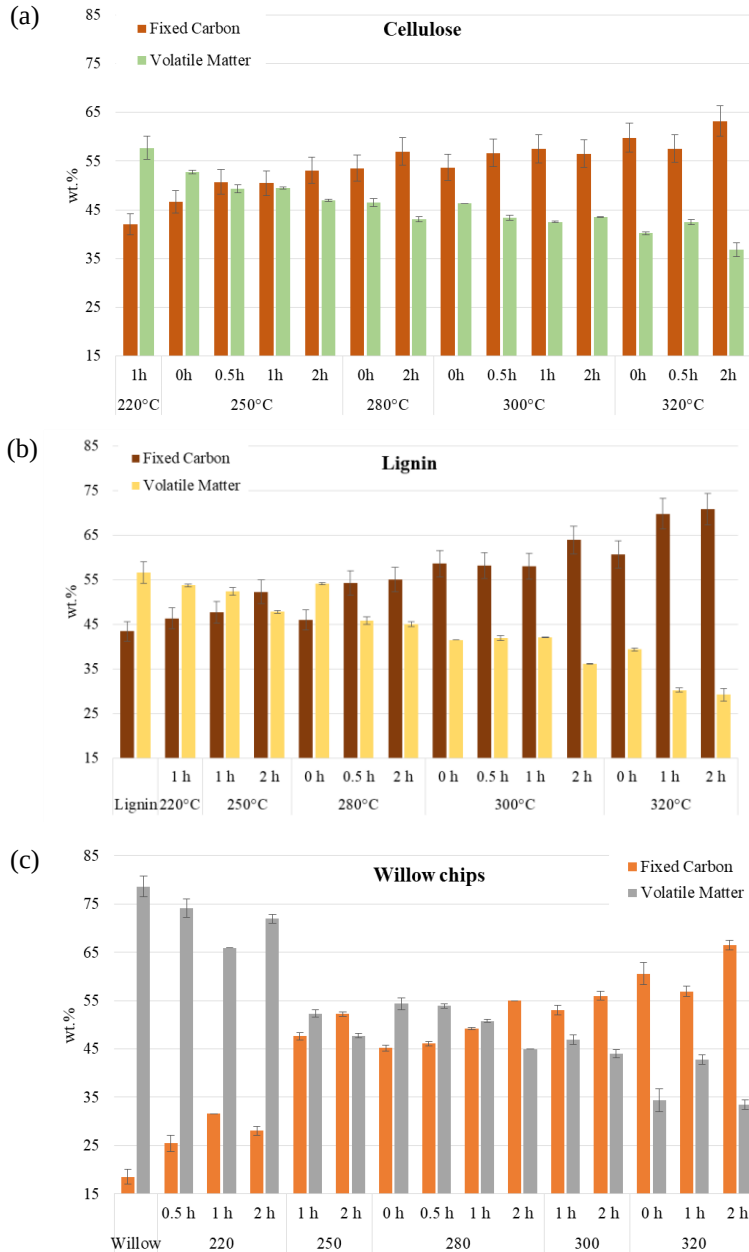
willow chips, respectively. At the same condition, C reaches 72.5, 76.5, and 78.2%, respectively. Indeed, it is well known that a higher process severity favors gasification and liquefaction reactions: temperature triggers decarboxylation/decarbonylation reactions and the formation of less polar compounds with less oxygen content (forming the oily phase), as well as the aromatization of the solid phase [75].

*Cellulose*, below 220°C, does not exhibit significant changes in its composition, as observed also by previous literature [76,77]. Indeed, before 180°C, its structure is practically unaffected [40], and then it slowly starts to depolymerize. This is due to the strong structure of cellulose, characterized by  $\beta$ -(1-4) glycosidic bonds between sugar monomers. Then, between 220 and 250°C, the effect of the HTC is drastic, with a FC that reaches 42.0% versus the initial one at 3.4% and 190°C of 6.2%. Then, it slightly grows oscillating around 56.5% at 300°C and 60% at 320°C. This clearly reflects in the carbon and oxygen contents (Figure 2.13), which, after 220°C, slightly vary at higher severity factors. The residence time has a certain effect on the FC carbon: moving from 0 to 2 h, it increments of 6.4% at 250°C, and between 2.8-3.5% above 250°C. Contextually to the release of VM, the oxygen content decreases (from 53.1 to 22.5%) as well the hydrogen one (from 5.9 to 4.2%), as expected from the formation of a more condensed structure. While oxygen progressively decreases with the severity, hydrogen remains nearly constant around 4.0%. The drastic change below 220°C, and the progressive carbonization of cellulose can find an interpretation in the mechanism proposed in the literature [40,78,79], according to which the hydrothermal conversion of cellulose decomposes through both a hydrolytic and a bulk-pyrolytic decomposition. At low temperatures, cellulose is disrupted into fragments with a nano/macro-size, which “envelope” the material reducing the free interface between the material and the surrounding water, and therefore the possibility of other hydrolysis reactions [61]. Meanwhile, the bulk part of cellulose (not exposed to water) is subjected to thermal degradation similarly to pyrolysis, forming an aromatic network. The presence of these solid-solid reactions and the minor presence of homogenous hydrolysis can explain the slow change in composition after 220°C and the effect of residence time: we can hypothesize that hydrolysis and condensation reactions, being sensitive to temperature, occur already during the heating phase and then let space to a bulk-decomposition, which is much slower to occur.

As cellulose, also *lignin* undergoes both homogenous and bulk reactions. Lignin has a highly aromatic structure, characterized by a complex network of phenolic polymers. This reflects in a high FC and carbon content already in its starting conditions (17.9 and 59.3%, respectively), and in a difficulty in decomposing at low temperatures. Indeed, its proximate composition is basically the same (with a FC of 45.9-47.7%) at 220, 250, and 280°C 0 h, and only longer residence times produce a significant difference. Harsher decomposition occurs above 300°C, and residence time significantly affects the composition (at 2 h, FC is 63.8 and 70.8% at 300 and 320°C, respectively). Also, at 320°C, there is a significant improvement in the carbon percentage (achieving 78.3%) and a certain decrement in the hydrogen content (4.5%), before undetected. The difficulty in decomposing lignin has been observed in the literature [77,80] and can be explained by the complex aromatic structure of lignin, which is difficult to hydrolyze and that needs harsher conditions to cleavage ether and carbon-carbon bonds into phenolic compounds [7,81]. Meanwhile, high temperatures favor solid-solid reactions, which increase the level of aromaticity. Together with these, high temperatures also favor the polymerization of phenolic compounds on the hydrochar surface, an aspect that could contribute to the high carbon content at 300 and 320°C with the solid-solid reactions, increasing the level of aromaticity. This re-polymerization phenomenon is well-known (and unwanted) in supercritical conditions, at which the liquefaction is fast but also the re-polymerization of phenolic compounds [81].

Meanwhile, *willow chips* show an intermediate behavior, containing around 36% of cellulose, 25% of lignin, and 33% of hemicellulose [82,83]. 220°C already causes certain carbonization of the feedstock (C passes from 47.8 to 56-59% and a FC from 18.5 to 25-30%), but with a lower degree than cellulose and lignin. This can be explained by the presence of hemicellulose that, due to its random non-crystalline structure and lack of repeating  $\beta$ -(1-4) glycosidic bonds, easily degrade into sugars already at low temperatures [84]. Previous studies [85–88] reported that hemicellulose completely or mostly degrades even at low hydrothermal conditions (with no evidence of unreacted hemicellulose inside the hydrochars), indicating that it mostly decomposes in solution in sugars derivatives that partially re-polymerize on the hydrochar. Therefore, these re-polymerized products, characterized by a high VM content, decrease the average “potential” FC carbon content and limit

the carbonization of cellulose and lignin by enveloping the material. Then, from 250°C, the process severity starts to be more important, causing a progressive increment of FC and C content, with a similar composition of hydrochars from cellulose and lignin.



**Figure 2.14.** Fixed carbon (FC) and volatile matter (VM) contents of hydrochars from (a) cellulose, (b) lignin, (c) and willow chips. Data on raw cellulose are not reported for graphical reasons (FC=3.4%, VM=96.6%).

### *Thermal behavior of hydrochars*

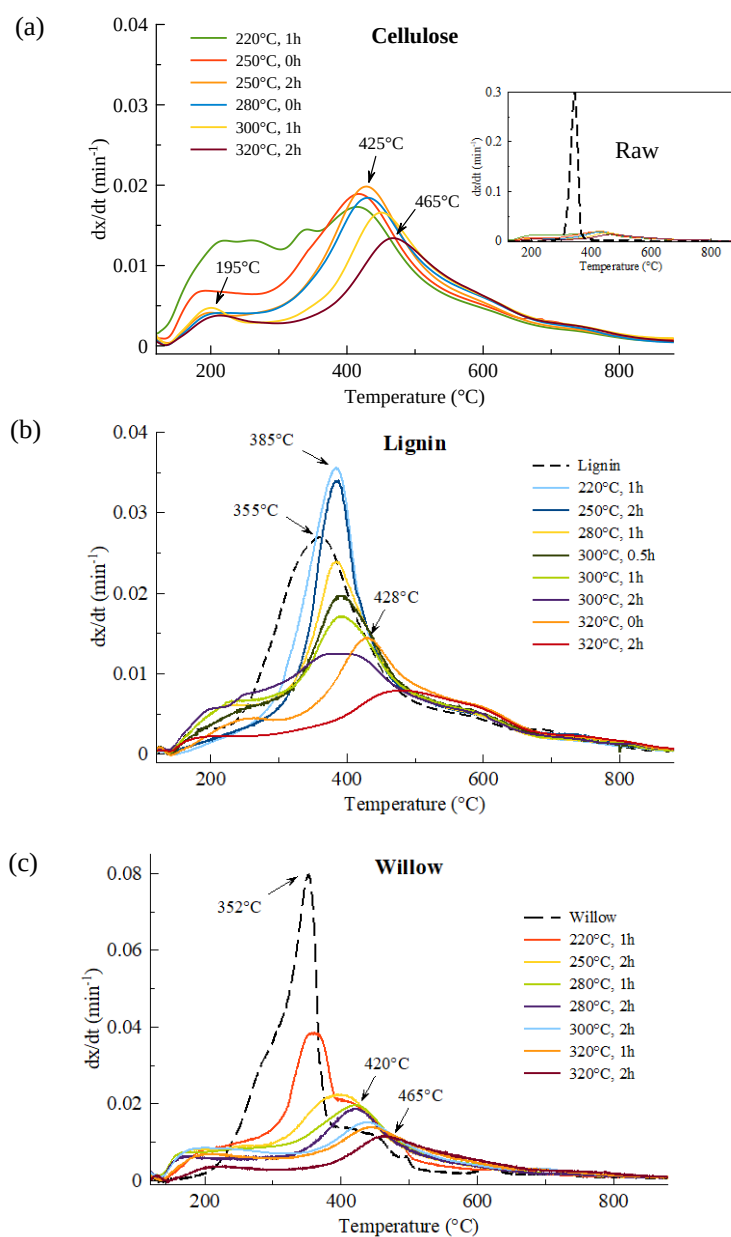
Figures 2.15(a-c) show the derivative conversion curves (DTG) under pyrolysis of some selected hydrochars. The high reactivity of raw willow compared to cellulose and lignin is due to the presence of hemicellulose.

As expected from their composition, hydrochars from lignin at 250°C present a similar profile to that of lignin alone and a higher reactivity than those from cellulose and willow. Indeed, 250°C is not harsh enough to significantly degrade the aromatic structure of lignin.

Then, thanks to the loss in VM and progressive aromatization, the severity improves the thermal stability of hydrochars, which exhibit a decomposition peak shifted towards higher temperatures and characterized by a lower intensity. For example, raw cellulose shows the main decomposition peak at 345°C, while its hydrochars are between 415 and 465°C. Similarly, lignin and willow decompose at 355°C, while their hydrochars up to around 460-500°C. Hydrochars from lignin and willow produced above 300°C show a flat profile, approaching that of low-rank coal.

Interestingly, all the profiles, even the most carbonized with a low VM content, show an early decomposition shoulder in a range of 160-220°C. Since the raw biomasses do not show this decomposition stage, we can attribute it to the volatilization of re-polymerized products derived from the aqueous phase (the so-called “secondary char”). In the case of cellulose, this secondary char derives from the polymerization and condensation of HMF produced from the dehydration of glucose produced from the cleavage of  $\beta$ -(1-4) glycosidic bonds forming cellulose [40]. Its presence is particularly noticeable for hydrochars obtained under the mildest conditions (220°C and 250°C 1h), at which aromatization phenomena are weaker than harsher conditions. Meanwhile, the secondary char present in lignin hydrochars derives from the re-condensation/crosslinking of fragments deriving from the hydrolysis of ether bonds and alkylation of the aromatic structures [89,90]. Similar considerations can be done for hydrochars from willow, with the addition of polymerized products formed from

hemicellulose and those deriving from the interaction among the different compounds.



**Figure 2.15.** DTG curves of hydrochars from (a) cellulose, (b) lignin, and (c) willow chips, under pyrolytic conditions.

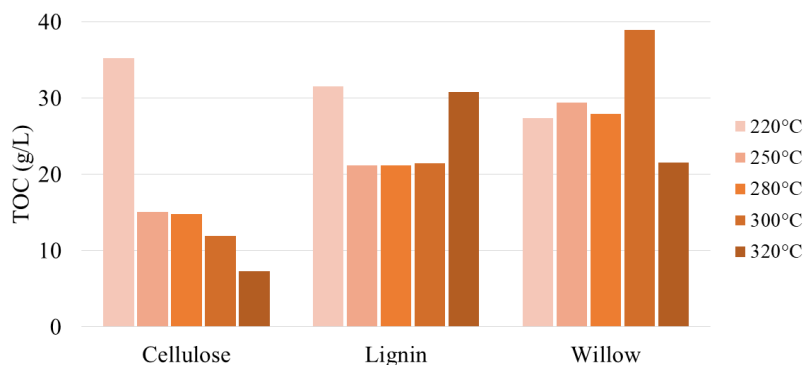


### Total organic carbon (TOC) in the process water

Figure 2.16 shows the TOC content in the aqueous phase from the different feedstock at different temperatures.

The TOC of cellulose decreases with the hydrothermal severity: it rapidly decreases from 35.2 to 21.0 g/L moving from 220 to 250°C, respectively, and then progressively decreases until 7.3 g/L at 320°C 2h. At 220°C, the carbon dissolved in the aqueous phase represents around 17.6% of cellulose ( $g_{\text{TOC}}/g_{\text{bio}}$ ), while above 250°C always less than 10.5%. A similar decreasing trend was observed by Becker et al. [91], and can be explained by the presence of soluble products (like HMF) resulting from the hydrolysis of cellulose, that polymerize under harsher conditions (forming the secondary char) mainly  $\leq 250^\circ\text{C}$ . Then, higher temperatures favor gasification reactions, which contribute to decrease the TOC.

Meanwhile, the TOC of lignin and willow is mostly stable with temperature, with ranges of 21.1–31.5 g/L (10.6–15.8% of lignin) and 21.5–38.9 g/L (10.8–19.5% of willow), respectively. Fluctuations are due to the balance between the release of organics in the liquid phase (favored at high temperature) due to the aromatic structure of lignin, re-polymerization of dissolved organics (phenolic compounds for lignin and phenolic plus sugar monomers for willow), and a progressive release of carbon in the gas phase.



**Figure 2.16.** Effect of temperature on TOC of the aqueous phase from hydrothermal conversion of cellulose, lignin, and willow (standard deviations < 2%), at 2 h.

*Evolution of compounds in the aqueous phase*

Looking at the evolution of the concentration of selected compounds in the aqueous phase can help to understand the process chemistry. The detected compounds were divided into four families: carboxylic acids, furans, ketones, and phenols. As shown in Figure 2.17, the aqueous phase from the hydrothermal processing of cellulose mainly consists of carboxylic acids, followed by furans (mainly HMF), and small fractions of ketones and phenols. Meanwhile, lignin shows a high presence of phenol derivatives (Figure 2.18), while willow a more variegate composition resembling its lignocellulosic nature, with higher concentration in carboxylic acids, furans, and phenols (Figure 2.19). Table 2.8 shows the distribution among the compounds divided by categories.

*Carboxylic acids* are produced from the degradation of monomers deriving from the hydrolysis of cellulose and hemicellulose, like fructose, galactose, mannose, and xylose (this last dominant in hemicellulose) [7]. At the end of the reaction (2 h), as reported in Table 2.8, they reach 1.8 g/L for cellulose and 5.6 g/L for willow. In particular, in willow chips, carboxylic acids occupy the highest fraction of detected compounds of the TOC, with values in a range of 11.0-25.8% (vs cellulose 5.2-23.4% and lignin 1.9-3.0%). This aspect is due to the overlapping and interaction between sugars deriving from cellulose and hemicellulose, both present in willow. Meanwhile, in lignin, carboxylic acids occupy a very small concentration, always lower than 0.7 g/L. This is due to the poor presence of sugar derivatives from lignin. Glycolic acid is highly present in both cellulose and willow around 0.6-1.3 g/L and 2.5-3.9 g/L, respectively. Glycolic acid derives from the oxidation of intermediates (like pyruvaldehyde) produced through the dehydration of fructose derived from the cleavage of carbohydrates from cellulose/hemicellulose [92] (indeed, it is absent in lignin compounds). Glycolic acid decreases with temperature due to its decarboxylation into sub-products like methanol. Similarly, lactic acid degrades to ethanol through decarboxylation at high temperatures and dehydrates to propionic acid [93], which indeed slightly increases with temperature. Butyric acid is present in all the feedstocks, especially in willow chips (arriving up 4.2 g/L at 320°C 2h). Its presence was also measured by Becker et al. [91]. Formic acid derives from the rehydration of HMF, and it was observed both in cellulose and lignin,

while it is mostly absent in willow, where it probably degrades to acetic acid, instead highly present. Similarly, levulinic acid derives from the rehydration of HMF and decreases with temperature, indicating that is consumed for other reactions. These could include that with hydrochar: indeed, it is known that levulinic acid can take part in the formation of the solid phase by being embedded inside it as a co-polymerized compound [48]. Acetic acid was measured in all the feedstock, and can derive from (1) fructose fragmentation, involving decarboxylation or decarbonylation of lactic acid produced from the oxidation of acetaldehyde derived from the glucose and (2) oxidation of HMF and furfurals [37,42,69]. Even if the oxidation of HMF was demonstrated to play a minor role compared to fructose fragmentation, its partial oxidation could contribute to the decrement in temperature together with decarboxylation and decarbonylation reactions.

*Furan derivatives* were measured in all the feedstocks and are mainly composed of HMF, furfural, and 5-methylfurfural. Being intermediates of reaction, their concentration is higher at low temperature and they show an increasing-decreasing trend during the time (in the case of cellulose, they reach their maximum at 1 h), as clearly shown in Figure 2.17c. HMF forms through direct dehydration of glucose or aldose-ketose isomerization of glucose to fructose and subsequent dehydration [43,44,94]. Then, it can re-polymerize in the solid phase, forming the “secondary char” and rehydrate into levulinic and formic acid. Meanwhile, furfural is produced by dehydration of pentose sugars [84]. The decline through the process is slower compared to HMF and indicates slightly higher stability. The absence of furan derivatives in willow could be due to their re-condensation into nuclei (then forming the secondary char), catalyzed by formic acid. Anyway, at 320°C, they are absent also for cellulose. Similar behavior was observed by Becker et al. [91], with no evidence of furfural above 270°C 6 h.

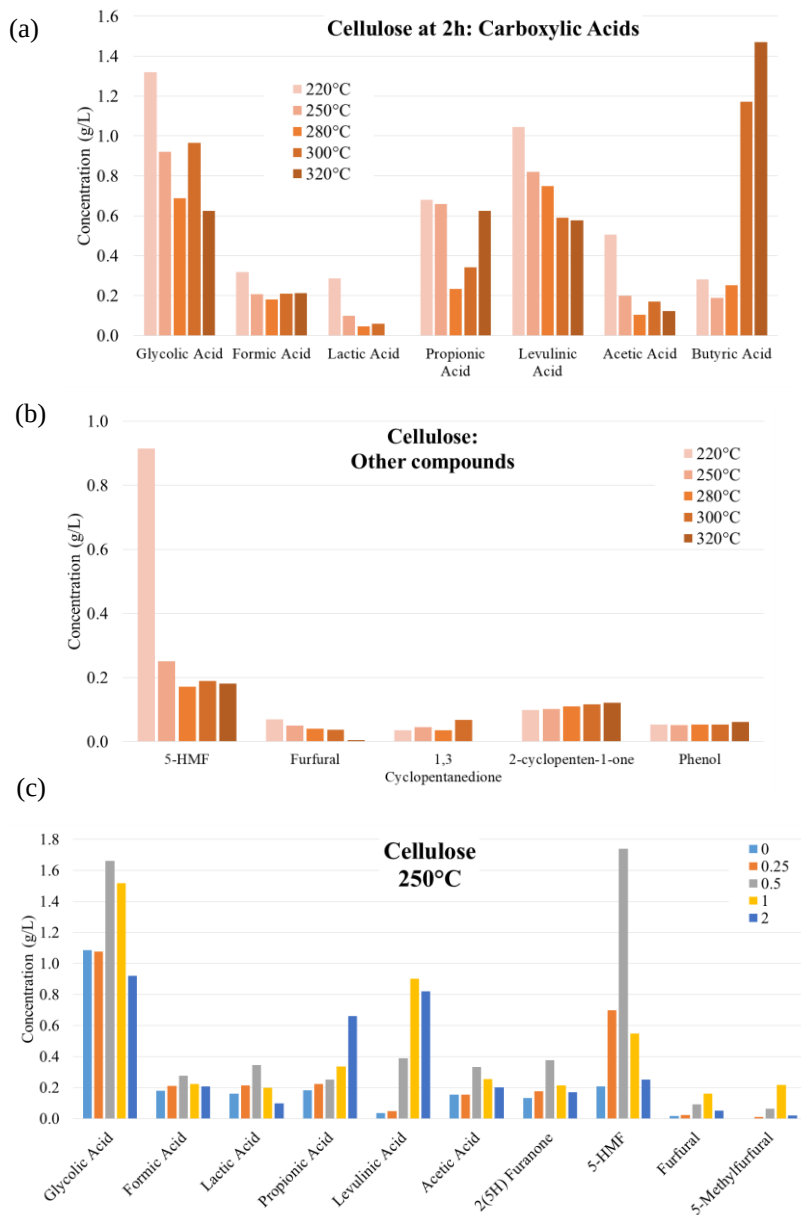
*Phenolic compounds* are the main products of the decomposition of lignin, with concentrations ranging from 0.9 to 2.2 g/L, representing up to 9.7% of the TOC. This is in clear agreement with the nature of lignin, which is characterized by a complex network of phenolic polymers, three-dimensional and non-crystalline [81]. All the phenol compounds concentrations increase as the temperature increases, which triggers the cleavage of the aromatic structure of lignin. In particular, phenol derives from the hydrolysis of lignin through the cleavage of  $\beta$ -O-4 ether bonds and C-C bond, while other phenolic compounds through

demethoxylation and alkylation reactions [81]. Phenol derivatives are present in small quantities also in willow chips and cellulose, deriving from the degradation of furfurals and aldehydes/unsaturated intermediates [93,95,96].

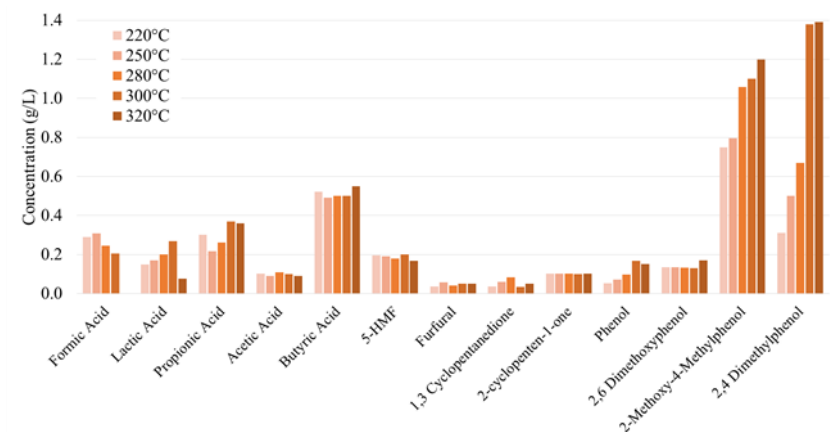
Even with smaller fractions, *ketones* were also detected in all the feedstock, showing similar concentrations (around 0.11 g/L, occupying always less than 1.3% of TOC). 1,3 cyclopentanedione and 2-cyclopenten-1-one (C<sub>5</sub>-C<sub>6</sub>) were the most present, mostly constant with temperature and time.

**Table 2.8.** Contribution of main detected families of compounds at 2h for the different feedstocks. Data are expressed as g/L and percentage of the total organic carbon (TOC) of the aqueous phase.

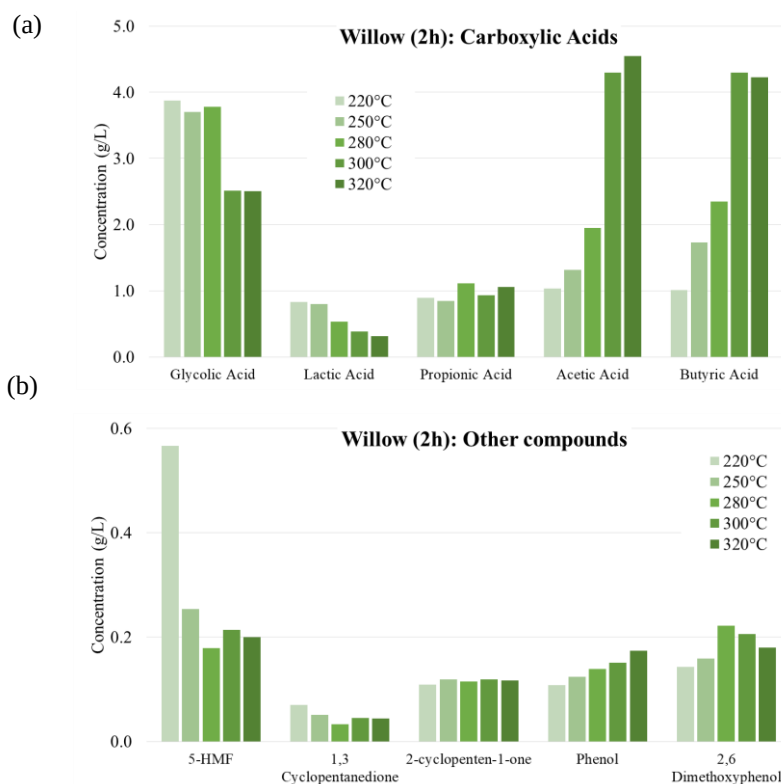
|                  | Carboxylic Acids |      | Furans |     | Ketones |     | Phenolic compounds |     |
|------------------|------------------|------|--------|-----|---------|-----|--------------------|-----|
|                  | g/L              | %    | g/L    | %   | g/L     | %   | g/L                | %   |
| <i>Cellulose</i> |                  |      |        |     |         |     |                    |     |
| 220°C            | 1.84             | 5.2  | 0.97   | 2.8 | 0.11    | 0.3 | 0.32               | 0.9 |
| 250°C            | 1.31             | 8.7  | 0.27   | 1.8 | 0.12    | 0.8 | 0.04               | 0.3 |
| 280°C            | 0.96             | 6.5  | 0.12   | 0.8 | 0.11    | 0.8 | 0.04               | 0.3 |
| 300°C            | 1.56             | 13.1 | 0.21   | 1.8 | 0.15    | 1.3 | 0.36               | 3.0 |
| 320°C            | 1.70             | 23.4 | 0.11   | 1.5 | 0.09    | 1.3 | 0.28               | 3.8 |
| <i>Lignin</i>    |                  |      |        |     |         |     |                    |     |
| 220°C            | 0.61             | 1.9  | 0.22   | 0.7 | 0.11    | 0.3 | 0.91               | 2.9 |
| 250°C            | 0.56             | 2.6  | 0.24   | 1.1 | 0.14    | 0.7 | 1.10               | 5.2 |
| 280°C            | 0.59             | 2.8  | 0.13   | 0.6 | 0.15    | 0.7 | 1.44               | 6.8 |
| 300°C            | 0.65             | 3.0  | 0.40   | 1.9 | 0.10    | 0.5 | 2.08               | 9.7 |
| 320°C            | 0.64             | 2.1  | 0.17   | 0.6 | 0.12    | 0.4 | 2.17               | 7.0 |
| <i>Willow</i>    |                  |      |        |     |         |     |                    |     |
| 220°C            | 3.02             | 11.0 | 0.55   | 2.0 | 0.14    | 0.5 | 0.19               | 0.7 |
| 250°C            | 3.37             | 11.4 | 0.14   | 0.5 | 0.13    | 0.5 | 0.22               | 0.7 |
| 280°C            | 4.01             | 14.3 | 0.10   | 0.4 | 0.13    | 0.5 | 0.54               | 1.9 |
| 300°C            | 5.45             | 14.0 | 0.12   | 0.3 | 0.13    | 0.3 | 0.64               | 1.6 |
| 320°C            | 5.56             | 25.8 | 0.11   | 0.5 | 0.13    | 0.6 | 0.27               | 1.3 |



**Figure 2.17.** Evolution of main compounds in the aqueous phase from cellulose: (a-b) effect of temperature at 2 h and (c) of time at 250°C.



**Figure 2.18.** Evolution of main compounds in the liquid phase from the hydrothermal conversion of lignin, at 2 h.



**Figure 2.19.** Evolution of main compounds in the aqueous phase from the hydrothermal conversion of willow chips, at 2 h.

### 2.2.4 General considerations on the engineering of hydrochars

Overall, hydrochars show an improvement in their thermal stability with the severity of the process (time and temperature), with a shift towards higher temperatures and an intensity reduction of the decomposition, and carbon content. Since the ash content is approximately constant, hydrochars result in more energy-dense, appearing more appealing for possible coal co-combustion applications. However, thermogravimetric curves highlight a not negligible high reactivity at moderate temperatures (160-250°C), attributed to the secondary char phase resulting from re-polymerization and condensation reactions from the liquid phase back onto the solid. Its presence diminishes at higher processing temperatures, raising a question mark on the effective usage of hydrochar as a fuel. Indeed, this high reactivity makes the hydrochar deviate from standard coal, which exhibits a stable and constant thermal decomposition behavior. Moreover, secondary char can have several origins: it can from carbohydrates or phenolic compounds (as in our study), but in oily-rich substrates (like organic wastes), it can consist of fatty acids, which still confer high reactivity to the material. Therefore, it seems very rare to have a substrate immune from its presence. Therefore, to move towards fuel co-combustion applications, we should probably consider post-treatments on the hydrochar, like thermal desorption or chemical extractions. We will expand this concept in chapter 5, post-treating hydrochars from organic waste through post-treatments.

### 2.2.5 Conclusions and next steps

This work is part of a bigger project dealing with the kinetics involved during the transition from carbonization to liquefaction of three typical biomasses (cellulose, lignin, and wood chips). The experimental method adopted approaches a “real-time” kinetic study, with analyses performed on the solid and liquid phases (both aqueous and oily), spilled during hydrothermal operation. To investigate kinetics, a wide range of temperatures and residence times (220, 250, 280, 300, and 320°C, at 0, ¼, ½, 1, and 2 h) were investigated.

Sampling at points during the reaction with immediate liquid analysis through HPLC and TOC techniques enabled us to follow the evolution

of a variety of carboxylic acids, furans, phenol derivatives, and ketones to provide insight into existing hydrothermal reaction pathways. For example, as temperature increases, organic acids and furans progressively give way to a multitude of ketones and phenol-derivatives, while HMF reaches its maximum at around ½ h of residence time at 250 and 280°C.

As abovementioned, this report part of a bigger project. Next stages include:

- (1) Analysis of the collected bio-crude through GC-MS techniques. This will ensure a complete overview of the decomposition and the fate of liquid compounds.
- (2) Building reaction pathways starting from the collected data and previous literature, as well as computing kinetic parameters (like the reaction rate) on the base of time-evolution data.

## 2.2.6 Appendix

**Table A1.** Effect of operating conditions on proximate and ultimate analysis of hydrochars from (a) cellulose, (b) lignin, (c) and willow chips.

(a) Cellulose

| T<br>(°C) | Time<br>(h) | Proximate<br>Analysis (wt.%) |      |     | Ultimate Analysis<br>(wt.%) |     |     |      |
|-----------|-------------|------------------------------|------|-----|-----------------------------|-----|-----|------|
|           |             | FC                           | VM   | Ash | C                           | H   | N   | O*   |
| Cellulose |             | 3.4                          | 96.6 | 0.0 | 40.1                        | 6.3 | 0.3 | 53.3 |
| 190       | 1           | 6.2                          | 93.8 | 0.0 | 43.1                        | 6.2 | 0.3 | 50.5 |
| 220       | 1           | 42.0                         | 57.7 | 0.0 | 65.1                        | 4.8 | 0.2 | 29.9 |
| 250       | 0           | 46.7                         | 52.8 | 0.0 | 66.8                        | 4.4 | 2.1 | 26.7 |
|           | 0.5         | 50.7                         | 49.3 | 0.0 | 67.3                        | 4.1 | 0.3 | 28.3 |
|           | 1           | 50.5                         | 49.5 | 0.0 | 68.8                        | 4.3 | 0.2 | 26.7 |
|           | 2           | 53.1                         | 46.9 | 0.0 | 69.3                        | 4.4 | 2.1 | 24.2 |
| 280       | 0           | 53.5                         | 46.5 | 0.0 | 67.4                        | 3.7 | 0.2 | 28.6 |
|           | 1           | 53.1                         | 46.9 | 0.0 | 66.9                        | 3.6 | 0.3 | 29.2 |
|           | 2           | 57.0                         | 43.0 | 0.0 | 68.5                        | 3.7 | 0.3 | 27.5 |
| 300       | 0           | 53.7                         | 46.3 | 0.0 | 70.7                        | 4.3 | 2.3 | 22.7 |
|           | 0.5         | 56.6                         | 43.4 | 0.0 | 70.5                        | 3.8 | 0.3 | 25.4 |
|           | 1           | 57.5                         | 42.5 | 0.0 | 70.7                        | 3.8 | 0.3 | 25.2 |
|           | 2           | 56.5                         | 43.5 | 0.0 | 71.5                        | 4.0 | 0.2 | 24.3 |
| 320       | 0           | 59.8                         | 40.2 | 0.0 | 75.6                        | 4.5 | 0.3 | 19.6 |
|           | 0.5         | 57.6                         | 42.4 | 0.0 | 69.5                        | 3.7 | 0.2 | 26.5 |
|           | 2           | 63.2                         | 36.8 | 0.0 | 72.5                        | 4.2 | 0.8 | 22.5 |



## (b) Lignin

| T<br>(°C) | Time<br>(h) | Proximate<br>Analysis (wt. %) |      |     | Ultimate Analysis<br>(wt. %) |     |     |      |
|-----------|-------------|-------------------------------|------|-----|------------------------------|-----|-----|------|
|           |             | FC                            | VM   | Ash | C                            | H   | N   | O*   |
|           | Lignin      | 43.4                          | 56.6 | 0.0 | 59.3                         | 5.9 | 1.0 | 33.8 |
| 220       | 1           | 46.3                          | 53.7 | 0.0 | 64.0                         | 5.4 | 0.9 | 29.7 |
| 250       | 1           | 47.7                          | 52.3 | 0.0 | 69.0                         | 5.3 | 0.6 | 25.1 |
|           | 2           | 52.3                          | 47.7 | 0.0 | 71.1                         | 5.4 | 0.7 | 22.7 |
| 280       | 0           | 45.9                          | 54.1 | 0.0 | 72.4                         | 5.2 | 0.8 | 21.6 |
|           | 0.5         | 59.4                          | 40.6 | 0.0 | 73.5                         | 4.9 | 0.9 | 20.7 |
|           | 2           | 55.0                          | 45.0 | 0.0 | 72.1                         | 5.2 | 0.8 | 22.0 |
| 300       | 0           | 58.5                          | 41.5 | 0.0 | 72.7                         | 5.5 | 2.4 | 19.4 |
|           | 0.5         | 54.3                          | 45.7 | 0.0 | 72.8                         | 5.2 | 0.8 | 21.2 |
|           | 1           | 58.0                          | 42.0 | 0.0 | 73.8                         | 5.0 | 0.8 | 20.3 |
|           | 2           | 63.8                          | 36.2 | 0.0 | 72.3                         | 5.2 | 0.7 | 21.9 |
| 320       | 0           | 60.6                          | 39.4 | 0.0 | 76.0                         | 4.6 | 2.0 | 17.4 |
|           | 1           | 69.8                          | 30.2 | 0.0 | 78.3                         | 4.6 | 3.8 | 13.4 |
|           | 2           | 70.8                          | 29.2 | 0.0 | 76.5                         | 4.5 | 1.0 | 18.0 |

## (c) Willow chips

| T<br>(°C) | Time<br>(h)  | Proximate<br>Analysis (wt. %) |      |     | Ultimate Analysis<br>(wt. %) |     |     |      |
|-----------|--------------|-------------------------------|------|-----|------------------------------|-----|-----|------|
|           |              | FC                            | VM   | Ash | C                            | H   | N   | O*   |
|           | Willow Chips | 18.5                          | 78.6 | 2.9 | 47.8                         | 5.9 | 1.4 | 42.0 |
| 220       | 0.5          | 25.5                          | 74.2 | 0.3 | 56.2                         | 5.2 | 1.5 | 37.2 |
|           | 1            | 31.6                          | 65.9 | 2.4 | 74.3                         | 3.9 | 2.4 | 19.4 |
|           | 2            | 28.0                          | 72.0 | 0.0 | 59.8                         | 5.6 | 3.0 | 31.6 |
| 250       | 1            | 43.6                          | 56.4 | 0.0 | 67.7                         | 5.2 | 0.6 | 26.5 |
|           | 2            | 42.8                          | 55.9 | 1.2 | 71.1                         | 5.4 | 0.7 | 22.7 |
| 280       | 0            | 45.2                          | 54.4 | 0.4 | 71.0                         | 5.2 | 2.3 | 21.5 |
|           | 0.5          | 46.1                          | 53.9 | 0.0 | 73.5                         | 4.9 | 0.9 | 20.7 |
|           | 1            | 46.2                          | 53.8 | 0.0 | 73.3                         | 5.2 | 0.5 | 20.9 |
|           | 2            | 55.0                          | 45.0 | 0.6 | 73.9                         | 5.1 | 0.5 | 20.4 |
| 300       | 0.5          | 54.3                          | 45.7 | 0.0 | 72.8                         | 5.2 | 0.8 | 21.2 |
|           | 1            | 53.1                          | 46.9 | 0.3 | 75.1                         | 5.1 | 4.6 | 15.2 |
|           | 2            | 56.0                          | 44.0 | 0.3 | 75.7                         | 5.1 | 0.6 | 18.6 |
| 320       | 0            | 60.6                          | 34.4 | 0.0 | 73.8                         | 5.2 | 0.6 | 20.4 |

|   |      |      |     |      |     |     |      |
|---|------|------|-----|------|-----|-----|------|
| 1 | 56.9 | 42.8 | 0.3 | 76.3 | 4.8 | 2.5 | 16.3 |
| 2 | 70.8 | 29.2 | 0.0 | 76.5 | 4.5 | 1.0 | 18.0 |

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## Section 2.3

# Exothermicity of hydrothermal carbonization: determination of heat profile and enthalpy of reaction via high-pressure differential scanning calorimetry

This section is based on the article that we published in *Fuel* (2022, vol. 310, 122312, by G. Ischia, M. Cazzanelli, L. Fiori, M. Orlandi, A. Miotello, <https://doi.org/10.1016/j.fuel.2021.122312>). The work approaches the thermodynamics of hydrothermal carbonization by determining the entity of exothermicity involved in the reactions. To achieve this, we used a high-pressure differential scanning calorimetry. The research was performed at the University of Trento. I would like to thank the precious support of prof. Claudio Della Volpe (University of Trento) for the fruitful discussion, Nicola Bazzanella and Claudio Cestari (University of Trento) for the technical support, and Marco Baratieri's group (Free University of Bolzano) for the fruitful discussions.

### 2.3.1 Abstract

Hydrothermal carbonization (HTC) is known for being exothermic, an aspect that can lower the external energy demand necessary to face thermal losses and therefore be conducive to the overall HTC process thermal efficiency. Also, heat profiles can help explore kinetics and generally provide an insight into the HTC process. Despite the potential of the HTC exothermicity, only a few studies examined it and with different methodologies. This work aims to deepen this aspect by exploring heat profiles, enthalpy of reaction, and comparing two experimental methods by performing an experimental campaign using a power-compensated high-pressure differential scanning calorimeter (DSC). Three substrates (glucose, cellulose, and grape seeds) were hydrothermally carbonized at 180, 200, 220, and 250°C at a biomass to water mass ratio of 0.25.

To determine net heat profiles and enthalpies, water thermal effects were removed using two approaches, consisting of running a thermal program on the water alone or the exhaust, i.e. the products after the reaction. An instrument baseline considering both a zero-line and a time-drift was considered. Results show that both methods allow inferring similar enthalpy values at low operating temperature, while at high temperature a difference in the heat capacity of the liquid phase (due to the production of compounds dissolved in the liquid phase and absorbed on the solid) can affect the results related to the exhaust approach. In both cases, higher temperatures favor the exothermic peak intensity, the curve sharpening, and the increase (in absolute value) of reaction enthalpies. These are in the ranges  $-0.20 \div -0.86$  MJ/kg,  $-0.61 \div -1.33$  MJ/kg, and  $-0.42 \div -1.53$  MJ/kg for glucose, cellulose, and grape seeds, respectively.

### 2.3.2 Introduction

Hydrothermal carbonization (HTC) is an attractive and increasingly popular thermochemical process to valorize biomasses into carbonaceous materials, energy, or chemical products [97,98]. HTC occurs in liquid water at 180-250°C and at self-generated pressure of 10-50 bar, which are relatively mild operating conditions compared with other thermochemical processes like pyrolysis or gasification. Thanks to the solvent action of water at mild temperature and pressure,

biomass undergoes a complex network of reactions (including hydrolysis, dehydration, decarboxylation, polymerization, and condensation) that mimic the natural coalification process [99–101]. Since it occurs in water, HTC particularly suits the treatment of wet substrates with a water mass content up to 70-90% that would otherwise require expensive pre-treatment stages [102]. Literature spans a variety of biomass waste like organic wastes [23], sewage sludge [103], food residues [104,105], algae [106], digestate [107,108], and cow manure [109], but also niche substrates like grape seeds [110], coconut shells [111], tobacco or banana stalks [112,113]. The carbonaceous phase, the so-called hydrochar, can be used as fuel in co-combustion thanks to its high carbon content, high heating value, and good grindability and homogeneity [19,20]. Besides, it can be used for a variety of applications, like as adsorbent [114], soil improver [115], or base for activated carbons [116]. Meanwhile, HTC has been investigated also for the synthesis of advanced carbonaceous materials from simple saccharides, like glucose or starch [7,117].

Until now, research mainly focused on understanding the effects of process parameters on the final products and their possible applications based on results obtained using lab-scale reactors [52]. Overall, there is a certain lack of research dealing both with the fundamentals and the modeling of HTC, whose complete knowledge is hampered by the complexity of the reaction mechanisms involved. In particular, a little-studied field is that of heat profiles and reaction enthalpies. HTC is known to be slightly exothermic, involving the release of a certain quantity of heat. The knowledge of the magnitude and the trends of the heat profiles can be beneficial both for the overall understanding of the HTC mechanisms and for accurate energy analysis of the process. Moreover, trends of heat profiles can be a base to develop kinetic models [118]. First evaluations of the enthalpy of reactions were based on the application of Hess's law using the yields and enthalpies of formation of reactants and products (calculated from the higher heating values) [119–122]. However, to reach a high precision, this method requires knowing all the reaction products, which is a challenging task especially for the liquid phase, and their relative yields. These points are not immediate to define and often are addressed by making strong simplifying assumptions. Therefore, this method provides only a general indication of the enthalpy of reactions. Meanwhile, several authors used a lab-scale reactor to measure the enthalpy change, but the

apparatus lacks precision and does not allow the time-evolution measure [123,124].

In this context, the usage of a differential scanning calorimeter (DSC) can provide more reliable and detailed measures. In particular, Funke and Ziegler [125,126] performed the first measures using a DSC coupled with high-pressure crucibles, used as a traditional reactor where HTC occurs. The authors also proposed an in-depth study on the experimental methodology, but high uncertainties affect the results. Then, other authors performed similar measures [127,128] but always recording high deviations. Only recently, Pecchi et al. [118,129] systematically investigated HTC through a high-pressure DSC and obtained reproducible data that were used to calibrate a higher-order kinetic model. Overall, these studies differ from each other in the methodology adopted. Differences involve the method used to remove the instrument baseline and the sensible heat of water, which occupies a relevant fraction in the overall energy balance. As explained by Pecchi et al. [129], three methods can be used to remove the water effect: (1) using the reference crucible filled with water (and therefore obtaining immediately a net curve); (2) running an independent run on water alone performing the same thermal program; (3) running a second run on the exhaust material and considering this instead of the water one. Until now, authors have independently adopted only one method at a time, without comparing them using the same instrument.

In this framework, this work proposes the HTC of different biomasses (glucose, cellulose, and grape seeds) and operating temperatures (180, 200, 220, and 250°C) using a high-pressure DSC. Two methods were adopted to remove the effects of water, consisting of running a second thermal program (1) on a crucible containing water alone and (2) on the exhaust material consisting of residues of the first HTC run. Then, net heat profiles and reaction enthalpies were determined using a baseline consisting of a zero-line and a time-drift contribution. The two methods were compared with each other and with results available in the literature. Therefore, this work presents a critical examination of what is available in the literature and proposes which is the best method for measuring thermal profiles and HTC reaction enthalpy. Then, it also presents the reaction enthalpy values for less or more complex biomasses: from simple glucose to its cellulose polymer (key biopolymer in plant biomass), to a "real" lignocellulosic biomass

represented by grape seeds, thus covering a relatively wide range of biomass.

### 2.3.3 Materials and methods

#### *Starting feedstock*

The biomasses investigated are glucose, cellulose, and grape seeds. Pure glucose and microcrystalline cellulose (Sigma Aldrich 50  $\mu\text{m}$ ) were chosen as model compounds of biomass. Meanwhile, grape seeds, deriving from the wet sieving of grape marc and provided by an alcohol producer company in Italy, represent agro-industrial waste. Biomasses were dried in a 105°C ventilated oven overnight, and grape seeds were also ground. Table 2.8 shows the sum of their main properties.

**Table 2.8.** Biomass properties, from literature, per unit mass of dry biomass.

| Analysis    |                       | Glucose<br>[130] | Cellulose<br>[76] | Grape Seeds<br>[110,131] |
|-------------|-----------------------|------------------|-------------------|--------------------------|
| Proximate   | Volatile Matter (wt%) | -                | 97.7              | 69.8                     |
|             | Fixed Carbon (wt%)    | -                | 2.4               | 29.2                     |
|             | Ash (wt%)             | 0                | 0                 | 1.1                      |
| Elemental   | Carbon (wt%)          | 40.0             | 45.1              | 53.2                     |
|             | Hydrogen (wt%)        | 6.7              | 5.5               | 6.7                      |
|             | Oxygen* (wt%)         | 53.3             | 49.4              | 36.2                     |
|             | Nitrogen (wt%)        | 0                | 0                 | 1.7                      |
|             | Sulfur (wt%)          | 0                | 0                 | 0.3                      |
| HHV (MJ/kg) |                       | 15.7             | 16.9              | 22.5                     |

\* computed by difference

#### *DSC tests*

DSC tests were performed using a PerkinElmer DSC 8500, double furnace power compensation type that directly measures the heat flow difference between two independent furnaces. Four HTC temperatures were investigated (180, 200, 220, and 250°C) at a fixed residence time of 3 h. The instrument was calibrated with standard metals having a melting point near to the required temperature range, which is indium and tin (presenting a melting point of 156.6, and 231.9°C, respectively).

Before each test, the starting material was placed into 30  $\mu\text{L}$  stainless steel capsules (resisting at a maximum pressure of 150 bar) sealed with gold-plated disks (changed after each test) and closed using a torque wrench to a tightening torque of 1.6 Nm.

During the runs, the capsule with the material and a reference capsule were placed into two different sections of the DSC, consisting of two independent furnaces performing instantaneously the same thermal program. Each sample consists of 0.50-1.05 mg of material (biomass plus water or water alone), properly weighted with a high-precision balance of  $\pm 0.01$  mg. This amount of material was chosen consistently with the volume of the capsule (to leave a proper free space to avoid overpressures) and after having carried out preliminary tests with a higher quantity of material, during which non-negligible leakages were detected. The total weight (capsule + sealing + material) was measured before and after each test, and those with material weight losses higher than 12% discarded. This value, due to both the instrument error (due to the very low amount of material used) and a certain mass loss of gas occurring during the operation due to the high pressure, was chosen according to preliminary tests performed on the apparatus. The time between the sample preparation and the starting of the DSC run was less than 3 minutes.

Basing on the procedure proposed by Pecchi et al. [129], the performed thermal program consists of three phases performed under a protective and purging atmosphere of 50 mL/min. In particular: (1) isothermal phase at 50°C kept for 15 minutes necessary to stabilize the instrument and set a common starting point for the runs; (2) heating phase from 50°C to the set-point temperature (180, 200, 220, and 250°C) at 5°C/min; (3) isothermal phase at the set-point temperature maintained for 3 h. Once the residence time elapsed, the instrument automatically rapidly cools down at room temperature. Tests were repeated at least twice. Two methods were explored and compared to remove the effect of the water from the heat curves and obtain the net flows. Both involve a first run on the mixture biomass plus water, in which the total heat flow is recorded. Then, the contribution of water was accounted for by applying the “water” and “exhaust” methods. The “water method” involves independently measuring the effect of water by performing the same thermal program of the first run on a crucible filled with water. Meanwhile, the “exhaust method” involves a second run on the same crucible used for the first run, containing the residues



of HTC (hydrochar + liquor + gas) [129]. Practically, this translates in running three types of DSC runs (the subscript “raw” stands for the heat before the baseline removal):

(1) First run on a mixture of dry biomass and distilled water (at a biomass to water ratio of 0.25). The raw measured heat flow  $q_{1,raw}$  is necessary to heat ( $q_{heat}$ ) the system biomass plus water from the initial to the set-point temperature and to evaporate water to achieve saturation conditions i.e. liquid-vapor equilibrium ( $q_{heating,1}$ ) and to make the HTC reaction occurring ( $q_{HTC,1}$ ), as expressed in equation 2.16.

$$q_{1,raw} = q_{heating,1} + q_{HTC,1} \quad (2.16)$$

(2) Second run on the exhaust material obtained after the first run, consisting of hydrochar plus liquor and gas produced. The capsule was maintained in its position after the first run, without moving it. Therefore, the heat measured  $q_{2,raw}$  (equation 2.17) is required to heat up the system from the initial to the set point temperature and to evaporate water to achieve saturation conditions ( $q_{heating,2}$ ) and to possibly make residual reactions occurring ( $q_{HTC,2}$ ).

$$q_{2,raw} = q_{heating,2} + q_{HTC,2} \quad (2.17)$$

(3) Third independent run on distilled water alone. Since no reactions occur, the total heat required  $q_{water,raw}$  consists only of the sensible heat necessary to take water from the initial to the set-point temperature and to evaporate water to achieve saturation conditions (equation 2.18).

$$q_{water,raw} = q_{heating,water} \quad (2.18)$$

In all the cases, heat losses do not appear in the energy balance because the instrument already measures the net heat deprived by them thanks to the reference sample.

### *Baseline determination and curve processing*

The determination of a correct baseline is crucial to obtain high levels of accuracy. In literature, the baseline is also referred to as “thermokinetic background” and, through the years, authors proposed several methods to improve its accuracy (such as linear interpolation

and tangential area-proportional function [132–134]). In this work, the baseline was assumed consisting of two contributions: a zero-line and a time-drift correction. The zero-line ( $q_{\text{base,zero}}$ ) is the heat measured by the instrument with an empty crucible and is peculiar of the calorimeter itself. Indeed, as explained by Hohne et al. [133], every DSC produces, even without loads, a measured curve deriving from the residual temperature arising from a temperature difference between the sample furnace and the reference one. The zero-line depends on temperature. Therefore, it varies during the heating and remains constant during the isothermal phase. The time-drift baseline is the variation between the zero-line and the real curve. It is common for long DSC tests (in the order of hours) [125,133,134]. Indeed, during DSC experiments, a certain slope remains after the subtraction of the zero-line. Since the slope intensity is peculiar of each run, a general drift cannot be defined, and each test requires its proper evaluation. Therefore, the total baseline ( $q_{\text{base}}$ ) was computed as equation 2.19:

$$q_{\text{base}}(t) = q_{\text{base,zero}}(t) + q_{\text{base,drift}}(t) \quad \left( \frac{\text{mW}}{\text{mg}_{\text{bio}}} \right) \quad (2.19)$$

Composed by: (1) A zero-line ( $q_{\text{base,zero}}$ ), obtained by running the first two phases of the thermal program (stabilization and heating from 50°C to the set-point temperature) on an empty crucible. This was performed before and after every run to observe the daily variation. The zero-line was assumed to equal the measured value for the heating phase, while constantly equal to the last point of the heating phase for the isothermal 3 hours phase. (2) A drift ( $q_{\text{base,drift}}$ ) in the isothermal phase computed by linearly interpolating the zero-lines over the complete timeframe. In the case of biomass, since the second zero-line was measured after the second run, the time length considered for the interpolation was the sum of two runs (heating phases plus 6 isothermal hours). In the case of water, the time frame was that of one run.

Then, the total baselines were subtracted from the raw heats ( $q_{1,\text{raw}}$ ,  $q_{2,\text{raw}}$ , and  $q_{\text{water,raw}}$ ) to compute the final heats ( $q_1$ ,  $q_2$ , and  $q_{\text{water}}$ ). Signal was removed using a Savitzky–Golay filter, a low pass filter based on local least-squares polynomial approximation having the advantage of maintaining the shape and height of the peaks while smoothening the noise [135]. To implement it, the “savgol\_filter” function (with a 3rd

order polynomial) present in the “Scipy” library of Python 3.7 was used.

#### *Net heat curves and enthalpy computation*

The net heat curves associated with the HTC process were obtained using both the exhaust and the water approach. The net heats are referred to as  $q_{\text{HTC},w}$  (water method, equation 5) and  $q_{\text{HTC},ex}$  (exhaust method, equation 6), computed by firstly removing the baseline from the raw data. Since data refer to the amount of starting biomass, the  $q_{\text{HTC},w}$  was referred to the actual mass multiplying for the water mass ( $m_w$ ) and dividing for the initial biomass ( $m_{\text{bio},0}$ ).

$$q_{\text{HTC},w}(t) = [q_{1,\text{raw}}(t) - q_{\text{base},1}(t)] - [q_{\text{water},\text{raw}}(t) - q_{\text{base},w}(t)] \cdot \frac{m_w}{m_{\text{bio},0}} \quad (2.20)$$

$$q_{\text{HTC},ex}(t) = [q_{1,\text{raw}}(t) - q_{\text{base},1}(t)] - [q_{2,\text{raw}}(t) - q_{\text{base},2}(t)] \quad (2.21)$$

For both cases, the enthalpy of reaction  $\Delta H$  was computed by integrating the curves between 32 min (corresponding to 140°C) and the final time, as expressed in equation 2.22. It was assumed that no HTC reactions occur before 140°C, as supported by literature [120].

$$\Delta H = \int q_{\text{HTC}}(t)dt \quad \left( \frac{\text{MJ}}{\text{kg}_{\text{bio}}} \right) \quad (2.22)$$

### 2.3.4 Results and discussion

#### *Water runs and comparison with theory*

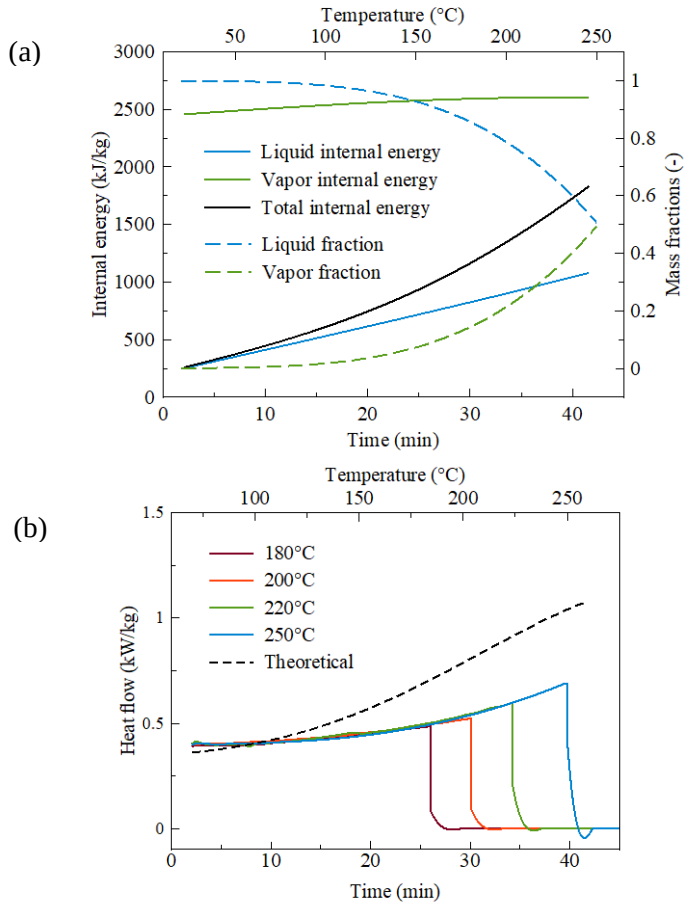
Figure 2.20b shows a comparison between the water heats at different set-point temperatures during the heating phase. Curves present a common starting point at around 0.40 kW/kg. Then they grow progressively during the heating phase up to the point immediately preceding the beginning of the isothermal phase. This absorbed growing heat is necessary to increase the water temperature according to its specific heat capacity under saturation conditions, in which condensation equilibrates vaporization. They reach maximum values of 0.48, 0.52, 0.59, and 0.70 kW/kg at 180, 200, 220, and 250°C, respectively. Once reached the set-point temperatures, curves rapidly decrease to zero for all the isothermal phases, as expected by the absence of water reactions and heat losses towards the environment (already excluded by the instrument). The small negative peaks at the beginning of the isothermal phase do not have a physical meaning and are due to the adjustment of the control system. The different curves at the same temperature present an average standard deviation lower than 1.1%.

Overall, curves at different temperatures present good repeatability. To quantify the repeatability, Pearson's correlation coefficients (extended to multiple variables) were computed by fixing a common final time until the specific operating temperature (26, 30, and 34 min, at 180, 200, and 220°C, respectively). This coefficient measures the linear correlation among the data set, ranges between -1 and 1 (minimum versus maximum correlation level), and it was computed using the Python function `corr.coeff` over the considered time [136]. Results show a high data reproducibility, with minimum values of each Pearson matrix equal to 0.983, 0.990, and 0.994 for the 180, 200, and 220°C cases, respectively. The 250°C one was not evaluated since only one curve is present.

To verify the reliability of experimental data, theoretical water heat curves were evaluated using thermo-physical properties of water in saturation conditions from the NIST database [130]. Theoretical data were estimated by dividing the process into discrete temperature and time intervals ( $\Delta T$  and  $\Delta t$ , respectively). For each  $i$ -interval, theoretical heat flow  $q_{i,th}$  was computed by applying the first principle of

thermodynamics as  $q_{i,th} = (U_{i+1} - U_i) / \Delta t$  (net work produced equal to zero), where  $U_i$  is the internal energy of water in saturation conditions: more precisely, water with a certain steam content. Therefore, for each  $i$ -interval, the steam mass was computed, as a first approximation, applying the ideal gas law on the capsule free volume and considering the partial pressure of water equal to the vapor pressure (and therefore dependent on temperature), a hypothesis supported by the literature [36]. The capsule free volume was assumed constant during time: actually, the variation of the volume occupied by the gas phase in the capsule is negligible. Then, the mass of the (liquid) water was computed by subtracting the steam mass from the initial mass of water introduced inside the capsule. Figure 2.20a shows the mass fractions of water and steam calculated as stated above, and the internal energy  $U$  of water, steam (from the NIST database [130]), and the mixture of water-steam as a function of temperature and time. Water passes from the liquid to the vapor phase according to the liquid-vapor equilibrium curve and the available free volume. Figure 2.20b shows the theoretical heat flow (dotted line), which is the derivative of the total internal energy (Figure 1a), compared with the experimental heat flows measured by the DSC. As the experimental curves, the theoretical heat flow increases with time mainly due to the latent heat of evaporation necessary to transform a portion of water into steam.

Overall, theoretical and experimental curves present similar trends and similar absolute values, proving the consistency of the experimental data. In particular, curves slightly differ below 120-130°C, while they progressively mismatch at higher temperatures. This fact could be attributed to both the experimental procedure itself, not aimed to deduce exact parameters like the heat capacity (more accurate procedures should be performed), and a possible delay in establishing phase equilibrium in the experimental system. Indeed, for each increment, theoretical curves refer to a system at thermodynamic equilibrium while experimental curves follow the system dynamics, making them dependent on extensive parameters: in the present case, the high free volume available in the capsule (around 95%) available to accommodate steam could be to some extent responsible of the mismatching.



**Figure 2.20.** (a) Internal energy of saturated water and steam (NIST database [130]) and of the mixture water-steam (solid lines), whose theoretical mass fractions (dotted lines) are also reported, per unit mass of water loaded into the capsule. (b) Experimental water heat curves during the heating phase (solid lines), compared with theoretical data obtained from internal energy (dotted line), per unit mass of water loaded into the capsule

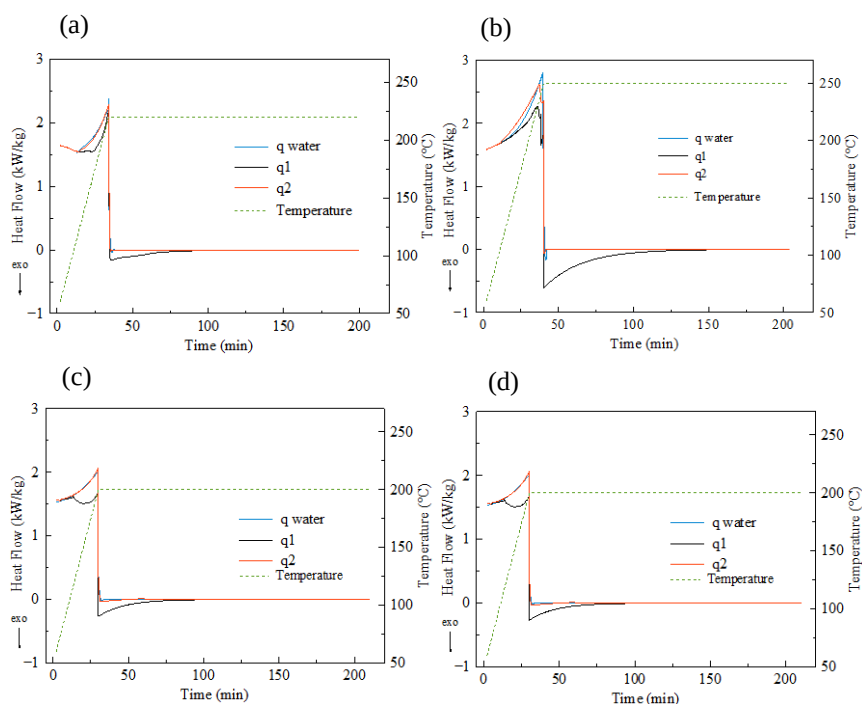
### DSC heat profiles

Overall, DSC curves of the different biomasses present similarities among them. Figure 2.21 shows four examples of DSC curves representative also for the other cases. Until 130-140°C (corresponding to 15-18 min), HTC reactions still do not occur and  $q_1$  and  $q_2$  indicate both the heat necessary to increase the temperature of the mixture biomass plus water and to evaporate water to maintain liquid-vapor

equilibrium. In this phase, since both the amount of biomass and its specific heat is considerably lower than those of water (mass ratio of 0.25 and  $c_p$  of 1.1-1.5 Jg<sup>-1</sup>K<sup>-1</sup> versus 4.02 Jg<sup>-1</sup>K<sup>-1</sup>, respectively [137]),  $q_1$  and  $q_2$  approximate the water curve. After 130-140°C, HTC reactions progressively trigger, releasing or producing an amount of heat that sums to the heating one. Indeed,  $q_1$  starts departing from  $q_2$  and  $q_{\text{water}}$  and its greater concavity already suggests a net heat release corresponding to an exothermic behavior. Indeed,  $q_1$  is lower than  $q_2$ , which means a negative net curve (which is defined as  $q_1 - q_2$ ), i.e. exothermicity. This behavior mostly continues up to the end of the heating phase. In particular, glucose and cellulose present a monotonic growing trend up to the set point when operating at 180, 200, and 220°C, as the cases reported in Figures 2a and 2c. Instead, with 250°C as a set-point temperature, an inverse trend variation is observed at around 225 and 230°C for glucose and cellulose (Figure 2b), respectively. Meanwhile, grape seeds present this inversion before, at around 200°C, which is evident in both the 220 and 250°C runs (Figure 2d). Therefore, these invert-trend temperatures represent precise points at which the exothermic behavior of HTC reactions assumes decisive importance regarding the heat required. This aspect makes the heating phase a non-negligible phase at all, during which HTC reactions occur, affecting the products and the energy balance. The occurrence of carbonization reactions during the heat-up phase at temperatures greater than 180°C was recently demonstrated by Lucian et al. [39], too. Once reached the isothermal phase, the fraction to heat up the mixture equals zero ( $\Delta T=0$ ), leaving the heat of reaction as the only one contributing to the energy balance. Therefore, for all the cases, once approached the isothermal phase,  $q_1$  suddenly shows up with a classical negative peak, clearly demonstrating the HTC exothermicity. During the isothermal phase,  $q_1$  remains in the negative region for a period and with an intensity function of both the substrate and the operating conditions. As the reaction time increases, HTC products form and reactions progressively conclude, causing  $q_1$  to approach gradually zero according to a mostly exponential decaying. Besides, the negative peaks intensity increases with the set-point temperature. Indeed, temperature favors the reaction kinetics, accelerating the occurrence of carbonization and causing a higher energy release at the beginning of the isothermal phase. For example, passing from 180 to 250°C, glucose passes from -0.16 to -0.59 kW/kg, cellulose from -0.20 to -0.61 kW/kg,

and grape seeds from -0.26 to -0.47 kW/kg. Meanwhile,  $q_2$ , representing the measured heat flow computed considering the exhaust products of HTC, effectively behaves as a “blank” sample for glucose under all the conditions, while for cellulose and grape seeds this is the case only at 180 and 200°C. In these cases, the curve follows the water curve, increasing during the heating phase and going to zero during the isothermal phase, as can be seen in Figures 2.21a and 2.21c. Different is the case at 220 and 250°C (like those in Figures 2.21b and 2.21d), in which  $q_2$  presents a trend inversion in the heating phase similar to  $q_1$ , indicating the presence of residual exothermic reactions. Then, during the isothermal phase,  $q_2$  is initially very slightly exothermic and then rapidly goes to zero. However, even if its contribution during the isothermal phase is zero, the heat released during the heating phase is enough to impact the net heat curves and the enthalpy of the reaction. This phenomenon can be explained by the different heat capacities of the mixture liquor plus hydrochar obtained at the end of the first run. Indeed, the discrepancy between the first and second run is observed only during the heating phase, where the heat capacity plays its major role, and not during the isothermal phase. This could be a valid alternative explanation instead of the hypothesis of “a still reacting phase”, which should indeed continue to react during the isothermal phase of both the first and second run. Meanwhile, temperature favors the production of volatile compounds dissolved in the liquid phase and absorbed in the solid, which affects the heat capacity of the mixture. Indeed, high temperatures lead to a higher presence of compounds dissolved in the liquid phase, affecting the heat capacity of the mixture. For example, propionic acid, formic acid, and phenol derivatives, which are highly present in the liquid phase, have a  $c_p$  of 158.6, 101.3, and 93.7 Jmol<sup>-1</sup>K<sup>-1</sup>, respectively, versus that of water of 75.4 Jmol<sup>-1</sup>K<sup>-1</sup> [91,92,138]. Grape seeds present also a higher presence of oily-derived compounds (like fatty acids), increasing the variety of compounds dissolved in the liquor [23]. Meanwhile, at low HTC temperatures, the composition and lower total concentration of the compounds dissolved in the liquid phase slightly affect the heat capacity. To explore the differences and evaluating the heat capacity of the liquor, future works could include running DSC on the HTC liquor only and some specific compounds (like fatty acids, organic acids, or phenols) typically obtained from an HTC run.





**Figure 2.21** Example of heat curves: (a) glucose at 220°C, (b) cellulose at 250°C, (c-d) grape seeds at 200 and 250°C, per unit mass of dry biomass introduced inside the capsule.

### Net heat flow curves

Figure 2.22 shows the net DSC heat flow for glucose, cellulose, and grape seeds at the different investigated temperatures.

Curves present a negative peak that progressively increases with the set-point temperature. This correlation agrees with the dependence of the process kinetics expected by the Arrhenius law. All the runs begin to release heat at around 130-140°C (18 min) when the first removal of extractives and hydrolysis reactions start [76,139]. Then, as the heating phase proceeds, the amount of heat increases to reach its maximum around the beginning of the isothermal phase. The only run that departs from this behavior is that of grape seeds at 250°C, which presents a slope variation (more significant in the case of the exhaust method, while mild for the water one) still during the heating phase. After

reaching the peak, the heat released diminishes following a shape function of the biomass (especially in the case of glucose and grape seeds) and then progressively reaches zero. Therefore, curves present a relatively simple trend, with one peak at the beginning of the isothermal phase and a characteristic tailing section lasting through the residence time. This trend is typical of isothermal measurements, where the reaction rate reaches its maximum as soon as the reaction temperature is reached, and then it becomes slower as the concentration of the initial biomass decreases, which consumes to form products [140,141]. These measures do not allow to show significant multiple peaks or shoulders. Therefore, they cannot identify the different reactions to develop a multi-step reaction model [142]. However, curves can be modeled through a single-step model with a reaction order higher than one, which, in the field of HTC, can provide more reliable results than the commonly adopted first-order models [52]. Indeed, HTC comprises also polymerization and condensation stages that are commonly characterized by second-order reactions. The modeling of DSC data was performed by Pecchi et al. [118], who also demonstrated the marginal contribution of autocatalytic reactions (induced by the presence of organic acids that catalyze the process [143]), such as hydrolytic reactions, compared to the higher-order ones.

For the three biomasses, the 180 and 200°C curves slightly differ, and they mostly overlap during the heating and beginning of the isothermal phase. Meanwhile, the difference is higher at 220 and 250°C. Therefore, according to these curves, passing from 180 and 200°C does not significantly impact the heat release. This fact could agree with the HTC mechanism: high temperatures favor aromatization reactions, which imply the formation of aromatic bonds and a subsequent higher heat release. In particular, glucose (Figures 2.22a and 2.22b) presents a monotonic steep curve during the heating phase. The water and exhaust methods lead to comparable curves, and their difference can be attributed to experimental deviations, making, in this case, the exhaust run an effective substitute for the water one. Therefore, 3 hours are enough to ensure the conclusions of exothermic reactions. The rapid decaying of the curves suggests a higher-order reaction mechanism, which agrees with the nature of the starting substrate and secondary char formation mechanism, derived by polymerization and condensation reactions [21,52]. The peaks slightly anticipate the end of the heating phase of 1-4 minutes. Then, after reaching their maxima,

curves progressively decrease and decay following an exponential curve. The width of the pre-decaying phase decreases with the temperature, passing from a rounded shape to a very sharp one moving from 180 to 250°C, as expected from the fastest kinetics to the highest temperatures. Thanks to its simple monosaccharidic nature, glucose reacts rather quickly.

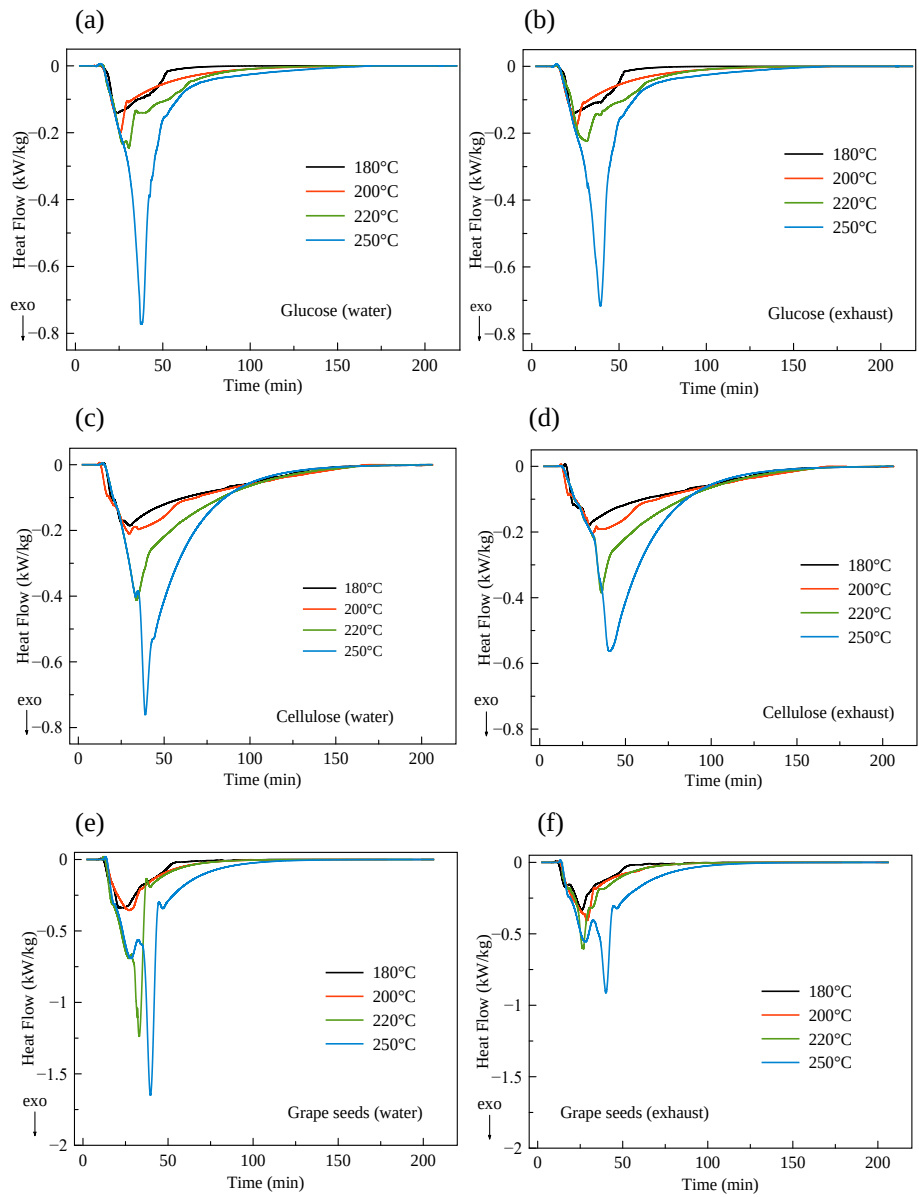
As shown in Figures 2.22c and 2.22d, the reaction kinetics of cellulose is slower than that of glucose, with peaks occurring after 1-3 min from the beginning of the isothermal phase. The height of the peaks is close for the 180 and 200°C runs (-0.19 and -0.21 kW/kg, with the water method) while they move at 220 and 250°C (-0.41 and -0.76 kW/kg). This fact agrees with the high stability of cellulose. Indeed, its structure is almost unaffected below 180°C and starts degrading at around 220°C due to the breaking of beta-glycosidic bonds [76,139,144]. The slower kinetics reflects in the long decaying stage for all 3 hours. Besides, even at 180 and 200°C and despite the low reactivity of cellulose at low temperature, it is clear that cellulose presents higher exothermicity than glucose. This fact could find an explanation in the reaction mechanisms of cellulose during HTC. As explained by Falco et al. [40], the HTC of cellulose is mainly dominated by a bulk pyrolysis-like degradation, while hydrolysis reactions are present only marginally. During HTC, cellulose rearranges and forms an intermediate polymeric compound, which then converts to aromatic structures that are favored by high operating temperature [40]. Bulk pyrolysis-like reactions, which start since the first reaction stages, lead to a high molecular re-organization into aromatic structures characterized by a high-energy state. This fact, added to hydrolysis reactions, which are present even if marginally on the surface, could lead to a huger re-organization than glucose. This translates into a higher energy difference between the feedstock and the products with a consequent higher exothermicity. Moreover, comparing the water and exhaust method, we can state that significant differences are observed only for the 250°C run, where the decomposition peaks are -0.76 and -0.56 kW/kg for the water and exhaust case, respectively.

Grape seed curves at different temperatures present a certain non-uniformity attributable to the intrinsic inhomogeneity of the biomass. Similar to glucose, curves obtained through the water method reach their maximum slightly before the end of the heating phase (around 0-3 minutes). This higher reactivity at low temperatures can be due to the starting compositions of grape seeds. Indeed, they contain lignin, hard

to decompose, but also hemicellulose and oil (in a fraction of 10-15% [110,139]), which have less tight bonds than cellulose. Therefore, being “easier to break”, these components confer to grape seeds a higher reactivity at lower temperatures. Besides, the difference between the water and exhaust method is remarkable, especially for the 220 and 250°C cases. Indeed, the exhaust material inflects as the first run at 218 and 220°C for the 220 and 250°C runs, respectively, as shown in Figure 2d. Interestingly, this value is highly close to the smoke point of grape seed oil, equal to 216°C [130], and similar to the degradation of fatty acids absorbed on the hydrochar conditions between 220 and 260°C under pyrolytic conditions [23].

Therefore, it seems reasonable to attribute this inflection point to the degradation of the oily phase, which is still present, probably in a derived form like fatty acids, at the end of the first run. The new mixture composition at the end of the first run heavily affects the peak intensity. For example, at 250°C, it equals -1.65 versus -0.91 kW/kg for the water and exhaust method, respectively.

As shown in Figures 2.22e and 2.22f, grape seeds present sharper peaks than glucose and cellulose even at 180°C, with intensities between 2 and 5 times huger. After reaching the maximum heat release, they decrease fast to zero in a similar way to glucose. Again, this behavior can be explained by the presence of hemicellulose and the oily phase, easy to degrade, and whose hydrolysis leads to acid intermediates that catalyze the degradation of cellulose and partially lignin [86,129].



**Figure 2.22.** (a-b) Net heat flows of glucose, (c-d) cellulose, and (e-f) grape seeds, computed through the water and exhaust methods, per unit mass of dry biomass introduced inside the capsule.

### *Enthalpy of reaction*

Table 2.10 shows the enthalpy of HTC reactions  $\Delta H$  obtained through the water and exhaust methods. Overall, data present order of magnitude and ranges that are fully comparable with the few presented in the literature [122].

Data resemble the trends of net heat flows shown in Figure 2. For all the biomasses, the operating temperature highly affects the data. Indeed, high temperatures favor carbon densification and the progressive reorganization of the carbonaceous structure of biomass, characterized by intra and intermolecular hydrogen bonds, into aromatic structures [144]. Thanks to HTC operating conditions, chemical bonds typical of carbohydrates like  $-\text{COO}$  and  $-\text{OCH}_3$  decrease to form aromatic structures characterized by  $-\text{C-H}$ ,  $-\text{C-C}$ , and  $-\text{C-O}$  bonds [144]. The passage from a material with low-energy states (the biomass) to one with high-energy states (the hydrochar) is accompanied by a heat release that increases with the strength of newly bonds formed. This aspect justifies both the increasing  $\Delta H$  with temperature and the higher values of cellulose and grape seeds compared to glucose. Indeed, as previously mentioned, cellulose and grape seeds undergo solid-solid reactions, which involve a high molecular reorganization into aromatic structures (and therefore to higher energy states). Solid-solid reactions add to the liquid-solid reactions, which are the only ones present in glucose, resulting in a consequent higher  $\Delta H$ . Besides, grape seeds present an even more complex situation, in which also the decomposition of the hemicellulose and the oily fraction plays a role in producing organic acids able to catalyze cellulose degradation. Moreover, during HTC, biomass partially dissolves through hydrolysis and dehydration into a variety of intermediates that further undergo degradation (for example, HMF degrades into formic and levulinic acid), involving their specific enthalpy of reaction. Depending on the biomass and the operating conditions, the reaction enthalpy contributed to the final value.  $\Delta H$  of glucose passes, with the water method, from  $-0.20$  to  $-0.86$  MJ/kg at  $180$  and  $250^\circ\text{C}$ , respectively. The  $180$  and  $200^\circ\text{C}$  cases present similar values, indicating that no significant reaction differences occur between the two temperatures. These values could result from the balance between isomerization (slightly endothermic,  $0.05$  MJ/kg), endothermic dehydration reactions to HMF (up to  $0.51$  MJ/kg), and

exothermic reactions of the degradation of HMF into lactic and formic acid (from -0.13 to -1.06 MJ/kg) [145]. Only one reference value for glucose is present in literature and equals -1.06 MJ/kg at 250°C [126], where however big standard deviations were observed. Meanwhile, data consistently differ from the theoretical -5.78 MJ/kg proposed by Titirici et al. [146] in one of the first studies of HTC, in which the exothermicity was attributed to the dehydration of carbohydrates. However, we should consider that the computation of the enthalpy of reaction through Hess's law should be managed carefully. Indeed, it requires some assumptions on the chemical reactions involved, including the number of water molecules formed, the knowledge of the yields, and the chemical formula of all the molecules formed and dissolved into the liquid phase.

Cellulose presents high enthalpy values already at 180°C, with -0.61 MJ/kg, which can be justified by the overlapping of reactions involved in the liquid phase and the beginning of solid-solid reactions. Similarly to glucose and due to the mild operating conditions, dehydration and degradation reactions of HMF and intermediates into organic acids could have a great role. Besides, the aromatization degree increases with severity, leading to -1.33 MJ/kg at 250°C. For cellulose, the difference between the water and exhaust method is significant at 250°C, with values of -1.33 and -1.10 MJ/kg. Values differ from the -0.88 MJ/kg value reported by Pecchi et al. [129]. The discrepancy can be due to both the method adopted (exhaust) and the removal of a baseline that considers the heat absorbed by the gas phase [129], which should be supported by the higher heat absorbed in the second run. This aspect was not observed in our case, where, on the contrary, the exhaust is expected to present a consistent difference in the heat capacity with respect to water. Similar considerations can be done on grape seeds. Their  $\Delta H$  mostly increases once overcome 220°C, reaching a maximum of -1.53 MJ/kg at 250°C. In this case, the difference between the water and exhaust method is evident and enthalpies hugely differ among them. Indeed, the exhaust of the 220 and 250°C hugely differ from water (Figure 2d) causing a contraction of the net curves. No  $\Delta H$  data relevant to grape seeds are present in the literature, but their values are fully comparable with those of other biomasses. For example, in a recent work, Rolland et al. [124] measured -1.7 MJ/kg for the HTC of industrial *Spirulina* at 260°C using a 300 mL HTC reactor. Meanwhile, Pecchi et al. [129] report -0.64 and -0.25 MJ/kg for wood and digestate

at 250°C, respectively, but after removing the heat absorbed by the gas phase.

**Table 2.10.** Enthalpy of HTC reactions at different conditions (standard deviations < 0.05 MJ/kg), per unit mass of dry biomass.

| Biomass     | Temperature (°C) | $\Delta H$ (MJ/kg) |         |
|-------------|------------------|--------------------|---------|
|             |                  | water              | exhaust |
| Glucose     | 180              | -0.20              | -0.21   |
|             | 200              | -0.27              | -0.27   |
|             | 220              | -0.41              | -0.40   |
|             | 250              | -0.86              | -0.67   |
| Cellulose   | 180              | -0.61              | -0.59   |
|             | 200              | -0.71              | -0.69   |
|             | 220              | -0.97              | -0.91   |
|             | 250              | -1.33              | -1.10   |
| Grape Seeds | 180              | -0.42              | -0.36   |
|             | 200              | -0.49              | -0.49   |
|             | 220              | -0.94              | -0.56   |
|             | 250              | -1.53              | -1.20   |

### *General considerations and recommendations for future works*

Considering the little research around the investigation of HTC heat profiles and enthalpy of reactions, it seems worth reporting some considerations and suggestions below.

(1) This work investigated and compared two experimental methods applied to a high-pressure DSC. The exhaust method, proposed originally by Pecchi et al. [129], suits the investigation at low temperatures where, according to our data, the exhaust run follows the water curves. However, this method appears less accurate at high temperatures or with complex biomasses, in which the high variety and presence of compounds dissolved in the liquid phase affect the heat capacity of the mixture. Meanwhile, the water method applies to all possible cases and, being dependent only on the temperature and not on the specific biomass, saves operating time. The counterpart is that it requires a more precise evaluation of the instrument baseline. Indeed, the water method requires comparing two tests run independently, and therefore to carefully evaluate baselines (for both the HTC and water run). With the exhaust method, since the pan is not moved between the



first and second run, a less precise baseline can be used, and offset errors avoided. Meanwhile, future works could include a comparison with the methodology that involves a reference crucible filled with water, as already proposed by other authors [126,128,147,148], through which the final curve is already at the net of the water effects.

(2) The experimental activity shows that the exhaust sample mostly resembles the water trend, making the heat absorbed by the gas phase almost negligible. This fact also agrees with the absolute low amount of gas produced during HTC, which has in general yields less than 10% at 250°C and less than 1% at 180-200°C [149]. For example, cellulose, which is one of the biomasses with the hugest gas production, presents gas yields between 0.1 and 7.8 % in the range of 180-260°C [76]. Besides, the gas phase has a much lower specific heat capacity compared to water ( $0.66 \text{ Jg}^{-1}\text{K}^{-1}$  for  $\text{CO}_2$  versus  $4.02 \text{ Jg}^{-1}\text{K}^{-1}$  of water [130]). Therefore, especially at low temperatures, it may seem reasonable to neglect the gas production for the exhaust.

(3) Being the heat released during the operation, HTC reactions have a positive effect on the energy balance of the reactor. However, as explained by Rolland et al. [124], we should keep in mind the small impact of this exothermicity on the overall balance, especially if compared to the heat required to warm the water-biomass mixture before entering the reactor. Indeed, the reaction exothermicity represents only a small contribution to the energy efficiency route of the process, and extra heat-recovery solutions are certainly required.

(4) Particular attention has to be paid to determine the instrument baseline. Indeed, especially when working with little amounts of materials and at the bottom scale of the DSC instrument, the baseline highly affects the result. From a technical point of view, we suggest running at least two baselines per day (before and after the test) so to quantify the drift of the instrument. Minimizing the usage hours of the instrument could be useful to reduce the drift. From this point of view, the water method can be helpful to minimize the number of continuous operating hours of the instrument.

(5) Leakages were an insidious technical issue faced during the preliminary experimental campaign [125]. To avoid this, we suggest using an amount of material that ensures a high void volume inside the crucible (less than 40% of occupancy of the capsule) and performing some thermal programs on the empty crucibles to thermally stabilize them.

(6) Particular attention has to be paid when comparing theoretical data computed applying Hess's law. A precise evaluation requires the knowledge of the complete HTC reaction, comprehensive mass yields, chemical formula, and higher heating values of all the products. In particular, it requires precise data on the liquid phase and amount of water produced, which are in general difficult to evaluate. Meanwhile, simplifying the reaction mechanism can provide a rough estimation and an indication of the orders of magnitude that could be useful as a starting point for further analysis or some preliminary technological considerations.

### 2.3.5 Conclusions

This work investigated heat profiles and reaction enthalpy of hydrothermal carbonization using high-pressure differential scanning calorimetry. Three biomasses (glucose, cellulose, and grape seeds) were tested experimentally at 180, 200, 220, and 250°C for an HTC test duration of 3 hours. Two experimental approaches were adopted to remove the water effects on the net heat curves and enthalpies. The first considers the water alone, while the second uses the exhaust material obtained after the first run. Results show that the exothermic peaks and enthalpies increase with the operating temperature, with values ranging from -0.20 to -0.86 MJ/kg, -0.61 to -1.33 MJ/kg, and -0.42 to -1.53 MJ/kg for glucose, cellulose, and grape seeds, respectively (water method). Peaks and enthalpies variation with temperature agrees with the dependence of chemical kinetics and aromatization reactions with it. Among the biomasses, cellulose presents a slower decomposition mainly due to its chemical structure characterized by tighter bonds than grape seeds, where hemicellulose and oil are present, and glucose, which undergoes a liquid-solid conversion avoiding the slower solid-solid stage. Comparing the two experimental approaches, no significant differences were observed at low temperatures, at which the exhaust curve mainly follows the water one. However, especially for cellulose and grape seeds, differences become more important at higher temperatures, where the exhaust reacting profile departs from that of water during the heating phase due to the different heating capacity of the liquid phase. Therefore, the water method seems more accurate for complex biomasses and/or at high HTC temperatures.

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## Chapter 3

# Hydrothermal carbonization of organic waste and biomass: a review on process, reactor, and plant modeling

This chapter is based on the article that we published in *Waste and Biomass Valorization* (2021, vol. 12, issue 6, by G. Ischia and L. Fiori <https://doi.org/10.1007/s12649-020-01255-3>). This work presents a comprehensive analysis of the current state of the art of models and simulations adopted for hydrothermal carbonization (HTC) of organic waste and biomass. Three different subsystems to which modeling has been applied are critically investigated: the modeling of the process itself (i.e. reaction kinetics and statistical correlations), the reactor (heat transfer and computational models), and the overall plant. A glance at life cycle assessment analyses applied to HTC is also provided. Strengths and gaps of each modeling system are highlighted to guide future research to further improvements.



#### 3.1 Abstract

Hydrothermal carbonization (HTC) is an emerging path to give a new life to organic waste and residual biomass. Fulfilling the principles of the circular economy, through HTC “unpleasant” organics can be transformed into useful materials and possibly energy carriers. The potential applications of HTC are tremendous and the recent literature is full of investigations. In this context, models capable to predict, simulate and optimize the HTC process, reactors, and plants are engineering tools that can significantly shift HTC research towards innovation by boosting the development of novel enterprises based on HTC technology. This review paper addresses such key-issue: where do we stand regarding the development of these tools? The literature presents many and simplified models to describe the reaction kinetics, some dealing with the process simulation, while few focused on the heart of an HTC system, the reactor. Statistical investigations and some life cycle assessment analyses also appear in the current state of the art. This work examines and analyzes these predicting tools, highlighting their potentialities and limits. Overall, the current models suffer from many aspects, from the lack of data to the intrinsic complexity of HTC reactions and HTC systems. Therefore, the emphasis is given to what is still necessary to make the HTC process duly simulated and therefore implementable on an industrial scale with sufficient predictive margins.

#### 3.2 Introduction

The conversion of biomass into bio-based products and bioenergy is an important innovative aspect to pursue the circular economy principles, aimed to increase the amount of renewable sources and to reduce the consumption of raw material and energy [1]. The usage of second-generation biomass as starting material for the valorization stage is a linkage between circular economy and waste management. Indeed, this category of biomass does not interfere with food production or land usage and, in addition, it requires a particular disposal strategy. Second-generation biomasses, which comprise agricultural and forest residues, and industrial and organic waste, are commonly heterogeneous and characterized by a high moisture content [2]. Besides, they require significant pretreatment stages for improving their handling and storage. For example, energy-consuming drying stages are often needed to make the feedstock suitable for traditional thermochemical conversion processes.

In this context, hydrothermal carbonization (HTC) could play a significant role in the valorization of organic waste and biomass residues. Even if the first concept of HTC dates back to the early 20<sup>th</sup> century to understand the mechanism of natural coalification, only in the past 10 years HTC has acquired significant attention inside the scientific community [3]. HTC is a thermochemical process occurring in subcritical water under relatively mild conditions, with a reaction temperature commonly maintained in a range of about 180-250 °C and pressure between 10 and 50 bar. Under these conditions, liquid water behaves both as a reactant and a solvent/reaction medium for the biomass, leading to its degradation through heterogeneous chemical reactions, which include dehydration, hydrolysis, aromatization, and polymerization. The result is a solid phase rich in carbon (the so-called hydrochar), a liquid by-product, and a gas phase rich in CO<sub>2</sub>. The hydrochar can be used in a variety of applications like agriculture (e.g. as a solid amendment), energy production (e.g. in co-combustion with traditional fossil fuels [4]), as adsorbent [5,6], and as a base for advanced carbon materials [7]. Occurring in water, HTC is particularly suitable for the upgrading of biomass with high moisture content and a wide variety of “wet” biomasses have been investigated, like lignocellulosic biomasses, organic wastes, sewage sludge, as well as basic compounds like pure cellulose and other carbohydrates [8–10]. Meanwhile, HTC has been successfully applied in the context of the third and fourth generation biomass, mainly represented by algae [11–13]. Even if algae are particularly attractive for the production of liquid biofuel through hydrothermal liquefaction (HTL), the adoption of HTC has been also investigated. Indeed, HTC has been suggested as an intermediate step for the production of biodiesel. Actually, the hydrochar adsorbs the fatty acids formed from the hydrolysis of the lipids naturally present in the algae and, if extracted, these fatty acids can be used as precursors for the production of biodiesel [14,15].

The hydrochar, the carbonaceous solid phase resulting after the process, presents interesting properties that make it suitable for various applications. Due to dehydration and decarboxylation reactions, it presents a higher carbon content than the biomass, with a contextual improved heating content (HHV) and lower O/C and H/C ratios [16]. This makes it similar to conventional solid fuels. It also presents an improved hydrophobicity [17]. Regarding surface properties, these are similar to those of the starting material: during the process, products resulting from polymerization deposit on the surface, blocking the pores

[18]. Nevertheless, modifying the hydrochar through activation can provide properties suitable for contaminants remediation [19].

A field of rising interest is the integration of HTC with other valorization techniques. Among these, combining HTC with anaerobic digestion (AD) for the production of biogas represents a promising strategy for valorization and usage of the HTC process water (the liquor), otherwise a by-product of the process [10,13]. Several integration strategies have been investigated for different feedstock at lab scale [13,20–24], showing that the combination is effective both to energetically sustain HTC and to reduce the environmental impact of the liquor disposal [10]. Moreover, some studies investigated the combination of HTC with conventional thermochemical processes, like gasification and pyrolysis, in which HTC represents the pretreatment stage to enhance the chemical and physical properties of the raw biomass [16]. In particular, the usage of the hydrochar for gasification can improve the quality of the syngas and reduce the tar formation [25–27], while pyro-hydrochar exhibits improved surface area and durability than the normal biochar [25,28]. Recently, renewable and alternative technological configurations have been also proposed, like the coupling between an HTC reactor and a solar concentrator [29,30].

Until now, significant experimental activity has been done to investigate HTC reaction mechanisms and the effects of process parameters (like reaction temperature, residence time, and solid load) on the final products, whose physical and chemical properties have been extensively characterized according to the desired application. Even if some pilot-scale and commercial continuous reactors are present worldwide, most of the research has been carried out on lab- or bench-scale reactors [3], typically batch reactors: certainly, more efforts are required to develop continuous reactors so to move towards the industrialization of HTC. Challenges must be faced to spread this technology on a commercial scale, and the development of reliable predictive models is considered by the research community as one of these challenges.

Despite the massive experimental activity performed to characterize the whole process, the modeling and simulation of HTC are lagging. Indeed, the development of validated models and tools is crucial for the optimization and design of the process, as well as for the understanding of the HTC mechanism dynamics. Through modeling, virtual scenario can be settled and the variations of process parameters investigated, so to provide useful information to obtain a product with the required characteristics. This is essential to increase the HTC competitiveness

and attractiveness with respect to traditional techniques [31]. Commercial parties need to easily explore the possibility of HTC, without investing significant resources [32]. Therefore, the scarcity of reliable models in the current state of the art for HTC makes modeling a challenge that should be faced in our immediate future for the diffusion of the HTC technology [33]. Moreover, the current literature mainly focuses on the modeling of properties related to biofuels applications (like carbon content and HHV), limiting the spread of the other applications.

Until now, modeling and simulation of HTC have been applied on different system levels. These pass from the description of the process itself through reaction kinetics models (which often use pseudo-first-order reactions) and statistical correlations, to the description of the reactor and the overall plant. Overall, the current reaction kinetics and statistical approaches are highly affected by the experimental data through which they are calibrated. Indeed, they propose relations among the properties of the resulting phases and the HTC operating conditions. Meanwhile, techno-economic analyses are often associated with the large-scale models and some life cycle assessment studies have been also performed. While many reviews present the general knowledge of HTC with focus on the chemistry of the process, physical and chemical properties of the products, and applications [3,16,34,35], few works address the current state of the art on HTC modeling and simulation, without however presenting a complete overview of all aspects related to modeling in the HTC field.

This work provides a review of the main aspects related to HTC modeling, highlighting strengths and aspects that should be improved to reach an appropriate knowledge level. The different models applied to the different HTC subsystems (process, reactor, and plant) are presented. We hope that this work could guide researchers to the development of new and more reliable models that could drive the shift of HTC from a laboratory to an industrial scale.

### **3.3 Overview on HTC process modeling**

HTC models and simulations that have been developed so far can be categorized according to the subsystem of analysis. Results and data of the previous subsystem are necessary for the definition of the subsequent subsystem (see below). The following categories (or subsystems), characterized by progressive broader boundaries, have been identified:

- (1) *Process*, i.e. reaction kinetics and statistical models. They focus on the reaction mechanisms and the correlations between process parameters (i.e. temperature, reaction time, and biomass load) and certain observed output variables. They are usually used to predict properties (like hydrochar yield, carbon content, and energy content) and are valid for a narrow range of experimental conditions and feedstock. Kinetics models are often based on a simplified reaction scheme and assume lumped first-order reactions following the Arrhenius law. Linear and non-linear correlations are widely adopted to statistically correlate HTC variables.
- (2) *Reactor*. These models are intended to predict variations in space and time considering a multitude of aspects that include reaction kinetics, mass and heat transport phenomena, as well as the geometry of the reactor. Validated computational models developed in proper software (like Comsol Multiphysics®) belong to this model category. Until now, few studies cover this category and the lack of precise kinetics and thermodynamic (e.g. the HTC heat of reaction) data negatively affects their implementation and diffusion.
- (3) *Plant*. They model the entire HTC process, considering the reactor and the various plant equipment (like pumps, heat exchangers, boiler, filters, and dryer). They are often coupled with an economic evaluation and use commercial software (like Aspen Plus® and VMGSim®) or software codes ad hoc developed in proper computer programming languages. The current plant simulation models lack of precise data on the reactor and presume that lab-scale data can be representative at a much larger plant-scale.
- (4) *Life Cycle Assessment (LCA)*. They evaluate the environmental impacts accounting for all the stages of the process, providing a holistic view of the process by applying a “cradle to grave” approach. Some examples of LCA applied to HTC are present in the literature, but more large-scale data are necessary to make them more reliable.

#### 3.4. Reaction kinetics models

The modeling of the reaction kinetics has a key role in the industrialization of the HTC technology. Indeed, a model capable to

predict the output on the base of input parameters is a fundamental requirement to design and optimize a process that could be reliable, competitive, and attractive when compared with other technologies. Mainly due to the complexity of the reactions involved and the difficulty of developing a general reaction mechanism, current kinetics models suffer from strong simplifying hypothesis and dependence on the experimental conditions under which they are calibrated and/or validated.

Nevertheless, significant improvements have been done in the last years. In the current state of the art, many reaction kinetics models adopt simplified lumped schemes based on the Arrhenius equation, which describe the process through pseudo-first- or higher-order reactions. In addition, simplified “all-inclusive” models, like the severity and coalification models, were developed to predict hydrochar properties using only one parameter condensing the more affecting factors. Recently, new approaches propose the description and corresponding modeling of HTC through probabilistic laws.

### **3.4.1 General concept**

Until now, the kinetics of the HTC of biomass has been primarily modeled adopting simplified Arrhenius kinetics schemes, in which the overall mechanism is simplified in a network of discrete reactions comprised of few elementary steps. Few attempts consider other approaches, like the stochastic or computational one in a three-dimensional space. Authors developed models on a wide variety of feedstock, ranging from pure components to complex biomasses, hypothesizing possible reaction pathways. In particular, the main components of lignocellulosic biomass (cellulose, hemicellulose, lignin, and extractives), have been investigated both individually and when enclosed into biomass. Generally, even on the pure compounds, it is observed a great variability in the results.

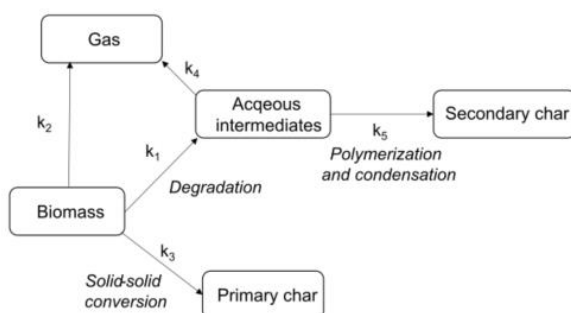
A standard procedure to develop a kinetics model for HTC consists in:

- (1) collecting data through HTC experimental campaigns at different key variables (i.e. temperature, time, and initial biomass load);
- (2) hypothesizing a reaction mechanism;
- (3) writing one or more mass balance equations with Arrhenius-based kinetics relations with some “free parameters” (e.g. activation energies);

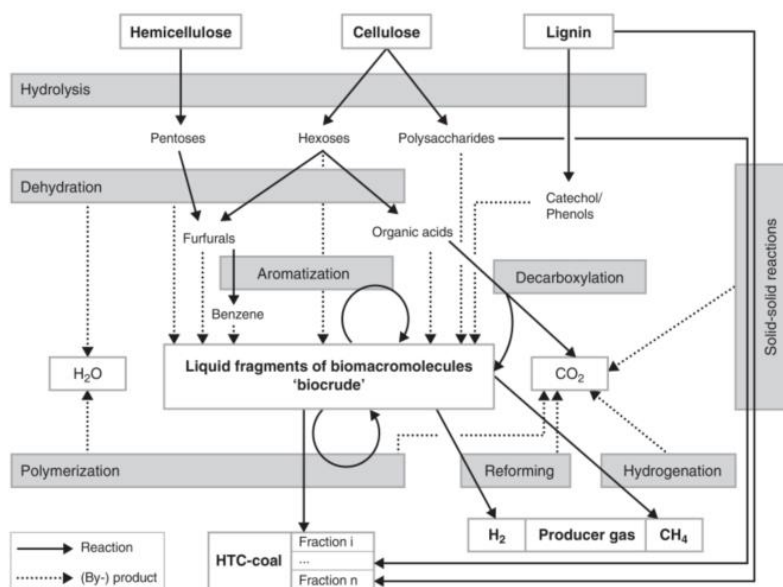
- (4) solving the equations by applying numerical methods and fitting the experimental data in order to obtain the best values of the free parameters.

The kinetics modeling of HTC is a demanding task. HTC occurs through a complex network of reactions, whose mechanisms are still not fully understood. The intrinsic complexity and variety of biomass do not facilitate the modeling. Indeed, each biomass component has its own structural and chemical properties, and interacts with the other components/chemical species during the process involving heterogeneous reactions. Therefore, a detailed and comprehensive reaction mechanism, comprised of reaction kinetics, thermodynamics, and transport phenomena, is still lacking [36].

(a)



(b)



**Figure 3.1.** Simplified (a) and detailed (b) HTC reaction pathway [37–39]. Figure 3.1a reprinted and adapted with permission from Keiller et al. Copyright (2019) American Chemical Society; and Lucian et al., published by MDPI, 2019. Figure 3.1b reprinted with permission from Kruse et al. Copyright © 2013, Elsevier.

To date, several reaction mechanisms have been identified and experimentally observed. During HTC, biomass undergoes several reactions that include hydrolysis, dehydration, decarboxylation, polymerization, and aromatization [40]. In particular, the solid residue resulting from HTC (the hydrochar) derives from two different reaction pathways, which lead to two distinct solid fractions. One is called “primary char” and results from a solid-solid conversion. Having a similar morphology and structure of the parent biomass, it represents somehow its unconverted fraction [41]. The second is known as “secondary char” and forms in the aqueous phase. Secondary char results from the condensation and re-polymerization of dissolved intermediates in the liquid phase [42]. Therefore, to model this complex mechanism, a sequence of steps of reaction is usually established. Schematic sketches of the simplified and detailed reaction mechanism are shown in Figure 3.1a and Figure 3.1b, respectively.

After having assumed a certain reaction pathway, mass balances have to be written for the various species involved in the HTC reactions: the biomass, the hydrochar(s), the gaseous molecules and the molecules present in the liquid phase. Some species can be grouped in “lumped-components” in order to simplify such a complex reacting system.

The approach adopted to model the HTC kinetics derives from typical homogenous phase reacting systems. This is a huge approximation of the reality: HTC occurs in a complex heterogeneous system, characterized by mass transfer phenomena and where reactions take place in the solid, liquid, and gaseous phases. The approximation of a homogenous system is to some extent justified by the much larger amount of water than biomass (common biomass to water ratio are around 0.10-0.15). Moreover, a constant volume system is considered. Despite the approximations, this approach is particularly useful for its simplicity and well suits an easy collection of experimental data. Therefore, a relation often found in the literature because of its simplicity and its direct link to the experimental practice is represented by Equation (3.1).

$$r = -\frac{dC_F}{dt} \quad (3.1)$$



Where  $r$  represents the reaction rate ( $\text{mol L}^{-1} \text{s}^{-1}$ ), in particular the degradation rate of a certain feedstock  $F$  whose concentration is  $C_F$  ( $\text{mol L}^{-1}$ ), and  $t$  is time (s). Through experimental activity, the variation in the concentration of the selected feedstock vs. time provides a direct indication of the reaction rate through which the same feedstock reacts. Actually, Equation (3.1) is valid only when a series of conditions are fulfilled: 1) the reactor is a closed system (no matter in, no matter out during the reaction process); 2) reactor perfectly mixed; 3) reacting mixture occupying a constant volume. These three conditions can be actually achieved in a lab HTC reactor. The mass balance of species  $F$  then turns into Equation (3.1) where the transient term on the right of the equal sign equals the generation rate (actually, the degradation rate) of  $F$ . Saying that the degradation rate of  $F$  coincides with the reaction rate contains the implicit simplification that species  $F$  reacts through a single “degradation reaction” where its stoichiometric coefficient is equal to 1. In the case of a degradation reaction, the reaction rate can be expressed through Equation (3.2).

$$r = k C_F^n \quad (3.2)$$

Where  $k$  is the reaction rate constant ( $\text{s}^{-1}$  if  $n=1$ ) and  $n$  is the reaction order. Equation (3.2) is strictly valid for an irreversible reaction occurring in a homogeneous phase. Moreover, according to Equation (3.2), the reaction rate depends only on the concentration of species  $F$ , while in the general case the concentration of other reactants can affect  $r$ . In the case of HTC, it is reasonable to consider that the concentration of water  $C_{\text{H}_2\text{O}}$  is really high and constant, so that  $C_{\text{H}_2\text{O}}$  can be avoided in the reaction rate expression.

Then, the reaction rate constant  $k$  is normally expressed through the Arrhenius Equation (3.3).

$$k = k_0 e^{\frac{-E_a}{RT}} \quad (3.3)$$

Where  $k_0$  is the pre-exponential factor ( $\text{s}^{-1}$ ),  $E_a$  the activation energy ( $\text{J mol}^{-1}$ ),  $R$  the universal gas constant ( $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ ), and  $T$  the temperature (K). The Arrhenius Equation follows from the collision theory developed for reacting fluid systems: its application to reacting solid system, in particular to solids undergoing degradation reactions, is not so straightforward.

Through modeling, values for the Arrhenius kinetic parameters are estimated. The common approach consists of: (1) estimation of the rate

constants  $k_i$  of each reaction by applying specific numerical methods starting from experimental data of concentration variations; (2) construction of the Arrhenius plot ( $\ln k_i$  versus  $1/T$ ) for each  $i$ -reaction; (3) computation of  $E_a$  and  $k_0$  from the Arrhenius plot.

As previously shortly mentioned, a step forward in respect to the very basic description offered by Equations (3.1) and (3.2) is considering the whole set of mass balances of all the species (or lumped-components) involved in the HTC reactions: reactants, intermediates and products. This would result in a system of differential equations where time  $t$  is the only independent variable. The generation term of the various species appears in the mass balances as a function of the various reaction rates. For example, assuming that all the reactions are first order apart from the secondary char formation, Figure (1a) can be described by the set of Equations (3.4-3.8):

$$\frac{dC_B}{dt} = -k_1 C_B - k_2 C_B - k_3 C_B \quad (3.4)$$

$$\frac{dC_L}{dt} = k_1 C_B - k_4 C_L - k_5 C_L^n \quad (3.5)$$

$$\frac{dC_{HC1}}{dt} = k_3 C_B \quad (3.6)$$

$$\frac{dC_{HC2}}{dt} = k_5 C_L^n \quad (3.7)$$

$$\frac{dC_G}{dt} = k_2 C_B + k_4 C_L \quad (3.8)$$

where  $C_B$ ,  $C_L$ ,  $C_{HC1}$ ,  $C_{HC2}$ , and  $C_G$  are the molar concentrations in  $\text{mol/m}^3$  of the  $i$ -species (like atomic carbon) respectively in the biomass (B), liquor (L), primary char (HC1), secondary char (HC2), and gas phase (G) derived from the biomass and the HTC liquor [38]. In addition to the description of HTC through an Arrhenius behavior and a lumped approach, which are however functional and hardly imaginable to be removable, the current modeling presents other limits like the following.

- (1) A common approach consists in considering *pseudo-first-order reactions* ( $n=1$ ) which, although its constructive simplicity, hides the effective behavior of the system and omits some important aspects. Indeed, secondary char formation is highly favored at high concentrations of the intermediate products, meaning that the kinetics model should be likely described by a higher reaction order ( $n>1$ ). This issue is pointed out also by Jung et al. [43], who

demonstrated that a first-order reaction is unsatisfactory for describing the growth of hydrochar from fructose and that computational results are affected by the initial fructose concentration. The formation of secondary char is thought to occur through typical second-order polymerization reactions.

- (2) Due to the complexity of the mathematical handling, authors often consider the *temperature independent of time*, so that Equations (3.1), (3.2) and (3.3) can be solved by separation of variables. This is a strong simplification. Indeed, a distinct heat-up phase and a constant temperature period characterize batch reactors, which have been the object of most kinetics studies [44,45]. During the heating period, a significant biomass conversion occurs and the transient phase could be neglected only in case of rapid heating times, which could occur only in batch micro-reactors [38,46].
- (3) Common kinetics models do not consider *mass and heat transport phenomena*, which instead can affect the overall reaction rate. Indeed, during HTC, water and products have to enter and leave the biomass pores. Therefore, mass transfer through pores can represent a limiting factor for the HTC process kinetics.
- (4) Models are commonly developed on the base of experimental activity performed on a certain *feedstock*, which affects the chemistry of the reactions involved. Therefore, their application is often limited to the type of biomass and operating conditions used.

#### 3.4.2 Pseudo-first-order reaction kinetics models

As abovementioned, most of the developed models are pseudo-first-order ( $n=1$ ), which implies that the reaction rate depends linearly on the reactant (e.g. the biomass) concentration. Even if these models neglect higher-order reactions (like polymerization), they have the advantage of being easily numerically implemented. Table 3.1 sums up the process conditions and the results of the main literature concerning these models.

It is important to underline that the hypothesized reaction pathway highly affects the values of the kinetic parameters: each reaction stage is characterized by its activation energy and pre-exponential factor. Therefore, the same feedstock (better: feedstock degradation reaction) can present different values of kinetic parameters in dependence of the reaction scheme adopted. Moreover, the models currently present in the

literature consider ashes as inert during the process, neglecting their slight decomposition/evolution during HTC and their possible catalytic effect.

Cellulose has been a widely investigated feedstock applying a pseudo-first-order approach. During HTC, due to temperature and pressure conditions, glycosidic bonds that connect glucose monomers forming cellulose are broken up to form glucose monomers [47]. However, competing reactions can also be present, which complicates the overall mechanism. One of the first kinetic studies on cellulose was performed by Schwald and Bobleter [48] who, assuming a pseudo-first-order reaction into glucose and degradation products, obtained an activation energy of cotton cellulose of  $129 \text{ kJ mol}^{-1}$ . Besides, Yousefifar et al. [49] considered a peculiar reaction pathway, in which the gaseous products derive from both the direct conversion of soluble intermediates and volatilization of intermediates, finding out three different kinetic parameters on the base of the chemical oxygen demand (COD). Dos Santos Rocha et al. [50] developed a more complex pathway to describe the decomposition of cellulose derived from sugarcane straw, considering cellobiose, glucose, formic acid, and hydroxymethylfurfural (HMF) in the reaction mechanism. The model fits well experimental data in a temperature range between 180 and 195 °C, while it overestimates the degradation at 210 °C since it does not consider the difference between crystalline and amorphous cellulose. In the same study, the degradation of hemicellulose into xylose, arabinose, acetic acid, glucuronic acid, and furfural was also investigated.

Other pure compounds have been investigated. Rogalisnki et al. [51] characterized the degradation of different biopolymers (starch, cellulose, and protein), focusing on the formation of glucose and its subsequent degradation. In the proposed reaction scheme, the substrate undergoes two possible parallel first-order reactions to form glucose and by-products (mainly maltose, oligomers, and oligopeptides). Then, glucose further decomposes into degradation products. Results showed that the degradation kinetics of protein was disadvantaged compared to that of cellulose and starch due to the presence of peptide bonds, which are more stable under hydrothermal conditions [52].

Moreover, several authors studied the kinetics of lipids degradation [53–55]. These, during HTC, due to the high temperature and pressure, become more miscible in water and degrade into fatty acids [47]. Fujii et al. [53] demonstrated that monoacylglycerides degrade into fatty acids following first-order reactions, with an activation energy of  $73.9 \text{ kJ mol}^{-1}$  and a pre-exponential factor of  $5.65 \times 10^4 \text{ s}^{-1}$  when subjected to

an isothermal program, while the value of the two kinetic parameters become equal to, respectively,  $77.5 \text{ kJ mol}^{-1}$  and  $1.01 \times 10^5 \text{ s}^{-1}$  in a constant heating program. In the same research, they tested several fatty acids (caprylic, capric, and lauric acid) under different constant heating rates, demonstrating that their degradation can be also described through a first-order reaction mechanism [53].

First-order reactions were also used to model other pure compounds that are representative of biomass. Bicker et al. [56] studied the degradation of glucose, fructose, and pyruvaldehyde between 200 and 300 °C using zinc as catalyst. Jing and Lu [57] characterized the hydrothermal decomposition between 180 and 220 °C of glucose, which degrades through a series of first-order reactions into 5-HMF and levulinic acid. Braghiroli et al. [58] investigated the hydrothermal decomposition of tannin. Sheehan et al. [59] carried out a study on the degradation of tetra-alanine between 170 and 230 °C, considering the effects of the pH on the kinetics.

Although the individual characterization of the biomass fractions is useful to understand their separate behavior, when enclosed inside the biomass each compound may kinetically behave differently, as demonstrated for instance for cellulose [60]. Single reactions can affect the others, causing a variation of the kinetics parameters. Therefore, research on the biomass itself is crucial for the development of accurate models predictive of the biomass behavior during HTC. Generally, however, for an understandable model simplification, the biomass is assumed to be composed of certain fractions of model compounds that decompose independently of each other and can lead to the formation of common species.

In this regard, Reza et al. [61] validated a simple kinetics model for cellulose and hemicellulose through HTC tests on loblolly pine (200-260 °C). In the proposed reaction mechanism, cellulose and hemicellulose degrade through two parallel first-order reactions, while lignin is considered as inert. In particular, they assumed that cellulose degrades into a solid phase, aqueous chemicals, and gases, while hemicellulose into aqueous chemicals and gases. The reduction of the solid concentration was used to validate the model, resulting in kinetic parameters (activation energy:  $73 \text{ kJ mol}^{-1}$  for cellulose and  $29 \text{ kJ mol}^{-1}$  for hemicellulose) lower than those computed for the separate compounds. Indeed, the degradation of aqueous extractives enhances the degradation of cellulose and hemicellulose by reducing the pH, while the degradation of hemicellulose into acids catalyzes the cellulose cleavage [61,62]. Besides, Basso et al. [63] and Baratieri et al. [64]

modeled the HTC reaction kinetics of grape seeds, a constituent of grape marc residual from the wine-making process, in a two-step first-order reaction scheme, between 180 and 250 °C. The scheme assumes that biomass degrades into intermediate products that further degrade into hydrochar, while volatiles form from both intermediates and hydrochar.

The HTC kinetics of sugarcane bagasse was studied by Iryani et al. [65], who adopted a shrinking core model according to which the reaction zone progressively moves from the external to the internal part of the reacting biomass particle. Then, they assumed that the degradation can be described by two parallel first-order reactions (hydrolysis and dehydration), characterized by their proper kinetics parameters.

First-order reactions have been also applied for the modeling of sewage sludge. Danso-Boateng et al. [66] modeled the decomposition of primary sewage sludge and synthetic feces with a simple first-order model based on solid yield conversion. They obtained values in the ranges of 70.4-77.8 kJ mol<sup>-1</sup> and 4.0 x 10<sup>6</sup>-1.5 x 10<sup>7</sup> min<sup>-1</sup> for the activation energy and pre-exponential factor, respectively. Still regarding sewage sludge, Yin et al. [67] developed a first-order model based on the decomposition of macromolecules and low molecular weight constituents. The reaction scheme includes two stages: a first decomposition of the biomass into its primary constituents (proteins and saccharides, present inside microorganism and the extracellular polymeric substances of the sludge) and their degradation in water into ammonia, acetic acid, and gaseous products. Besides, they included correction terms to fit the data. Results show that the decomposition into proteins and saccharides occurs with an activation energy of 51 kJ mol<sup>-1</sup>, while into acetic acid of 29.7 kJ mol<sup>-1</sup>.

Adopting a liquid-phase thermogravimetric technique, Mochidzuki et al. [68] investigated the HTC kinetics of biomass residues like old newspaper, rice husk, and spent malt from brewery.

A different approach was proposed by Luo et al. [69], who developed a first-order model for the degradation of water hyacinth by considering kinetics expressions commonly used for pyrolysis and combustion of solid feedstock. They adopted a non-isothermal method and, by introducing the heating rate ( $\beta = dT/dt$ ) constant during the tests and the conversion factor  $\alpha$ , obtained Equation (3.9).

$$\frac{d\alpha}{dT} = \frac{A}{\beta} \exp\left(-\frac{E_a}{RT}\right) \alpha \quad (3.9)$$

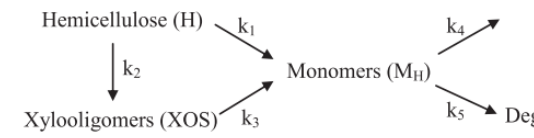
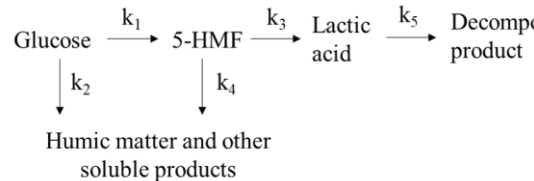
By applying the Coats-Redfern differential method, they computed the kinetic parameters in two temperature ranges (150-210 and 200-280 °C). Notably, there is a considerable different behavior between the two ranges, which is attributed to the subsequent decomposition of hemicellulose (lower temperature range,  $E_a = 147.11 \text{ kJ mol}^{-1}$  and  $k_0 = 1.74 \times 10^{16} \text{ min}^{-1}$ ) and cellulose (upper temperature range,  $E_a = 90.79 \text{ kJ mol}^{-1}$  and  $k_0 = 5.08 \times 10^8 \text{ min}^{-1}$ ). Using the same approach, Liu and Balasubramanian [44] performed a kinetic model study on coconut fiber and eucalyptus leaves, treated between 150 and 300 °C.

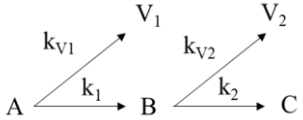
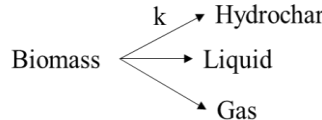
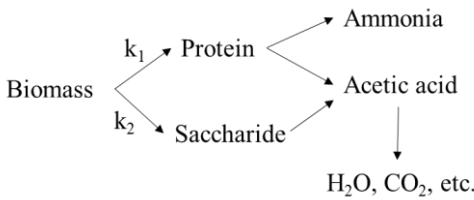
**Table 3.1.** Sum of first-order (n=1) reaction models presented in this work.

| Feedstock                                   | Mechanism and details of the process                                             | Process conditions      | Arrhenius kinetic parameters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Ref. |
|---------------------------------------------|----------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Cellulose                                   | Cellulose $\xrightarrow{k_1}$ Glucose $\xrightarrow{k_2}$ Decomposition products | 200-300 °C              | $E_{a,1}=129.1 \text{ kJ mol}^{-1}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | [48] |
| Cellulose                                   |                                                                                  | 180-260 °C              | $E_{a,1} = 226.5 \text{ kJ mol}^{-1}$ , $k_{0,1} = 2.3 \times 10^{23} \text{ s}^{-1}$<br>$E_{a,2} = 423 \text{ kJ mol}^{-1}$ , $k_{0,2} = 9.0 \times 10^{40} \text{ s}^{-1}$<br>$E_{a,3} = 12.9 \text{ kJ mol}^{-1}$ , $k_{0,3} = 5.7 \text{ s}^{-1}$                                                                                                                                                                                                                                                                                                      | [49] |
| (1) Cellulose<br>(2) Starch<br>(3) Proteins |                                                                                  | 240-310 °C              | (1) $E_{a,f} = 160.1 \text{ kJ mol}^{-1}$ , $k_{0,f} = 1.7 \times 10^{13} \text{ s}^{-1}$<br>$E_{a,d} = 168.7 \text{ kJ mol}^{-1}$ , $k_{0,d} = 4.6 \times 10^{14} \text{ s}^{-1}$<br>(2) $E_{a,f} = 134.4 \text{ kJ mol}^{-1}$ , $k_{0,f} = 1.1 \times 10^{11} \text{ s}^{-1}$<br>$E_{a,d} = 72.5 \text{ kJ mol}^{-1}$ , $k_{0,d} = 1.8 \times 10^5 \text{ s}^{-1}$<br>(3) $E_{a,f} = 157.2 \text{ kJ mol}^{-1}$ , $k_{0,f} = 7.4 \times 10^{11} \text{ s}^{-1}$<br>$E_{a,d} = 236.5 \text{ kJ mol}^{-1}$ , $k_{0,d} = 1.9 \times 10^{20} \text{ s}^{-1}$ | [51] |
| Cellulose                                   |                                                                                  | 180-210 °C              | $E_{a,1} = 105.14 \text{ kJ mol}^{-1}$ , $\ln(k_{0,1}) = 22.40$<br>$E_{a,2} = 216.44 \text{ kJ mol}^{-1}$ , $\ln(k_{0,2}) = 49.67$<br>$E_{a,3} = 105.60 \text{ kJ mol}^{-1}$ , $\ln(k_{0,3}) = 25.32$<br>$E_{a,4} = 181.35 \text{ kJ mol}^{-1}$ , $\ln(k_{0,4}) = 41.93$<br>$E_{a,5} = 118.98 \text{ kJ mol}^{-1}$ , $\ln(k_{0,5}) = 28.68$<br>$E_{a,6} = 266.17 \text{ kJ mol}^{-1}$ , $\ln(k_{0,6}) = 62.78$                                                                                                                                             | [50] |
| Cellulose                                   | Liquid-phase thermogravimetry, at constant heating rate                          | 250-280 °C,<br>< 10 Mpa | $E_a = 165 \text{ kJ mol}^{-1}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | [68] |



### 3. Modeling hydrothermal carbonization

|                                                  |                                                                                                                            |            |                                                                                                                                                                                                                                                                                                                                                                                                                             |      |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Hemicellulose                                    |                                          | 180-210 °C | $E_{a,1} = 62.68 \text{ kJ mol}^{-1}, \ln(k_{0,1}) = 10.90$<br>$E_{a,2} = 109.49 \text{ kJ mol}^{-1}, \ln(k_{0,2}) = 25.75$<br>$E_{a,3} = 220.23 \text{ kJ mol}^{-1}, \ln(k_{0,3}) = 53.66$<br>$E_{a,4} = 122.57 \text{ kJ mol}^{-1}, \ln(k_{0,4}) = 27.95$<br>$E_{a,5} = 146.47 \text{ kJ mol}^{-1}, \ln(k_{0,5}) = 34.71$<br>$E_{a,6} = 119.91 \text{ kJ mol}^{-1}, \ln(k_{0,6}) = 26.08$                                 | [50] |
| (1) Glucose<br>(2) Fructose<br>(3) Pyruvaldehyde | Zinc catalyzed reactions                                                                                                   | 200-300 °C | $E_{a,1} = 102 \text{ kJ mol}^{-1}, k_{0,1} = 8.25 \times 10^8 \text{ s}^{-1}$<br>$E_{a,2} = 96 \text{ kJ mol}^{-1}, k_{0,2} = 1.79 \times 10^8 \text{ s}^{-1}$<br>$E_{a,3} = 58 \text{ kJ mol}^{-1}, k_{0,3} = 5.21 \times 10^4 \text{ s}^{-1}$                                                                                                                                                                            | [56] |
| Glucose                                          |                                          | 180-200 °C | $E_{a,1} = 108.03 \text{ kJ mol}^{-1}, k_{0,1} = 5.06 \times 10^9 \text{ s}^{-1}$<br>$E_{a,2} = 135.72 \text{ kJ mol}^{-1}, k_{0,2} = 4.29 \times 10^{12} \text{ s}^{-1}$<br>$E_{a,3} = 95.40 \text{ kJ mol}^{-1}, k_{0,3} = 2.62 \times 10^6 \text{ s}^{-1}$<br>$E_{a,4} = 108.91 \text{ kJ mol}^{-1}, k_{0,4} = 3.57 \times 10^9 \text{ s}^{-1}$<br>$E_{a,5} = 31.29 \text{ kJ mol}^{-1}, k_{0,5} = 0.031 \text{ s}^{-1}$ | [57] |
| Tannin                                           | Dissolved tannin (aq) $\xrightarrow{k}$ Hydrochar                                                                          | 130-200 °C | $E_a = 91 \text{ kJ mol}^{-1}$                                                                                                                                                                                                                                                                                                                                                                                              | [58] |
| Monoacylglycerides                               | (1) Isothermal thermal program<br>(2) Constant heating rate program                                                        | 25-350 °C  | $E_{a,1} = 73.9 \text{ kJ mol}^{-1}, k_{0,1} = 1.40 \times 10^{11} \text{ s}^{-1}$<br>$E_{a,2} = 77.5 \text{ kJ mol}^{-1}, k_{0,2} = 1.21 \times 10^8 \text{ s}^{-1}$                                                                                                                                                                                                                                                       | [53] |
| Loblolly pine                                    | (1) Hemicellulose $\rightarrow$ Aqueous chemicals + Gas<br>(2) Cellulose $\rightarrow$ Hydrochar + Aqueous chemicals + Gas | 200-260 °C | $E_{a,1} = 73 \text{ kJ mol}^{-1}, k_{0,1} = 8.24 \times 10^5 \text{ s}^{-1}$<br>$E_{a,2} = 29 \text{ kJ mol}^{-1}, k_{0,2} = 58.6 \times 10^5 \text{ s}^{-1}$                                                                                                                                                                                                                                                              | [61] |

|                                                      |                                                                                      |                          |                                                                                                                                                                                                                                                                                                                                                        |      |
|------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Grape seeds                                          |     | 180-250 °C               | $E_{a,1} = 94.5 \text{ kJ mol}^{-1}, k_{0,1} = 3.34 \times 10^7 \text{ s}^{-1}$<br>$E_{a,v1} = 139.7 \text{ kJ mol}^{-1}, k_{0,v1} = 1.10 \times 10^{10} \text{ s}^{-1}$<br>$E_{a,2} = 93.7 \text{ kJ mol}^{-1}, k_{0,2} = 9.15 \times 10^6 \text{ s}^{-1}$<br>$E_{a,v,2} = 146.2 \text{ kJ mol}^{-1}, k_{0,v,2} = 1.55 \times 10^{10} \text{ s}^{-1}$ | [64] |
| Sugarcane bagasse                                    | Shrinking core model, with simultaneous hydrolysis (1) and dehydration (2) reactions | 200-300 °C               | $E_{a,1} = 88.1 \text{ kJ mol}^{-1}, k_{0,1} = 58.6 \times 10^5 \text{ s}^{-1}$<br>$E_{a,2} = 129.4 \text{ kJ mol}^{-1}, k_{0,2} = 2.9 \times 10^9 \text{ s}^{-1}$                                                                                                                                                                                     | [65] |
| (1) Primary sewage sludge<br>(2) Synthetic faces     |     | 140-200 °C               | $E_{a,1} = 70 \text{ kJ mol}^{-1}, k_{0,1} = 4.0 \times 10^6 \text{ min}^{-1}$<br>$E_{a,2} = 78 \text{ kJ mol}^{-1}, k_{0,2} = 1.5 \times 10^7 \text{ min}^{-1}$                                                                                                                                                                                       | [66] |
| Sewage sludge                                        |     | 180-300 °C               | $E_{a,1} = 51.0 \text{ kJ mol}^{-1}, k_{0,1} = 238.21 \text{ s}^{-1}$<br>$E_{a,2} = 50.89 \text{ kJ mol}^{-1}, k_{0,2} = 75.16 \text{ s}^{-1}$                                                                                                                                                                                                         | [67] |
| (1) Rice husk<br>(2) Old newspaper<br>(3) Spent malt | Liquid-phase thermogravimetry, at constant heating rate                              | < 300 °C and 10 MPa      | $E_{a,1} = 100 \text{ kJ mol}^{-1}, k_{0,1} = 7.4 \times 10^8 \text{ s}^{-1}$<br>$E_{a,2} = 79 \text{ kJ mol}^{-1}, k_{0,2} = 2.4 \times 10^6 \text{ s}^{-1}$<br>$E_{a,3} = 96 \text{ kJ mol}^{-1}, k_{0,3} = 4.4 \times 10^8 \text{ s}^{-1}$                                                                                                          | [68] |
| Water hyacinth                                       | Constant heating rate program                                                        | 150-210 °C<br>200-280 °C | $E_{a,1} = 110.36 \text{ kJ mol}^{-1}, k_{0,1} = 7.19 \times 10^{11} \text{ min}^{-1}$<br>$E_{a,2} = 116.92 \text{ kJ mol}^{-1}, k_{0,2} = 7.07 \times 10^{10} \text{ min}^{-1}$                                                                                                                                                                       | [69] |
| (1) Coconut fibers<br>(2) Eucalyptus leaves          | Constant heating rate program                                                        | 150-300 °C               | $E_{a,1} = 67.41 \text{ kJ mol}^{-1}, k_{0,1} = 4.08 \times 10^{12} \text{ s}^{-1}$<br>$E_{a,2} = 59.18 \text{ kJ mol}^{-1}, k_{0,2} = 4.38 \times 10^{11} \text{ s}^{-1}$                                                                                                                                                                             | [44] |

### 3.4.3 Higher-order reaction kinetics models

Higher-order reaction kinetics models ( $n > 1$ ) allow considering important aspects commonly absent in first-order models. Indeed, the formation of secondary char, the water-insoluble solid fraction favored under harsh HTC conditions [41], occurs through polymerization and polycondensation reactions. Polymerization reactions are often characterized by a second-order kinetics or, to be more generalist, their rate increases more than proportionally with respect to the increase in concentration of the reagents. Therefore, from the experimental activity and the intrinsic kinetics of the polymerization reactions, severe HTC conditions and high biomass to water ratios (i.e. high biomass concentrations) seem to increase the probability of having higher-order reactions. Overall higher-order reaction models are more sophisticated and reliable than the first-order models. The price for this precision is a more complicated numerical handling and the need of a very well-defined reaction pathway consistent with the experimental measures.

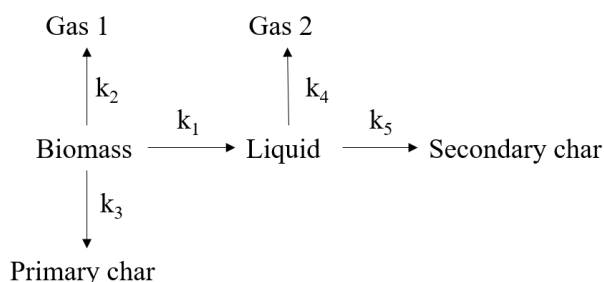
Thus, several authors developed HTC kinetics models considering higher reaction orders, as summarized in Table 3.2. It is worth noting that the handling of higher-order reaction models requires particular attention. Indeed, the values of the concentration of the starting feedstock can affect the reaction kinetics parameters. This is particularly true for biomasses, where the concept of molar concentration is not straightforward. Authors often try to adapt the concepts valid for homogenous systems (where the molar concentration is referred to as  $\text{mol/m}^3$ ) also to heterogeneous systems (insoluble biomass in water), and this introduces a certain subjectivity and dependence of the results on the choices made. For example, when the initial concentration is lower than one, the reaction rate is lower for models with  $n > 1$  than for those with  $n = 1$ , which may be counterintuitive. Intuitively, a higher order reaction ( $n > 1$ ) is faster than a first order reaction, which actually occurs if the initial concentration of the reagents is greater than one - therefore it becomes decisive as the concentration in biomass-water systems is defined. Generally, when  $n$  is different than 1, the pre-exponential factor  $k_0$  will assume the unit of measure  $\text{s}^{-1}(\text{m}^3)^{n-1}\text{mol}^{1-n}$ .

Knežević et al. [70] are the first authors who stressed the need for developing a kinetic model for HTC with a reaction-order higher than one. They investigated the decomposition rate of glucose between 250 and 350 °C. According to their proposed reaction model, glucose degrades into water-soluble products (WSS) that can further degrade

into water-solvent insoluble products (WSIS) through a second-order reaction. By measuring the yield of the products and glucose degradation at different residence times, temperatures, and initial concentrations, they obtained the reaction rate for each modeled reaction. In addition, after observing that lower initial glucose concentrations increase the overall glucose degradation rate, they considered that glucose degrades into WSS according to a reaction order smaller than one. In this regard, they fitted the data using a reaction order equal to 0.9, and they computed an activation energy of glucose degradation equal to  $114 \text{ kJ mol}^{-1}$ , and a pre-exponential factor of  $1.0 \times 10^9 \text{ g}^{1-n} \text{ s}^{-1} \text{ L}^{n-1}$ .

Later, Jatzwauck and Schumpe [71] modeled the HTC of soft rush using a simplified scheme, where the solid substrate undergoes degradation into dissolved intermediates ( $n=1$ ), which can form gaseous molecules ( $n=1$ ) or polymerize into hydrochar according to a higher-order reaction ( $n = 1.53$ ). Therefore, solid-solid reactions were not considered. The concentration was defined in terms of atomic carbon. Results show that the activation energy of the first degradation step (hydrolysis,  $141 \text{ kJ mol}^{-1}$ ) is much higher than that of the other subsequent reactions ( $75.0$  and  $74.3 \text{ kJ mol}^{-1}$ ).

A kinetics scheme that considers the pathways of primary and secondary hydrochar formation was developed by Lucian et al. [38,72]. The model predicts the carbon distribution among reaction intermediates and the products, and was validated on several biomasses: olive trimmings, grape marc, and *Opuntia Ficus Indica*. Even if do not performed by the authors, after a proper calibration, the model could be used to predict the elemental composition of the considered products. The reaction scheme is reported in Figure 3.2, where the secondary char is produced according to an  $n$ -order reaction and derives from the polymerization and condensation of the water soluble-species in the liquid phase. Results show that the formation of primary char is the most favored reaction pathway, while that of secondary char occurs with a reaction order between 1 and 2 (the higher the HTC temperature, the higher the order). The model comprises also the heating-up phase and both modeling and experimental results point out the importance of considering into the computations such a temperature transient phase.



**Figure 3.2.** Reaction pathway scheme proposed by Lucian et al. [38,72]. Reproduced with permission from Lucian et al., *Chemical Engineering Transactions*; published by AIDIC, 2018. And Lucian et al., *Energies*; published by MDPI, 2019.

Another higher-order model was presented by Jung et al. [43] for the decomposition of fructose. The authors point out the necessity of introducing a higher reaction order to account for the effect of the initial fructose concentration on the hydrochar yield. In the proposed scheme, fructose decomposes into degradation products and HMF, which can polycondensate into hydrochar (in a second-order reaction) or other degradation products. The model was calibrated using gravimetric data and predicts kinetics parameters at different feedstock concentrations.

Recently, Keiller et al. [37] proposed a model based on the relative amount variation of cellulose, hemicellulose, and lignin during HTC of Australian saltbush. Through HPLC analysis of key-compounds, kinetic parameters were computed assuming several reaction orders (0.5, 1, and 2) for the degradation of cellulose, hemicellulose, and lignin. By comparing the predicted parameters with experimental results, they showed that the first-order reaction models behave the best for hemicellulose and lignin, with activation energies of 61 and 66 kJ mol<sup>-1</sup>, respectively. Meanwhile, a 0.5-order model fits better data for cellulose degradation, leading to an activation energy of 127 kJ mol<sup>-1</sup> and suggesting that other mechanisms could be involved during the degradation of cellulose.

A novel methodological approach was recently proposed by Pecchi et al. [73], who used a high-pressure differential scanning calorimetry to assess the kinetic parameters of sludge and digestate degradation during HTC. The method adopts the concept of the extent of reaction  $\alpha$ , defined as the ratio between the instantaneous heat of reaction and the maximum heat released at the highest temperature. Experimental curves

were used to calibrate an  $n^{\text{th}}$  order and an autocatalytic kinetic model. Results show reactions orders of 2.68 (sludge) and 2.46 (digestate).

Overall, what emerges from the above analysis is that, even if modeling the degradation of the same biomasses, it is difficult to compare the values of the kinetics parameters among the different models and with the first-order models. Indeed, these values are highly dependent on the hypothesized reaction mechanism. Each reaction stage is characterized by its proper parameters and reflects to some extent the overall reaction pathway.

**Table 3.2.** Sum of higher-order reaction models presented in this work.

| Feedstock                                        | Mechanism and details of the process                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Process conditions and reaction orders                                           | Arrhenius kinetic parameters                                                                                                                                                                                                                                                                                                                                                 | Ref. |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Glucose                                          | <p>Glucose <math>\xrightarrow{k_G, n=1}</math> a WSS-A <math>\xrightarrow{k_A, n=1}</math> gas</p> <p>Glucose <math>\xrightarrow{k_G, n=1}</math> b WSS-B <math>\xrightarrow{k_B, n=2}</math> WSIS-A <math>\xrightarrow{k_D, n=1}</math> f gas</p> <p>Glucose <math>\xrightarrow{k_G, n=1}</math> c WSS-C <math>\xrightarrow{k_C, n=2}</math> WSIS-A <math>\xrightarrow{k_D, n=1}</math> (1-f) WSIS-B</p> <p>Glucose <math>\xrightarrow{k_G, n=1}</math> d WSS-D <math>\rightarrow</math> e H<sub>2</sub>O</p> <p>WSS = water-solvent phase<br/>WSIS = water-solvent insoluble phase</p> | 250-350 °C<br>$n_{G1} = 1$<br>$n_{G2} = 0.9$<br>$n_A, n_D = 1$<br>$n_B, n_C = 1$ | $E_{a,G1} = 114 \text{ kJ mol}^{-1}$ , $k_{0,G1} = 7.7 \times 10^8 \text{ s}^{-1}$<br>$E_{a,G2} = 114 \text{ kJ mol}^{-1}$ , $k_{0,G2} = 1.0 \times 10^9 \text{ g}^{1-n} \text{ L}^{n-1} \text{ s}^{-1}$<br>$k_A = 0.0042 \text{ s}^{-1}$<br>$k_B = 0.000083 \text{ g}^{-1} \text{ L s}^{-1}$<br>$k_C = 0 \text{ g}^{-1} \text{ L s}^{-1}$<br>$k_D = 0.00025 \text{ s}^{-1}$ | [70] |
| Fructose                                         | <p>Fructose <math>\xrightarrow{k_3}</math> R1</p> <p>Fructose <math>\xrightarrow{k_1}</math> HMF <math>\xrightarrow{k_4}</math> R2</p> <p>HMF <math>\xrightarrow{k_2}</math> Hydrochar</p>                                                                                                                                                                                                                                                                                                                                                                                               | 190-250 °C<br>$n_1, n_3, n_4 = 1$<br>$n_2 = 2$                                   | -                                                                                                                                                                                                                                                                                                                                                                            | [43] |
| (1) Hemicellulose<br>(2) Cellulose<br>(3) Lignin | <p>Hemicellulose <math>\xrightarrow{k_1}</math> Monosaccharides <math>\rightarrow</math> Secondary char + Gas</p> <p>Cellulose <math>\xrightarrow{k_2}</math> Oligoglucans + Glucose <math>\rightarrow</math> Secondary char</p> <p>Lignin <math>\xrightarrow{k_3}</math> Secondary char</p>                                                                                                                                                                                                                                                                                             | 200-260 °C<br>$n_1 = 0.5$<br>$n_2 = 1$<br>$n_3 = 1$                              | $E_{a,1} = 61 \text{ kJ mol}^{-1}$ , $k_{0,1} = 3.1 \times 10^5 \text{ s}^{-1}$<br>$E_{a,2} = 127 \text{ kJ mol}^{-1}$ , $k_{0,2} = 8.8 \times 10^{11} \text{ s}^{-1}$<br>$E_{a,3} = 66 \text{ kJ mol}^{-1}$ , $k_{0,3} = 9.5 \times 10^5 \text{ s}^{-1}$                                                                                                                    | [37] |

|                                                                |                                                                                                                                                                                                                           |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                              |      |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Soft rush                                                      | <pre> graph LR     Substrate -- k1 --&gt; DI[Dissolved intermediates]     DI -- k2 --&gt; Hydrochar     DI -- k3 --&gt; GDB[Gaseous and dissolved byproducts]           </pre>                                            | 180-240 °C<br>$n_1, n_3 = 1$<br>$n_2 = 1.53$                       | $E_{a,1} = 141.0 \text{ kJ mol}^{-1}, k_{0,1} = 1.89 \times 10^{14} \text{ h}^{-1}$<br>$E_{a,2} = 75.0 \text{ kJ mol}^{-1}, k_{0,2} = 1.15 \times 10^7 \text{ m}^{1.59} \text{ kg}^{-0.53} \text{ h}^{-1}$<br>$E_{a,3} = 74.3 \text{ kJ mol}^{-1}, k_{0,3} = 1.89 \times 10^7 \text{ h}^{-1}$                                                                                                | [71] |
| Olive trimmings,<br>Grape marc,<br><i>Opuntia Ficus Indica</i> | <pre> graph LR     Biomass -- k1 --&gt; Liquid     Liquid -- k5 --&gt; SC[Secondary char]     Biomass -- k2 --&gt; G1[Gas 1]     Biomass -- k3 --&gt; PC[Primary char]     Liquid -- k4 --&gt; G2[Gas 2]           </pre> | 180-250 °C<br>$n_1, n_2, n_3, n_4 = 1$<br>$n_5 = 1.10\text{-}2.01$ | $E_{a,1} = 22.03 \text{ kJ mol}^{-1}, k_{0,1} = 82.32 \text{ s}^{-1}$<br>$E_{a,2} = 42.93 \text{ kJ mol}^{-1}, k_{0,2} = 2.51 \times 10^3 \text{ s}^{-1}$<br>$E_{a,3} = 7.75 \text{ kJ mol}^{-1}, k_{0,3} = 7.99 \text{ s}^{-1}$<br>$E_{a,4} = 67.35 \text{ kJ mol}^{-1}, k_{0,4} = 5.38 \times 10^4 \text{ s}^{-1}$<br>$E_{a,5} = 10.37 \text{ kJ mol}^{-1}, k_{0,5} = 1.41 \text{ s}^{-1}$ | [38] |
| (1) Sludge<br>(2) Digestate                                    | -                                                                                                                                                                                                                         | 190-250 °C<br>$n_1 = 2.68$<br>$n_2 = 2.46$                         | $E_{a,1} = 139.16 \text{ kJ mol}^{-1}, k_{0,1} = 2.15 \times 10^{11} \text{ s}^{-1}$<br>$E_{a,2} = 161.68 \text{ kJ mol}^{-1}, k_{0,2} = 1.52 \times 10^{14} \text{ s}^{-1}$                                                                                                                                                                                                                 | [73] |



### 3.4.4 Severity and coalification models

Another approach of kinetic modeling is that adopted in severity and coalification models. These models assess and predict the effects of HTC parameters on hydrochar properties (like yields, chemical composition, and energy content) in a compact way. Indeed, a unique parameter with a particular form condenses the effect of the main influencing parameters (i.e. temperature and reaction time). Overall, these “all-inclusive” models have the advantage of having a relatively easy implementation and show the effect of the HTC process parameters at first glance. However, in addition to a certain dependence on the starting feedstock, they do not reflect HTC reaction mechanisms and do not include important parameters like the biomass load (often expressed as biomass to water ratio) and the initial feedstock carbon content. Among these, we can distinguish between the severity models and the coalification models.

The *severity models* use the so called “severity factor”. This concept was originally introduced by Geniesse and Reuter [74] to model the cracking of oils, and was developed later to characterize the hydrolytic depolymerization of lignocellulosic biomass [75,76]. The severity factor approach was applied successfully also to HTC. The model proposed by Abatzoglou et al. [76] assumes a first-order reaction with an Arrhenius dependence on temperature and defines the reaction coordinate ( $R_0$ ) as expressed in Equation (3.10).

$$R_0 = \int_0^t \exp\left(\frac{T-T_{\text{ref}}}{\omega}\right) dt \quad (3.10)$$

For an isothermal process, Equation (3.10) becomes Equation (3.11).

$$R_0 = t_R \exp\left(\frac{T-T_{\text{ref}}}{\omega}\right) \quad (3.11)$$

Where,  $t_R$  is the reaction time (in minutes) and  $T_{\text{ref}}$  the reference temperature.  $\omega$  is a parameter depending on the gas constant ( $R$ ), the activation energy ( $E_a$ ), and the reference temperature, as expressed in Equation (3.12).

$$\omega = \frac{RT_{\text{ref}}^2}{E_a} \quad (3.12)$$

Montané et al. [77] demonstrated that the reference conditions do not affect significantly the model and can be chosen at the middle of the data sets and  $T_{\text{ref}}$  is usually set at 100 °C and  $\omega$  at 14.75 [75,78]. Being an empirical parameter,  $\omega$  value has been sometimes varied when using other feedstocks. For example, Ko et al. [79] used an  $\omega$  equal to 4.6 when modeling the decomposition of glucose.

To graphically represent the process severity, the reaction coordinate is expressed through the logarithmic function ( $\log R_0$ ). This is referred to as severity factor (SF) and is expressed in Equation (3.13):

$$\text{SF} = \log(R_0) = \log \left[ t_R \exp \left( \frac{T - T_{\text{ref}}}{\omega} \right) \right] \quad (3.13)$$

Hoekman et al. [80] used the severity factor to compare products from fast HTC (reaction time of 20-30 seconds) of loblolly pine with those of typical HTC batch reactors. Starting from experimental data (relevant to 39 HTC tests), Heidari et al. [81] made an attempt to predict the effect of several severity factor values (3.83, 5.01, and 6.19) on the hydrochar mass yield, higher heating value (HHV), and carbon content. With the same aim, in a work dealing with the water recycling effect during the HTC of sawdust, Heidari et al. [82] used severity factors between 4.3 and 5.3. Similarly, Basso et al. [83] reported the solid, liquid, and gaseous phase yields resulting from HTC of grape marc constituents as a function of the severity factor. For the solid (hydrochar) mass yields, the maximum absolute error between predicted and experimental values was of 7.1 %, while for the liquid and gas phases yields was higher than 10 % [83]. In addition, Borrero-López et al. [84] used the severity factor to model the kinetics of furfural and 5-HMF derived from the degradation of hemicellulose, cellulose, and lignin, using two alternative models (one with and one without side reactions).

When fitting the parameters as a function of  $R_0$ , a proper function has to be chosen. Several functions have been reported in literature. For example, Suwelack et al. [85] demonstrated through statistical analysis that the solid, gaseous, and liquid yields follow the pattern expressed by Equation (3.14).

$$Y = a \ln(R_0) + b \quad (3.14)$$

Guo et al. [86] implemented the correlation between HTC output properties (hydrochar yield, carbon content, and HHV) and the severity factor by using a dose-response function as expressed in Equation

(3.15). By investigating the trends of the derivative of  $Y$  (representing the output property) as a function of  $\ln(R_0)$ , they observed that the maximum variation rate (correspondent to the first derivative peak) of the solid yield, carbon content, and HHV, occurs in a severity factor range of 5.8-6.4.

$$Y = a + \frac{b-a}{1+10^{(c-\ln(R_0))d}} \quad (3.15)$$

The coalification model was firstly developed by Ruyter [87], who performed a series of hydrothermal treatments on a variety of low-rank materials, from biomass waste to sub-bituminous coal. The model derives from the Arrhenius relation and describes the oxygen content variation of the feedstock during HTC. One of the main assumptions is that the oxygen of the feedstock is converted to  $\text{CO}_2$  and  $\text{H}_2\text{O}$  at a fixed ratio, regardless of pressure, pH and feedstock, while the complete conversion has been conventionally assumed to occur when the oxygen content is 6 wt.%. The model is based on the relation reported in Equation (3.16).

$$At^B \exp \left[ \frac{-C}{T} \right] = \frac{O_{\text{feed}} - O_t}{O_{\text{feed}} - 6} \quad (3.16)$$

Where  $t$  is the reaction time,  $T$  the reaction temperature,  $O_{\text{feed}}$  the feedstock oxygen content,  $O_t$  the oxygen content of the feedstock undergoing carbonization at reaction time  $t$ , and  $A$ ,  $B$ , and  $C$  are calibrating coefficients ( $A = 50$ ,  $B = 0.2$ , and  $C = 3500$ ).

The coalification model was used by Kruse et al. [88] to characterize the kinetics of brewer grains during HTC, in which  $A$ ,  $B$ , and  $C$  were adjusted on the basis of the experimental campaign performed.

Later, Jung and Kruse [89] evaluated the ability of the coalification model to predict carbon content, oxygen content, and hydrochar yield for a variety of feedstock (29 typologies). For doing this, they linearized Equation (3.16) in Equation (3.17) and determined the coefficients  $A$ ,  $B$ , and  $C$  by fitting experimental data relevant to every feedstock.

$$\ln A + B \ln(t) - \frac{C}{T} \quad (3.17)$$

Thus, to establish the accuracy of the model, results were compared with those obtained applying the severity model (linearized and with the dose-response curve). Results show that the coalification model can

be applied to predict carbon content, oxygen content, and hydrochar yield with reasonable low mean absolute errors. In particular, even if the best fit was achieved when every parameter was fitted individually to every feedstock, little accuracy loss arose when all the parameters were adjusted considering all the feedstock. Authors also proposed and validated a method to predict the hydrochar properties based on the lignin content of biomasses.

### 3.4.5 Other kinetics models

Although the majority of the research around HTC kinetics involves pseudo-first or higher order models, some new interesting approaches have been recently proposed.

In this regard, Gallifuoco [33] demonstrated that HTC can be modeled stochastically. Indeed, it is observed that important HTC properties (like hydrochar HHV and yield) follow a sigmoidal behavior with respect to the reaction time, whatever biomass is adopted [31]. In particular, they demonstrated that an HTC generic property  $y(t)$  can be described with the empirical law as for Equation (3.18).

$$y(t) = y_{\infty} + \frac{y_0 - y_{\infty}}{1 + \left(\frac{t}{t_h}\right)^p} \quad (3.18)$$

Where  $y_0$  is the initial value of the property,  $y_{\infty}$  is its final value,  $t$  is the process time,  $t_h$  the time corresponding at half of the increase of the studied property, and  $p$  an empirical parameter.  $Y(t)$  may describe any property function of time, like H/C and O/C ratios, HHV, solid yield, and the carbon content.

Equation (3.18) is connected to the Hill's equation, which was originally developed for biochemistry processes. In addition, Gallifuoco and Di Giacomo [31] suggested that the Hill's equation could describe HTC properties as a logical consequence of assuming that HTC is governed by probabilistic laws, which describe the average behavior of thousands of random detachments of particles from the parent biomass that simultaneously react between each other.

In another study, Gallifuoco [33] suggested that the formation of hydrochar and the evolution of its properties over time are connected to a Markov process. The study performed was supported by experimental data obtained from several biomasses and a proper mathematical handling. The Markov approach is well established in the literature for

the modeling of stochastic processes in several technological areas (like polymers science, biology, astrophysics, computer networks, etc.) and is the base for the general method of the Markov chain Monte Carlo [90]. According to the Markov process, the HTC can be described in a stochastic way, in which the dynamic of the reactions depends only on the current state, and does not depend on the previous history. Clearly, further research is needed to confirm this attractive approach to modeling, which could avoid the resolution of complicated systems of non-linear equations [33].

### 3.5 Statistical models

Statistical models are often used to correlate key process variables on the base of the experimental activity. Many research articles dealing with HTC contain a proper section dedicated to a statistical analysis in which the process conditions are correlated with output data (like temperature and time with the hydrochar yield, carbon content, and energy properties). A statistically model can be used both to provide a better knowledge highlighting correlations between process variables and to optimize the system [2]. These models are not directly related to the kinetics models: an in-depth modeling investigation around HTC certainly requires the integration of these tools.

In the statistical modeling process, a desired observed quantity (the dependent variable) is correlated with a process factor of interest (the independent variable) through proper statistical correlations, like linear and non-linear regressions, or best fittings derived from a design of experiment (DOE) approach [2]. To validate the model, additional experimental activity is commonly performed and the errors (e.g. standard mean deviation or root square mean error) computed.

Most statistical models highly depend on the specific experimental activity. Thus, they are a function of parameters like the type of feedstock, the HTC conditions (i.e. temperature, residence time, biomass load), the heating rate, the type of reactor used (continuous or batch), and the geometry of the reactor. This dependence limits the prediction capability of these models to a narrow range of experimental conditions and restricts their universal applicability. Nevertheless, statistical models are of great importance for several reasons. Indeed, they are useful:

- (1) to make predictions of specific properties (like mass yields, hydrochar carbon content, energy properties, etc.) within the original experimental range and set-up;
- (2) to understand the magnitude of the correlation between the different variables;
- (3) to individuate the type and optimal combination of process parameters that maximize the information derived from the experimental activity.

### 3.5.1 Linear and non-linear regression models

A widely adopted and easily implementable approach is the multiple linear regression. This method adopts linear functions to relate HTC outputs with the input variables of interest, in which the coefficients of the linear function are estimated from the experimental data. A generic observed variable  $y$  is written as a function of the  $i$ -th independent variable  $x$  as expressed in Equation (3.19).

$$y = a_0 + a_1x_1 + a_2x_2 + \dots + a_nx_n \quad (3.19)$$

Where  $a_i$  are the coefficients estimated through the linear regression and  $n$  the number of independent variables. The advantage of this method is that it bases on a simple function easy to manage and understand. However, linear modeling is limited to specific conditions and can be not sufficient to highlight important relations [91].

For example, Mumme et al. [92] linearly correlated data derived from the HTC of anaerobically digested maize silage. They used a linear function to correlate the carbon content with temperature, residence time, and pH. Even if the mean deviations computed in the validation phase are small (1.54 %), the authors stress the limitations of this kind of model. Indeed, important factors like biomass concentration were not included and the correlation results from - and thus is valid in - a narrow investigated experimental range.

Multiple linear regression was adopted also by Volpe et al. [93] to characterize hydrochars from *Opuntia ficus-indica* cladodes obtained at different carbonization conditions. In particular, they individuated that the hydrochar yield is influenced by three process parameters (temperature, reaction time, and biomass to water ratio), while the energy yield is driven by the biomass to water ratio. However, they highlight that a multivariate model is not able to correlate the fixed

carbon and volatile matter contents and a non-linear model should better represent these data.

An extensive study on the data available in the literature was performed by Li et al. [94], who continued a previous work carried out by the same research group [91]. These papers are of particular interest because they aggregate data derived from a multitude of papers, for a total of 313 in [91] and 138 in [94], thus providing a model based on a variety of process conditions, feedstock, and hydrochar properties. In the first work, Li et al. [91] analyzed data using two statistical approaches, which are the multiple linear regression and the regression tree models. The regression tree approach consists of producing binary trees for each observed variable by splitting them into nodes by applying a recursive portioning [95]. With respect to the linear regression model, it has the advantage of not assuming any a priori relation between the variables. Results show that both the linear and the tree regression approaches can be used to fit the data. Anyway, the authors point out that, as suggested by the highly branching of the tree model, a significantly high level of interactions between the independent variables occurs and a non-linear model is more suitable for the fitting. Moreover, the models highlight that process conditions (temperature and time) mainly affect the solid yield, while the type of feedstock mainly influences the hydrochar carbon and energy content. The regression tree model was expanded to a random forest model by Li et al. [94,96]. Indeed, it consists of a large collection of trees, and its performance is the average between the tree models considered. The studies show that the random forest models have an improved predictive capacity with respect to the linear regression and tree models. In particular, these models were used to represent all the combinations of model parameters to describe the hydrochar yield. Results show that the relation between hydrochar yield and HTC parameters (feedstock and process conditions) is non-linear.

The multiple linear regression model and regression tree developed by Li et al. [91] were adopted by Ro et al. [97] to predict the properties of hydrochars derived from animal manure.

#### **3.5.2 Design of experiments - Response surface methodology**

The design of experiment – response surface methodology (DoE-RSM) is a statistical approach used to investigate the interaction between independent and response variables of a process. The concept implies

the usage of a set of designed experiments to obtain an optimal response (the output) [81]. This approach aims to identify the optimal conditions that lead to the optimal response so to optimize the experimental activity. The variations of the responses are often represented graphically, thus improving the understanding of the connection among variables [35]. Several DoE-RSM approaches have been adopted in the HTC literature.

For example, Heilmann et al. [11] performed two-level-factorial experiments with replicated center points to investigate the correlation between three variables (temperature, time, and solid load) for the HTC of microalgae. After developing an orthogonal factorial design, a linear regression function was used to predict the percentage of carbon recovered. The same model was presented by Heilmann et al. [98] to observe the effects of temperature, time, and solid load on the carbon content and mass yield of the hydrochar from distiller's grains.

Mäkelä et al. [99] investigated the statistical effect of temperature, residence time, and solid load for the HTC of paper sludge residue by constructing individual response models. The responses are the solid yield, the carbon content, the O/C ratio, the energy densification, and the energy yield. These were computed by applying the multiple regression equation as represented by Equation (3.20).

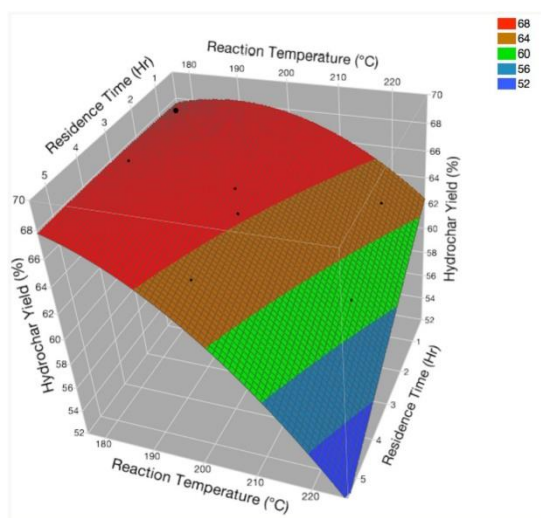
$$\mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{e} \quad (3.20)$$

Where  $\mathbf{y}$  is the vector of measured or calculated responses,  $\mathbf{X}$  the design matrix of design factors,  $\mathbf{b}$  a vector of coefficients, and  $\mathbf{e}$  a vector of residuals.  $\mathbf{b}$  was solved separately for each response, while the confidence of intervals (i.e. the range of certainty of the unknown parameter) by applying equations as proposed by Montgomery [100]. Results show that hydrochar properties are mainly affected by temperature, while poorly by the solid load.

Recently, Afolabi et al. [101] presented a DoE-RSM method to investigate and optimize the combined effect of time and temperature (the “factors”) on the hydrochar yield and HHV (the “responses”) of spent coffee grounds. They adopted a faced-center central composite factorial design to fit the two responses with the two factors. Results show that in the investigated range (180-220 °C and 1-5 hours), the best combination of hydrochar yield and HHV (63.9 % and 31.6 MJ kg<sup>-1</sup>, respectively) is achieved at a temperature of 216.4 °C and residence time of 1 hour. This optimum conditions corresponds to a severity factor equal to 5.2, which belongs to the interval of values reported in



section 3.4. Some results of the analysis are graphically reported in Figure 3.3, where the simultaneous effect of residence time and temperature on hydrochar yield is reported. The hydrochar yield at different times and temperatures forms a curve-shaped surface, in which the effect of the temperature is more significant than the residence time.



**Figure 3.3.** Response surface plot of the effects of temperature and time on hydrochar yield. Reprinted with permission from Afolabi et al. [101]. Copyright © 2020, Elsevier.

Similarly, Heidari et al. [81] used RSM to find relations between biomass constituents and HTC severity factor, considering as responses the mass yield, HHV, carbon content, and energy recovery factor. To achieve this, they performed 39 HTC experiments on a mixture of pure hemicellulose, cellulose, and alkali lignin. The resulting relations consist of polynomial equations in which the output response is function of the factors. For each model, the statistical significance was determined by performing an analysis of variance (ANOVA). Results show that the optimal point for the mixture in terms of solid yield and HHV (corresponding to 67 % and 25 MJ kg<sup>-1</sup>, respectively) is achieved at a severity factor of 4.5. The authors tested the model also on 10 different biomasses, demonstrating that it is reliable at low ash and extractives content.

The Central Composite Design (CCD) approach was used by several authors to implement the DoE in the RSM to determine the optimum process variables and the number of experimental runs. The CCD builds

a second-order polynomial for the RSM without using a full factorial design of experiments, thus avoiding the usage of all the possible combinations levels of the experimental factors [102]. It bases on a two-level factorial design, where the total number of experiments (N) using k number of factors (like HTC temperature and residence time) is computed as the sum of  $2^k$ , 2k axial runs, and n centre runs, as shown in Equation (3.21).

$$N = 2^k + 2k + n \quad (3.21)$$

The CCD method was adopted to observe the effects and optimize the variables of the HTC of different biomasses (e.g. olive stone, tomato peels, and water hyacinth) by several authors, like Álvarez-Murillo et al. [103], Sabio et al. [104], and Román et al. [105]. Authors observed the influence of temperature, residence time and biomass to water ratio (which implied a number of random experiments equal to 18). Then, the investigated target functions Y (the solid yield and HHV) were modeled using a second-order function as Equation (3.22).

$$Y = A_0 + A_1R + A_2T + A_3t + A_4RT + A_5Rt + A_6tT + A_7R^2 + A_8T^2 + A_9t^2 \quad (3.22)$$

where  $A_i$  are coefficients (computed after the fitting of the experimental data), R is the biomass to water ratio, T the temperature, and t the residence time. Results show this statistical approach fits the data well and enables identifying important correlations which would not be highlighted with classical methods.

Moreover, the CCD approach was successfully adopted in other studies to observe the combined effect of each factor during the HTC of several biomasses (primary sludge, fish waste, poultry litter, sunflower stalk, and algae) [106–108].

### 3.5.3 Other statistical models

Other statistical models were also developed to understand the correlation between process parameters and product properties.

An alternative approach was proposed by Ismail et al. [109], who used the artificial neural network (ANN) coupled with a Kriging interpolation approach. The model was used to generate more data points from the collected data to understand the effect of temperature

and residence time on the capacity of recovering carbon and inorganic phosphorus. The ANN model is useful for complex systems and is inspired by biological neural networks, while the Kriging is an interpolation method based on stochastic relations. Results show that temperature has the highest impact on carbon recovery.

Besides, Xu et al. [110] applied the Taguchi statistical method, in which, adopting orthogonal arrays to perform experiments, control factors are identified to obtain the optimum process results. In particular, they investigated the HTC of microalgae, identifying as control factors reaction temperature, time, initial solid load, particle size, and catalyst amount (citric, acrylic, and sulfuric acid).

## 3.6 Reactor models

The modeling of the HTC reactor is an essential step towards the design of a process that can be energetically efficient, economically feasible, and competitive with the current technologies. The reactor modeling should include several aspects, like heat transfer phenomena and the fluid-dynamics of the system as a function of the selected geometry and process conditions.

Until now, little research has been carried out around this topic. Indeed, while many efforts have been done to understand the HTC reaction mechanism itself, few studies investigate the design of the reactor and its modeling. This shortage can be attributed to both the complexity of the process itself, still not fully understood and highly dependent on process conditions, and to the few data available concerning the larger-scale systems (like pilot and industrial).

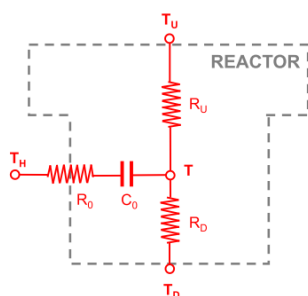
### 3.6.1 Heat transfer models

Developing a heat transfer model is of great importance to evaluate and minimize energy consumptions. Currently, as highlighted by Heidari et al. [16], these models have been poorly investigated mainly due to a certain lack of research around:

- (1) the thermodynamics of HTC, i.e. the enthalpy of reaction (endothermic or exothermic process);
- (2) the change of physical properties (like porosity, density, and permeability) of the mixture of biomass and water during the process.

A simple model was developed by Baratieri et al. [64] to simulate the thermal behavior (also the transient phase) of an HTC lab apparatus

previously designed by Fiori et al. [111]. To do this, the HTC system was reduced to a resistance-capacitance network considering thermophysical properties of the system like the reactor shell, the gaseous and liquid phases. A sketch of the system is shown in Figure 3.4, where the HTC system at the set-point temperature  $T$  exchanges heat with the external environment ( $T_U$  and  $T_D$  are the upper and lower temperatures of the system, while  $T_H$  is the temperature of the external electrical band heater).  $C_0$  is the sum of heat capacities of pure water, stainless steel, and marble support,  $R_0$  is the overall thermal resistance,  $R_U$  and  $R_D$  are the thermal resistances of the upper and lower areas, respectively. Then, heat transfer equations were written and used to simulate the thermal behavior of HTC starting from the reactor temperature profile experimentally measured. Although its easy implementation, this model is limited since only water is considered inside the HTC reactor and the reactor is dimensionless.



**Figure 3.4.** Network of the resistance-capacity model. Reproduced with permission from Baratieri et al. [64], Chemical Engineering Transactions; published by AIDIC, 2015.

### 3.6.2 Computational models

Computational models can provide a detailed description of the process and a multitude of phenomena (e.g. heat transfer, transport phenomena, and reaction kinetics) considering variations in both time and space. Indeed, they consider important parameters like the specific geometry of the reactor, the porous structure of the biomass, and possibly the fluid-dynamics of the reacting mixture inside the HTC reactor. Despite their potential, they require heavy numerical computations as well the usage of proper software and properties databases, limiting their diffusion. Actually, the properties of a reactive heterogeneous system comprising biomasses are quite difficult to calculate or correlate. As

also explained by Román et al. [35], a common procedure to develop computational models consists of:

- (1) specifying the reactor geometry;
- (2) establishing a proper mesh to divide the domain into cells;
- (3) setting of the initial and boundary conditions;
- (4) defining the properties of the different phases (e.g. state equations, kinetic equations);
- (5) defining the proper solvers and models to compute the system behavior in each cell.

Until now, according to author's knowledge, the only computational model has been developed by Álvarez-Murillo et al. [45]. They used COMSOL Multiphysics® to characterize the decomposition of cellulose during HTC, using as inputs of the model the reactor geometry, materials, and reaction medium. Figure 3.5 shows the system geometry (reactor placed in an oven) as used in the program. Only water was considered to affect thermal properties of the reaction medium. The total heat transfer  $\dot{Q}_{wall}$  (in W) through the HTC walls was computed as for Equation (3.23).

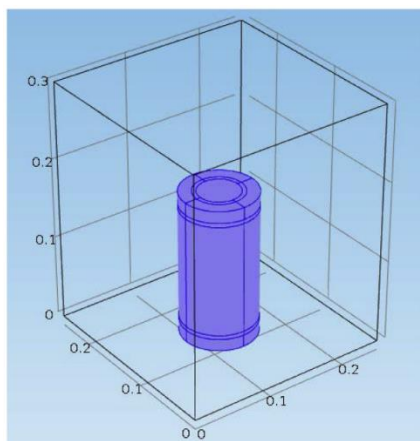
$$\dot{Q}_{wall} = (h_{conv} + h_{rad}) \cdot A_S \cdot (T_{\infty} - T_S) \quad (3.23)$$

Where  $h_{conv}$  and  $h_{rad}$  are the convective and radiative heat transfer coefficients ( $\text{W m}^{-2}$ , computed by the program through traditional heat transfer relations),  $A_S$  the surface of the external wall,  $T_{\infty}$  the bulk temperature,  $T_S$  the temperature of the external wall. The kinetics of the process was evaluated by applying a first-order Arrhenius mechanism. Cellulose with a biomass to water ratio equal to 0.1 was chosen as the feedstock. Temperatures were investigated between 170 and 245 °C, with residence times between 2 and 11.8 hours. The simulated reactor was a stainless-steel cylinder with the following geometry: 18 cm height, external diameter of 9 cm, and wall thickness of 0.5 cm. The internal volume of the oven was defined as a rectangular prism of 25 x 25 x 30 cm.

Through the simulation, the variations of the investigated parameters (like relative mass loss of cellulose, HHV, H/C ratio) with time were computed at the different process conditions. The good fitting with experimental data confirms the reliability of the model. This presents also several limitations. Indeed, it considers only water affecting the thermal properties of the reaction medium and neglects the exothermic behavior of the HTC reactions. Moreover, the porosity of the cellulose

is neglected and the heating method is uncommon for industrial and lab scale reactors. Anyway, despite the limits, this work represents the first computational model developed for HTC and then, it can provide the basis for further works.

Therefore, significant efforts should be done to develop this kind of models, which could lead to a better understanding and optimization of the process. A future step could be considering higher-order reactions and, as suggested by Heidari et al. [16], the investigation of the thermal effect of the HTC reactions on the heat fluxes.



**Figure 3.5** Schematization of the HTC reactor placed in the oven. Reprinted with permission from Álvarez-Murillo et al. [45]. Copyright © 2016, Elsevier.

### 3.7 Large scale modeling

The development of a validated plant simulation can provide important insight on the process dynamics [32]. Through a process simulator, operating parameters can be varied so to optimize the process considering both energy and economic aspects. Commercial suites (like Aspen Plus® or UniSim Design®) and software environments (like Matlab) allow the development of such type of models. The current process simulators solve the process equations using proper databases and solving methods, which require accurate equilibrium and kinetics input data [32]. However, few data are currently available in literature regarding HTC thermodynamic and kinetics, limiting the reliability of the implemented models. Therefore, the inclusion of these aspects requires first an extensive availability and consistency of these data.

Then, they can be integrated inside the specific software or suite using specific subroutines where kinetics and thermodynamics data (or equations) are specified.

#### 3.7.1 HTC plant simulations

The available models on HTC are mainly techno-economic analyses, where both the energy and economic aspects are investigated. Even if important steps have been done, there are still several limitations. Indeed:

- (1) no reaction kinetics effects are considered in the HTC reactor, which is often modeled using equilibrium relations or yield data experimentally measured;
- (2) the thermodynamics of the HTC reactions (i.e. the quantification of the exothermic behavior of the process) is often not considered or over simplified using the few data available in the literature but relevant to lab-scale experiments;
- (3) mass transport and fluid-dynamics phenomena are always neglected;
- (4) large-scale and continuous processes are often modeled with experimental data obtained with batch reactors through laboratory experiments.

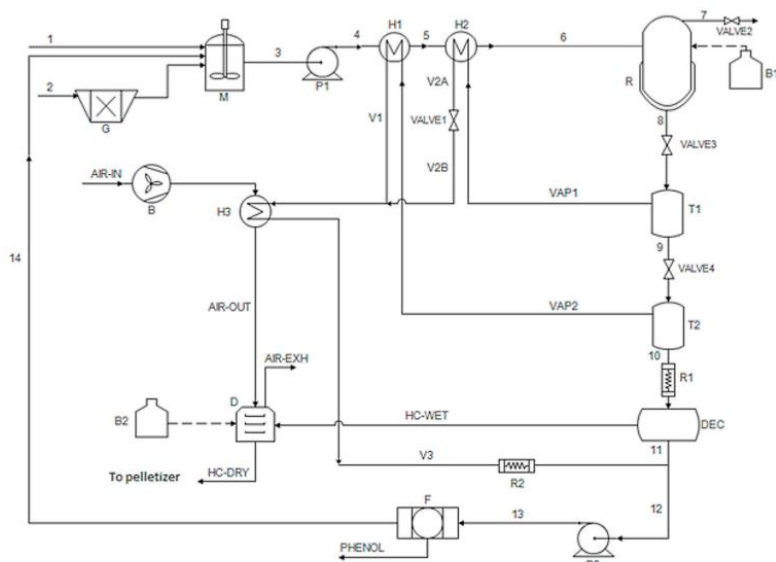
A steady-state model was developed by McGaughy and Reza [112] using Aspen Plus<sup>®</sup> to simulate the HTC of food waste. The plant is thought to treat continuously 1 ton per day of biomass. The model bases on experimental data derived from an experimental campaign performed on a batch reactor and equations are forced to satisfy carbon content and solid yield according to experimental results. No reaction kinetics is considered and HTC reactions are modeled in a “Gibbs reactor”, i.e. a continuous stirred tank reactor (CSTR) working at thermodynamic equilibrium which is calculated by minimizing the system Gibbs energy. In the model, food waste converts into hydrochar (that does not participate in the equilibrium), a liquid phase consisting of organic acids, and carbon dioxide. Utilities (pump, heater, filter, and dryer) are also considered, while the reaction is considered exothermic (1 kJ kg<sup>-1</sup>). The outputs of the model are the energy consumptions and the carbon dioxide produced by HTC. Even if results confirm the effect of temperature and moisture content of the starting feedstock on the process output parameters, the model suffers from over simplification





equipment. Both plants work with 160 tons per day of yard waste. The reactor was modeled using a “RYield reactor”, in which the yield of the different phases is specified as input data on the base of experimental data. Such data were obtained in previous works. Results show that the optimal configuration leads to a cost of production of the hydrochar of 13.1 euro/GJ.

A techno-economic analysis was performed by Lucian and Fiori [115] to model the continuous treatment of organic wastes (20000 tons per year) to pelletized hydrochar. All the plant equipment were considered working at steady state, and mass yield data were assumed as those obtained experimentally in a batch HTC reactor. A sketch of the adopted scheme is shown in Figure 3.7. The model computes the plant efficiency (equal to 78 %) and the minimum selling price of the pelletized hydrochar (equal to 157 euro/ton) which makes the whole plant economically sustainable.



**Figure 3.7.** Sketch of the process model for the continuous treatment of organic wastes. Reproduced with permission from Lucian and Fiori [115], Energies; published by MDPI, 2017.

Saba et al. [116] performed a techno-economic analysis for the co-processing of coal and miscanthus for producing a solid fuel. A blend of coal and miscanthus (50:50, about 65000 kg h<sup>-1</sup>) was treated in an HTC reactor. The analysis was limited to an economic evaluation, which leads to a breakeven selling price of 117.7 \$/ton hydrocar.

Recently, Gómez et al. [117] modeled a continuous HTC plant adopting an alternative approach to previous works. Indeed, they assumed the HTC to occur in separated reaction stages. The model was developed with the UniSim Design<sup>®</sup> process simulator and calibrated using operational data derived from a commercial company (Grenol GmbH). Thus, after the pre-heating and pumping stages, the stream of biomass (10-50 ton/day) enters several reactors, placed in sequence, and dedicated to:

1. hydrolysis (170 °C): biomass degrades into sugars;
2. intermediate compounds degradation (220 °C): sugars react to form aqueous and gaseous products (acetic acid and carbon dioxide);
3. aromatics formation (220 °C): the aqueous phase reacts to form aromatic compounds (HMF and furfurals);
4. polymerization process (220 °C): organic compounds re-polymerize while the unreacted biomass undergoes a solid-solid reaction.

Given the scarcity of reaction kinetics data, each reactor was characterized according to a hypothesized stoichiometry of each theoretical reaction. Once developed, the model was adjusted considering operational parameters from Grenol GmbH and fractional conversions. The model predicts the hydrochar composition and yield.

### **3.7.2 New horizons: integrated plants and continuous reactors**

Even if integrated plants and continuous reactors have been little or nothing investigated through modeling, we believe that these are fundamental topics for guiding authors towards future innovative modeling schemes.

Coupling HTC with other processes is of particular interest to maximize the efficiency of the whole plant. In particular, significant efforts have been done around the integration of HTC with anaerobic digestion (AD), a consolidated technology for the valorization of wet substrates into biogas. The process exploits the HTC process water (the “liquor”), which is an inevitable HTC by-product, to generate biogas. During the operation, the process water resulting from HTC feeds an AD plant to generate biogas through a series of bacterial and metabolic pathways [10,13]. Then, the biogas can be directly used for the production of electricity and heat through combustion in internal

combustion engines or turbines or, alternatively, it can be injected in the natural gas grid after a purification stage to remove CO<sub>2</sub> and other minor constituents such as H<sub>2</sub>S. Several feedstock have been investigated, like digestate [21,118], sewage sludge [20,119], microalgae [12,13], orange pomace [22], spent coffee grounds [120], biowaste [121], and the organic fraction of municipal solid waste [24].

As explained by Merzari et al. [10], the chemical composition of the process water is fundamental to maximize the production of biomethane and avoid any inhibition of the AD process. Regarding the HTC process water to be recycled to AD, important parameters to control are those typical of the wastewater treatment field, like the chemical oxygen demand (COD), the total and suspended solids (TS and SS), the volatile fatty acids (VFAs), the ammonia, and the pH [10]. For AD, control parameters are the organic loading rate (OLR) and the hydraulic retention time (HRT).

Until now, promising research has been performed on the lab-scale to individuate the optimal parameters to avoid the formation of toxic or inhibiting compounds [122]. For example, the process water from sewage sludge HTC seems particularly suitable for AD if HTC temperatures and residence times are kept lower than 180-200 °C and 15-30 minutes, respectively, leading to methane yields between 0.022 and 0.277 L<sub>CH4</sub>gCOD<sup>-1</sup> [10]. Besides, Paul et al. [121], dealing with the HTC process water from biowaste, found that residence times longer than 30 minutes (at 240 °C) cause the formation of inhibitory chemicals for AD. Recently, Lucian et al. [24] tested the biomethane potential from the coupling of AD with HTC of organic wastes in two different scenarios: AD of the HTC liquor and AD of the HTC slurry (containing both the hydrochar and the liquor). Besides, an energy evaluation of the two scenarios was reported. On the one hand, results show that the highest biomethane potential is obtained by the slurry rather than the liquor (+363% and +37% compared to the raw biomass, respectively). On the other hand, in the hypothesized plants, the use of the sole HTC liquor represents the most energy efficient pathway as the energy content of the hydrochar can be efficiently exploited through combustion.

Interestingly, Aragon-Briceno et al. [118] modeled different scenarios of an HTC-AD plant coupled with a cogeneration stage by the use of Aspen Plus<sup>®</sup>. The analysis focuses on the energy duties of the plant, in which both hydrochar and biogas were considered as combustion resources. RYield reactors were used to model both the HTC and the AD reactors on the base of experimental data. Results show that

scenarios with high solid loads lead to a lower heat production from the combustion of the biogas, but more electricity and heat are obtained from the combustion of the hydrochar. At the net, large solid load scenarios (30 % the highest) lead to the highest electrical and thermal efficiencies, equal to 25.8 and 38.9 %, respectively. Furthermore, Heidari et al. [23] compared an HTC-AD scenario with a direct combustion scenario for power production. The plants comprise a Rankine and a Bryton cycle to generate electricity and the thermodynamic modeling was performed through EES (Engineering Equation Solver). Results show the HTC-AD case has higher efficiencies only under high moisture contents of the starting biomass (sawdust in this case).

Regarding the integration of HTC with other technologies, our group recently developed and investigated a renewable solution consisting of a hybrid HTC-solar apparatus. The system consists of an HTC reactor directly coupled with a solar concentrator, which was experimentally demonstrated to completely cover the HTC energy needs [29]. Through this system, a “zero-energy technology” can be realized. Besides, our group also performed a techno-economic analysis of a large-scale plant consisting of a continuous HTC reactor coupled with a solar concentrator complete of thermal energy storage tanks: results show that this apparatus ensures a minimum fuel selling price of the hydrochar equal to 37.4 €/ton [30].

The development of a continuous reactor is a mandatory challenge for the industrialization of HTC and moving towards large-scale plants. In addition to the intrinsic benefits of a continuous configuration (greater automation and better control, less use of manpower), continuous reactors enable to work with a wide range of HTC residence times when compared to batch reactors, instead characterized by useless long times to charge and discharge the reactors. Consistent research around this topic is still lacking and significant efforts are required. In particular, the biomass conveying system requires particular attention in the design phase [12]. Indeed, an effective and non-trivial pumping system is required to continuously feed the biomass-water stream to the high pressure HTC reactor [123]. Among the few authors that faced this topic, Hoekman et al. [80] adopted a modified two-screw extruder to realize a fully continuous reactor. The reactor consists of an initial rotating portion (with increasing pitch) for increasing the pressure of the biomass, a central part at a constant pitch, and a final part with another rotating screw. Water is injected at the end of the first section. Experimental activity was successfully performed on loblolly pine with

very short residence times (20-30 seconds) and high temperature (290 °C). In another work, Hoekman et al. [124] designed a semi-continuous reactor for the conversion of lignocellulosic biomass. Operating with 2-3 kg of biomass per run and with short residence times, the system consists of a V-shaped reactor. The reactor consists of a feeding inclined auger and a second auger to convey the material out of the system. Moreover, Stemman and Ziegler [125] simulated a continuous HTC system through EES (Engineering Equation Solver), accounting for the exothermic behavior of HTC.

### 3.8 Life cycle assessment

The development of a Life Cycle Assessment (LCA) is an important step for the assessment of the environmental impacts of a technology. By investigating the whole life cycle, from the extraction to the end of life, LCA provides a holistic evaluation of the whole process. Until now, mainly due to the lack of large-scale data, few studies were performed on the LCA of HTC.

Recently, Gievers et al. [126] carried out a LCA of the HTC of sewage sludge derived from anaerobic digestion for four possible applications (agriculture, horticulture, co-firing in waste incineration, and co-firing in a lignite power plant). These were compared with the results of traditional processing, which implies incineration and landfilling of the ash. The global warming potential (in kg CO<sub>2</sub> equivalent) was assessed. According to the results obtained by the authors, total CO<sub>2</sub> emissions are due for the 63% to the electricity required to run the HTC plant and auxiliary process, for the 28% to the natural gas used for the heating, and for about the 9% to the treatment of the process water. Besides, the analysis demonstrates that using hydrochar for substituting fossil fuel has the highest potential to reduce the global warming potential of the sewage sludge treatment. This LCA could be improved considering more life cycle impact categories and more data for the HTC plant.

Liu et al. [127] performed a LCA to assess the impacts of the production of briquettes of hydrochar with coal (with a hydrochar percentage from 10 to 100%) to be used as a solid fuel for electricity production in a power plant. The hydrochar is produced starting from loblolly pine, and a fast HTC (30 s of residence time and temperature between 260 and 290°C) was considered. The HHV of the hydrochar resulted equal to 24.4 MJ/kg. The stages considered in the LCA are the harvesting of the loblolly pine, the pre-processing, the HTC apparatus (a twin-screw extruder reactor), and the production of electricity in a

power plant. The investigated parameters are the GHG emissions and life cycle energy use. Results show that the GHG impact of the hydrochar-coal briquettes is much lower than that of coal (from 1.033 to 0.149 kgCO<sub>2,e</sub>/kWh from pure coal to 100% of hydrochar), while the difference in the energy use is relatively small (ranging between 10.64 and 12.63 MJ/kWh).

Benavente et al. [128] assessed the environmental impacts of the HTC of olive mill wastes compared with traditional management strategies (composting, anaerobic digestion, oil extraction, and incineration). HTC is combined with an incineration stage for heat and power production. Results show that this solution is less environmentally impacting than composting and anaerobic digestion, while no significant improvements are observed compared with incineration and oil extraction. This gap is mainly due to the water disposal stage required for HTC. However, as pointed out by Heidari et al. [16], important aspects like the impact of ash in the incineration phase were neglected in this LCA.

Other interesting LCA studies were performed by Ahamed et al. [129] to assess a combined plant HTC-oil system, and by Fornes et al. [130] regarding hydrochar derived from forest waste for horticultural applications.

Overall, the current LCA studies applied to HTC suffer from the lack of large-scale data. Moreover, future studies should include the valorization stages (like the coupling with anaerobic digestion) of the liquid phase.

### 3.9 Conclusions

Hydrothermal carbonization is acquiring a rising interest in the research community and industry for the valorization of organic waste and biomass residues. Through hydrothermal carbonization, biomasses with high moisture content often considered unpleasant can be upgraded into useful bio-based products, usable for energy applications, in agriculture or as a basis for advanced carbon materials. Several challenges have to be tackled to shift HTC from the laboratory activity to industry. The development of modeling and simulation tools is one of these. Through modeling, virtual scenarios can be implemented and the effects of process variations investigated. In this way, the process can be optimized so to improve its attractiveness to commercial parties.

This work presents the different modeling strategies present in the current state of art. These have been divided according to the system of

analysis. Three main system are proposed: (1) HTC process itself, involving reaction kinetics and statistical models; (2) the reactor, with heat transfer and computational models that account for a variety of physical phenomena (like mass transfer and biomass porosity); (3) the HTC plant, with techno-economic analyses considering the equipment and auxiliaries. Besides, life cycle assessment models applied to HTC are also presented. Even if significant improvements have been done in the last years, more efforts are required to develop appropriate and reliable models able to drive a shift to the industrialization of HTC. Indeed, the current models suffer from the general complexity of the reaction mechanism (HTC involves heterogeneous reactions in a three-phase system), still not fully understood, and the lack of some important kinetics and thermodynamic data. Moreover, models are often limited to a narrow range of experimental conditions (i.e. type of feedstock, reaction temperature, residence time, and starting biomass load) and pilot plant data are often not available to develop reliable large-scale models.

Therefore, further research is required to develop more credible and reliable models. In this regard, the production of more reaction kinetics and thermodynamic data could be an important starting point. Then, the development of reactor simulations by using computational tools, accounting for mass and heat transfer phenomena, could drive to important improvements. Moreover, it is worth mentioning that, ideally, models should be generalized and user-friendly. In this way, they could increase their attractiveness for commercial parties, who could easily explore the possibility of adopting hydrothermal carbonization as a valuable and competitive technology for waste and biomass valorization.

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## **Chapter 4**

# **Towards sustainability: coupling hydrothermal processes with solar energy**

### **Overview**

The potential of hydrothermal processes is well recognized. Capable to convert biomass into bio-based products, from advanced carbons to chemicals and potential liquid biofuels, and to valorize unpleasant and unconventional substrates: hydrothermal processes are full-fledged green routes for a bio-based future.

However, hydrothermal carbonization (HTC) and liquefaction (HTL) generally occur at relatively harsh conditions (180-400°C and 10-220 bar), making the process energy-intensive if compared to other processes [1]. Indeed, energy has to be provided to:

(1) Heat both the biomass and the water, which is present in a high percentage (from 70 to 95%, in mass) and is characterized by a higher heating capacity than the solid phase;

(2) Sustain the process during the constant-temperature phase, compensating for heat losses and possible endothermic reactions. Even if HTC is slightly exothermic (as shown in section 2.3), the system has to face heat losses.

Despite the extensive amount of research performed on characterizing the mechanisms of hydrothermal processes and the products, few studies involve the energy optimization of the entire process. Indeed, the actual plants adopt traditional sources and/or solutions involving heat recovery from internal streams. For example, in integrated plants, heat derives from the combustion of biogas produced through anaerobic digestion or a fraction of the hydrochar or produced bio-oil [1,2]. Certainly, providing energy through traditional strategies decreases the efficiency of the process and lowers their appeal as green routes: if from one side hydrothermal processes concretize biorefinery principles, from the other side we have to remember that they are energy-intensive. The sustainable solution comes naturally: providing heat through renewable sources.

Therefore, to pursue sustainability and realize a “zero-energy” process, we propose the coupling of hydrothermal systems with concentrated sunlight. Through this solution, biomass can be converted to biofuels or biomass-based products through a zero-energy route, in which the needs are covered by concentrated sunlight and the use of traditional energy sources is avoided. Indeed, to reach temperatures like those of HTL, a concentrated sunlight technology, like linear and point focus concentrators, is needed [1]. In recent years, several groups studied the possibility of incorporating concentrated sunlight with hydrothermal carbonization and liquefaction, developing a few proofs of concepts [3–6] and techno-economic studies [7–9]. More efforts are needed to make the solar solution a valuable and feasible alternative to conventional systems.

Therefore, we developed the concept of a hydrothermal-solar-driven apparatus through two main projects consisting of the direct or indirect coupling between the two technologies. Both the solutions have advantages and drawbacks, but we believe that on a large scale, the passage to an indirect apparatus is an obliged step for independence from weather conditions and easier operation and control. Therefore, we move from a more simple solution consisting of a direct system, to a more complex solution consisting of an indirect one.

The first project (section 4.1) is a proof-of-concept of a hybrid solar-HTC apparatus, where a hydrothermal carbonization reactor is directly coupled with a solar concentrator apparatus. We realized and tested the

system at different conditions, proving its effectiveness as zero-energy technology for valorizing biomass waste. The second project, divided into two sections, consists of a techno-economic study involving an indirect system, in which a standard hydrothermal reactor is coupled with a solar concentrator using a heated fluid (molten salts or thermal oil). The fluid is also stored in a tank to disengage from weather fluctuations. The first part (section 4.2) is a techno-economic analysis for a solar HTC or HTL plant, and the second part (section 4.3) is the technical design of a possible lab-scale plant to be realized at the University of Trento.

## Section 4.1

# Realization of a solar hydrothermal carbonization reactor: A zero-energy technology for waste biomass valorization

This section is based on the article that we published in *Journal of Environmental Management* (2021, vol. 259, 110067, by G. Ischia, M. Orlandi, M.A. Fendrich, M. Bettonte, F. Merzari, A. Miotello, L. Fiori, <https://doi.org/10.1016/j.jenvman.2020.110067>). The work consists of a proof-of-concept for the realization of an apparatus consisting of a hydrothermal reactor directly coupled with a solar concentrator. The apparatus was realized and tested at the Physics Department of the University of Trento. Thanks to Nicola Bazzanella (University of Trento) for technical support with SEM images acquisition, and Vittoria Benedetti and Marco Baratieri (Free University of Bolzano) for ultimate analysis.

### 4.1.1 Abstract

Research around hydrothermal carbonization (HTC) has seen a huge development in recent years, materializing in the first pilot and industrial plants. Even though HTC reactions are slightly exothermic, the overall process entails energy consumption to both reach operating conditions and tackle heat losses. With the purpose of facing this issue and developing a zero-energy process, this work proposes an innovative solution: the coupling of an HTC reactor with a solar concentrator,

designed to fully cover the HTC energy needs. As a method, a 300 ml stainless steel HTC reactor was constructed and positioned on the focus of a parabolic dish concentrator (PDC), consisting of one parabolic mirror of 0.8 m<sup>2</sup>. To maximize the light absorption, the illuminated side of the HTC reactor was coated with a thin layer of nanostructured copper oxide. This was realized via electron beam deposition and, with a maximum absorbance of 95.6% (at 300 nm wavelength), confers excellent absorbing performances to the system. Then, the effectiveness of the hybrid solar-HTC solution was demonstrated by carrying out an experimental campaign on a residual agro-biomass (grape seeds), which was treated at 180, 220, and 250°C for 2 hours. Results show that heating times, yields, composition, and energy properties of “solar hydrochars” resemble those of studies performed in traditional HTC systems, thus proving the feasibility of the solar-HTC apparatus.

#### **4.1.2 Introduction**

Climate change and fossil fuel depletion call for the development of technologies and processes based on renewable energy. Biorefinery facilities represent a smart strategy leading to a wide spectrum of products (such as biofuels, bioenergy, biomaterials, and biochemicals) starting from biomass residues, which are abundant nature and renewable [10]. However, biorefineries need to improve their profitability to be competitive in the global market and face the current low price of oil [11]. A way to achieve progress would be developing a highly efficient facility, in which energy consumption for the enhancement of biomass is minimized.

Conversion of biomass into valuable carbon products can be achieved through hydrothermal carbonization (HTC), a thermochemical process that is acquiring a rising interest. HTC occurs in water and exploits the unique properties of liquid water under elevated temperatures to carbonize biomass into a carbonaceous material, referred to as hydrochar. Since the process is wet, HTC is well suited to enhance biomass with high moisture content, avoiding any energy-intensive drying stage. The process occurs under subcritical conditions, with temperatures usually ranging between 180 and 250 °C, self-generated pressure from 10 to 50 bar and reaction times from a few minutes to several hours [12]. A complex network of reactions takes place during HTC, involving solid-solid and solid-liquid conversions and including



reactions such as hydrolysis, dehydration, decarboxylation, and polymerization [13,14].

Even if the HTC reactions are slightly exothermic [15–18], the energy release is lower than thermal requirements (i.e. the energy necessary to reach operating conditions and to tackle heat losses), making external energy sources mandatory.

Until now, many efforts have been made to understand the fundamentals of HTC (e.g. reaction pathways, effect of operating parameters, characterization and optimization of hydrochars [14,19,20] and towards scalability, bringing to the first pilot and industrial-scale plants [21,22]. Because of its coal-like properties, hydrochar is particularly suitable for combustion applications [23,24]. Furthermore, tunable surface properties of hydrochars make them usable as precursors for high-value carbon-based materials [25]. In that respect, post-treatments have been recently investigated to enhance hydrochar as an adsorbent for water remediation applications [26,27] and as an electrode material for batteries and fuel cells [14].

Nevertheless, there is a certain lack of research around the optimization of the technological process itself. In particular, despite the huge and established potential of concentrated solar energy, the usage of concentration solar systems combined with hydrothermal processes has been little investigated. A commercial solution involving a concentrating solar power system combined with hydrothermal liquefaction was proposed by Pearce et al. [28], while conceptual studies were carried out both on hydrothermal liquefaction [9,29] and hydrothermal gasification [30]. Until now, according to the authors' knowledge, only a few works dealing with solar HTC were reported in literature. A pilot plant made by an HTC reactor coupled with solar panels was simulated by Bertolucci et al. [31] while an HTC reactor working with a parabolic trough collector was tested by Khongkrapan and Chinpensawat [32]. Recently, Xiao et al. [33] proposed a direct solar-driven hydrothermal apparatus to pretreat microalgae slurry through hydrolysis (operating temperature lower than 180°C). Despite the scarcity of research around solar hydrothermal systems, the adoption of solar concentration to sustain thermochemical processes has seen a rising development from the mid-2000s [34], driven by lower costs and improved design. These solutions conceptually derive from the consolidated concentrated solar power (CSP) systems used for electrical production. In particular, the adoption of solar systems for producing hydrogen and syngas (through methane reforming and gasification of carbonaceous feedstock) and for industrial commodities

(such as zinc production) has been demonstrated successfully on plant scale [34]. Among these, a pilot plant of supercritical water gasification of biomass using concentrating solar energy was recently realized for hydrogen production [35,36]. Within the solar thermal systems, the parabolic dish technology is attractive for its high conversion efficiency and for remote grids [37,38].

Thus, to address the lack of research concerning solar HTC and propose a novelty in the current state-of-the-art, we propose the realization of an innovative hybrid solution consisting of an HTC reactor coupled with a parabolic dish concentrator (PDC). Through this system, concentrated sunlight is converted into heat, which fully covers the thermal energy requirements of HTC and avoids the need for any external energy source. The coupling between the two technologies is direct, meaning that the reactor is directly illuminated and heated by the concentrated solar radiation. The experimental apparatus consists of a 300 ml HTC reactor placed on the focus of a PDC prototype, equipped with a mirror of 0.8 m<sup>2</sup>, while a copper oxide coating allows the maximization of the solar radiation absorption.

In this work, we report details on the solar-HTC experimental apparatus and the first experimental campaign conducted to assess its effectiveness. Since the absorption process of solar light is a key factor for the overall performance, particular attention was given to the design of an efficient optically absorbent coating. Besides, HTC tests were carried out at 180, 220 and 250°C (residence time of 2 hours) on grape seeds, an agro-biomass residue resulting from the winemaking industry. Given their high moisture content of about 60-70 % [39], grape seeds are excellent candidates for HTC. Properties of “solar hydrochars” obtained through our hybrid solution were compared with those from conventional HTC systems to provide a proof of concept for the solar-HTC.

### **4.1.3 Materials and methods**

#### *Description of the system*

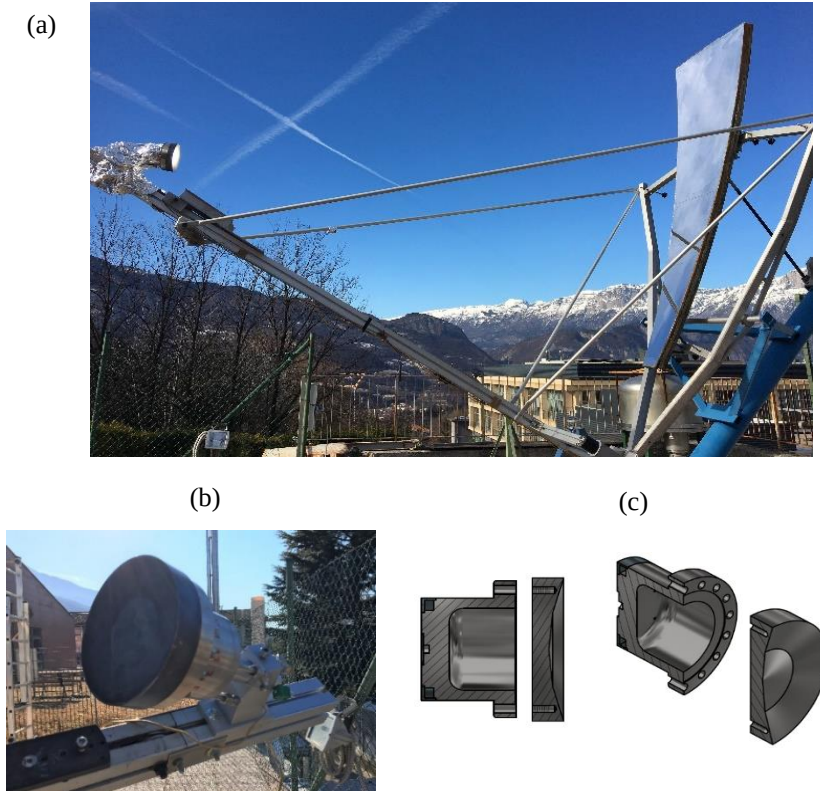
The demonstrator system realized at the University of Trento consists of an HTC reactor coupled with a parabolic dish concentrator (PDC), both developed in-house. The coupling between the two technologies is direct: the HTC reactor was positioned on the focus of the parabolic dish, as shown in Figures 4.1a and 4.1b. This configuration allows the

achievement of high concentration ratios and the minimization of heat losses in comparison with indirectly heated systems. Moreover, it presents a good constructive simplicity and versatility.

The solar concentrator is a prototype and before this work was used for various applications, including wastewater treatment through photo-Fenton reactions [40,41], integrations with a molten salt storage system and with photovoltaic modules [42]. It is a parabolic dish type consisting of solar reflector modules constructed by applying a specific manufacturing procedure based on the sandwich approach, whose details are explained by [43]. The full parabolic dish can assemble up to 24 modular parabolic sectors, each one having a  $15^\circ$  aperture and  $0.76 \text{ m}^2$  surface, up to a total diameter of 5 m and an overall projection area of about  $20 \text{ m}^2$  [42]. The system is equipped with a polar-equatorial axis solar tracker, which ensures alignment with the sun throughout the day. It consists of a framework driven by a step motor coupled with a gearbox (resulting in an angular speed of  $0.004167^\circ/\text{s}$  [42]) and has its main rotation axis parallel to that of Earth. The solar declination angle is compensated via manual regulation. In this work, one single parabolic module was used. Previous tests proved that one module leads to a maximum peak concentration ratio of 870 X and ensures a Gaussian distribution of the light focus, with 95 % of the power contained in a spot diameter of 5.8 cm [44]. A pyrliometer (Kipp & Zonen CHP1) with an aperture con of  $5^\circ$  was mounted on the tracker to measure the direct component of the solar irradiance.

The HTC reactor was designed to match the solar concentrator specifications, according to the European standard on unfired pressure vessels (UNI EN 13455-3:2014), and was manufactured at the machine workshop of the Physics Department of the University of Trento. The HTC reactor is a batch type, is made of stainless steel AISI 316 and has an internal volume of 300 ml. It was designed to withstand pressures and temperatures up to 190 bar and  $280^\circ\text{C}$ , respectively. Figure 4.1c shows the reactor schematic drawing. A Teflon gasket guarantees the sealing between the main body of the HTC reactor and its closing flange. This faces the focus of the concentrator and aims both to absorb the concentrated sunlight and transfer heat to the internal side of the reactor. To maximize the sunlight absorption, a thin layer of copper oxide was deposited on the flange surface. Two thermocouples pass through the top side of the HTC reactor body for the measurement of temperature at different heights (1 cm from the top and 1 cm from the bottom, respectively). From the same side, a rupture disk (calibrated at 120 bar) was positioned for safety reasons. Two pipes were predisposed

to the purging with nitrogen. During the HTC tests, the internal temperature of the reactor was maintained at the set point value (accuracy of  $\pm 5\text{ }^{\circ}\text{C}$ ) by raising and lowering a shield positioned between the reactor and the mirror.



**Figure 4.1.** Solar-HTC system: (a) HTC reactor placed on the solar apparatus, made up of one parabolic module and a sun tracker; (b) HTC reactor: the coated closing flange; (c) technical drawings of the HTC reactor.

#### *Realization and characterization of the coating*

To maximize both the sunlight absorption and the heat transfer, the side of the HTC reactor on which sunlight is focused (i.e. the flange) was sandblasted and treated with a copper oxide coating. This consists of a  $1.78\text{ }\mu\text{m}$  thin layer of copper oxide  $\text{CuO}_x$  realized by combining electron beam deposition (EBD) with thermal oxidation in air. The

coated flange can be appreciated in Figure 4.1b, in which the HTC reactor is mounted on the solar collector.

Before the coating deposition, the flange was cleaned with acetone in an ultrasonic bath and sandblasted using silica particles of diameter in the range 90-150  $\mu\text{m}$  for a duration of 120 s. Sandblasting allows to increase the roughness of steel, which influences the infrared emittance [45].

The deposition was performed through a laboratory-built evaporator system equipped with a 10 kV electron gun with a 10 mA average beam current. Copper powder (Alfa Aesar, 99+%) was placed in a graphitic crucible on a water-cooled holder. Background pressure was maintained lower than  $2.66 \times 10^{-5}$  Pa before operation and at  $9.33 \times 10^{-4}$  Pa during evaporation. Deposition was performed at room temperature at a rate varied between 3.0 and 5.2  $^\circ\text{A/s}$ , controlled with a quartz crystal transducer. After that, the coated steel was annealed in air at 450  $^\circ\text{C}$  for 12 hours to ensure the oxidation of copper.

For characterization purposes, the same deposition procedure followed by thermal oxidation was repeated on stainless steel and glass slides (1.5 mm x 25 mm x 75 mm). Total reflectance ( $R_{\text{tot}}$ , specular plus diffuse), diffuse reflectance ( $R_d$ ) and transmittance ( $T$ ) of the coated samples were measured through a UV-Vis-NIR spectrophotometer (Varian Cary 5000), equipped with an integrating sphere. A Labsphere Spectralon SRS 99 white standard was used as reference material, while the data acquisition was carried out over a wavelength range between 300 and 800 nm. For comparison purposes, measures were performed also on an uncoated sandblasted stainless steel sample.

$T$  of the blasted steel was assumed equal to zero, both with and without the coating. Indeed, the fraction of light transmitted through the coating is likely absorbed by the steel substrate or reflected back and absorbed by the coating. Furthermore, the  $T$  of the coating itself was assumed equal to that of the coated sandblasted glass sample. In this case, the effect of a sandblasted substrate is approximated by the underlying glass.

Absorptance ( $A$ ) and specular reflectance ( $R_s$ ) were determined by difference at each wavelength, according to Equations 4.1 and 4.2.

$$A (\%) = 100 - R_{\text{tot}} - T \quad (4.1)$$

$$R_s (\%) = R_{\text{tot}} - R_d \quad (4.2)$$

The composition of the coating was established by X-ray Diffractometer (XRD, Rigaku ultima IV) using the Cu K $\alpha$  radiation in Bragg-Brentano configuration. The morphology of the coating was investigated by scanning electron microscopy (FE-SEM, JSM-7001F), at an electron beam energy of 15 keV.

### *Experimental campaign*

To show the effectiveness of the solar-HTC reactor, an experimental campaign was carried out for the treatment of grape seeds. The substrate derives from the wet sieving of grape marc and was provided by an alcohol producer company in the North-East of Italy (province of Trento), already dry and with a residual moisture content of about 5 wt.%. Solar HTC tests were performed at temperatures of 180, 220 and 250°C, while residence time and dry biomass to water weight ratio (B/W) were maintained fixed at 2 hours and about 0.3, respectively. Each test was performed twice to address reproducibility.

For each run,  $28.6 \pm 0.1$  g of dried grape seeds and  $95.6 \pm 0.1$  g of distilled water were introduced inside the reactor. The amounts of materials were chosen to guarantee a B/W of about 0.3 and to leave a free volume in the reactor of 60%. After positioning the Teflon gasket and the rupture disk, the reactor was sealed and mounted on the frame of the solar concentrator at a relative angle of 45°, with its coated side facing the parabolic mirror. To improve insulation and protect cables from the solar radiation, the reactor was wrapped with an aluminum foil. Thus, the parabolic mirror was aligned with the sun to have the concentrated light spot contained into the coated side of the reactor. Figure 4.1a shows a picture of the system under operation. The heating phase began and the counting of residence time started when the internal temperature reached the set point value. The internal temperature was maintained at the set point value through a shield, which was manually operated to ensure a maximum temperature variation of  $\pm 5^\circ\text{C}$  to the set point value. During the tests, direct radiation was recorded through the pyrliometer. After the residence time elapsed, the shield was raised, the system misaligned with the sun and the reactor cooled down naturally until ambient temperature. Thereafter, the system was disassembled: the reactor was opened, the solid fraction recovered by filtration and dried in an oven at 105 °C overnight.

The hydrochar solid yield (SY) was computed as the ratio between the mass of dry hydrochar ( $m_{HC,dry}$ ) and the mass of dry biomass ( $m_{bio,dry}$ ), as reported in Equation (4.3):

$$SY (\%) = \frac{m_{HC,dry}}{m_{bio,dry}} \cdot 100 \quad (4.3)$$

#### *Characterization of hydrochars*

Ultimate analyses were performed by means of a Vario MACRO Cube (CHNS-O version) and applying the ASTM D-5373 standard method. The ash content of hydrochars was determined according to the ASTM D3174 -12 standard at 900 °C in a muffle, while the oxygen content was computed by difference ( $O = 100 - C - H - N - \text{Ash}$ ). All compositional data refers to a weight percentage (wt.%), on a dry basis.

An IKA C 2000 calorimeter was used to measure the higher heating values (HHVs) of dry hydrochars and dry grape seeds according to the CEN/TS 14918 standard. The energy yield (EY) was computed as shown in Equation 4.4.

$$EY (\%) = \frac{HHV_{HC,dry}}{HHV_{bio,dry}} \cdot SY(\%) \quad (4.4)$$

For each sample, analyses were performed at least twice and the average values with their relative errors reported.

### **4.1.4 Results and discussion**

#### *Properties of the coating*

The coating confers to the blasted steel good absorbing properties. Indeed, it leads to a maximum absorptance of 95.5% in the UV region and values always higher than 90.7% over the observed wavelength range. The effect of the coating on the substrate is significant, as shown in Figures 4.2(a) and 4.2(b). In correspondence of the solar radiation peak (at around 500 nm), the substrate total reflectance and absorptance pass from 45.0 to 5.7% and from 55.0 to 94.2%, respectively. Besides, the coating reduces the impact of the wavelength variation, flattening the curves. Either with or without the coating, reflectance mainly consists of the diffuse fraction, which occupies more than 92.6% of the

total reflectance over the entire observed spectrum. In this regard, sandblasting is decisive because it increases the steel macroscopic roughness and therefore the light scattering. Moreover, previous studies demonstrated that roughness positively impacts the emittance and adhesion of the film with the substrate [45,46], making sandblasting an effective pre-treatment for the coating.

Table 4.1 summarizes the value ranges of the variables indicative of optical performances over the observed wavelength range (300-800 nm), for both the coating deposit on blasted steel and the coating itself. The difference among them is mainly due to transmittance, which is equal to zero for the coating deposited on the steel substrate. In the case of the coating itself, the amount of radiation transmitted is extremely low (maximum value of 0.1%). Interestingly, with an absorbance ranging between 95.6% (at 300 nm wavelength) and 89.8% (at 800 nm wavelength), the performances of the coating are comparable with commercial coatings. These are used in a variety of light absorption applications, such as in spacecraft, sensors and analyzer instruments for plasma and energy [45]. A list of commercial coatings with high absorbance (88.0 - 97.0%) can be found in literature [47].

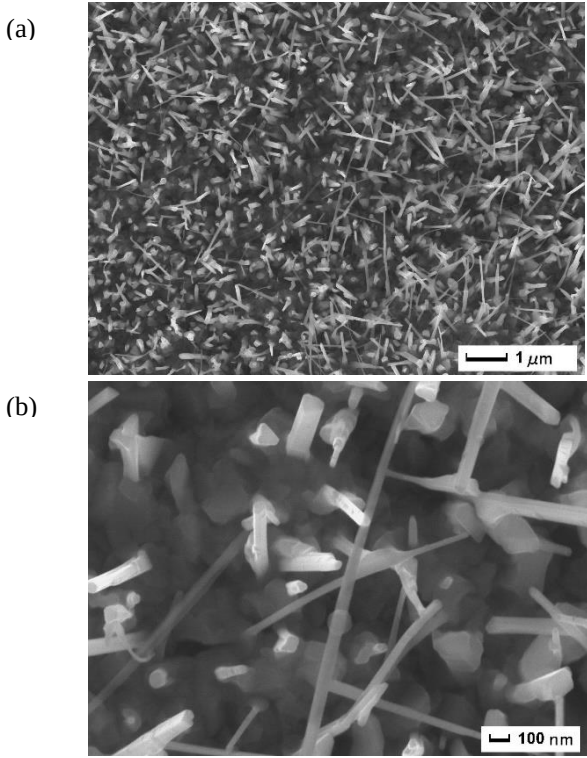
XRD analysis demonstrates that the coating is a sub-stoichiometric mixture of copper oxides, consisting of CuO (cupric oxide) and a smaller fraction of Cu<sub>2</sub>O (cuprous oxide). The XRD pattern of the coating deposit on blasted steel is shown in Figure 3, where the presence of steel phases (austenite and ferrite) of the substrate can be also appreciated. During oxidation, metallic Cu oxidizes to Cu<sub>2</sub>O ( $4\text{Cu} + \text{O}_2 \rightarrow 2\text{Cu}_2\text{O}$ ) which becomes the precursor to CuO ( $2\text{Cu}_2\text{O} + \text{O}_2 \rightarrow 4\text{CuO}$ ), whose slow formation ensures a continuous and uniform growth for CuO nanowires [48]. As shown in SEM images reported in Figs. 4(a) and (b), CuO nanowires cover all the surface and have diameters between 20 and 100 nm. This kind of nanostructure is often found in transition metal oxides synthesized by physical vapor deposition methods followed by thermal annealing in oxidative conditions [48–50]. These have been employed in a broad field of applications in which a high surface is required, such as medical devices, sensors, solar cells and catalysis [51].

Overall, the ordered nanostructured surface confers good absorption properties. The probability of the coating of trapping photons is positively affected by copper oxide nanowires, which allow multiple reflections of light inside the material. Moreover, sandblasting provides more sites for nucleation and then enhances the density and length of nanowires [51].

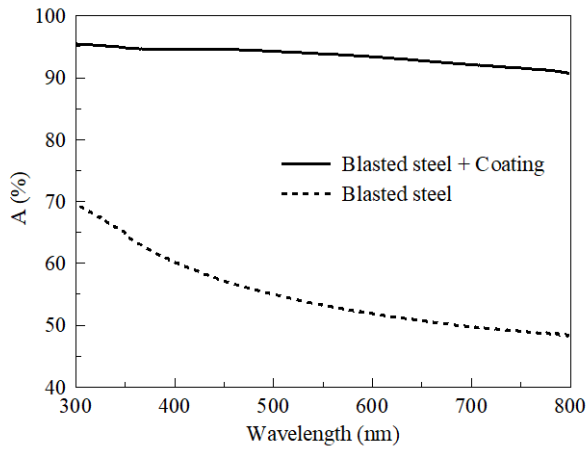


**Table 4.1.** Range of parameters values of coating deposited on blasted steel and of coating itself, in the wavelength range 300 – 800 nm (A: absorptance,  $R_{\text{tot}}$ : total reflectance,  $R_d$ : diffuse reflectance,  $R_s$ : specular reflectance, T: transmittance).

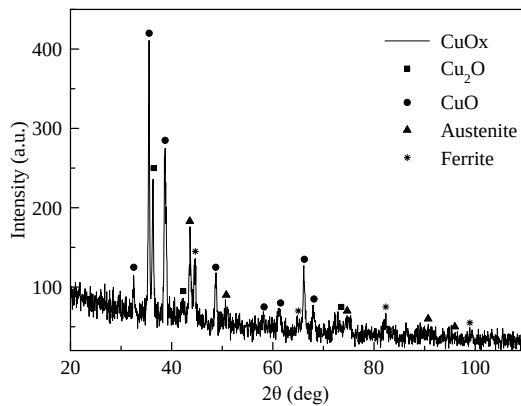
|                      | Blasted Steel + Coating | Coating     |
|----------------------|-------------------------|-------------|
| A (%)                | 95.5 - 90.7             | 95.6 - 89.8 |
| $R_{\text{tot}}$ (%) | 4.5 - 9.3               | 4.4 - 10.1  |
| $R_d$ (%)            | 4.6 - 8.6               | 4.1 - 9.7   |
| $R_s$ (%)            | 0.1 - 0.7               | 0.2 - 0.4   |
| T (%)                | 0                       | 0.0 - 0.1   |



**Figure 4.2** SEM images (a-b) of CuO<sub>x</sub> film deposited by EBD on blasted steel.



**Figure 4.3.** Trends of absorbance (A) of blasted steel, with and without the coating, at different wavelengths.



**Figure 4.4.** XRD pattern of the coating deposited on blasted stainless steel ( $\theta$ : diffraction angle).

#### *Effect of process parameters on heating times*

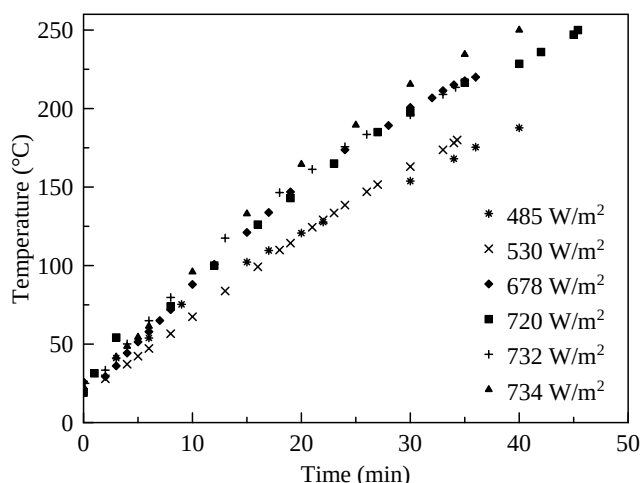
As shown in Table 4.2, as direct irradiance increases, the time required to reach the set point temperature decreases since more thermal power is supplied to the HTC reactor. In this respect, tests performed at 485 and 734 W/m<sup>2</sup> require the same time (40 min) to reach the set point HTC temperatures of 180 and 250°C, respectively. Heating times are comparable with those involving non-renewable heating sources, such as band heaters electrically powered, which amount to approximately 30 min [12,17,52]. However, a certain lack of investigation around

heating times involved during HTC is observed in the current literature [53].

Figure 4.5 shows the trends of the HTC reactor internal temperature versus time during reactor heating-up. For all the investigated irradiances, trends are almost linear and can be approximated by two straight lines with two different slopes ( $R^2 > 0.99$ ). Temperatures increase steeper in the first region (from 20 to 100°C) than in the second one (from 100°C to the set point value). Angular coefficients ( $m$ ), computed through interpolation and summarized in Table 4.2, mark this difference. Up to about 100°C, energy is used to heat water and biomass, without any reaction occurring, and the amount of vaporized water is limited. In the second region, the heating phase slows down because HTC reactions progressively trigger, thermal losses to the environment increase, and the amount of water which evaporates increases (the vapor pressure of water at 180, 220 and 250°C is equal to, respectively, 10, 23 and 40 bar [54]. As shown by curve trends and angular coefficient, tests performed under higher irradiances show steeper slopes. Indeed, higher irradiance means a higher amount of heat rate provided to the system and then a faster heating phase. Moreover, with the relative difference between angular coefficients lower than 10%, slopes measured at similar irradiances (720, 732, and 734  $\text{W/m}^2$ ) ensure a good reproducibility of the tests.

**Table 4.2.** Angular coefficients ( $m$ ) and  $R^2$  values of temperature-heating time interpolating lines in the intervals 0-100 °C and 100 °C-set point temperature ( $I_d$ : direct irradiance).

|                               | $I_d$ ( $\text{W/m}^2$ ) |      |      |      |      |      |
|-------------------------------|--------------------------|------|------|------|------|------|
|                               | 485                      | 530  | 678  | 720  | 732  | 734  |
| $m_{0-100^\circ\text{C}}$     | 5.50                     | 4.92 | 6.55 | 6.79 | 7.48 | 7.43 |
| $R^2_{0-100^\circ\text{C}}$   | 0.99                     | 1.00 | 0.99 | 0.99 | 1.00 | 1.00 |
| $m_{100^\circ\text{C-set}}$   | 3.39                     | 4.17 | 4.85 | 4.02 | 4.38 | 4.73 |
| $R^2_{100^\circ\text{C-set}}$ | 1.00                     | 1.00 | 0.99 | 0.99 | 0.99 | 0.99 |



**Figure 4.5.** Trends of HTC reactor internal temperature as a function of time, at different direct irradiances.

#### *Properties of “solar hydrochars”*

Higher HTC temperatures enhance the conversion degree of biomass into hydrochar, as demonstrated by the decreasing trend of solid yields reported in Table 3. Indeed, harsher conditions favor the mass transfer of the starting biomass into the gas and liquid phases [55]. Biomass directly decomposes into gas molecules (mainly  $\text{CO}_2$  and  $\text{CO}$ ) through decarboxylation and decarbonylation reactions, whilst into organic molecules in the liquid phase [26,56]. Meanwhile, some of these organics further decompose to  $\text{CO}_2$ , react and possibly polymerize in the liquid phase.

Table 4.3 also reports the elemental composition of the raw biomass and hydrochars, in terms of carbon (C), hydrogen (H), oxygen (O), nitrogen (N) and sulfur (S). As expected, the HTC operating temperature has a positive effect on the carbonization process. Elemental C increases along with a decrease in O, which arrives at 69.2 and 19.4% (at  $250^\circ\text{C}$ ), respectively. H and N do not vary significantly between the tests, while the presence of sulfur is almost negligible (less than 0.4%).

H/C and O/C atomic ratios were computed and visualized on a Van Krevelen diagram (Figure 4.6), in which raw biomass and “solar

hydrochars” are compared with typical coals (anthracite, bituminous coal, lignite, and peat). HTC lowers both the H/C and O/C ratios, making hydrochars close to low-rank coals, like lignite and peat, and indicating that dehydration is a dominant reaction during the process.

The ash relative content of hydrochars slightly varies with temperature and concerning the raw biomass (Table 3). However, their absolute amounts decrease compared to biomass and exhibit loss ratios ranging between 4 and 28%. This confirms the HTC capability of dissolving some inorganics (especially  $\text{Na}^+$  and  $\text{K}^+$ ) from the parent biomass into water [57,58].

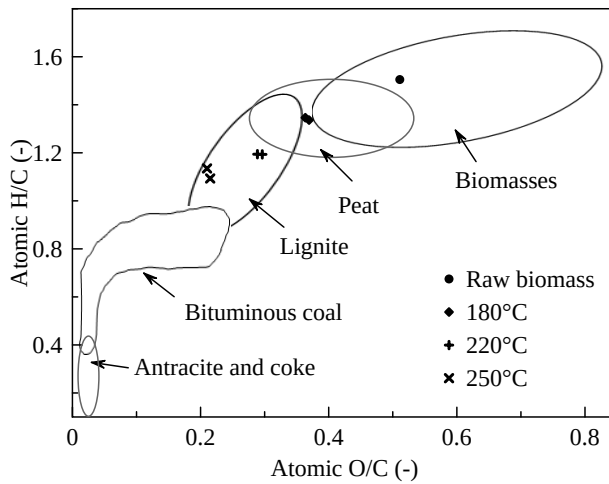
The effect of temperature on the carbonization degree is reflected in the energy properties of hydrochars (Table 4.3). Through HTC, the higher heating value (HHV) of grape seeds passes from 23.5 up to a 30.5 MJ/kg (at 250°C). The energy yield (EY) slightly decreases with temperature mainly due to the counterbalance between the mass loss and the HHV enhancement.

Properties of hydrochars obtained at the same operating conditions and different irradiances are fully comparable: the differences in solid yields, composition, and energy properties (Table 3) are very small. Irradiance variations in the investigated range (from 485 to 734  $\text{W/m}^2$ ) do not affect hydrochar properties, proving the reproducibility of the experimental procedure. Moreover, hydrochar data resemble those obtained in previous works through traditional HTC procedures and starting from the starting feedstock. In particular, Basso et al. [59] report a carbon content of hydrochars of 60.2, 63.6, and 69.5% (180, 220, and 250°C, respectively) that imply a percentage difference with solar hydrochar lower than 1.4%. This is valid also for carbon and hydrogen contents, as well for solid yields. Besides, the ash content of solar hydrochar is slightly lower than the conventional one (ranging from 3.5 to 4.5%) due to the different temperature adopted during the measuring procedure (900°C here against 550°C). Therefore, the similarity in the composition of solar and conventional hydrochars provides proof of the effectiveness of the proposed apparatus.

**Table 4.3.** Solar-HTC test details, ultimate analyses and HHVs of “solar hydrochars” ( $S < 0.4\%$ . Relative errors  $\leq 0.5\%$  for ultimate analysis,  $\leq 0.5\%$  for ash,  $\leq 0.6\%$  for HHVs). All data are on a dry basis. Heating time: 34-40 min.

| Sample | $I_d$<br>(W/m <sup>2</sup> ) | SY<br>(%) | Ultimate analysis (wt.%) |     |      |     | Ash<br>(wt.%) | HHV<br>(MJ/kg) | EY<br>(%) |
|--------|------------------------------|-----------|--------------------------|-----|------|-----|---------------|----------------|-----------|
|        |                              |           | C                        | H   | O*   | N   |               |                |           |
| Raw    | -                            | -         | 53.2                     | 6.7 | 36.2 | 1.7 | 1.9           | 22.5           | -         |
| 180-1  | 485±2                        | 78.8      | 59.7                     | 6.7 | 29.4 | 1.5 | 2.3           | 24.9           | 84        |
| 180-2  | 530±8                        | 80.1      | 60.1                     | 6.7 | 29.1 | 1.6 | 2.2           | 25.1           | 86        |
| 220-1  | 678±5                        | 72        | 64                       | 6.4 | 25.3 | 1.7 | 2.2           | 27.2           | 83        |
| 220-2  | 73±9                         | 70.1      | 64.5                     | 6.4 | 24.8 | 2   | 2             | 27.7           | 83        |
| 250-1  | 720±17                       | 61.5      | 69.2                     | 6.5 | 19.4 | 2   | 2.5           | 30.5           | 80        |
| 250-2  | 734±14                       | 61.1      | 69.1                     | 6.3 | 19.8 | 2   | 2.7           | 30             | 78        |

\*Computed by difference.



**Figure 4.6.** Van Krevelen diagram of raw biomass and “solar hydrochars” [60].

#### 4.1.5 Conclusions

This work provides a successful proof-of-concept for a hybrid solar-HTC technology. This solution represents a zero-energy process used to enhance waste biomass into valuable carbonaceous materials. The HTC energy needs are fully covered by concentrated sunlight and the use of traditional energy sources is avoided. This system represents an innovative biorefinery concept in an actual context in which the combination of HTC with renewable energy sources has been little investigated, thus providing a step-forward towards an environmental sustainable HTC process.

Technical details on the components forming the system (i.e. the HTC reactor and the parabolic dish concentrator) were provided. A particular attention was given to characterize the coating, which was realized to maximize the absorption of the solar radiation. In particular, it showed very satisfactory light absorption properties (absorptance up to 95.6%) and an interesting nanostructured surfaced made up of copper oxide nanowires. An experimental campaign performed on grape seeds (treated at 180, 220, and 250°C, for 2 hours) proved the operational effectiveness of the technology. Indeed, heating time, solid yields, chemical and thermal properties (elemental composition and heating values) of “solar hydrochars” obtained through our hybrid solution resemble well those of traditional HTC systems.

Future developments include the realization of a sophisticated control system, able to handle the shield as a function of the internal HTC temperature. The authors see as a chance for the spreading of the technology, the development of techno-economic evaluation on an industrial-scale plant, in which collected biomass waste of municipalities are collected. This should be integrated with a proper life cycle assessment and a parametric analysis.

#### 4.1.6 Appendix

Economic and energy considerations were performed on a scaled-up prototype, consisting of a full parabolic dish (projected area of 20 m<sup>2</sup>) and an HTC reactor with a volume of 20 l.

The power collected by the PDC was computed assuming an average irradiance of 700 W/m<sup>2</sup> and a PDC collection efficiency of 86 %, while the available power to the HTC reactor considering an overall efficiency of 65% (considering transmission, reflection, and heat losses). The specific energy consumption ( $q_{\text{heat}}$ ) required to heat the slurry (biomass plus water) from the initial temperature ( $T_0$ , equal to 20 °C) to the set point temperature ( $T_{\text{set}}$ , equal to 250 °C), was computed according to Equation (4.A). The specific heat of biomass ( $c_{p,\text{bio}}$ ) was assumed constant and equal to 1.3 kJ/(kgK), while the specific heat of water ( $c_{p,w}$ ), was determined according thermodynamic data. The power required during the steady state phase and the B/W were assumed equal to 0.5 kW and 0.3, respectively.

$$q_{\text{heat}} = c_{p,\text{bio}} \cdot (T_{\text{set}} - T_0) + W/B \cdot \int_{T_0}^{T_{\text{set}}} c_{p,w}(T, P) \cdot (T_{\text{set}} - T_0) \quad (4.A)$$

Then, the amount of usable biomass per test was determined dividing  $q_{\text{heat}}$  by the required energy to heat the slurry in 30 minutes, while the HTC reactor volume considering an additional 20 % free volume. The initial costs of the prototype were estimated on the base of the incurred expenditures and commercial quotations. Costs were compared with those of a commercial HTC system consisting of a reactor heated by thermal oil electrically heated, whose price was determined by averaging two commercial quotations, scaled up from reference to design conditions using an exponent of 0.8, as shown in Equation (4.B).

$$\text{Cost (Euro)} = \text{Cost}_{\text{ref}} \cdot \left( \frac{\text{Capacity}}{\text{Capacity}_{\text{ref}}} \right)^{0.8} \quad (4.B)$$

Table 4.A reports the main results. The HTC reactor has a volume of 20 l and can treat 18.5 kg of slurry per test. The total initial cost is 23500 Euro, similar to the cost of the commercial system, which is equal to 23000 Euro. The energy required to heat the HTC reactor and sustain the 2 hours steady-state phase is 4.9 kWh per test. Through the solar-HTC apparatus, the concentrated solar power covers the energy need, avoiding the usage of any external energy source. In the case of the



traditional system, the thermal oil provides the thermal energy after being heated by an electrical resistance. Therefore, about 6.1 kWh<sub>e</sub> (assuming a heat exchanger efficiency of 80%) has to be purchased from the grid or self-produced.

**Table 4.A.** Technical details and initial costs of the solar-HTC apparatus.

|                                      |       |
|--------------------------------------|-------|
| PDC projected area (m <sup>2</sup> ) | 20.0  |
| Reactor volume (L)                   | 20.0  |
| Slurry capacity (kg)                 | 18.5  |
| Power PDC (kW)                       | 12.0  |
| Power HTC (kW)                       | 7.8   |
| Energy HTC (kWh)                     | 4.9   |
| Total solar-HTC cost (Euro)          | 23500 |
| PDC (Euro)                           | 8500  |
| HTC reactor (Euro)                   | 15000 |
| Commercial HTC cost (Euro)           | 23000 |

## Section 4.2

# Organic waste to biofuels through a zero-energy hydrothermal solar plant: process design

This section is based on the article that we published in *Chemical Engineering Transaction* (2020, vol. 80, 7-12, by G. Ischia, D. Castello, M. Orlandi, A. Miotello, L.A. Rosendahl, L. Fiori <https://doi.org/10.3303/CET2080002>). The work is a result of our collaboration with Daniele Castello and Lasse Rosendahl at Aalborg University (Denmark), experts in hydrothermal liquefaction. The work is a techno-economic analysis of a plant consisting of two possible hydrothermal plants (HTC or HTL) indirectly coupled with a solar concentrator.

### 4.2.1 Abstract

Hydrothermal processes enable an effective conversion of waste biomasses into fuels and carbonaceous materials. Covering the heat requirements with concentrated solar energy is a clever strategy to increase the plant efficiency and pursue the principles of circular economy. With the purpose of producing liquid and solid biofuels through zero-energy routes, this work presents two conceptual designs for integrating a concentration solar system (CSS) with a hydrothermal liquefaction (HTL) and a hydrothermal carbonization (HTC) plant. The solar configuration used to cover hydrothermal heat requirements consists of a set of parabolic trough collectors operating with molten salts, which are used as both thermal carrier fluid and thermal energy

storage medium. Two different scenarios were modelled to continuously process wood and organic wastes. In the first scenario, the CSS is coupled with a continuous HTL reactor (operating at 400°C and 300 bar) followed by thermal cracking and hydroprocessing for upgrading bio-crude to a saleable liquid biofuel. The second scenario considers the CSS operating with a continuous HTC reactor (working at 220°C and 24 bar) for the conversion of organic wastes into a solid fuel (hydrochar). The CSS and both the hydrothermal plants were modelled based on experimental data. Energy consumptions and techno-economic aspects were investigated.

#### 4.2.2 Introduction

Climate change and resource depletion require the development of new strategies to concretize positive impacting actions. In this context, the valorisation of waste biomass has a key role in a circular economy vision. In particular, hydrothermal processes enable the upgrade of biomass into liquid, solid and gaseous biofuels. Occurring in water, these processes can treat a multitude of non-conventional biomass, as sewage sludge, agricultural residues, and organic wastes. Operating conditions vary according to the desired product. Hydrothermal carbonization (HTC) occurs at mild operating conditions (180-250 °C and 10-40 bar) and lead to a solid phase, known as hydrochar, which can be used as solid fuel [61]. Under harsher conditions (300-400°C and 100-300 bar), hydrothermal liquefaction (HTL) occurs, leading to the production of a crude oil. This finds applications in the production of liquid fuels [62].

One of the main drawbacks of hydrothermal processes (especially HTL due to the harsher operating conditions) is the large amount of heat required to reach operating conditions [8]. In this regard, the usage of renewable energy to cover energy needs is a possible strategy to accomplish the principles of circular economy and develop a zero-energy plant.

This work investigates two possible scenarios in which two hydrothermal plants (a HTC and a HTL) are coupled with a solar concentrator system (CSS). The CSS, equipped with solar collectors and thermal energy storage systems, is designed to cover thermal energy requirements of the hydrothermal plants. An economic evaluation was performed to show the feasibility of these sustainable options.

### 4.2.3 Materials and methods

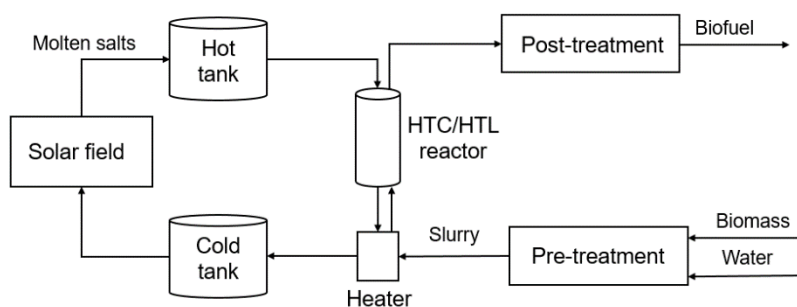
#### *General layout*

Figure 4.7 shows the general layout of the investigated conceptual plants. The hydrothermal plant schemes are based on previous research articles [21,63], while the configuration of the CSS is based on commercial installations [64,65]. The HTL plant aims to produce a liquid drop-in fuel starting from biomass (500 tons per day of dry wood). The HTC plant treats 200 tons per day of organic wastes on a dry basis, corresponding to about 400 ton per day of as-received matter, (about 3 millions of equivalent inhabitants). The context of the two systems is different: HTL focuses on the production of a high-quality product, while HTC addresses waste management issues.

Inside the hydrothermal plants, the raw biomass is converted into the desired product. The feedstock of the HTL system is a slurry of wood and water, processed at 400°C and 300 bar. Regarding HTC, organic wastes are continuously treated at 220°C and 24 bar (residence time of 3 hours). In both cases the model inputs (e.g. product composition and yields) are based on previous works [19,66]. The CSS is sized to cover the thermal energy required to perform a continuous thermochemical conversion. To this purpose, a heat fluid carrier (consisting of a mixture of molten salts) is heated up inside the parabolic trough collectors and releases heat to the hydrothermal plant. The solar field is composed of modular solar collectors connected in parallel rows, known as Solar Collector Assemblies (SCA). Each row consists of a 2x100 m collectors characterized by an area of 1113.6 m<sup>2</sup> [29]. To ensure a continuous operation of the plant, thermal energy storage (TES) tanks are used. These have the advantage of mitigating weather fluctuations and extending the heat supply during overnight and cloudy periods. The molten salt mixture adopted is a commercial one (HITEC) and consists of a mixture of NaNO<sub>2</sub>, NaNO<sub>3</sub>, and KNO<sub>3</sub> [42]. In particular, under the chosen operating conditions, corrosion effects with stainless steel are almost negligible [67].

As shown in Figure 4.7, biomass is initially pre-treated with water to obtain a slurry. Then, the slurry is compressed and heated up to the design conditions by a pump and a counter-current heat exchanger. Here, the mixture of molten salts, previously heated inside the solar collectors, provides the required thermal energy. The outlet temperature of the molten salts stream is maintained constant by varying the flow rate according to the direct solar radiation. A hot TES was positioned

downstream the solar field. When the thermal energy from the sun is higher than the process demand, a fraction of heating fluid is stored inside the tank for being used in absence of solar radiation. Then, hot molten salts are conveyed to the heat exchanger, where they heat up the slurry. Afterwards, they are stored in a cold TES and pumped back to the solar field. When the direct radiation is not sufficient to guarantee a proper molten salt exit temperature, the stream is pumped back to the cold tank. TES has a storage capacity of 24 hours. Inside the HTL reactor, wood is converted into bio-crude with a yield of 40%. This value has been chosen on a conservative basis, since recent studies have demonstrated that higher yields can be reached [63]. The gas phase (mainly consisting of  $\text{CO}_2$ ) represents the 13%, while solids and the aqueous phase with dissolved organics the remaining 47%. The solid phase is removed inside a cyclone positioned downstream the HTL reactor. The pressurized hot stream is then cooled down, depressurized and sent to a vessel to separate the three phases (biocrude, water and gas phases), where 90% of the water phase is recirculated back. The bio-crude is upgraded through thermal cracking ( $600^\circ\text{C}$  and 4 bar) and hydrotreating (at  $360^\circ\text{C}$  and 77.5 bar). Regarding the HTC reactor, organic waste is converted into the desired solid phase (hydrochar) with a yield of 77.2%. After the reaction, the slurry is depressurized, cooled down and the liquid phase separated in a decanter. Part of the water is recirculated back.



**Figure 4.7.** General layout of the CSS-hydrothermal plants.

### *Modeling of the process*

Energy and mass balance computations were performed using Matlab. Thermodynamic properties of water were obtained by using the

software Cantera. Energy balances were performed to size the different components. The same procedure was adopted for the HTL and HTC design.

The reactor is tubular and in stainless steel AISI 316L. A free volume of 30% was considered. The reactor volume was chosen to guarantee the nominal volumetric flow rate of the slurry and a residence time of 15 min for HTL and 3 hours for HTC.

The specific heat demand ( $q_{\text{heat}}$ ) to rise up the slurry stream from the initial temperature ( $T_0$ , equal to 25°C) to the set point one ( $T_{\text{set}}$ ) was computed according to Equation (4.5).  $c_{p,\text{bio}}$  is the specific heat capacity of dry biomass, assumed constant and equal to 1.2 kJ/(kgK).  $c_{p,w}$  is the specific heat capacity of water, function of temperature and pressure according to available thermodynamic data. B/W is the ratio between dry biomass and water, equal to 0.30 for HTL and 0.25 for HTC.

$$q_{\text{heat}} = c_{p,\text{bio}} \cdot (T_{\text{set}} - T_0) + \frac{W}{B} \cdot \int_{T_0}^{T_{\text{set}}} c_{p,w}(T, P) \cdot (T_{\text{set}} - T_0) \quad (4.5)$$

The heat required by the reactor was computed considering both the heat of reaction ( $\Delta H_r$ ) and heat losses ( $q_{\text{loss}}$ ). For HTC, the gas was assumed composed entirely of CO<sub>2</sub>, while for HTL the composition proposed by Hansen et al. [68] was considered. For both HTC and HTL, the liquid phase was assumed composed by water and acetic acid.  $\Delta H_r$  was computed by applying the Hess's law, as shown in Equation (4.6) where the subscript p indicates the products. The molar enthalpy of formation ( $\Delta H$ ) of biomass, bio-crude and hydrochar was computed starting from their experimental higher heating values (on a dry ash free basis), on the base of the combustion reactions. This approach has been widely adopted for modeling biomass processes (like torrefaction, gasification and pyrolysis) [69]. For conventional components, thermodynamic data were used. Water was assumed non participating to the reaction. MY and MW are the mass yield and the mean molecular weight of the i-compound, respectively.

$$\Delta H_r = \sum MY_p \frac{\Delta h_p}{MW_p} - \frac{\Delta h_{\text{bio}}}{MW_{\text{bio}}} \quad (4.6)$$

$q_{\text{loss}}$  were computed as the sum between conduction, convection and radiation contributions. Reactors are in stainless steel AISI 316L and are thermally isolated by 25 cm of rock wool. Heat transfer relations of a cylinder were applied by using thermal properties of the used materials [70]. The power required by the pump ( $P_{\text{pump}}$ ) to increase the

pressure of the slurry from  $P_0$  (set equal to 1 bar) to  $P_{\text{set}}$  was computed by applying Equation (4.7), where  $\dot{V}$  is the slurry volumetric flow rate and 0.8 the efficiency of the pump.

$$P_{\text{pump}} = \frac{\dot{V}(P_{\text{set}} - P_0)}{0.8} \quad (4.7)$$

The overall thermal requirements of hydrothermal plants ( $P_{\text{tot}}$ ) were used to compute the respective molten salt flow rate ( $\dot{m}_{\text{HTF}}$ ), assuming a heat exchanger efficiency ( $\eta_{\text{hx}}$ ) equal to 90 %. The temperature of the HTL tanks was fixed at 420 and 380 °C, while for HTC at 230 and 210°C. Then, the CSS was sized. The efficiency of the solar collectors ( $\eta_{\text{solar}}$ ) is equal to 72 % and comprises the optical and receiver efficiencies [29]. The projected area of the solar collectors ( $A_{\text{solar}}$ ) was computed from Equation (4.4), where  $I_d$  is the direct solar radiation. This last was assumed equal to 700 W/m<sup>2</sup>, which corresponds to the peak average hourly irradiance in the center of Italy, in summer [71]:

$$A_{\text{solar}} = \frac{P_{\text{tot}}}{\eta_{\text{hx}} \eta_{\text{solar}} I_d} \quad (4.8)$$

Once the plant component were sized, an economic evaluation was carried out. A plant lifetime of 10 years and a continuous operation (24/24) were assumed. Regarding the HTL plant, its capital and operating costs (CAPEX and OPEX) were estimated on the base of results obtained by Pedersen et al. (2017). For the HTC plant, reference is done to the work of Lucian and Fiori (2017), from which the costs of methane burner were removed. After the actualization, CAPEX and OPEX were scaled by applying Equation (5), where a scaling exponent equal to 0.65 was used [73]. The same procedure was applied for the operating costs, making reference to those reported by Lucian and Fiori [21], reduced by methane costs.

$$\text{Cost}_{\text{plant}} = \text{Cost}_{\text{plant},2019} \left( \frac{\text{Plant capacity}}{\text{Plant capacity, ref}} \right)^{0.65} \quad (4.9)$$

Cost of installation of the solar plant (solar collectors, molten salts and TES) were computed by averaging and actualizing data reported in previous studies [64,65,74,75]. In particular, a CAPEX of 205 €/m<sup>2</sup> and of 28.0 €/kW<sub>th</sub> was assumed for the solar field (comprised of solar collectors and molten salts) and TES, respectively. Operating costs of the solar field and TES were assumed equal to 2.2% of the CAPEX [65]. Then, the final production cost was computed as a minimum fuel

selling price (MFSP). This represents the fuel selling price at which the net present value of the process is equal to zero [9]. A cash flow analysis was carried out considering the overall plant life of 10 years and it was assumed an operative plant time of 328 days per year, 5% of annual interest loan with a payback time of 10 years, and a taxation rate of 37%.

#### 4.2.4 Results and discussion

##### *Thermal energy requirements*

Table 4.4 reports a summary of specific thermal energy requirements. As expected from the harsher working conditions of HTL, the specific heat required to raise the slurry temperature is much higher than that required by HTC. Regarding the heat of reactions, with a positive value of  $\Delta H_r$ , the HTL of wood has an endothermic behavior. Unlike HTL,  $\Delta H_r$  of HTC is negative, indicating that the reaction is exothermic. This behavior is well-known in the literature and was already measured through analytical methods (like differential scanning calorimetry, DSC [18,76] and lab-scale reactors [17,76–78]). For the sizing of the solar plant, the heat released during HTC was not considered, in order to adopt a conservative approach. Heat losses are much more important for HTC, representing about 1.5% of the total thermal energy demand. For HTL, they account for about 0.5% of the total, meaning the size of the reactors has a higher impact than the operating temperature.

**Table 4.4.** Specific thermal energy requirements (MJ/kg<sub>bio</sub>) of hydrothermal reactors (B/W of 0.30 for HTL and 0.25 for HTC).

|     | Heating ( $q_{heat}$ ) | Heat of reaction ( $\Delta H_r$ ) | Heat loss ( $q_{loss}$ ) |
|-----|------------------------|-----------------------------------|--------------------------|
| HTL | 7.09                   | 3.00                              | 0.05                     |
| HTC | 3.60                   | -0.32                             | 0.05                     |

##### *Design parameters and economic analysis*

Table 4.5 summarizes the design parameters of the hydrothermal plants and respective solar fields. The difference between the two reactors is evident. Due to the longer residence time required for the reactions, the HTC reactor exhibits a much bigger volume than the HTL one. Regarding the required power, the process operating conditions are



decisive, leading to 58.7 and 8.4 MW for the HTL and HTC reactors, respectively. Similarly, with a value of 852.1 kW, the nominal power of the HTL pump is much higher than the HTC one.

Overall, solar collectors have a huge impact on the overall plant surface. Indeed, to guarantee a continuous operation for the HTL plant, 114 parabolic trough collectors (each of 1136 m<sup>2</sup>) are required. 17 solar collectors are enough for the HTC plant, which requires a lower amount of thermal power. This reflects on the TES volumes, equal to 15.6 and 4.5 m<sup>3</sup>.

**Table 4.5.** Design parameters of the hydrothermal plants and respective solar plants.

|                                        | HTL    | HTC   |
|----------------------------------------|--------|-------|
| <b>Hydrothermal plant</b>              |        |       |
| Biomass flow rate (ton/day)            | 500    | 200   |
| Total input flow rate (ton/day)        | 2167   | 1000  |
| Power (MW)                             | 58.7   | 8.4   |
| Reactor volume (m <sup>3</sup> )       | 29     | 162   |
| Reactor surface area (m <sup>2</sup> ) | 78     | 166   |
| Pump power (kW)                        | 852.1  | 28.9  |
| <b>Solar plant</b>                     |        |       |
| Power (MW)                             | 58.7   | 8.4   |
| Collector area (m <sup>2</sup> )       | 129456 | 18620 |
| Number of collectors                   | 114    | 17    |
| TES volume (m <sup>3</sup> )           | 15.6   | 4.5   |

Table 4.6 shows the results of the economic analysis. The installations and operating costs of the hydrothermal plants were computed independently of the solar field and TES. In particular, CAPEX of the HTC plant includes the costs for the on-site equipment (like the reactor, pumps, agitator, crusher, filter, dryer, and the pelletizer) as well as depreciation costs, land use, utilities, start-up of the plant, and working capital. Meanwhile, OPEX comprises annual costs like electricity, labour, maintenance, and waste-water treatment. For HTL, costs related to the post-treatment phase are also considered. Costs for the thermal cracker and the hydrotreater are included in the total, while the hydrogen purchase is included in the operating costs. CAPEX of the solar field includes the costs for the solar collectors, molten salts, back up heaters, instrumentations, civil works, installation, balance of plant, and contingency. The costs for the replacement of molten salts are included in the OPEX.

The model predicts a CAPEX of 157.1 and 5.6 M€ for the HTL and HTC apparatus, respectively. The investment cost of the hydrothermal plant is predominant for HTL (69.1% of the total), while it represents a much lower fraction in the case of HTC (35.7%). This difference is due to the much higher complexity of the HTL plant, which comprises also the post-treatment stage. TES costs dominate the solar plant investment. In both cases, operating costs involved in the hydrothermal plant represent up to 90% of the total.

The minimum selling prices of bio-oil and hydrochar were computed starting from total costs, considering a life-time period of 10 years. The production accounts for 200 tons of bio-oil and of 120 tons of hydrochar per day.

Bio-oil exhibits a MSFP value equal to 1.39 €/kg, while pelletized hydrochar equal to 163 €/ton. This value is competitive with wood pellets, whose prices are usually in a 150-200 €/ton range and does not differ too much from coal prices, currently equal to 20.2 €/ton (lignite) and 59.0 €/ton (bituminous) [79]. Interestingly, considering a gain of 60 €/ton for the as-received organic waste (about 400 ton/day) the hydrochar MFSP becomes as low as 37.4 €/ton.

Therefore, according to the obtained results, coupling HTL or HTC with a solar concentration apparatus appears as a reasonable strategy towards the development of a zero-energy plant. Solar energy covers the thermal energy process demand, leading to an improved sustainability of traditional systems. Certainly, for a more accurate evaluation, taxation, subsidies and policies should be considered.

Finally, it should be noted that all computations were performed in a conservative fashion and with the aim of optimizing the process, certain measures could be adopted. For example, the hot stream exiting from the hydrothermal reactors could be used to pre-heat the slurry in a heat exchanger and a part of the off-gases exiting from the HTL reactor could be burnt to cover part of the heat requirement.

**Table 4.6.** Economic parameters of the integrated CSS-hydrothermal plants (in M€).

|                    | CSS-HTL plant |      | CSS-HTC plant |      |
|--------------------|---------------|------|---------------|------|
|                    | CAPEX         | OPEX | CAPEX         | OPEX |
| Hydrothermal plant | 157.1         | 43.7 | 5.6           | 3.0  |
| Solar field        | 26.5          | 0.6  | 3.8           | 0.1  |
| TES                | 43.8          | 1.0  | 6.3           | 0.1  |
| Total              | 227.4         | 45.3 | 15.8          | 3.2  |

#### 4.2.5 Conclusions

This work investigates the coupling of a solar concentration plant with an HTL and an HTC plant. Energy requirements necessary to perform liquefaction and carbonization reactions were provided through a stream of molten salts heated inside the solar collector. A TES apparatus enables the continuous operation of the plants. Reasonable values of MFSP show that this solution could constitute an effective strategy to improve the sustainability of the hydrothermal processes. Indeed, thermal energy requirements are fully covered by solar energy and sub-products derived from the process, like the gas phase rich in  $H_2$  produced during HTL [63], which could be used for other purposes (e.g. in the chemical industry).

## Section 4.3

# Design of a solar-hydrothermal liquefaction apparatus

This section describes a preliminary design of a hybrid apparatus consisting of a hydrothermal liquefaction reactor indirectly coupled with a solar concentrator. The system aims to convert biomass to bio-crude through a zero-energy pathway and represent a natural transition from the directly-coupled prototype (described in section 4.1) to an indirect system. The apparatus is intended for lab-scale applications and uses the already existing solar concentrator placed at the University of Trento (Physics Department). I am really grateful for the precious support of Marco Bettonte (University of Trento), who helped and followed me through all the design.

### 4.3.1 Introduction

In the last years, hydrothermal liquefaction (HTL) has seen a rising interest for its potential to produce second-generation biofuel. Indeed, HTL is a green pathway to convert biomass into a liquid oily phase, called bio-oil or bio-crude, having properties similar to petroleum-derived products [80,81]. As with the other hydrothermal processes, HTL has the great advantage of being flexible in terms of feedstock and suitable for “difficult” substrates like wet biomass. Compared to fast pyrolysis, which is the other class of thermochemical processes capable to convert biomass into liquid fuel, the bio-crude from HTL has the advantage of having a considerably lower oxygen content [62]. Indeed, oxygen causes instability and enhances an unwanted polymerization of

the liquid phase [62]. The bio-crude then can be upgraded to “drop-in” fuel through post-processing, like hydrodeoxygenation [81].

HTL is generally performed at relatively harsh conditions, 280-370°C and 100-220 bar. Despite its potential, HTL has the drawback of being energy-intensive compared to other processes. Indeed, heat has to be provided to heat both the biomass and the water, characterized by a higher heating capacity than the solid phase. Currently, systems with heat recovery solutions have been implemented using internal streams, like recovering the heat produced from the combustion of the same produced bio-oil [1]. Therefore, using renewable strategies, as concentrated sunlight is a sustainable solution to supply the external energy and increase plant efficiency. To reach temperatures like those of HTL, a concentrated sunlight technology, like linear and point focus concentrators, is needed [1].

The coupling between HTL and concentrated sunlight can be realized in two ways: direct, in which the receiver is modified into a “solar reactor” (like that proposed in section 4.1), and indirect, where the energy is provided to a conventional chemical reactor by a heating fluid. The direct solution has the advantage of avoiding any heat exchanger to supply the heat, increasing the efficiency of the process. This approach requires the design of a proper reactor, increasing the complexity of the research. Also, the system suffers from thermal gradients that can cause non-homogeneity inside the reactor and mechanical stresses. Meanwhile, indirect plants have the advantage of using standard reactors and being integrated with storage systems, decoupling the process from weather variations. The plant is complicated by the presence of other components, like heat exchangers (operating at high temperature) and the heat carrier. However, indirect systems can be easily scaled, making them appealing for industrial applications [1,7].

With this work, we propose the design of an indirect apparatus in which a parabolic dish solar concentrator (PDC) is coupled with a batch hydrothermal reactor using thermal oil. Through this system, the concentrated sunlight heats the thermal oil, which is stored in a tank and conveyed to the HTL reactor. The thermal storage enables mismatch sun fluctuations from the HTL heat demand. The apparatus is thought for lab-scale applications and uses the already existing solar concentrator realized at the University of Trento (details in section 4.1). The work focuses on the plant design, with particular attention to the sizing of the HTL reactor according to the available energy and HTL requirements.

### 4.3.2 Overview of the system

The proposed system consists of a batch HTL reactor indirectly coupled with a solar concentrator. The apparatus is sized for lab-scale applications, but the concept can be scaled for larger dimensions.

Figure 4.8 shows a general scheme of the apparatus. During the operation, biomass undergoes decomposition inside a batch HTL reactor to form a bio-crude. Meanwhile, a parabolic dish solar concentrator provides the energy necessary to sustain the process. The coupling between the two technologies is indirect: the solar concentrator heats up a thermal oil operating at high temperatures, which is conveyed and stored in a high-temperature tank to minimize the effect of weather fluctuations. When needed, the hot oil is sent to the HTL reactor through a pump. A water-cooling system cools down the reactor. The main components of the plant are:

- *HTL reactor.* Here, biomass reacts with water at high temperature (300–374°C) and pressure (85–220 bar), forming a liquid phase consisting of a water-insoluble fraction (bio-crude) and a water-soluble one (rich in organics), a solid phase (hydrochar), and gas phase. The reactor is a batch type and positioned on the ground. It is positioned inside a concentric tank where the hot oil coming from the high-temperature tank is conveyed.

The sizing of the reactor was performed by following four stages: (1) Preliminary sizing according to the input net thermal power derived from the solar apparatus; (2) Choice of the proper reactor geometry; (3) Verification according to the standard EN 13445-3: 2019; (4) Stress dynamic simulation with Inventor.

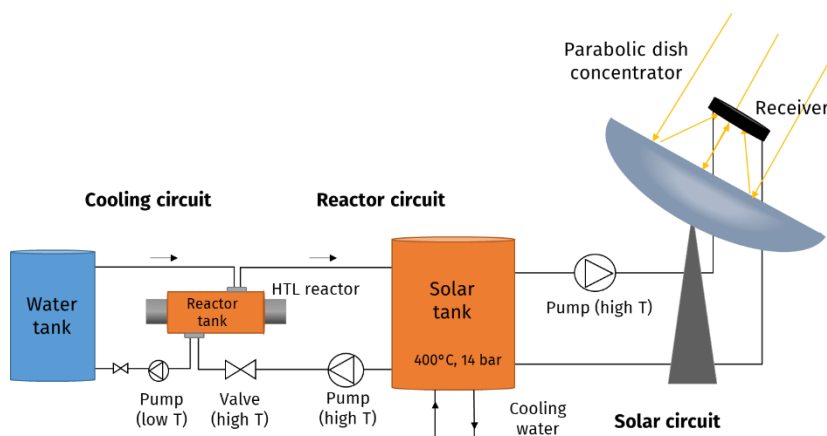
- *Parabolic dish concentrator.* It concentrates the sunlight into a receiver positioned on its focus, where thermal oil flows and is heated. It is the prototype already present at the University of Trento (described in section 4.1 and several articles [42–44,82]): it consists of a modular parabolic sector and is equipped with a solar tracker to ensure alignment with the sun throughout the day. Also, the receiver was built in-house and reference is done to the work by Turrini et al. [42].
- *Thermal oil.* It is the heat carrier to decouple into two processes. It operates at 400°C to ensure the internal HTC conditions.
- *High-temperature tank (solar tank).* It is used to store the oil coming from the solar concentrator. It decouples the operation of

the HTL reactor from weather conditions. The temperature of the oil inside the tank is limited by flowing cooling water.

- *Solar and reactor circuits*, aimed to convey the thermal oil from the solar receiver to the solar tank and from the solar tank to the reactor jacket.
- *Water tank*. It is used to cool down the HTL reactor from operating conditions to ambient temperature.
- *A control system* that opens/closes the valves and turns on/off the pumps.

During the operation, the following phases can be individuated:

1. *Pre-heating phase* ( $T_{oil} < T_{oil,set}$ ): the thermal oil is pumped from the solar tank to the receiver in the solar circuit.
2. *Heating phase* ( $T_{reactor} < T_{reactor,set}$  and  $T_{oil} = T_{oil,set}$ ): as soon the oil reaches the set point temperature, the oil is sent to the HTL reactor tank to heat the slurry (biomass plus water). Simultaneously, the solar circuit still operates to ensure  $T_{oil} = T_{oil,set}$ . This phase lasts for the residence time necessary to ensure the conversion.
3. *Cooling phase*  $T_{reactor} = T_{ambient}$ . Once the residence time elapses, the high-temperature pumps are turned off while the cooling circuit remains active until the reactor reaches ambient conditions.



**Figure 4.8.** Scheme of the solar-HTL apparatus.

### 4.3.2 Preliminary sizing and energy balance

The HTL reactor was preliminary sized using data from Pedersen et al. [72], reported in Table 4.7, referring to the conversion of aspen into the bio-crude.

**Table 4.7.** Properties of the wood and bio-crude used to size the reactor. Data from Pedersen et al. [72].

|                           | Aspen Wood | Bio-crude |
|---------------------------|------------|-----------|
| <i>Proximate analysis</i> |            |           |
| Fixed carbon (%)          | 20.6       | 42.1      |
| Volatile matter (%)       | 77.0       | 57.0      |
| Ash (%)                   | 2.4        | 0.9       |
| <i>Ultimate analysis</i>  |            |           |
| C (%)                     | 50.5       | 75.4      |
| H (%)                     | 6.2        | 10.1      |
| O (%)                     | 43.3       | 14.5      |
| HHV (MJ/kg)               | 18.8       | 35.2      |

#### *Solar circuit*

The maximum available power provided by the solar concentrator was computed considering the average incoming sunlight inside the system, the capacity of the parabolic mirrors to concentrate the sunlight (quantified by a concentration factor), and the conversion efficiency of the solar receiver.

The concentration factor (CF) was defined as the ratio between the mirror area ( $A_M$ ) and useful receiver area ( $A_R$ ), corresponding to the area of the light spot. Under ideal conditions, CF corresponds to the ratio between the irradiance on the receiver ( $I_R$ ) and the incoming one ( $I_0$ ), as expressed in Equation (4.10).

$$CF = \frac{A_M}{A_R} \approx \frac{I_R}{I_0} \quad (4.10)$$

The maximum power available to heat up the thermal oil ( $P_{oil}$ ) can be expressed as a function of the sun power ( $P_{sun}$ ) as expressed in Equation (4.11).

$$P_{oil} = \eta_{rec} P_{sun} = \dot{m}_{oil} c_{p,oil} (T_{oil,in} - T_{oil,out}) \quad (4.11)$$

Where  $\eta_{rec}$  is the thermal efficiency of the receiver,  $\dot{m}_{oil}$  is the oil flow rate inside the solar circuit (imposed by the pump),  $c_{p,oil}$  is the specific



heat of the oil,  $T_{oil,in}$  and  $T_{oil,out}$  are the inlet and outlet oil temperature of the receiver, respectively.

Table 4.8 summarizes the parameters describing this first preliminary stage. Technical data were taken from previous studies performed the same solar concentrator [42,83]:  $A_M$  corresponds to the useful area of one module, while  $A_R$  was computed considering a light spot having a diameters of 5.8 cm, and  $\eta_{rec}$  set equal to 0.9. A  $P_{sun}$  of  $835 \text{ W/m}^2$  was chosen.

**Table 4.8.** Parameters of the solar apparatus.

|                             |         |
|-----------------------------|---------|
| $P_{sun} (\text{W/m}^2)$    | 835     |
| $A_M (\text{m}^2)$          | 0.76    |
| $A_R (\text{m}^2)$          | 0.00264 |
| CF (-)                      | 287.7   |
| Spot Area ( $\text{cm}^2$ ) | 26.4    |
| P spot ( $\text{W/cm}^2$ )  | 21.62   |
| $P_{oil} (\text{W})$        | 571.1   |

#### *Energy requirements for the HTL reactor*

The total energy demand of the HTL reactor ( $q_{HTL}$ ) can be expressed as the sum between the energy required during the heating ( $q_{heat}$ ) and the constant-temperature phase ( $q_{const}$ ). Computations were performed considering a temperature of  $373^\circ\text{C}$  and a pressure of 220 bar.

$$q_{HTL} = q_{heat} + q_{const} \quad (4.12)$$

In particular:

- $q_{heat}$  is the heat required to rise the mixture (biomass plus water) from the initial temperature ( $T_0$ , equal to  $25^\circ\text{C}$ ) to set point one ( $T_{set}$ ). It was computed by applying Equation (4.13)

$$q_{heat} \left( \frac{\text{kJ}}{\text{kg}_{bio}} \right) = c_{p,bio} \cdot (T_{set} - T_0) + \frac{W}{B} \cdot \int_{T_0}^{T_{set}} c_{p,w}(T, P) \cdot dT \quad (4.13)$$

Where  $c_{p,bio}$  is the specific heat capacity of dry biomass, assumed constant and equal to  $1.2 \text{ kJ/(kgK)}$ ,  $c_{p,w}$  is the specific heat capacity of water, a function of temperature and pressure according to available thermodynamic data [54], while  $B/W$  is the mass ratio between dry biomass and water (equal to 0.30).

- $q_{const}$  was computed is equal to the heat of reaction  $\Delta H_r$ . Heat losses were neglected, while the liquid phase was assumed composed of

water and acetic acid.  $\Delta H_r$  was computed by applying Hess's law, as shown in Equation 4.14, where the subscript p indicates the products. The molar enthalpy of formation ( $\Delta h$ ) of biomass and bio-crude was computed starting from their experimental higher heating values (on a dry ash-free basis), on the base of the combustion reactions. For conventional components, thermodynamic data were used. Water was assumed as a non-participating component inside the reaction. MY and MW are the mass yield and the mean molecular weight of the i-compound, respectively.

$$\Delta H_r \left( \frac{\text{kJ}}{\text{kg}_{\text{bio}}} \right) = \sum \text{MY}_p \frac{\Delta h_p}{\text{MW}_p} - \frac{\Delta h_{\text{bio}}}{\text{MW}_{\text{bio}}} \quad (4.14)$$

Once computed  $q_{\text{heat}}$  and  $q_{\text{const}}$ , the power involved in the process was computed by considering the time length for each phase.

**Table 4.9.** Specific thermal energy requirements of the HTL reactor (B/W=0.3).

|         | $q_{\text{heat}}$<br>(kJ/kg <sub>bio</sub> ) | $q_{\text{const}}$<br>(kJ/kg <sub>bio</sub> ) |
|---------|----------------------------------------------|-----------------------------------------------|
| Water   | 6044.3                                       | -                                             |
| Biomass | 423.6                                        | -                                             |
| Total   | 6467.9                                       | 3002.9                                        |

### 4.3.3 Sizing of the HTL reactor

#### *Geometry and type of reactor*

The proposed reactor has an internal volume of 40 mL, with maximum operating temperature and pressure of 374°C and 220 bar, respectively. The chosen material is stainless steel (AISI 316L), which is capable to resist the highly corrosive phenomena occurring in the HTL environment.

The reactor was designed using a commercial medium-pressure pipe with its fitting instead of realizing a reactor directly from the raw material. This choice is advantageous for several reasons, like (1) saving time consumed for the machining of the raw steel and proper fittings; (2) avoiding the realization of the bottom of the reactor, which requires particular attention for the sizing and welding phase; (3) usage of commercial fittings, already tested to resist to the HTL operating conditions. Therefore, medium-pressure stainless steel pipes of the Swagelok® catalog (Figure 4.9) were compared. Since the working

pressure of stainless steel decreases with the temperature due to thermal expansion, a decrement factor (Figure 4.10), was used to account for this phenomenon: a temperature factor of 0.79 (at 426°C) was considered.

The internal volume of the reactor was chosen at 40 mL. To contain the height of the reactor, a pipe with an outside diameter of 20 mm and a maximum allowable working pressure of 380 bar was chosen. An extra height of 20 mm from both sides was added for the proper positioning of the fittings. Table 4.10 summarizes the final parameters.

| Tube<br>OD<br>mm | Tube Wall Thickness, mm                                                                                                                |                  |     |     |     |     |     |                  |     |     |     |     |     |     |     | Swagelok<br>Fitting<br>Series |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------|-----|-----|-----|-----|-----|------------------|-----|-----|-----|-----|-----|-----|-----|-------------------------------|
|                  | 0.3                                                                                                                                    | 0.8              | 1.0 | 1.2 | 1.5 | 1.8 | 2.0 | 2.2              | 2.5 | 2.8 | 3.0 | 3.5 | 4.0 | 4.5 | 5.0 |                               |
|                  | Working Pressure, bar<br>Note: For gas service, select a tube wall thickness outside of the shaded area.<br>(See Gas Service, page 2.) |                  |     |     |     |     |     |                  |     |     |     |     |     |     |     |                               |
| 1                | 430 <sup>Ⓢ</sup>                                                                                                                       |                  |     |     |     |     |     |                  |     |     |     |     |     |     |     | 1M0                           |
| 2                | 210                                                                                                                                    | 660 <sup>Ⓢ</sup> |     |     |     |     |     |                  |     |     |     |     |     |     |     | 2M0                           |
| 3                |                                                                                                                                        | 670              |     |     |     |     |     |                  |     |     |     |     |     |     |     | 3M0                           |
| 4                |                                                                                                                                        | 500              | 660 |     |     |     |     |                  |     |     |     |     |     |     |     | 4M0                           |
| 6                |                                                                                                                                        | 310              | 420 | 540 | 710 |     |     |                  |     |     |     |     |     |     |     | 6M0                           |
| 8                |                                                                                                                                        |                  | 310 | 390 | 520 |     |     |                  |     |     |     |     |     |     |     | 8M0                           |
| 10               |                                                                                                                                        |                  | 240 | 300 | 400 | 510 | 580 |                  |     |     |     |     |     |     |     | 10M0                          |
| 12               |                                                                                                                                        |                  | 200 | 250 | 330 | 410 | 470 |                  |     |     |     |     |     |     |     | 12M0                          |
| 14               |                                                                                                                                        |                  | 160 | 200 | 270 | 340 | 380 | 430              |     |     |     |     |     |     |     | 14M0                          |
| 15               |                                                                                                                                        |                  | 150 | 190 | 250 | 310 | 360 | 400              |     |     |     |     |     |     |     | 15M0                          |
| 16               |                                                                                                                                        |                  | 170 | 230 | 290 | 330 | 370 | 400 <sup>Ⓢ</sup> |     |     |     |     |     |     |     | 16M0                          |
| 18               |                                                                                                                                        |                  |     | 150 | 200 | 260 | 290 | 320              | 370 |     |     |     |     |     |     | 18M0                          |
| 20               |                                                                                                                                        |                  |     | 140 | 180 | 230 | 260 | 290              | 330 | 380 |     |     |     |     |     | 20M0                          |
| 22               |                                                                                                                                        |                  |     | 140 | 160 | 200 | 230 | 260              | 300 | 340 |     |     |     |     |     | 22M0                          |
| 25               |                                                                                                                                        |                  |     |     |     | 180 | 200 | 230              | 260 | 290 | 320 |     |     |     |     | 25M0                          |
| 28               |                                                                                                                                        |                  |     |     |     |     | 180 | 200              | 230 | 260 | 280 | 330 |     |     |     | 28M0                          |
| 30               |                                                                                                                                        |                  |     |     |     |     | 170 | 180              | 210 | 240 | 260 | 310 |     |     |     | 30M0                          |
| 32               |                                                                                                                                        |                  |     |     |     |     | 160 | 170              | 200 | 220 | 240 | 290 | 330 |     |     | 32M0                          |
| 38               |                                                                                                                                        |                  |     |     |     |     |     | 140              | 160 | 190 | 200 | 240 | 270 | 310 |     | 38M0                          |
| 50               |                                                                                                                                        |                  |     |     |     |     |     |                  |     |     | 150 | 180 | 210 | 240 | 270 | 50M0                          |

Ⓢ Rating based on repeated pressure testing of the Swagelok tube fitting with a 4:1 design factor based upon hydraulic fluid leakage.

Figure 4.9. Types of pipes compared for the designing phase, from the Swagelok® catalog [84].

| Temperature |     | Tubing Materials |        |                           |                            |                            |                            |                  |                  |
|-------------|-----|------------------|--------|---------------------------|----------------------------|----------------------------|----------------------------|------------------|------------------|
| °F          | °C  | Aluminum         | Copper | Carbon Steel <sup>Ⓢ</sup> | 304, 304/304L <sup>Ⓢ</sup> | 316, 316/316L <sup>Ⓢ</sup> | 317, 317/317L <sup>Ⓢ</sup> | 321 <sup>Ⓢ</sup> | 347 <sup>Ⓢ</sup> |
| 200         | 93  | 1.00             | 0.80   | 0.95                      | 1.00                       | 1.00                       | 1.00                       | 1.00             | 1.00             |
| 400         | 204 | 0.40             | 0.50   | 0.87 <sup>Ⓢ</sup>         | 0.93                       | 0.96                       | 0.96                       | 0.96             | 0.96             |
| 600         | 315 |                  |        |                           | 0.82                       | 0.85                       | 0.85                       | 0.85             | 0.85             |
| 800         | 426 |                  |        |                           | 0.76                       | 0.79                       | 0.79                       | 0.79             | 0.79             |
| 1000        | 537 |                  |        |                           | 0.69                       | 0.76                       | 0.76                       | 0.76             | 0.76             |

Figure 4.10. Temperature factors, from Swagelok® catalog [84].

**Table 4.10.** Geometrical parameters of the HTL reactor.

|                                 |       |
|---------------------------------|-------|
| Volume (mL)                     | 40    |
| Outside diameter (mm)           | 20    |
| Thickness (mm)                  | 2.8   |
| Internal diameter (mm)          | 14.4  |
| Total height (mm)               | 285.6 |
| Working pressure at 20°C (bar)  | 380   |
| Working pressure at 426°C (bar) | 300.2 |
| Useful height (mm)              | 245.6 |
| Free height for fittings (mm)   | 20    |

### *Stress analysis*

A load-stress analysis was performed to verify the effectiveness of the HTL reactor to withstand its working conditions, set at 374°C and 220 bar. In particular, the considered parameters are the yield strength, 0.2% offset yield ( $R_{p0.2}$ ), 1.0% offset yield ( $R_{p1.0}$ ), and ultimate tensile strength ( $R_m$ ).

The stress analysis was performed on the HTC reactor, using the geometry proposed (Table 4.10) and the properties of stainless steel reported in Table 4.11. The analysis was performed using two approaches: (1) using a finite element method (FEM) through Inventor Autodesk and (2) the EN 13455-3 standard (Design of unfired pressure vessels).

**Table 4.11.** Mechanical properties of stainless steel AISI 316L.

|                               |       |
|-------------------------------|-------|
| Density (g/cm <sup>3</sup> )  | 8     |
| Yield strength (MPa)          | 250   |
| Tensile strength (MPa)        | 540   |
| Young module (GPa)            | 193   |
| Poisson (su)                  | 0.3   |
| Decrement factor at 426°C (-) | 0.79  |
| Shear module (GPa)            | 74.23 |
| $R_{p0.2}$ (MPa)              | 220   |
| $R_{p1.0}$ (MPa)              | 260   |
| $R_m$ (MPa)                   | 520   |

### (1) Verification via FEM

This test was performed by the use of Inventor, the suite present in Autodesk. The analysis is based on FEM, which is a method for the

evaluation of stresses through the solution of partial differential Equations. To set the analysis, the amount and direction of the applied pressure, the mesh size, and the constrained position have to be specified. After drawing the pipe with its geometry (Table 4.10), a load of 25 MPa (corresponding to the pressure exerted by the hydrothermal process) was applied on the internal face (from inside to outside) and the constraints were positioned on the lateral and superficial points. The effect of the temperature is not considered in the simulation. The safety factor (defined as the ratio between the maximum allowable stress to the Von-Mises stress) was used as a parameter to check the capacity of the pipe to withstand (less than 1 means there is some permanent deformation). Table 4.12 reports the results of the simulation: a minimum safety factor of 2.1 and 8.8 was obtained when considering a lateral or superficial constraint, confirming the capacity of the reactor to withstand a load of 25 MPa.

**Table 4.12.** Results of the mechanical stress simulation from Inventor.

|                        | Lateral Constraint |       | Superficial Constraint |       |
|------------------------|--------------------|-------|------------------------|-------|
|                        | Min                | Max   | Min                    | Max   |
| Von Mises Stress (MPa) | 38.2               | 117.8 | 9.6                    | 28.4  |
| First stress (MPa)     | -14.8              | 108.7 | -9.2                   | 7.93  |
| Third stress (MPa)     | -75.09             | 3.47  | -27.9                  | -17.1 |
| Movement (mm)          | 0                  | 0.026 | 0                      | 0     |
| Safety Factor (-)      | 2.1                | 6.5   | 8.8                    | 15.0  |

## (2) Verification via the standard

The minimum thickness and corresponding maximum operating pressure were verified by applying the standard for high-pressure vessels (UNI EN 13445:2019). Once specified a certain geometry, the standard computes the maximum permissible pressure that can withstand the reactor ( $P_{\max}$ ) by using as Equation 4.15.

$$P_{\max} = \frac{2 \cdot f \cdot e_a}{D_m} \quad (4.15)$$

where  $e_a$  is the chosen thickness,  $D_m$  is the average diameter, and the design resistance  $f$ , shown in Equation 4.16), which considers the tensile strengths ( $R_{p,T}$ ) at the operating temperature (400°C). The coefficients are safety factors from the standard.

$$f = \max \left[ \left( \frac{R_{p1.0,T}}{1.5} \right); \min \left( \frac{R_{p1.0,T}}{1.2}; \frac{R_{m,T}}{3} \right) \right] \quad (4.16)$$

Then, the minimum standard thickness ( $e_{\min}$ ) was computed as Equation (4.17)

$$e = \max\left(\frac{P \cdot D_i}{2fz + P}; \frac{P \cdot D_i}{2fz - P}\right) \quad (4.17)$$

where  $P$  is the operating pressure (25 MPa),  $D_i$  is the internal diameter, and  $z$  is the joint coefficient (set equal to 1).

Table 4.13 shows the parameters computed through the standard. Since,  $P < P_{\max}$  and  $e > e_{\min}$ , the chosen pipe satisfies the requirements.

**Table 4.13.** Parameters computed through the standard.

|            |           |
|------------|-----------|
| $f$        | 116.7 MPa |
| $P_{\max}$ | 41.2 MPa  |
| $e$        | 0.840 mm  |

#### 4.3.4 Other components

##### *Thermal oil*

The thermal oil works both as a heat carrier and thermal energy storage medium. To guarantee an operating temperature of the reactor of at least 373°C, a commercial oil that can withstand high temperatures was chosen. In particular, a silicone-based fluid (Syltherm 800 [85]) capable to operate between -40 and 400°C was chosen.

##### *Solar and reactor tanks*

The system involves two high-temperature tanks:

- The solar tank stores the oil coming from the solar concentrator to decouple the operation from the weather conditions. It has a 2 L volume.
- The reactor tank contains the oil that flows around the solar.

The two tanks are both in stainless steel (AISI316L) and have to withstand the conditions of the thermal oil (400°C and 15 bar).

##### *Reactor jacket*

A jacket was designed to be placed between the HTL reactor and its tank. Through the jacket, the heat is transferred from the oil to the reactor. The jacket is a circular area in stainless steel, with an external diameter of 42 mm and an internal diameter of 38 mm. The diameter of

the jacket is such to contain the reactor with the bottom fitting (cap). To improve the thermal conduction, we considered the usage of sand between the internal wall of the jacket and the external wall of the reactor (removed and cleaned through a vacuum cleaner).

### *High-temperature pumps*

Two high-temperature pumps with two respective electric motors are used. The choice of the pumps requires peculiar attention since they have to withstand a temperature of around 400°C. Two commercial pumps already used in previous work from our group were chosen, whose specifications are shown in Table 4.14.

**Table 4.14.** Specifications of the pumps.

|            | Pump 1                                                     | Pump 2         |
|------------|------------------------------------------------------------|----------------|
| Pump name  | Viking C32                                                 | MVV ASCO VL805 |
| Material   | cast iron (body), carbon steel, carbon graphite, nitralloy | grafoil        |
| Flow range | 0.04÷1 mL/min                                              | 0.04÷1 mL/min  |

### **4.3.5 Conclusions and next steps**

This work is a technical report showing the preliminary design of an HTL reactor coupled with a parabolic dish concentrator. The apparatus is a prototype intended for lab-scale applications.

Through the apparatus, the concentrated sunlight covers the energy needs necessary for the conversion. The coupling is indirect: during the operation, the sunlight heats a thermal oil that is stored inside a high-temperature tank at 400°C. When needed, the oil is sent to the reactor, where the conversion of biomass into bio-crude occurs.

A particular focus is given to size the reactor according to the HTL needs and the available energy provided by solar energy. The reactor was sized to ensure an operating temperature of 373°C and a heating time of 2 min. Two load-stress analyses (through a finite element method and the standard) were performed to ensure the withstanding of the reactor to the operating conditions.

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## Chapter 5

# Valorizing municipal solid waste via integrating hydrothermal carbonization and downstream extraction for biofuel production

This chapter is based on the article that we published in *Journal of Cleaner Production* (2021, vol. 289, 125781, by G. Ischia, L. Fiori, L. Gao, and J.L. Goldfarb, <https://doi.org/10.1016/j.jclepro.2021.125781>). It bases on our collaboration with prof. Jillian Goldfarb and was partially carried out at Boston University (USA). The aim is to investigate how hydrothermal carbonization can be a pivot for the realization of a biorefinery aimed to treat organic wastes and valorize by-products. In particular, we focused on combining the hydrothermal process with downstream post-treatments (thermal and chemical extractions) to make hydrochar effective as a solid biofuel and recover chemicals.



## 5.1 Abstract

Hydrothermal carbonization (HTC) is a thermochemical process that can reduce the environmental burdens of wet, heterogeneous biomasses such as the organic fraction of municipal solid waste (OFMSW). Whilst the effect of processing parameters on hydrochar properties is well known, post-treatments to valorize hydrochars are infrequently investigated. Moreover, more severely carbonized hydrochars have a reactive species present on their surface that may limit hydrochars' use as a solid fuel or soil amendment/environmental adsorbent. To address these potential limitations, a low-temperature (180°C) thermal treatment and a chemical extraction (1:4 methanol: dichloromethane) were performed on OFMSW hydrochars. The thermal extraction removed up to 12% of this reactive volatile matter, comprised of alkanes, furans, ketones, and fatty acids. Chemical extraction removed up to 61% of the hydrochar, and the extract comprised mostly fatty acids, suggesting a potential pathway for recovery of fatty acids and condensation of fuel molecules in the solid hydrochar. The higher heating values of the extracts were much greater than the non-extractable solid hydrochar. The non-extractable primary char showed similar oxidative and pyrolytic behavior to a standard bituminous coal. The results indicate that HTC could valorize OFMSW by converting this wet waste into a dry solid fuel, soil amendment or environmental adsorbent while simultaneously extracting valuable biodiesel and biofuel precursors. Given increasing legislative pressure to divert OFMSW from landfills, this new pathway offers an alternative to traditional anaerobic digestion management strategies that produce only methane as a green energy product.

## 5.2 Introduction

CO<sub>2</sub> emissions from fossil fuel combustion and industrial processes comprise a majority of total global greenhouse gas emissions [1]. Explosive global population and economic growth over the past few decades have moved us to the precipice of a climatic catastrophe. Mitigating anthropogenic climate change requires the development of carbon recovery and conversion strategies and the production of net CO<sub>2</sub>-neutral biofuels from waste materials. Many biological and thermochemical processes could address the critical carbon recovery component of the circular economy [2]. However, the majority of currently employed waste-to-energy (WTE) technologies are based

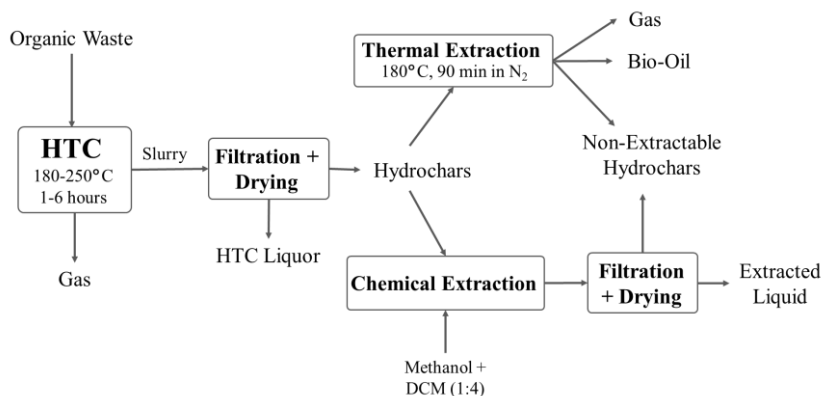
either on incineration with heat recovery [3], which produce copious amounts of ash, or relatively slow anaerobic digestion which requires downstream digestate management [4]. A growing consensus among biofuel researchers recognizes the need to develop integrated processes for the conversion of heterogeneous carbonaceous wastes to biofuels to maximize carbon recovery, energy efficiency and greenhouse gas reductions [4]. Recent advances in WTE technology development range from electrochemical regulation of anaerobic digestion to enable tuning of fermentation products [5], to calcium looping processes for CO<sub>2</sub> capture and storage from waste incineration [6], to considerable advances in plasma-based gasification technologies [7]. Despite these advances, carbonaceous waste remains an underutilized carbon source. While some localized exceptions exist, we have yet to develop both the market and the infrastructure required for widespread WTE practices [8]. By maturing technically and economically feasible solutions, we may be better positioned to develop successful WTE commercial pathways [9].

Sustainable management of the organic fraction of municipal solid waste (OFMSW) plagues countries around the globe. In Italy, OFMSW represents the highest fraction of separately collected wastes (around 41%) with an annual production of over 6.5 million tons [10]. The average American produces just over 2 kg of total municipal solid waste per day (for an annual production of over 260 million tons), of which almost 70% is non-plastic carbonaceous waste [11]. Typical OFMSW management strategies employ composting and anaerobic digestion [12]. While these processes are simple with relatively low capital cost [13], they require long residence times, and the conversion of carbon into gaseous species is incomplete [14]. Other conventional waste management processes include incineration, pyrolysis and gasification, but they are energetically unfavorable due to the substrate's high moisture content [15]. Hydrothermal carbonization (HTC) has received considerable attention in recent years for its potential to process heterogeneous moist carbonaceous wastes. Performed in aqueous conditions at moderate temperatures (ranging from 180 to 250°C) under autogenous pressure, HTC produces a carbon rich solid phase, referred to as hydrochar [16]. Hydrochar can be used as a solid fuel on its own or co-combusted with low rank fossil fuels such as coal [17]. Its condensed carbon structure and moderate porosity [18] makes hydrochar attractive as a feedstock for the production of materials such as catalysts, adsorbents, and activated carbons [19] and potentially for the production of nanocomposites [20]. Unlike thermochemical

processes such as combustion, pyrolysis and gasification, hydrothermal carbonization does not require drying of the feedstock as a pre-treatment. This makes HTC applicable to a wide variety of nontraditional carbonaceous wastes with varying but high moisture contents, including sewage sludge [21] as well as manure [22]; algae [23]; wet agricultural residues [24] and the organic fraction of municipal solid waste (OFMSW) [25].

Work by Berge et al. [26] suggests that OFMSW undergoes a similar carbonization pathway to many other carbonaceous sources during HTC: hydrolysis / dehydration / decarboxylation / condensation / decomposition of intermediates. During this processes, at certain intermediary (moderate) carbonization conditions, multiple researchers have observed that intermediates and their decomposition products can re-condense and/or repolymerize on the primary char for feedstocks as varied as olive mill waste [27], to prickly pear cactus grown on arid land [28] and aqueous plants such as waterweed [29]. For example, Lucian et al. investigated the effects of processing parameters on hydrochars produced from as-received OFMSW [30]. They demonstrated the existence of an amorphous secondary char having a spherical geometry produced under moderately severe HTC conditions (others have referred to this phase as a “coke,” [31] though it lacks a graphitic, condensed aromatic structure). While this secondary char is partially responsible for the higher energy content of hydrochars versus raw biomass, Gao et al. [32] suggested that this secondary char may lead to fuel segregation during hydrocharcoal combustion, limiting large scale co-combustion potential.

The present work explores the potential to recover secondary char e problematic in combustion scenarios e as a biofuel precursor. In the proposed integrated process, hydrothermal carbonization is used to reduce the intrinsic moisture content of OFMSW and increase the carbon density. This is followed by either low temperature pyrolysis (thermal desorption) or solvent extraction to recover biofuel precursors from the hydrochar. The residual heating content of the extracted solid and its oxidation properties were investigated to determine the potential for solid fuel combustion. Finally, an ASPEN simulation was performed to explain the exothermicity observed during the pyrolytic thermal extraction. Figure 5.1. shows a flowchart of the integrated pathways investigated in this work.



**Figure 5.1.** Integrated OFMSW thermochemical conversion pathways explored in this work.

### 5.3 Materials and methods

OFMSW was supplied by the municipal wastewater and OFMSW treatment plant in Rovereto, Italy; in the Trentino-Alto Adige region of Italy, OFMSW is collected separately from other MSW such as plastic and paper waste. This, combined with new regulations coming online across the United States requiring separate collection and treatment of OFMSW from households and larger generators alike [33], ensure an adequate supply of OFMSW for the foreseeable future and new commercial opportunities for OFMSW management. The material received was shredded, homogenized, separated from foreign matter (e.g. metals) and mixed with water. The average moisture of the as-received biomass, equal to  $87.6 \pm 0.5$  wt.%, was computed in triplicate by drying between 50 and 60 g of material in an oven at 105 C until it reached constant weight. The OFMSW was stored in 2 L bottles at 24 C and defrosted before each HTC run. A sample of Illinois No. 6 coal (a well-characterized bituminous coal from the Argonne Premium Coal Bank [34]), was used as a solid fuel comparison for thermal reactivity studies.

#### 5.3.1 Hydrothermal carbonization of OFMSW

Hydrothermal carbonization was carried out in a 2 L stainless steel (AISI 316L) batch reactor, built at the University of Trento. A detailed description of the experimental apparatus is reported in prior work [35].

Temperature and reaction time were chosen to represent a range of potential industrial processing conditions; 180°C for 1 h and 3 h (low temperature, short/long residence time to explore minimal energy input), 220°C for 3 h to represent a mid-point and what is currently touted as a “standard” maximum reaction temperature for pilot reactors, and 250°C for 6 h to gauge the impact of harsh carbonization conditions on the extraction of potential biofuels. To mimic industrial conditions, OFMSW was used as-received. The biomass to water (B/W) ratio, ranging between 12.9 and 15.5%, accounts for the OFMSW’s moisture and the water added by the plant. Prior work demonstrated that B/W ratios lower than 25% do not have a statistically significant effect on the resulting energy properties of hydrochars obtained up to 250°C [28]. For each experiment, the reactor was loaded with  $1.5 \pm 0.1$  kg of OFMSW, leaving approximately 25% free volume. Before sealing, the reactor was flushed with high purity nitrogen (N<sub>2</sub>, AirLiquide). The system was heated between 8 and 10°C/min until it reached the set point temperature, at which point the reaction time started. Pressure was allowed to increase autogenously based on production of HTC gas; maximum pressures reached were recorded. Afterwards, the system was cooled using compressed air blown through the reactor’s outer walls and stainless steel discs (at 24°C) placed on the bottom and top of the reactor. After the reactor reached ambient temperature, the exhaust gases flowed into a graduated cylinder to determine the volume produced. We assume that the gas was composed entirely of CO<sub>2</sub>, an assumption supported by considerable literature findings [36]. The solid fraction was collected, gravity filtered, and oven-dried at 105°C to constant mass. The solid yield (SY) was computed as the ratio between hydrochar and raw biomass (on a dry basis). The liquid yield was determined by difference.

**Table 5.1.** Process parameters and mass yield of hydrothermal carbonization; T: reaction temperature; t: reaction time; B/W: dry biomass to water ratio (yields based on average of at least 2 runs, less than 2% difference in each run).

| Sample     | Process parameters |       |         | Mass yield (wt.% on a dry basis) |        |     |
|------------|--------------------|-------|---------|----------------------------------|--------|-----|
|            | T (°C)             | t (h) | B/W (%) | Solid                            | Liquid | Gas |
| 180°C, 1 h | 180                | 1     | 15.2    | 66.9                             | 32.1   | 1.0 |
| 180°C, 3 h | 180                | 3     | 15.5    | 72.5                             | 26.0   | 1.5 |
| 220°C, 3 h | 220                | 3     | 12.9    | 65.8                             | 29.4   | 4.7 |
| 250°C, 6 h | 250                | 6     | 12.9    | 39.5                             | 55.0   | 6.0 |

### 5.3.2 Solid sample characterization

The elemental composition of solid samples was determined via a LECO 628 Elemental Analyzer equipped with sulfur module via ASTM D-5375 (CHN) and ASTM D-1552 (S) (oxygen by difference). Each test was performed twice; if values differed by more than 2% then a third trial was run. Proximate analysis was performed on a Mettler-Toledo Thermogravimetric Analyzer Differential Scanning Calorimeter (TGA/DSC-1;  $\pm 0.1$  K and  $\pm 0.1$  mg). Samples (between 8 and 14 mg) were placed into a 70 mL alumina crucible. The samples were heated in high purity nitrogen (50 mL/min, 20 mL/min balance protective gas flow) at  $15^\circ\text{C}/\text{min}$  up to  $110^\circ\text{C}$  and held for 15 min to remove moisture, then at  $50^\circ\text{C}/\text{min}$  from 110 to  $900^\circ\text{C}$  and held for 30 min to determine volatile matter. Samples were then heated from 900 to  $950^\circ\text{C}$  under dry air and held for 30 min with loss attributed to fixed (oxidizable) carbon.

Residual mass (non-oxidizable inorganics) was loosely termed “ash”. The pyrolytic step, as well as a separate oxidative run (110 to  $950^\circ\text{C}$  at  $50^\circ\text{C}/\text{min}$ , held 60 min in 50 mL/min of dry air) were used to assess the thermal stability and oxidative reactivity of the samples, respectively. The extent of sample converted at any time  $t$ ,  $x(t)$ , was computed as:

$$x(t) = \frac{m_{\text{dry}} - m_t}{m_{\text{dry}} - m_{\text{pyr}}} \quad (5.1)$$

where  $m_{\text{dry}}$  is the mass after the sample is held at  $110^\circ\text{C}$  (the start of pyrolysis),  $m_t$  the mass at any time  $t$ , and  $m_{\text{pyr}}$  the residual mass left at the end of the 30 min pyrolysis at  $900^\circ\text{C}$ . Derivative conversion curves (DTG) were constructed by plotting  $dx/dt$  versus temperature.

In Equation (5.1),  $m_o$ ,  $m_f$  and  $m(t)$  represent the initial sample mass (dry basis), final sample mass, and mass at a given time,  $t$ . Derivative thermogravimetric (DTG) curves were plotted as  $dx(t)/dt$  versus temperature. The energy content of the samples was gauged by measuring the higher heating values (HHVs) using an IKA C 200 calorimeter (CEN/ TS 14918 standard, dry basis). The morphology of the solid phases was probed using scanning electron microscopy on a Zeiss Supra 40 VP field emission scanning microscope (accelerating voltage of 2.00 kV, distance 8.4 mm, aperture size 30.00 mm).

### 5.3.3 Extraction and analysis of secondary char

To remove, recover and investigate the nature and potential use of the reactive species deposited on the surface of hydrochars, both a thermal and chemical extraction were performed. For thermal desorption, a mild pyrolysis (torrefaction) was performed at 180°C for 90 min. The goal of this thermal desorption was to remove the secondary char (while not pyrolyzing primary char) at the lowest temperature possible (to minimize energy input). From the DTG curves, 180°C was the lowest temperature at which we saw meaningful mass loss. Approximately 2 g of dried hydrochar was placed in a porcelain boat inside a 2" MTI GSL-1100 X quartz tube furnace. Each sample was heated at 10°C/min in 100 mL/min flowing high purity nitrogen from 25°C to 110°C, held for 10 min (to ensure moisture removal), and then up to 180°C, held for 90 min. The condensable gas released from the dry hydrochar was 0°C by the use of ethylene glycol and ice, respectively, positioned in trapped in two 50 mL glass cold traps, maintained at -18°C and series downstream of the tube furnace. After the 90 min extraction time, the solid sample was cooled under nitrogen and then removed from the furnace. The cold traps were rinsed with dichloromethane (DCM) to collect the condensed devolatilized matter.

For the chemical extraction, hydrochars were treated at room temperature with a 1:4 by volume mixture of methanol:DCM. Methanol and DCM, which are respectively protic with high polarity and aprotic with low polarity, are suitable in combination to extract a variety of compounds from the hydrochar surface [30]. For each extraction, 2 g of dried hydrochar and 40 mL of solvent were mixed in a 150 mL beaker at room temperature at 250 rpm for 15 min. The solid fraction was vacuum filtered using a medium porosity cellulose filter paper and the filtrate was collected, centrifuged and filtered again with fine porosity cellulose paper. The solid residue was dried at 105°C while the filtrate was collected and stored in sealed glass vials at 4°C.

The composition of the extracts were investigated using gas chromatography-mass spectrometry (GC-MS) on an Agilent 7890 B GC e 5977 A MS using electron ionization (70 eV). The GC was equipped with a 30 m long, 0.25 mm HP-5 column. Samples were diluted in 5 mL DCM to contain between 20 and 500 ppm per an- alyte. 1 mL sample volumes were auto-injected into the GC with an injector temperature of 250°C, operating with a split ratio of 25:1 using ultra

high purity helium as the carrier gas with a set flow of 29 mL/min. The oven program started at 50°C, held for 10 min, followed by heating at 3°C/min up to 250°C and held for 5 min. The MS interface temperature was 325°C. The MS was operated with a 6 min solvent delay in scan mode over a mass to charge ratio between 29 and 300. The compounds reported had a NIST library match greater than 90% (and were checked against common compounds identified in the biofuel literature). For each compound reported, the peak area is normalized to the grams of starting material and dilution factors are accounted for in the reported area. To determine the contribution of the secondary char to the overall heating value of the hydrochar, the theoretical higher heating value (HHV) of the extracted phase ( $\text{HHV}_{\text{extr}}$ ) was computed by applying the equation as proposed in prior work [30]:

$$\text{HHV} = \text{HHV}_{\text{non-extr}} \cdot w_{\text{non-extr}} + \text{HHV}_{\text{extr}} \cdot w_{\text{extr}} \quad (5.2)$$

where  $w_{\text{non-extr}}$  and  $w_{\text{extr}}$  are the mass fractions of the non-extractable and the extractable phases, respectively.

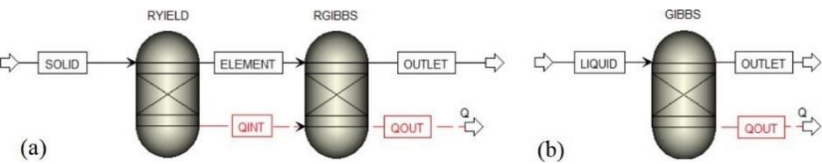
### 5.3.4 ASPEN simulation

During pyrolysis runs in the TGA-DSC, exothermic behavior was noted. A semi-quantitative interpretation of this behavior was investigated by building a thermodynamic model using Aspen Plus. It enables the comparison of the amount of heat produced or required during the pyrolysis of hydrochars (raw and post-treated) to a commercial coal and two fatty acids (palmitic and oleic acid) as constituent standards. The model is based on an equilibrium method, in which the product composition and other reaction parameters (e.g. the heat of reaction) are computed by minimizing the Gibbs energy.

Temperature was varied between 200 and 900°C, while pressure was held at 1 bar. Thermodynamic properties of the conventional components were computed by using a Peng-Robinson equation of state [37]. Table 5.2 summarizes the parameters used in the simulations. Two flow sheets were developed to model the pyrolysis solid and liquid phases, as shown in Figure 2. Regarding the one used for the solid phase, the feed SOLID was specified as a non-conventional stream, characterized by ultimate and proximate analyses and HHV. Hydrochar (raw and post-treatments) was characterized by means of experimental



data obtained here, while coal (Illinois n.6) by data from the Phyllis2 database [38]. The stream is fed into a RYield reactor, used to split the solid feed into its constituent elements on the basis of the input mass fractions (with the atomic species distribution specified according to the ultimate analysis). Ash also was assumed to be a non-conventional component. The stream ELEMENT enters the RGibbs block, an equilibrium reactor model that minimizes Gibbs energy. CO<sub>2</sub>, CO, H<sub>2</sub>, O<sub>2</sub>, H<sub>2</sub>O, CH<sub>4</sub>, C<sub>2</sub>H<sub>2</sub>, C<sub>2</sub>H<sub>4</sub>, C<sub>2</sub>H<sub>6</sub> and carbon graphite were included as possible products. N<sub>2</sub>, S<sub>2</sub> and ash were assumed to be inert compounds. The stream OUTLET consists of the products obtained by minimizing the Gibbs energy, while the heat flow stream QINT is used to correct the energy balance between the two blocks. The heat stream QOUT represents the real heat developed/required during the process. Regarding the flow sheet adopted for the pyrolysis of fatty acids, RYield was not necessary since the input feedstock consists of conventional components (i.e. chemical species available in the ASPEN database): the stream LIQUID consists only of a liquid phase (namely, fatty acids) and enters directly the Gibbs reactor.



**Figure 5.2.** Solid (a) and liquid phase (b) pyrolysis flow sheet developed for ASPEN Plus model.

**Table 5.2.** Parameters used in ASPEN simulation.

|                           |                                                                                                                                                                         |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactants                 | Hydrochars, Coal, Palmitic acid, Oleic acid                                                                                                                             |
| Products (at equilibrium) | C (graphite), CO <sub>2</sub> , CO, H <sub>2</sub> , O <sub>2</sub> , H <sub>2</sub> O, CH <sub>4</sub> , C <sub>2</sub> H <sub>2</sub> , C <sub>2</sub> H <sub>4</sub> |
| Equation of state         | Peng-Robinson                                                                                                                                                           |
| Pyrolysis Temperature     | 200-900°C                                                                                                                                                               |
| Pyrolysis Pressure        | 1 bar                                                                                                                                                                   |

## 5.4 Results and discussion

OFMSW was carbonized at several processing conditions to determine the relative feasibility to integrate hydrothermal carbonization and pyrolysis for the holistic recovery of solid and liquid biofuels.

### 5.4.1 Effect of HTC variables on hydrochar yield and composition and thermal behavior

The formation of liquid and especially gas phases is favored under more severe HTC conditions (higher temperatures and longer residence time) at the expense of a decrease in solid yield, as detailed in Table 2. This is not surprising: as demonstrated by prior studies, higher temperatures favor HTC reactions such as dehydration and decarboxylation, which increase solid conversion to produce more liquid and gas [39]. That the solid yield increased at 180°C with time with a small increase in gas yield at the expense of liquid is more likely attributable to the heterogeneity of the starting feedstock; this temperature is not sufficient to decompose compounds with great structural complexity, such that if the mixture had more recalcitrant compounds they would remain in the solid phase.

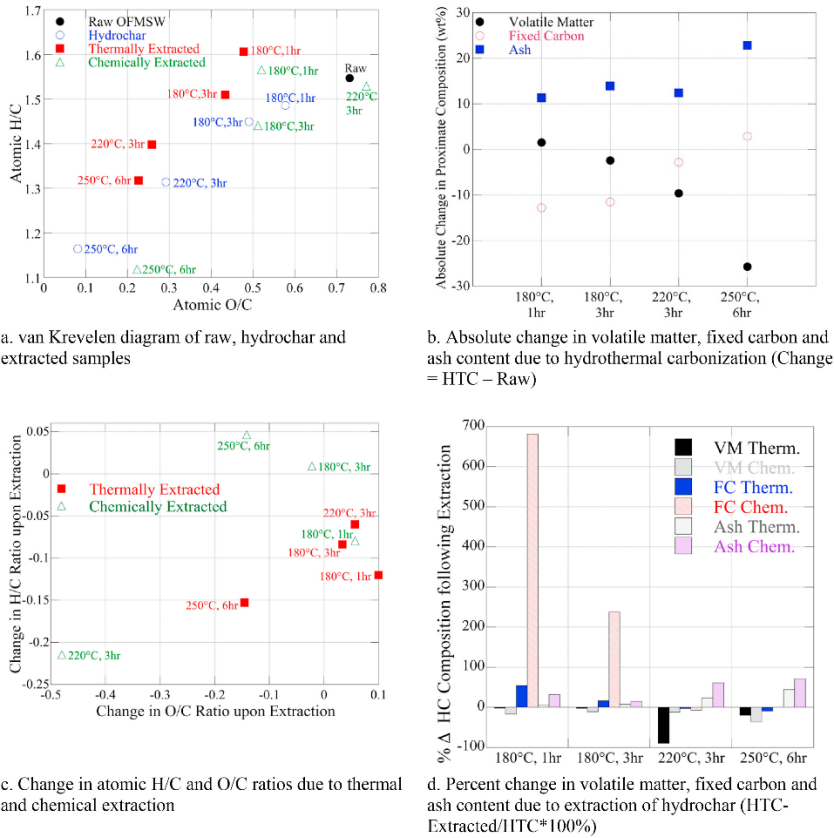
An additional explanation for the increase in SY from 180°C 3 h would be formation of larger amounts of secondary char at longer time. However, the volatile matter content decreased with time, suggesting that this is not a viable explanation (Table 5.3). One final insight from the solid yield data suggests that under the most severe HTC conditions (250°C, 6 h), dehydration and decarboxylation reactions appear to be more effective than the re-polymerization reactions that lead to secondary char formation. This is because the solid yield comprises both primary and secondary char, and the solid yield (Table 5.4) and volatile matter content (Table 5.3) drop precipitously at this harsh processing condition. Beyond impacts on yield, as the harshness of HTC increases, the concentration of elemental carbon increases, indicated by the convergence of atomic O/C and H/C ratios towards the origin of the van Krevelen diagram. This naturally leads to a reduction in the volatile matter (VM) content of the hydrochars (Figure 5.3). The thermal behavior of the hydrochars sheds further light into the loss and transformation of VM at harsher conditions.

**Table 5.3.** Proximate and ultimate composition of hydrochars pre- and post-extraction. Average values are reported (standard deviations < 0.3 wt% for proximate, <0.1wt% for ultimate analysis).

| Sample                                 | Proximate Analysis<br>(wt% dry basis) |      |      | Ultimate Analysis (wt%<br>dry basis) |     |     |      |
|----------------------------------------|---------------------------------------|------|------|--------------------------------------|-----|-----|------|
|                                        | FC                                    | VM   | Ash  | C                                    | H   | N   | O*   |
| Raw (OFMSW)                            | -                                     | -    | -    | 43.9                                 | 5.7 | 2.3 | 48.1 |
| <i>Hydrochars (as-carbonized)</i>      |                                       |      |      |                                      |     |     |      |
| 180°C, 1h                              | 1.1                                   | 82.2 | 16.7 | 43.3                                 | 5.4 | 1.4 | 33.3 |
| 180°C, 3h                              | 2.4                                   | 78.3 | 19.3 | 44.4                                 | 5.4 | 2.0 | 29.0 |
| 220°C, 3h                              | 11.1                                  | 71.1 | 17.8 | 53.5                                 | 5.9 | 2.0 | 20.8 |
| 250°C, 6h                              | 16.8                                  | 55   | 28.2 | 57.3                                 | 5.7 | 2.7 | 6.2  |
| <i>Thermally extracted hydrochars</i>  |                                       |      |      |                                      |     |     |      |
| 180°C, 1h                              | 1.7                                   | 80.9 | 17.5 | 41.2                                 | 5.6 | 1.9 | 33.8 |
| 180°C, 3h                              | 2.8                                   | 76.4 | 20.8 | 45.0                                 | 5.7 | 2.0 | 26.5 |
| 220°C, 3h                              | 10.8                                  | 67.3 | 21.9 | 52.0                                 | 6.1 | 2.1 | 17.9 |
| 250°C, 6h                              | 15.2                                  | 44.2 | 40.7 | 40.7                                 | 4.5 | 1.8 | 12.3 |
| <i>Chemically extracted hydrochars</i> |                                       |      |      |                                      |     |     |      |
| 180°C, 1h                              | 8.6                                   | 68.9 | 22.6 | 41.1                                 | 5.4 | 2.4 | 28.5 |
| 180°C, 3h                              | 8.1                                   | 69.7 | 22.1 | 42.2                                 | 5.1 | 1.8 | 28.8 |
| 220°C, 3h                              | 10.3                                  | 62.7 | 28.6 | 32.5                                 | 4.2 | 1.4 | 33.4 |
| 250°C, 6h                              | 16.8                                  | 34.9 | 48.3 | 35.7                                 | 3.3 | 2.1 | 10.6 |

**Table 5.4.** Solid yields and HHVs of hydrochars pre- and post-extraction. Average values are reported (standard deviations<0.1 MJ/kg for HHV).

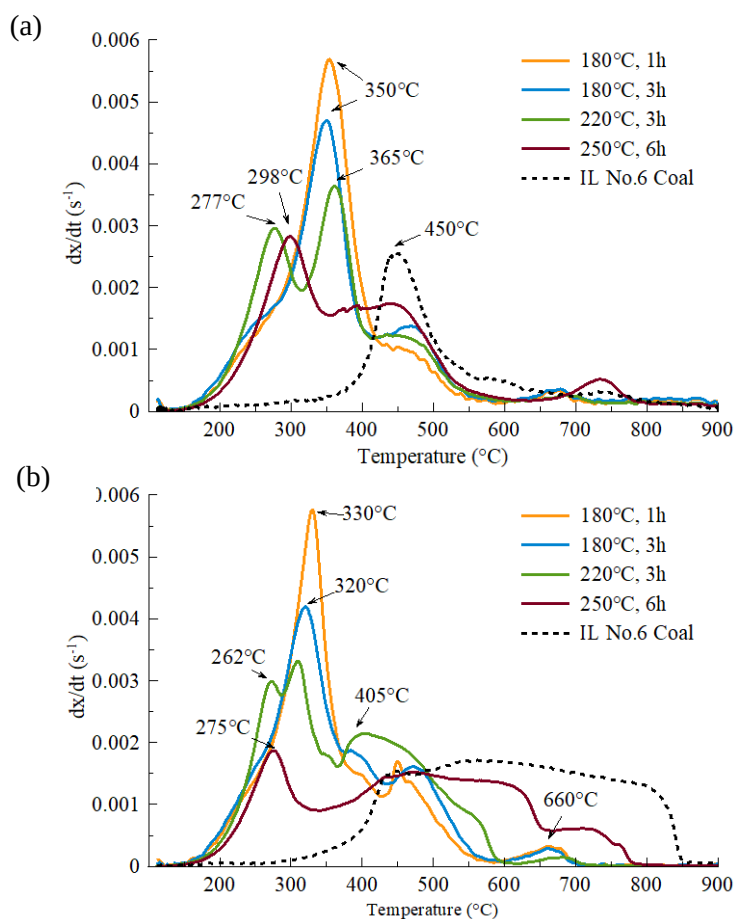
| Sample                                 | Solid yield*<br>(%) | HHV<br>(MJ/kg) | HHV daf<br>(MJ/kg) |
|----------------------------------------|---------------------|----------------|--------------------|
| Raw (OFMSW)                            |                     | 17.4           |                    |
| <i>Hydrochars (as-carbonized)</i>      |                     |                |                    |
| 180°C, 1h                              | 66.9                | 19.5           | 23.4               |
| 180°C, 3h                              | 72.5                | 18.9           | 23.4               |
| 220°C, 3h                              | 65.8                | 22.2           | 27.0               |
| 250°C, 6h                              | 39.5                | 25.6           | 35.6               |
| <i>Thermally extracted hydrochars</i>  |                     |                |                    |
| 180°C, 1h                              | 94.0                | 18.7           | 22.6               |
| 180°C, 3h                              | 91.9                | 18.3           | 23.1               |
| 220°C, 3h                              | 89.5                | 19.9           | 25.4               |
| 250°C, 6h                              | 88.5                | 18.3           | 30.8               |
| <i>Chemically extracted hydrochars</i> |                     |                |                    |
| 180°C, 1h                              | 75.7                | 17.1           | 22.0               |
| 180°C, 3h                              | 74.0                | 16.0           | 20.5               |
| 220°C, 3h                              | 44.9                | 15.7           | 22.0               |
| 250°C, 6h                              | 39.4                | 13.6           | 26.2               |



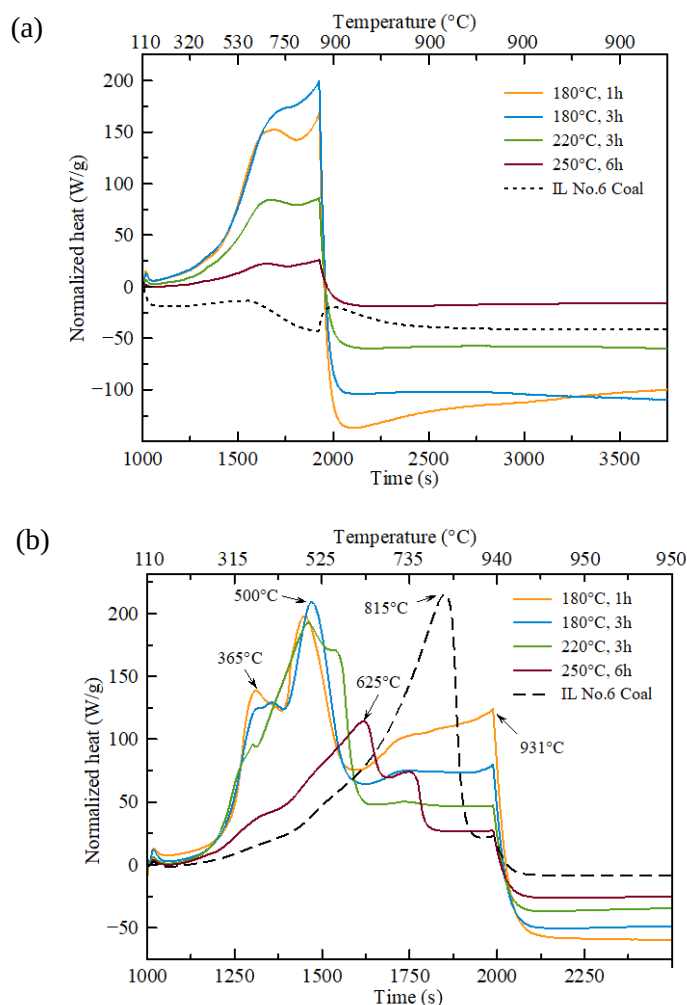
**Figure 5.3.** Impact of hydrothermal carbonization and post-carbonization on char composition.

Turning to the pyrolysis DTG curves (Figure 5.4), the samples carbonized at 220 and 250°C show an initial DTG peak at 277 and 298°C, respectively, considerably lower than the peaks at 350°C for the samples carbonized at 180°C. Initially, this seems counterintuitive: the VM content of the 220 and 250°C samples is lower, yet it is more reactive? This suggests that (at least a portion of) the VM present in the 220 and 250°C samples is comprised of re-polymerized compounds from the aqueous phase. Conversely, at 180°C only a fraction of the primary volatile matter is carbonized, and as such the HTC 180°C chars do not contain appreciable amounts of secondary char. The two-peak shape of the 220°C sample shows a combination of these behaviors. For the first DTG peak, we see a slightly faster reaction rate at lower temperature than for the 250°C sample, and for peak 2 a slower rate and higher temperature than the 180°C samples. This suggests that a portion

of the VM present in the 220°C carbonized sample is secondary char, and a portion is remaining primary char that did not release from the OFMSW matrix during carbonization. The oxidation DTG curves (Figure 5.4b) suggest that the secondary char present on the sample carbonized at 220°C is more reactive than the 250°C sample, as seen by a faster reaction rate and slightly lower initial peak temperature. The larger shoulder at temperatures above 450°C on the DTG curve for the 250°C carbonized sample, and a smaller but observable hump for the 220°C sample, is indicative of the larger amount of fixed carbon (FC) present. The more harshly carbonized samples have an order of magnitude higher FC than the 180°C carbonized samples (Table 5.3).



**Figure 5.4.** Derivative thermogravimetric curves of OFMSW hydrochars and bituminous coal under pyrolytic (a) and oxidative atmosphere (b) at 50°C/min.



**Figure 5.5.** Normalized heat flow of OFMSW hydrochars and bituminous coal under pyrolytic (a) and oxidative atmosphere (b) at 50°C/min.

Though the composition of the more harshly processed hydrochars begins to approach that of the Illinois No. 6 coal (recently reported as 65.3% C, 4.5% H, 9.2% O with an HHV of 24.2 MJ/kg [32], the hydrochars are still considerably more reactive than the coal at lower temperatures both in terms of devolatilization and oxidation behavior. The pyrolytic DSC curves are presented in Figure 4c. The exothermic nature is highest for the 180°C samples, and decreases for the 220 and 250°C samples. After the initial exothermic period, all samples show endothermic behavior above 850°C while removing more recalcitrant,

less reactive species. This behavior is investigated further via ASPEN modeling, the results of which are presented in Section 5.4. Not surprisingly, all samples show exothermic behavior during oxidation (Figure 4d). The 250°C carbonized sample exotherm was shifted to considerably higher temperatures than the other samples. This, combined with the lower overall solid yield, lower VM, higher FC and lower oxygen content support the hypothesis that there is considerably less reactive secondary char present on the surface of the 250°C than the 220°C sample, and that a larger portion of the volatile matter in the feedstock was transitioned to the aqueous and gas phases. As discussed throughout the literature, the choice of reaction conditions is a balancing act between increasing elemental carbon and HHV of the hydrochar versus the relative loss of volatile matter and corresponding increase in ash content. This analysis confirms the trends elucidated by prior work and provides a basis for the ensuing analysis of the potential to recover both biofuel precursors and a more thermally stable char for solid fuel combustion [28].

#### **5.4.2 Hydrochar properties following thermal and chemical extraction**

Overall, the chemical extraction removed more mass than the thermal method. The solid yields following thermal extraction are quite high (up to 94 wt%), indicative of minimal devolatilization and no real pyrolytic decomposition, and thus accomplishing minimal thermal extraction at this low temperature. Torrefaction of the 180°C, 1 h and 3 h hydrochars removed 6% and ~8% of the total mass, whereas chemical extraction of the same samples resulted in ~25% removal for both low-temperature hydrochars (Table 3). The minimal amount thermally extracted from these hydrochars aligns with common HTC knowledge that such weak “wet pyrolysis” conditions break down only the most labile volatiles, similar to dry torrefaction [40]. As such, the low temperature thermal extraction step does not pyrolyze much remaining “primary” char, nor is there secondary char available to remove. The extracted and as-carbonized chars produced at 180°C show similar DTG profiles (Figure 5), which suggest different amounts of VM present, but that both samples, overall, are similar in nature given the narrow temperature range over which the samples devolatilize. The 220°C, 3 h sample lost more volatile matter upon thermal treatment than chemical treatment

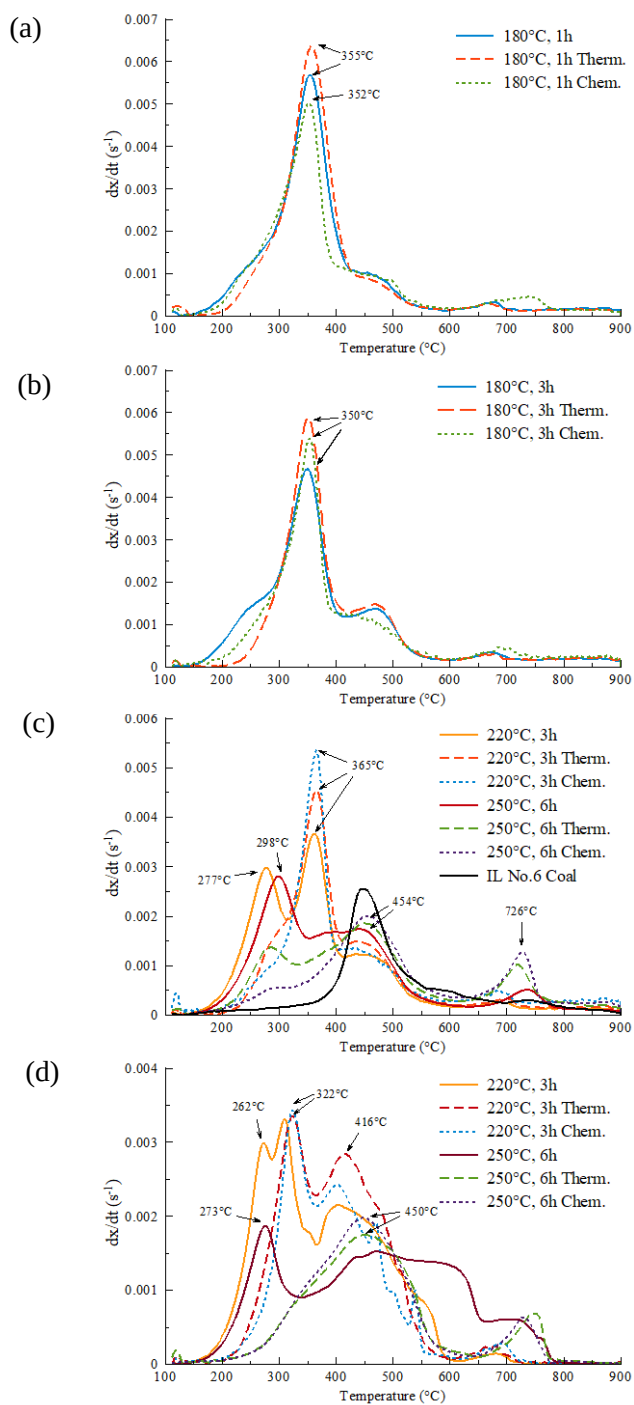
(Figure 3d). This indicates the presence of a volatile mixture on the surface of the 220°C, 3 h hydrochar that is vulnerable to low-temperature thermal desorption. This sample also saw the largest decrease in both O/C and H/C ratio of all samples (Figure 3c), suggesting that more reactive organic compounds were removed (tabulated data available in SI). The pyrolysis DTG curves of these samples (Figure 5c) indicate that both extraction techniques remove only the “most volatile” of the volatile matter the first DTG peak observed for the 220°C hydrochar is absent for the extracted samples, but the second DTG peak at 365°C remains after extraction. We attribute this second peak to “primary” volatile matter that does not decompose during HTC and is not removed via the low-grade thermal or chemical extractions. The first, larger peak is likely the secondary char, which is removed via extraction. Lucian et al. observed that for hydrochars obtained under moderately severe HTC temperatures, secondary char was characterized by a C/O atomic ratio much larger than that of the corresponding primary char, and was responsible for the high DTG reactivity over a lower peak temperature range [30]

We note that in several prior studies, secondary char was present in the form of a sphere [27,31,41] a feature that was not observed in SEM images of these samples, as reported in SI. However, the pyrolysis and oxidative (Figure 5d) DTG curves suggest similar behavior to the secondary char laden hydrochars observed previously in the literature. While the 250°C, 6 h carbonized sample lost only 11.5% of mass upon thermal treatment, the chemical treatment removed over 60%. This runs counter to the supposition that because of low volatile matter content that there was minimal extractable material. Given the minimal amount of thermally extractable secondary char and weakened carbon structure, it is possible that either the solvent mixture “washes” away fixed carbon and/or mineral matter [42], or hydrolyzes a portion of the remaining organic matter [43]. Given the decrease in VM and increase in both FC and ash content for this sample (Figure 3b), it is more likely a combination of hydrolysis and/or dissolution of the volatiles responsible for this finding, rather than removal of inorganic components. The oxidative DTG curves (Figure 5) support this hypothesis; the fixed carbon for both as-carbonized and extracted samples oxidizes at the same temperature, but with a higher peak (due to higher relative concentration) for the extracted samples.

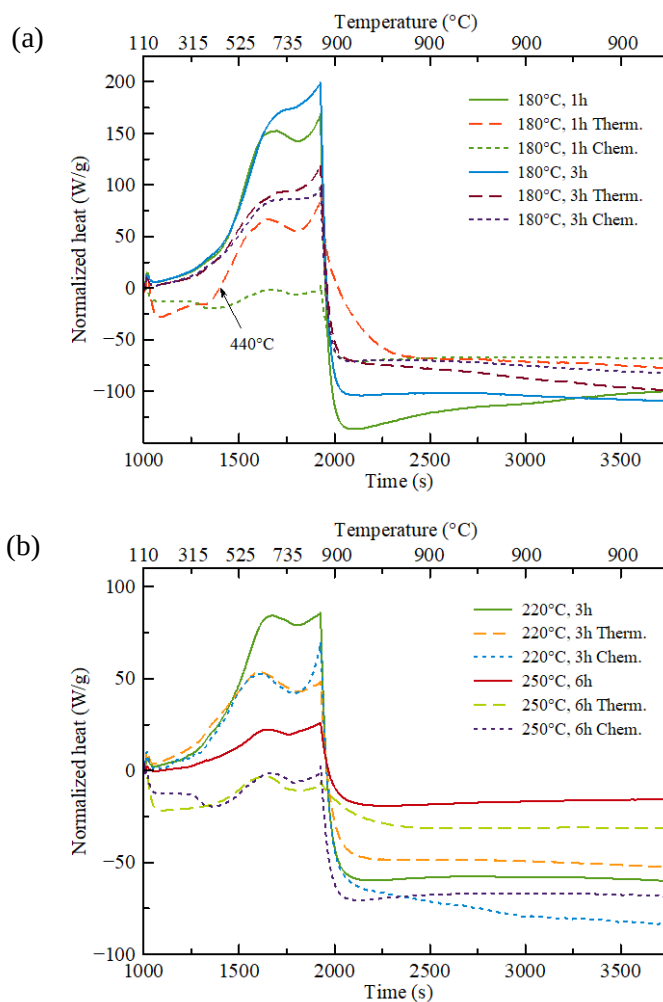


While the relatively high HHVs of the extracted samples and the removal of the low-temperature oxidation peak oxidation in the DTG curves suggest the potential for using the extracted hydrochars for energy recovery via combustion, the high ash content especially of the 250°C sample, maybe require co-firing with a low-ash solid fuel to prevent slagging and fouling issues [34]. Beyond combustion applications, this extracted hydrochar given its relatively high thermal stability but high elemental carbon content e could be used as a soil amendment [44] or an adsorbent for water treatment [45]. Given the high ash content, it is unlikely that this material would find use in applications where more “pure” carbon is required, such as electrodes for supercapacitors [46] and/or microbial fuel cells [47], as previously demonstrated with biomass-based carbons.

The HHVs of the extracted samples are less than the as- carbonized chars (Table 3); the extracted material is more energy-dense than the as-carbonized or primary biomass. We attribute this, partially, to a function of the higher ash content of the extracted solids, which leads to a relatively lower elemental carbon content (Table 3) and, as carbonization temperature increases, a lower O:C ratio (Figure 3). The differential scanning calorimetry curves in Figure 4 show a decrease in pyrolytic exothermicity upon extraction of the hydrochars, with several samples exhibiting endothermic behavior. The endotherms may be due to the high ash content of the materials, which require significant energy to heat, especially as temperature increases. We explore this further through ASPEN modeling and by analyzing the chemical composition of the recovered secondary char to shed light on the potential for recovery of these energetic components from the hydrochars.



**Figure 5.6.** Derivative conversion curves during pyrolysis of as-carbonized and extracted hydrochars.



**Figure 5.7.** DSC curves for the pyrolysis of as-received and extracted hydrochars.

### 5.4.3 Analysis of recovered secondary char

The thermally and chemically extracted secondary chars were analyzed via gas chromatography-mass spectrometry (Tables 5.5-5.7). Consistent with the heterogeneity of OFMSW, a large number of different compounds were detected, including furans, alkanes, ketones, and fatty acids. Furfuryl alcohol was detected in the greatest amount from the 180°C and 220°C thermal extracts, though not in the 250°C hydrochar extract. This compound likely originates from the cleavage

of  $\beta$ -1,4-glycosidic bonds present in carbohydrates. Given the low pyrolysis temperature, modest exothermic behavior observed at low temperature (Figure 5.5), and lack of furfuryl alcohol in the 250°C char extract, it is unlikely that this compound formed during the low-temperature thermal extraction step [48,49]. Rather, we surmise that it formed during HTC, and in fact it was identified previously as a product in low-temperature HTC studies [50]. At harsher HTC conditions, furfuryl alcohol, and also furfural, will undergo degradation and polymerization [51], which explains why we do not detect either of these compounds in the 250°C, 6 h hydrochar thermal extracts. Interestingly, neither furfuryl alcohol nor furfural nor many low-molecular weight compounds were detected in the chemically extracted samples. At harsher HTC conditions the thermally extracted samples do show a shift towards recovery of fatty acids, the predominant compounds detected in the chemical extracts. Despite the low yield of extracted secondary char via low-temperature desorption, this extraction method appears more suitable to remove low molecular weight compounds such as alkanes, furans, and ketones (Table 5.5) from mildly carbonized samples.

The chemical extraction, despite the high amount of mass removed, did not show the presence of any of these compounds. The only compounds detected in the chemical extracts were phenol, styrene and long chain fatty acids, including, for the 180°C hydrochars, the monoacylglycerol of glycerol 1-palmitate, often seen in olive and vegetable oils. The 1:4 ratio of methanol:DCM used as the extraction solvent clearly favored the removal of these compounds over those detected in the thermal extracts. While we would expect the methanol to solubilize compounds such as furfural and furfuryl alcohol, the greater percentage of fatty acids presents suggests that they preferentially dissolved in the solvent mixture. From this, we conclude that a coupled thermal-chemical extraction method may be a possible recovery scheme for biofuel precursors of varying volatilities and solubilities. The fatty acids present in the hydrochars likely derive from the lipids already present in the starting feedstock. During HTC, fatty acid ester linkages, especially those of triglycerides, phospholipids and glycolipids, hydrolyze and cleave into fatty acids. Prior work has shown that lipids do not contribute to the formation of solid hydrochar, but rather fatty acids formed during HTC may be adsorbed on the solid surface and retained on the hydrophobic char [52]. The chemically

extracted concentration (directly proportional to chromatogram area) of saturated and unsaturated fatty acids increases under more severe carbonization conditions, suggesting a higher yield and adsorption to char surface. Such high, relatively selective yields of lipids suggest that HTC followed by chemical extraction is a potential pathway to (1) reduce landfilling of OFMSW; (2) recover biodiesel precursors and (3) produce a combustible solid for energy generation and/or, given the capacity to adsorb fatty acids, a hydrophobic adsorbent for water remediation applications.

Future work will explore the potential for sequential recovery of low molecular weight biofuel components via thermal extraction followed by chemical extraction to remove these fatty acids and recover the solid hydrochar. The tendency of these compounds to selectively adsorb to the char surface, combined with their higher enthalpy of vaporization (as compared to compounds like furfuryl alcohol and butanoic acid) [53], also explains why the low- temperature thermal desorption step did not remove a large portion of them from the char.

**Table 5.5.** Percent (identified) compounds by chromatogram area from thermal and chemical extraction of OFMSW hydrochars.

|                            | % Chromatogram Area |              |              |              |
|----------------------------|---------------------|--------------|--------------|--------------|
|                            | 180°C,<br>1h        | 180°C,<br>3h | 220°C,<br>3h | 250°C,<br>6h |
| <i>Thermal extraction</i>  |                     |              |              |              |
| Fatty acids                | -                   | 14.0         | 36.3         | 54.0         |
| Alkanes                    | 22.9                | 33.2         | 5.5          | 23.0         |
| Furans                     | 61.9                | 44.8         | 34.0         | -            |
| Ketones                    | 3.8                 | 4.4          | 2.8          | -            |
| Fatty acids, methyl ester  | -                   | -            | 5.4          | -            |
| Phenols                    | 0.9                 | 3.6          | -            | -            |
| Other                      | 10.5                | -            | 15.9         | 23.0         |
| <i>Chemical extraction</i> |                     |              |              |              |
| Saturated fatty acids      | 34.84               | 32.04        | 30.49        | 40.92        |
| Unsaturated fatty acids    | 53.52               | 46.09        | 49.82        | 51.67        |
| Fatty acids methylester    | 10.39               | 18.56        | 19.70        | 6.99         |
| Other                      | 1.25                | 3.31         | 0.00         | 0.43         |

**Table 5.6** Compounds detected in secondary char from thermally extracted hydrochars by total chromatogram area and percent area.

| Compound                                     | Total Chromatogram Area/g |           |           |           |
|----------------------------------------------|---------------------------|-----------|-----------|-----------|
|                                              | 180°C, 1h                 | 180°C, 3h | 220°C, 3h | 250°C, 6h |
| <i>Thermally extracted</i>                   |                           |           |           |           |
| Butanoic acid/Butanoic acid, 3 methyl        | -                         | 1.31E+04  | 2.46E+04  | 1.51E+04  |
| 2-Propyl-1-pentanol                          | 7.80E+03                  | -         | 5.80E+03  | 4.20E+03  |
| Furfural                                     | 6.83E+03                  | -         | 4.34E+04  | -         |
| 2-Cyclopenten-1-one                          | 5.61E+03                  | -         | -         | -         |
| 2-Furanmethanol                              | 8.52E+04                  | 3.90E+04  | 8.09E+04  | -         |
| Acetic acid, ethenyl ester                   | 7.74E+03                  | -         | 3.08E+04  | -         |
| 2-Methyl-2-cyclopenten-1-one                 | -                         | 4.15E+03  | 1.03E+04  | -         |
| Ethanone, 1-(2-furanyl)-                     | -                         | 3.12E+03  | -         | -         |
| Butane, 2,3-dimethyl-                        | 2.28E+04                  | 2.99E+04  | 1.32E+04  | -         |
| 3-Ethylpentane                               | 1.12E+04                  | 1.32E+03  | 6.83E+03  | 6.45E+03  |
| Phenol                                       | 1.40E+03                  | 3.35E+03  | -         | -         |
| Propanoic acid, 2-methyl-                    | -                         | -         | 4.57E+03  | -         |
| n-Hexadecanoic acid                          | -                         | -         | 7.30E+04  | -         |
| Heptanoic acid, 2-ethyl-                     | -                         | -         | 1.69E+04  | -         |
| 9,12-Octadecadienoic acid, methyl ester      | -                         | -         | 6.67E+03  | -         |
| Z-9-Octadecenoic acid                        | -                         | -         | 3.49E+04  | -         |
| 9-Octadecenoic acid (Z)-, methyl ester       | -                         | -         | 1.31E+04  | -         |
| 1,2-Benzenedicarboxylic acid, diheptyl ester | -                         | -         | -         | 2.24E+03  |
| <i>Chemically extracted</i>                  |                           |           |           |           |
| Styrene                                      | -                         | 6.95E+03  | -         | -         |
| Phenol                                       | -                         | -         | -         | 5.65E+03  |
| n-Decanoic acid                              | -                         | 1.67E+03  | 7.44E+03  | 1.47E+04  |
| Tetradecanoic acid                           | 8.40E+03                  | -         | 2.99E+04  | 3.83E+04  |
| Cis-9-Hexadecenoic acid                      | 3.66E+03                  | 4.00E+03  | 1.58E+04  | 7.94E+03  |
| n-Hexadecanoic acid                          | 1.46E+05                  | 1.47E+05  | 3.23E+05  | 4.88E+05  |
| 9,12-Octadecadienoic acid, methyl ester      | 3.54E+04                  | 4.73E+04  | 1.75E+05  | 6.05E+04  |
| Z-9-Octadecenoic acid                        | 2.21E+05                  | 1.80E+05  | 5.33E+05  | 6.25E+05  |
| Octadecanoic acid                            | 1.26E+04                  | 3.02E+04  | 3.98E+04  | 5.10E+04  |
| 9-Octadecenoic acid (Z)-, methyl ester       | 1.07E+04                  | 3.89E+04  | 5.73E+04  | 3.20E+04  |
| Glycerol 1-palmitate                         | 5.53E+03                  | 8.45E+03  | -         | -         |

**Table 5.7.** Compounds detected in secondary char from chemically extracted hydrochars by total chromatogram area and percent area.

| Compound                                     | % Chromatogram Area |              |              |              |
|----------------------------------------------|---------------------|--------------|--------------|--------------|
|                                              | 180°C,<br>1h        | 180°C,<br>3h | 220°C,<br>3h | 250°C,<br>6h |
| <i>Thermally extracted</i>                   |                     |              |              |              |
| Butanoic acid/Butanoic acid, 3 methyl        | -                   | 14.0         | 6.7          | 54.0         |
| 2-Propyl-1-pentanol                          | 5.2                 | -            | 1.6          | 15.0         |
| Furfural                                     | 4.6                 | -            | 11.9         | -            |
| 2-Cyclopenten-1-one                          | 3.8                 | -            | -            | -            |
| 2-Furanmethanol                              | 57.3                | 41.5         | 22.2         | -            |
| Acetic acid, ethenyl ester                   | 5.2                 | -            | 8.5          | -            |
| 2-Methyl-2-cyclopenten-1-one                 | -                   | 4.4          | 2.8          | -            |
| Ethanone, 1-(2-furanyl)-                     | -                   | 3.3          | -            | -            |
| Butane, 2,3-dimethyl-                        | 15.3                | 31.8         | 3.6          | -            |
| 3-Ethylpentane                               | 7.6                 | 1.4          | 1.9          | 23.0         |
| Phenol                                       | 0.9                 | 3.6          | -            | -            |
| Propanoic acid, 2-methyl-                    | -                   | -            | 1.3          | -            |
| n-Hexadecanoic acid                          | -                   | -            | 20.0         | -            |
| Heptanoic acid, 2-ethyl-                     | -                   | -            | 4.6          | -            |
| 9,12-Octadecadienoic acid, methyl ester      | -                   | -            | 1.8          | -            |
| Z-9-Octadecenoic acid                        | -                   | -            | 9.6          | -            |
| 9-Octadecenoic acid (Z)-, methyl ester       | -                   | -            | 3.6          | -            |
| 1,2-Benzenedicarboxylic acid, diheptyl ester | -                   | -            | -            | 8.0          |
| <i>Chemically extracted</i>                  |                     |              |              |              |
| Styrene                                      | -                   | 1.5          | -            | -            |
| Phenol                                       | -                   | -            | -            | 0.4          |
| n-Decanoic acid                              | -                   | 0.4          | 0.6          | 1.1          |
| Tetradecanoic acid                           | 1.9                 | -            | 2.5          | 2.9          |
| Cis-9-Hexadecenoic acid                      | 0.8                 | 0.9          | 1.3          | 0.6          |
| n-Hexadecanoic acid                          | 32.9                | 31.7         | 27.3         | 36.9         |
| 9,12-Octadecadienoic acid, methyl ester      | 8.0                 | 10.2         | 14.8         | 4.6          |
| Z-9-Octadecenoic acid                        | 49.9                | 38.7         | 45.1         | 47.2         |
| Octadecanoic acid                            | 2.8                 | 6.5          | 3.4          | 3.9          |
| 9-Octadecenoic acid (Z)-, methyl ester       | 2.4                 | 8.4          | 4.9          | 2.4          |
| Glycerol 1-palmitate                         | 1.2                 | 1.8          | -            | -            |

#### 5.4.4 Thermodynamic considerations of secondary char via ASPEN modeling

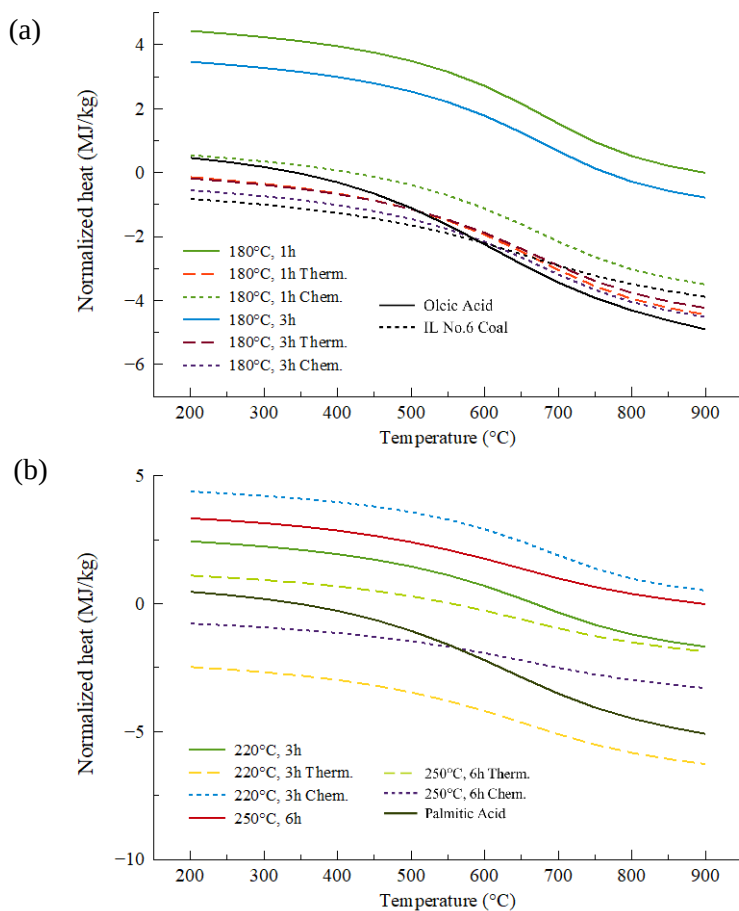
As noted in the differential scanning calorimetry results during pyrolytic TGA-DSC runs of the hydrochars (Figure 5.5), and as seen by other researchers studying pyrolysis of raw biomasses [54], the hydrochars exhibit both endothermic and exothermic behavior. The thermodynamic simulations (Figure 5.8) confirm the experimentally observed hydrochar exothermicity. Such exothermic behavior - in a process that requires heat input to cause devolatilization - is often attributed to heterogeneous reactions between the devolatilized compounds and solid biomass/ char [55], specifically via secondary reactions [56]. This mechanism explains the observation in Figures 5.4 and 5.5 that normalized heat released decreases as pyrolysis temperature increases; the ratio of surface reactive carbon to fixed carbon and ash decreases as more volatiles are driven off, making such heterogeneous reactions less likely to occur as pyrolysis progresses.

What is perhaps most interesting here is the almost complete disappearance of exothermic behavior for the 250°C, 6 h extracted hydrochar following both thermal and chemical extraction, and the significant reduction in exothermicity - with a shift to endothermicity observed for extracted chars as the temperature increases. As HTC becomes harsher, a greater percentage of the original biomass carbonizes, shifting the balance to primary (solid) and secondary (amorphous) char, the latter of which is removed during extraction. If we look at the ASPEN simulations of the chars, oleic acid and palmitic acid, we see similar exothermic decompositions shifting to endothermic behavior. This suggests that the exothermic behavior cannot be explained entirely by the heterogeneous reaction mechanism, as there are no such reaction sites - at least not initially - upon which the pure fatty acids would react. Rather, the fatty acids themselves undergo decarbonylation and decarboxylation, with cracking at the allylic C-C and conversion to lighter weight hydrocarbons [57]. At lower temperatures, the evolution of CO gas and ensuing partial oxidation is likely responsible for the mildly exothermic nature of the fatty acids.

As such, the pyrolytic exothermic nature of hydrochar is more strongly attributed to the presence of secondary char compounds (e.g. fatty acids) and unconverted raw biomass than the primary (solid) char. The primary char behaves similarly to coal, whose pyrolysis is



endothermic in the whole temperature range investigated. While the accuracy of the model to predict observed exotherm values is high at some temperatures, it underpredicts the enthalpy of the raw hydrochars, especially the 180°C chars, while improving for the extracted chars especially in the 400-600°C range. However, the key takeaway from this modeling exercise is the ability to explain the observed exothermicity by the presence of representative secondary char components. Future work could tune this predictive model to improve its accuracy and extend it to other hydrochar systems.



**Figure 5.8.** Normalized (per kg of starting material) heat produced (>0) and required (<0) across pyrolysis temperatures for as-carbonized and extracted hydrochars as compared to coal, oleic acid, and palmitic acid.

### 5.5 Recommendations for future work

The present work demonstrates a new integrated thermochemical conversion pathway for the valorization of OFMSW into liquid fuel precursors and thermally stable solid residues. To implement such a process, there are a few key areas of work remaining that were beyond the scope of this conceptual demonstration. In terms of the solid product (thermally or chemically extracted hydrochar), we suggest further characterizations and application testing to determine the optimal valorization pathway(s). As noted, the relatively high ash content may limit the extracted hydrochar's use as a solid fuel (though it could be co-fired with a low ash fuel) or electrochemical application. However, the extraction step may make these hydrochars valuable soil amendments (as leachable organics have been removed) or as adsorbents for water treatment. To gauge the economic potential of the solid product, such information would help determine a target market and selling price; biochars commonly used for soil amendments (prior to any upgrading) sell from between \$100 to \$2000 U.S. per ton, with activated carbons selling upwards of \$1500 per ton [58]. From a computational standpoint, we suggest development of a more rigorous thermodynamic model to predict the pyrolysis behavior of the hydrochars. The model presented here is a key "first step," and its overall behavior is validated by our experimental results. Nevertheless, it could be made more robust with further data collection to enable a full validation, verification and uncertainty quantification [59].

Additional experiments would also make it possible to perform a sensitivity analysis to determine the process variables controlling product characteristics (such as HTC process temperature and time). Overall, these recommendations for future work would enable full process mass and energy balances, necessary for reaction sizing, plant design, and a full life cycle and techno-economic analysis.

## 5.6 Conclusions

The conversion of organic wastes into fuels and by-products plays a key role in meeting circular economy objectives. Recent legislation across the United States that prohibits landfilling of such waste streams offers new opportunities for commercialization of waste valorization technologies. With its high moisture content, the organic fraction of municipal solid waste (OFMSW) is often seen as an appropriate feedstock for conventional techniques such as anaerobic digestion. However, there exist other pathways to extract higher value products than digestion. While hydrothermal carbonization (HTC) has previously been proposed as an alternative OFMSW management strategy, the HTC conditions required to significantly reduce the waste's volume (220-250°C, upwards of 1 h) result in the formation of a reactive secondary char on the surface of the hydrochar, comprised mainly of alkanes, furans, ketones and fatty acids.

In the present work we demonstrate that this reactive char, while potentially detrimental to combustion applications, can be extracted by low-temperature thermal desorption (to recover mostly alkanes, furans and ketones) or by chemical extraction, resulting in high and rather selective yields of fatty acids. Using thermodynamic simulations, this paper further demonstrates that these fatty acids are at least partially responsible for the exothermic nature of the hydrochars observed during pyrolysis, rather than the heterogeneous reaction mechanism usually ascribed to such behavior in the literature.

Overall, HTC is a viable option to convert OFMSW to a solid hydrochar, from which we can extract biofuel precursors. Given the relatively high residual heating value, the extracted hydrochar may serve as a solid fuel. Future work should probe the potential to use the extracted hydrochar as a potential soil amendment or environmental adsorbent (given its relatively high and thermally stable carbon content as a result of the removal of labile organics). Given the residual ash content of the extracted hydrochars, we do not foresee its upgrading into advanced materials such as electrodes, but rather intend its use as an upconverted waste for energy or environmental applications in and around communities where it is generated.

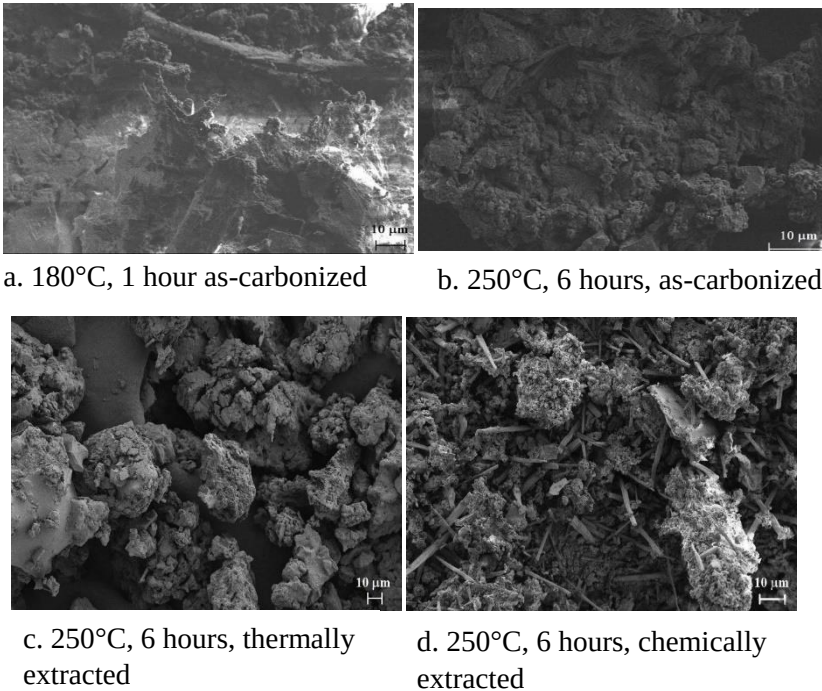
## 5.7 Supplemental material

**Table 5.S1.** Change in proximate analysis, ultimate analysis and HHV upon chemical and thermal treatment of hydrochars

| Treatment | Sample    | Loss ratio (%) |      |       |      |      |       |       |      | HHV  | HHV <sub>daf</sub> |
|-----------|-----------|----------------|------|-------|------|------|-------|-------|------|------|--------------------|
|           |           | FC             | VM   | Ash   | C    | H    | N     | O     |      |      |                    |
| Thermal   | 180°C, 1h | -45.3          | 7.5  | 1.5   | 10.5 | 1.4  | -29.9 | 4.5   | 10.0 | 9.1  |                    |
|           | 180°C, 3h | -7.2           | 10.3 | 1.0   | 6.8  | 2.3  | 7.5   | 16.0  | 11.3 | 7.5  |                    |
|           | 220°C, 3h | 12.9           | 15.3 | -10.1 | 13.0 | 7.8  | 6.0   | 23.0  | 20.0 | 11.5 |                    |
|           | 250°C, 6h | 19.9           | 28.9 | -27.7 | 37.2 | 29.2 | 39.9  | -74.5 | 36.7 | 18.6 |                    |
| Chemical  | 180°C, 1h | -491.8         | 36.5 | -2.4  | 28.1 | 23.2 | -36.1 | 35.3  | 33.8 | 28.8 |                    |
|           | 180°C, 3h | -149.8         | 34.1 | 15.3  | 29.7 | 29.2 | 32.5  | 26.6  | 37.6 | 35.3 |                    |
|           | 220°C, 3h | 58.3           | 60.4 | 27.9  | 72.7 | 68.5 | 69.8  | 27.8  | 68.2 | 63.4 |                    |
|           | 250°C, 6h | 60.6           | 75.0 | 32.5  | 75.5 | 76.6 | 69.5  | 32.8  | 79.1 | 71.0 |                    |



**Figure 5.S1.** Photograph of hydrothermal reactor used in this work.



**Figure 5.S2.** SEM images of as-carbonized and extracted OFMSW hydrochars.

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## Chapter 6

# Conclusions and future developments

In a historical moment in which the severity of impacts of climate change on the Earth and human health is increasing at dizzying levels, the worldwide community has to make giant leaps forward to accelerate the transition to a zero-carbon economy. As researchers, we have the responsibility to conceive and engineer new green processes. Biomass can play a crucial role in the transition from a fossil-based to a biomass-based economy, in which *sustainably produced, renewable, and natural sources* in the form of biomass can replace fossil fuel for the production of energy and materials. Among the strategies to valorize biomass, belonging to the family of thermochemical conversion techniques, hydrothermal processes have acquired a significant interest for the production of biofuels and bio-based materials, as well as for waste management.

In this context, this thesis investigated hydrothermal processes, touching on several areas, from research around the thermochemical fundamentals to the engineering of sustainable and biorefinery concepts. A particular focus is given to hydrothermal carbonization.

**Fundamental research** (chapter 2) included the study of insights into the chemistry, kinetics, and thermodynamics involved in the hydrothermal processes. The complexity behind the hydrothermal conversion is enormous, spreading from the variability of the feedstock to the interaction of the tons of compounds released from biomass. To face the intricacy and try to clarify what is happening, we studied the conversion of several biomasses through proper experimental activities and analysis techniques (like liquid chromatography, solid spectroscopy, microscopy, and calorimetry) over a broad hydrothermal spectrum, including carbonization and liquefaction.

Particular attention was given to the mechanisms behind the formation of the solid-phase deriving from the self-condensation and polymerization of the soluble products, the so-called “*secondary char*”. Some attractive properties (like functionalization, tunability, and photoluminescence) make secondary char a good candidate as a base for advanced carbon materials. Therefore, we investigated secondary char from glucose holistically, from the formation pathways to the kinetics and material properties. In particular, studying the evolution of intermediates in the liquid phase enabled the development of a kinetic model, while spectroscopy and microscopy permit the observation of luminescent properties of the material (thanks to the presence of nanocarbon) as well as the presence of amorphous and graphitic carbon. These properties make secondary char particularly appealing as a base for advanced carbon for several applications (like photocatalysis, batteries, supercapacitors): we believe that future works could focus on their characterization and coupling with other techniques (like templating, functionalization, pyrolysis) according to the optimized product.

To explore more in-depth the hydrothermal mechanisms and kinetics, we investigated the pathways behind the *transition from hydrothermal carbonization to liquefaction* of biomasses (cellulose, lignin, and wood). This focus can contribute to the general understanding of the process and the optimization. Following the evolution of main compounds over temperature and time, we focused on the characterization of the liquid phase trying to approach a “real-time” analysis by sampling during the operation. It mainly consists of carboxylic acids, furan and phenol derivatives, and ketones, whose ratios vary with the substrate and the operating conditions. Future works

will focus on the characterization of the bio-oil and computation of kinetic parameters.

Then, we faced the *thermodynamics of hydrothermal carbonization*, which is a little-studied topic in the field. Indeed, the process is exothermic, meaning that energy is released during decomposition. To quantify the amount of heat release, we measured through a high-pressure differential scanning calorimeter, the enthalpy of reactions involved during hydrothermal carbonization of different substrates at different temperatures. Also, we gave particular attention to comparing two experimental methodologies to remove the effects of water to compute the net heat flow. Knowing the magnitude and trends of the heat profiles can be beneficial for the overall understanding of the HTC mechanisms and accurate energy analysis of the process. Future works could focus on computing kinetic parameters on the base of experimental data, as well as compare results with reactor-size data observing the scalability effects.

As a junction between fundamentals and application, we reviewed the current ***modeling techniques*** of hydrothermal carbonization. Modeling is an unavoidable step for technological industrialization, and many efforts have to be addressed to develop some useful tools. Therefore, to provide a base for future works, we explained the current state of the art of this field, highlighting strengths and gaps to guide future research to further improvements. To achieve this aim, we divided the study into three sub-systems: the process (involving kinetics and statistical models), the reactor itself (with heat transfer and computational models), and the plant (with techno-economic analysis and the life cycle of the overall process).

Regarding ***technological aspects of hydrothermal processes*** (chapter 4), the focus was given on how to improve their sustainability trying to develop solutions leading to a zero-energy process. Indeed, despite the high-valuable aims of hydrothermal processes, there is a certain lack of research around the optimization of the technological process itself: aspects like the energy balance and efficiency are often neglected, forgetting that hydrothermal processes are energy-intensive. Therefore, the proposed solution consists of coupling hydrothermal processes with concentrated solar energy, aimed to cover the energy needs necessary to make the process chemically occurring and facing thermal losses.

Therefore, we developed two concepts of solar-driven hydrothermal systems through two main projects consisting of the direct or indirect coupling between these two technologies. In particular, we developed and experimentally tested an apparatus consisting of a *solar concentrator directly coupled with a hydrothermal carbonization reactor*. Through an experimental campaign, we demonstrated the effectiveness of this solution, which represents an effective solution to perform hydrothermal carbonization through a zero-energy process. This direct configuration can be a starting point for a more complex design. In particular, the passage to an indirect apparatus can help to move towards large-scale systems. Therefore, we performed a techno-economic study on an *indirect system* consisting of a solar concentrator indirectly coupled with continuous hydrothermal carbonization or liquefaction reactor and adopting a thermal storage solution. Since the solution seems promising, we designed a *lab-scale apparatus* consisting of the indirect coupling between a parabolic dish concentrator and a hydrothermal liquefaction reactor by using thermal oil. The next stages involve its realization and testing. Then, for future developments, moving to continuous hydrothermal systems will be mandatory to expand the possibility of scalability.

Finally, expanding the observation scale, we investigated the integration of **hydrothermal carbonization with subsequent treatments** (chapter 5). Indeed, hydrothermal carbonization can be a pivot to concertize a biorefinery concept, merging the management of biomass waste and the valorization of by-products. In particular, we focused on subsequent thermal and chemical treatments to make the hydrochar effective as a solid biofuel and recover by-products. Indeed, by itself, the hydrochar suffers from fuel segregation, exhibiting a too high reactivity at low temperatures. Results seem promising: thermal and chemical post-treatments are effective for removing the highly volatile solid fraction. The next mandatory stages include investigating green solvents.

# The end

Intense, extravagant, joyful, exciting, turbulent, meaningful, never boring, lovely, creative. Science, so complex as fascinating, has the unique ability to trigger curiosity and to spark lights inside us, creating connections and opening minds. So, like many of us, I found myself wandering through unexplored worlds, I got lost in the chaos of our temporary suppositions and I have been bewitched by the mysterious beauty of the world. I discovered the power of curiosity, capable of opening the doors to wrong theories, correct theories, stupid considerations, new visions, originality, irreverence, and non-conformity. So, thanks to all the humans that tried to understand something of how this weird world and life move on and brought together the scientific system: if our society proceeds with a decent level of humanity and does not get lost in hypocrisy, it is also thanks to the ideals of science.

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But finally, thanks to you, dear reader, the only one who had the courage of arriving at the end of this sermon. I hope you did not fall asleep too many times, you understood something, you didn't start to hate science, biomass, renewable energy, movies, books, stupid quotes, or me. I promise, no more thesis. In case, as Groucho Marx was saying, quote me as saying I was misquoted.

Fine.

Post Scriptum

*Dr. Frankenstein: "You know, I'm a rather brilliant surgeon. Perhaps I can help you with that hump"*

*Igor: "What hump?"*

Gene Wilder to Marty Feldman in *Young Frankenstein*

In a worldwide context where the community has to make giant leaps forward to contain the catastrophic consequences of climate change, we need to face the discordant “How do we power our economies?” with green and circular solutions instead of hiding behind the hypocrisy of fossil fuels. Biomass, renewable, abundant, and cheap, can trigger a shift towards a zero-carbon emission economy, in which it substitutes fossil fuels for the production of energy and materials. Among the strategies to valorize biomass, hydrothermal processes are green pathways for producing biofuels and bio-based materials. However, research has yet to fill several gaps to make these processes ready for industrial scaling and spreading.

Therefore, along with this PhD thesis, we provide new insights around hydrothermal processes, touching several scientific areas: from in-depth research around the thermochemical fundamentals to the engineering of new sustainable and biorefinery concepts. Through fundamental research, we try to answer “What’s happening during hydrothermal processes?” facing the enormous complexity of the process investigating chemical pathways, kinetics, and thermodynamics. Facing sustainability, we explore the coupling of hydrothermal conversion with concentrated solar energy to develop a zero-energy process and the integration of hydrothermal carbonization with subsequent treatments to valorize by-products.

**Giulia Ischia** is a passionate researcher who tries to find some solutions to tackle climate change through bioenergy and thermochemical processes. She has a Master's and a Bachelor's in Energy Engineering (University of Trento - Free University of Bolzano, and Polytechnic of Turin), and she has been visiting researcher in the United States (Cornell University and Boston University). During her Ph.D., she received several international fellowships and awards, like a Fulbright scholarship, the best young researcher award at the 2021 WasteEng conference, the Joseph Breen Memorial Fellowship from the American Chemical Society, and the fellow to the EIT Climate-KIC Journey on climate change. She is a reviewer of several international journals like Carbon, Waste Management, and Renewable Energy. She believes in innovation and humor.